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**Toward understanding the diversification of
microorganisms: methodological progress and
empirical applications**

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Résumé

Les micro-organismes constituent la forme de vie la plus diversifiée actuellement présente sur Terre. Néanmoins, les dynamiques à l'origine de cette diversité restent méconnues. Dans cette thèse, nous étudions la dynamique de spéciation (émergence de nouvelles espèces) et d'extinction de groupes de procaryotes et micro-eucaryotes à partir de phylogénies d'espèces actuelles, à l'aide de nouvelles approches de modélisation. Dans les deux premiers chapitres, nous proposons des développements méthodologiques dans le domaine des approches phylogénétiques de la diversification. Ces développements permettent de contourner certaines limitations actuelles et de proposer des outils davantage appropriés à l'étude de la diversification des micro-organismes. Ces outils méthodologiques sont accessibles en Open Source aux développeurs et utilisateurs. Le troisième chapitre vise à étudier la dynamique de diversification d'un riche groupe de micro-eucaryotes en exploitant de larges jeux de données génétiques et écologiques.

Le premier développement vise à estimer les taux de spéciation et d'extinction pour des groupes d'organismes dont la diversité globale reste encore inconnue ou incertaine. Nous proposons plusieurs approches, et comparons leurs performances sur des données simulées. Nous montrons que les estimateurs des taux de diversification sont moins précis que lorsque la diversité globale est connue, mais qu'il est néanmoins possible d'obtenir des estimateurs pour la plupart non biaisés. Nous appliquons ces approches à la plus large phylogénie reconstruite de bactéries disponible dans la littérature, en analysant séparément les différents groupes bactériens qui ont diversifié depuis l'Évènement Oxydant Néo-protérozoïque jusqu'à aujourd'hui. Nous estimons des taux net de diversification d'une moyenne de 0.01 événements par lignée par million d'années, soit d'environ un ordre de grandeur inférieur à ceux connus pour les macroorganismes. Nos analyses suggèrent également que ces taux varient seulement d'un facteur 2, et seraient donc beaucoup plus homogènes qu'entre grands groupes de macroorganismes.

Le deuxième développement propose une approche d'inférence innovante pour l'estimation phylogénétique des taux de diversification, qui s'appuie sur un algorithme d'apprentissage profond. Nous illustrons cette approche avec deux modèles de diversification simples, dont l'un permet d'estimer des taux de diversification qui dépendent des traits associés aux espèces. Les résultats sur données simulées montrent que la précision des estimateurs obtenus par cette nouvelle approche est comparable à celle des estimateurs obtenus par la méthode couramment utilisée basée sur le calcul de vraisemblance. De plus, cette nouvelle technique d'inférence permet d'une part de réduire drastiquement le temps d'inférence (l'inférence est plus rapide de plusieurs ordres de grandeur), et d'autre part, de rendre possible le développement de nouveaux modèles plus flexibles.

Dans le troisième chapitre, nous explorons les différentes stratégies écologiques des diatomées, l'un des groupes d'eucaryotes les plus diversifiés du plancton, afin d'expliquer

l'hétérogénéité de leurs taux de diversification. Précisément, nous examinons la position et la largeur de la niche écologique de chaque espèce et montrons leur importance dans la diversification des diatomées.

Cette thèse s'inscrit dans le champ de la macroévolution des microorganismes et apporte de nouvelles méthodologies et connaissances empiriques en s'appuyant sur des modèles probabilistes préexistants, en innovant grâce à de nouvelles approches computationnelles et en mettant à profit les avancées descriptives de la biodiversité.

Abstract

Microorganisms currently represent the most diversified form of life on Earth. Nonetheless, the dynamics at the origin of this diversity remain largely unknown. In this PhD, we study the dynamics of speciation (the emergence of new species) and extinction in groups of prokaryotes and microeukaryotes from phylogenies of extant species, using newly developed approaches. In the first two chapters, we propose new methodological developments in the field of phylogenetic approaches of diversification. These developments enable to overcome some current limitations and propose tools that are better suited to the study of microorganisms diversification. These methodological tools are available for users and developers in Open Source. The third chapter aims at studying the diversification dynamic of a large group of microeukaryotes by benefiting from extensive genetic and ecological datasets.

The first development aims at estimating speciation and extinction rates for groups of organisms with unknown or uncertain global diversity. We propose several approaches and compare their performances on simulated data. We show that diversification rates estimates are less precise than when the global diversity is known, but that it is nevertheless possible to obtain mostly unbiased estimates. We apply our approaches to the largest available reconstructed bacterial phylogeny, analyzing separately the different bacterial groups that have diversified from the Neoproterozoic Oxygenation Event until the present. We estimate a mean net diversification rate of 0.01 event per lineage per million years, which is around one order of magnitude below those known for macroorganisms. Our analyses also suggest that these rates vary by only a doubling factor, and could thus be much more homogeneous than across large groups of macroorganisms.

The second development introduces an innovative likelihood-free inference approach for the phylogenetic estimation of diversification rates based on deep learning. We illustrate our approach using two simple diversification models among which one enables the estimation of diversification rates dependent on species associated traits. The resulting estimates on simulated data obtained from this new approach exhibit comparable accuracy than the ones from the gold standard approach based on likelihood. Moreover, this new inference technique enables a drastic reduction of inference time (several orders of magnitude of differences) and provide a flexible framework for model development.

In the third chapter, we explore the different ecological strategies of diatoms, one of the most diversified groups of eukaryotic plankton, to explain the heterogeneity of their diversification rates. Precisely, we investigate the ecological niche position and breadth of each species and demonstrate their involvement in the diversification of diatoms.

This PhD contributes to the field of the macroevolution of microorganisms and provides new methodological and empirical knowledges by building on preexisting probabilistic models, innovating through the use of numerical approaches and benefiting from efforts in biodiversity description.

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Introduction

Microbial organisms encompass all domains of life. Together, they represent the vast majority of past and present diversity. Yet, the dynamic by which this diversity has emerged remains scarcely studied. How fast do microbial lineages accumulate through geological times? Is extinction a crucial component of their diversification? Why are some groups more diverse than others?

Understanding the dynamic at which species arise – through speciation – and go extinct – through extinction – as well as factors influencing this balance is a major challenge in microbial macroevolution. This thesis proposes several tools particularly adapted to study microbial diversification and investigates this question in two large microbial clades for which extensive data is available.

This chapter is intended as an introduction to microbial evolution as well as macroevolution concepts and approaches. I start by introducing the microbial world. I then detail the current knowledge and theories on how and why microbial diversity has fluctuated over geological time scales. I describe the data commonly used for investigating diversification dynamics and the techniques to assemble this data. I continue by defining some common and widely used statistical models developed for the study of species' diversification. I present several inference frameworks used in the field of macroevolution. Lastly, I describe the empirical datasets analyzed in the present manuscript. Rather than an exhaustive report of the evolution of microbial lineages and of methods used in macroevolution, I aim to introduce the key information and concepts that will be crucial to the understanding of the different chapters of my PhD.

0.1. The Diversity of Microorganisms

“ From my part I judge, from myself (howbeit I clean my mouth like I’ve already said), that all the people living in our United Netherlands are not as many as the living animals that I carry in my own mouth this very day. ”

Antoni van Leeuwenhoek from Clifford Dobell, Antoni van Leeuwenhoek and his little Animals” New York: Harcourt, Brace and Company, 1932 (Dobell 1932).

0.1.1. Brief History

Microorganisms are defined by their microscopic size. They are comprised of prokaryotic organisms (archaea and bacteria), micro-eukaryotes – algae, fungi, protozoa – and viruses. One of the first discovery and observation of microbial organisms was made by Robert Hooke and Antoni van Leeuwenhoek in 1665-83 ([Fig. 1](#)). Microorganisms were referred by Antoni van Leeuwenhoek as “animalcules” (Dobell 1932). Later on, the field of microbiology was essentially directed towards human health and parasitology in which microorganisms were mainly seen as pathogenic. This bias is still widely present in our contemporary medicine which takes its roots in the early stage of microbiology.

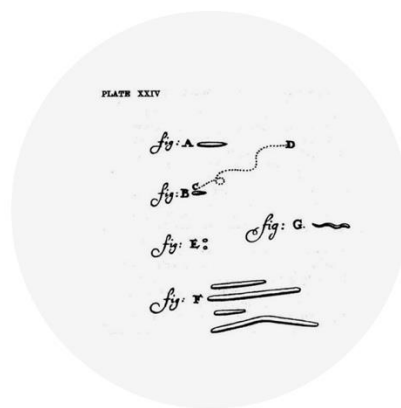


Figure 1: Original drawing from Antony van Leeuwenhoek's notes from 1683 of bacteria that he observed in a dental plaque sample from (Lux and Shi 2004).

Since then, the field has been enriched with many disciplines such as ecology, genetics, engineering and evolution amongst others. In ecology, microorganisms have been studied for their capacity of decomposition and nutrient recycling, as well as for their importance in the ecology of other organisms, from which the term *holobiont* has emerged. In genetics, they have been used to understand heritability, gene expression and more recently antibiotic resistance. In engineering, they have been used to synthesize specific compounds and more recently to create new compounds. As the most ancient form of life was microbial, microorganisms have also provided insights into the apparition, maintenance, and evolution of life on Earth.

0.1.2. Which Diversity?

Microorganisms are paraphyletic organisms which span all domains of life. Their history goes back to the origin of known life. Several attempt has been made to reconstruct this history. The reconstruction of all the main evolutionary events from a single molecular marker was initiated in the late 1960s by (Woese and Fox 1977) who started to assemble a catalogue of rRNA sequences. Recently, substantial efforts have been made to get a comprehensive Tree of Life (Hug et al. 2016) ([Fig. 2](#)). Despite being far from an exhaustive inventory of life on Earth, the

sequencing revolution have enabled the collection of extensive genome-resolved metagenomics data (Parks et al. 2022) and the discovery of members of large unknown phylum such as the Candidate Phyla Radiation (CPR) (Brown et al. 2015; Elie-Fadrosh et al. 2016) or the Diapherotrites, Parvarchaeota, Aenigmarchaeota, Nanoarchaeota and Nanohaloarchaea (DPANN) (Rinke et al. 2013).

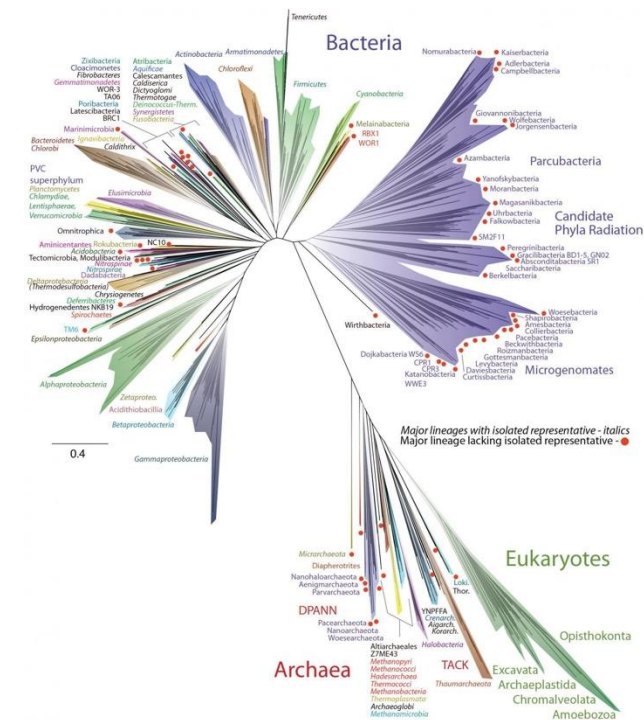


Figure 2: A current view of the tree of life, encompassing the total diversity represented by sequenced genomes from (Hug et al. 2016). The tree includes 92 named bacterial phyla, 26 archaeal phyla and all five of the Eukaryotic supergroups. Major lineages are assigned arbitrary colors and names, with well-characterized lineage names in italics. Lineages lacking an isolated representative are highlighted with non-italicized names and red dots.

Microorganisms also cover a wide range of ecological and physiological properties. Their metabolic capabilities (Fig. 3), biochemistry, and heat and pH tolerance cover the entire range of environments suitable for life on Earth. This is specifically true for prokaryotes, and to a lesser extent for eukaryotes which inherited their biochemistry from prokaryotes (Jelen et al. 2016).

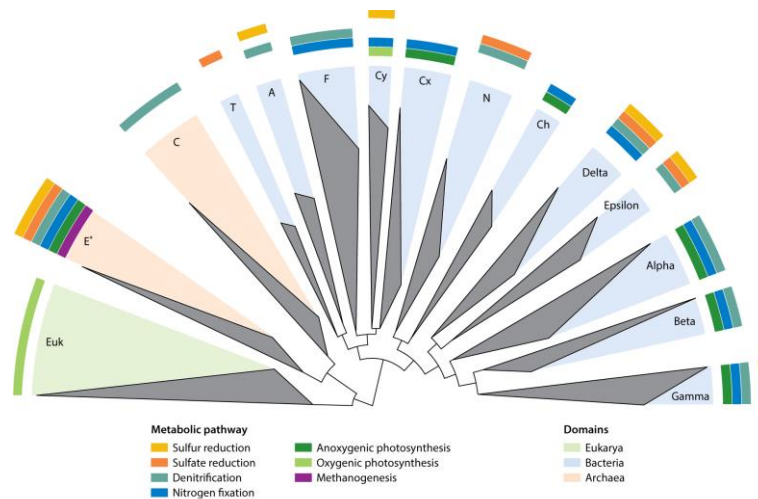


Figure 3: Distribution of selected metabolic pathways on the 16S rRNA tree of life from (Jelen et al. 2016). The tree was edited for clarity and shows selected bacterial and archaeal taxa. The area of each branch is proportional to the total number of 16S rRNA sequences present in the database. Abbreviations: A, Aquificae; Alpha, Alphaproteobacteria; Beta, Betaproteobacteria; C, Crenarchaeota; Ch, Chlorobi; Cx, Chloroflexi; Cy, Cyanobacteria; Delta, Deltaproteobacteria; E, Euryarchaeota; Epsilon, Epsilonproteobacteria; Euk, Eukarya; F, Firmicutes; Gamma, Gammaproteobacteria; N, Nitrospirae; T, Thermodesulfobacteria.

Despite their wide representation across the tree of life, microorganisms share common characteristics besides their small size, such as large populations, high dispersal rates and high lineage diversity.

0.1.3. Species Definition and Operational Taxonomic Units

How to define species is the matter of intense debate that we do not delve into here ((Doolittle and Papke 2006), [Box 1](#)). We adopt the common practice of defining instead “Operational Taxonomic Units” (OTUs) constructed by clustering sequence data with some criteria of sequence similarity. Conserved marker genes such as the ribosomal RNA (16S rRNA for prokaryotes and 18S rRNA for eukaryotes) are most often used, but clustering is also sometimes performed using whole genomes along with appropriate sequence similarity thresholds (Chun et al. 2018; Ciufo et al. 2018). Several clustering software have been developed specifically for dealing with extensive *de novo* microbial genetic data (Li and Godzik 2006; Li et al. 2012; Westcott and Schloss 2015; Rognes et al. 2016). Those tools are based on global pairwise sequence comparison to cluster sequences according to a given threshold. They rely on a centroid-based algorithm for which the predefined sequence order matters – which can be a limit (Koeppel and Wu 2013). Typically, the sequences would be ordered according to their abundance for sequences of similar length. Other methods without stringent predefined global thresholds exist. One is based on unsupervised single-linkage-clustering algorithm and fast pairwise global alignment (Mahé et al. 2014). The rationale is to cluster sequences as long as they differ by less than d nucleotides – a user-chosen local clustering threshold – from another sequence of the cluster.

Box 1: Species Concept and Definition.

How to define a species consistently for all living entities inhabiting the Earth is a challenging task. According to (de Queiroz 2005), a species must satisfy the following conditions: (i) be a cohesive unit suggesting the existence of a mechanism limiting intraspecific genetic diversity, (ii) be monophyletic, (iii) be irreversibly separated and (iv) ecologically distinct from other species. The last condition was brought by Hutchinson for whom a necessary property in systematics is to ensure that species names and clusters hold information on the organisms' ecology, physiology and biochemistry (Hutchinson 1968). While intraspecific phenotypic diversity tends to be limited in macro-organisms, microorganisms share different phenotypes among cohesive groups. One of the first attempt to define prokaryotic species based on their ecology was (Cohan 2001) who proposed the concept of *ecotype*. Here we instead define *species* based on genetic data; they represent groups of individuals that are genetically similar.

0.1.4. Estimates of Diversity

The pace of species discovery is still fast for microorganisms, only a small fraction of which has been characterized. Despite the long-lasting attempts to quantify the diversity of life on Earth (May 1988; Mora et al. 2011; Vitorino and Bessa 2018), huge uncertainty surrounds the estimation of microbial species richness. The estimates span several orders of magnitudes already for prokaryotic specific diversity alone ranging from 10^6 to 10^{12} (Locey and Lennon 2016; Louca et al. 2019) with high controversy (Willis 2016a). A main source of data for the estimation of microbial diversity comes from metabarcoding, which consists in sequencing DNA from environmental samples using high throughput sequencing. Metabarcoding allows the collection of a very large number of microbial sequences belonging to different taxonomical groups, which, clustered into OTUs, provide a measure of the observed diversity. Next, several statistical methods exist to estimate total diversity.

Extrapolation method. – This method is based on the Species Accumulation Curve (SAC) which displays the cumulative number of species over the number of individuals collected among samples. As sampling increases, the pace of species discovery slows down thus the accumulation curve is expected to ultimately reach an asymptote. The estimation method consists of extrapolating the estimated richness from the accumulation curve assuming a functional form to model the accumulation of newly observed species as sampling increases (Soberon and Llorente 1993). One of the first study to estimate richness from microbial sequences was for diatoms and cyanobacteria using extrapolation methods (Nübel et al. 1999). This method reaches its limit when only a small fraction of the total diversity has been sampled, which is the case for vast majority of microorganisms. Indeed in this situation, several

functional forms can fit the accumulation curve equally well but predict very different asymptotes.

Parametric models. – The parametric method is based on the relative abundances of sampled species, also known as the Species Abundance Distribution (SAD). The method fits a model for the SAD, such as the lognormal (Preston 1948; Pielou 1975) or Poisson lognormal (Bulmer 1974) to sample data. Often the low abundant species are highly numerous (left skewed distribution) and this is where the unsampled species lie as well (metaphorically under the veil in the literature – see (Preston 1948; Nee et al. 1991; Dewdney 2000; Chisholm 2007) for debate on sampling theory) (Fig. 4). The major drawback of this parametric method relies in choosing the appropriate model for the SAD, which can be challenging (Hughes 1986; Hughes et al. 2001; Williamson and J. Gaston 2005). Several software have been developed to estimate global diversity using this method in a frequentist (Bunge et al. 2012) or Bayesian framework (Quince et al. 2008; Barger and Bunge).

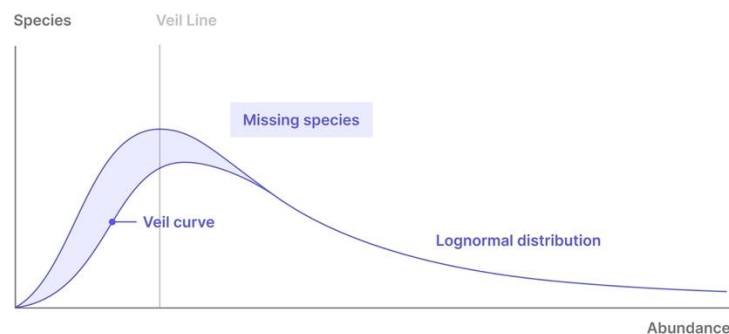


Figure 4: Veil line and veil curve for the lognormal distribution adapted from (Dewdney 2000).

Non parametric models. – This type of model is inspired from the Capture-Marking-Recapture (CMR) approach used to estimate the size of a population. It relies on the proportion of species seen – from one to several times – to estimate the proportion of species that have not yet been observed. The rationale behind this is that species will be collected more often if the diversity is low in a community than if it is high. These methods can be based on either abundance data (also referred as frequency counts) from a single sample such as the Abundance Coverage Estimation (ACE) method (Chao 1984; Chao and Lee 1992), or on incidence data from multiple samples such as the Incidence Coverage Estimation (ICE) (Chao 1987; Chao et al. 1992). In the latter case, an estimate of the proportion of species seen in none of the multiple samples is obtained. This non parametric method is particularly well suited when rare species are preponderant (Colwell and Coddington 1994) which is often the case with microorganisms and amplification-based methods. Several other variants of the model described above have been developed (Burnham and Overton 1979; Heltshe and Forrester 1983; Smith and van Belle 1984) and have different performances depending on the type of data used. Yet, the incidence-based method seems to outperform the others (Colwell and Coddington 1994). Therefore, this method has since received a lot of attention from the scientific community (Bunge et al. 2012; Colwell and Elsensohn 2014; Chao and Chiu 2016; Louca et al. 2019) and forms the basis of

the best currently available diversity estimate method, breakaway (Willis and Bunge 2015). Breakaway can be used on frequency or incidence ratios (Willis and Bunge 2015) and was extended to account for discarded singletons due to sequencing error (Willis 2016b).

0.2. The Diversification of Microorganisms

Diversification can refer to microevolutionary processes happening at the intraspecific level. We refer here to the macroevolutionary process of species diversification, the generating process of diversity above the species level, by which species emerge – through speciation events – and go extinct – through extinction events. This process typically occurs over geological time scales (on time scales on the order of the millions of years) and shapes large-scale patterns of species diversity. Species diversity drastically fluctuates across the taxonomy. This fluctuation can be attributed to difference in time of species origination or to difference in the dynamic of species accumulation. The latter factor has happened to explain a myriad of diversity patterns observable across the taxonomy. The most striking example inhabits East African Great Lakes and represents an exceptional adaptive radiation resulting in the species-rich cichlids family.

0.2.1. The Speciation and Extinction Processes

Speciation occurs from a process where a permanently distinct group of organisms emerges from closely related individuals. It ensues from one group diverging sufficiently enough from individuals of the former species, that genetic exchange (i.e. gene flow) is not sufficient to maintain a cohesive unit among them. While this process takes time, it is often assumed to be instantaneous over geological time scales (see (Etienne and Rosindell 2012) for a model with non-instantaneous splits). It is also commonly assumed that the process is dichotomic whereby two species are obtained from one speciation event. While speciation is not observable in short time microorganisms offer the opportunity to observe divergence among strains in real time due to their short generation time. Even if performed under a controlled and constant environment, these experiments have enabled to unravel some evolutionary mechanisms of rapid adaptation and the implication of genetic and phenotypic novelties (Lenski and Travisano 1994; Good et al. 2017).

Extinction is a process consisting of the death of a lineage. It occurs when the population size of the species is not sufficient enough for its maintenance. This takes place under conditions improper for the survival of individuals. It has been hypothesized that microbial lineages should undergo less extinctions than macro-organisms, based on their population size, clonal reproduction, capacity of resistance to unfavorable environmental conditions – i.e. dormancy – among other characteristics (Dykhuizen 1998; Morono et al. 2020). Meanwhile others have

observed microbial evolutionary stasis across deep time (Schopf et al. 2015) questioning them as being living fossils.

0.2.2. The Tempo of Diversification

Diversification rates – Speciation and extinction events happen at a given pace, the “tempo” of diversification, typically quantified in number of events per lineage per million year, and referred to as speciation and extinction rates. The rate of speciation and extinction in microorganisms have been suggested by some (Cohan 2001; Lawrence 2002; Scholl and Wiens 2016; Hernández-Hernández et al. 2021) to differ from the rate of macro-organisms. Since microorganisms harbor high population size and clonal reproduction, they are typically associated to high speciation rate (e.g. prokaryotes (Cohan 2001)). Moreover, their high population size have encouraged to expect low extinction (e.g. prokaryotes (Cohan 2001)). Empirical findings have not necessarily abounded in the same direction (Martin et al. 2004; Louca et al. 2018). Besides their overall values, temporal variations in diversification rates can inform on the diversification process. Often tested hypotheses in the field are whether diversification operates with constant rates, by “pulses”, and whether diversification rates decline through time as an effect of diversity-dependent processes.

Constant Diversification. – In the 1960s, Leigh Van Valen found striking the persistent relationship between clade ages and their probability of extinction in the geological records. To explain this apparent constant extinction rate, he formulated the widely known Red Queen Hypothesis (Van Valen 1973) in which species must constantly evolve to survive to their evolving environment. Some evidences of constant diversification rates have been found in prokaryotes using fossil-independent approaches mainly focusing on net diversification (*speciation – extinction*) rate (Martin et al. 2004; Lorén et al. 2014; Marin et al. 2016; Sanglas et al. 2017; Straub and Zhaxybayeva 2017; Louca et al. 2018), ([Fig. 5a,b](#)). According to (Straub and Zhaxybayeva 2017), this finding supports the potential randomness of the diversification process in prokaryotes as opposed to a selection-driven hypothesis.

Pulses of Diversification. – While constant diversification rates predict a steady accumulation of diversity through time, diversity curves drawn from the fossil record often suggest that diversity has fluctuated (Sepkoski 1984; Benton and Emerson 2007). In addition, diversification rates often appear to be variable across time, with periods of diversification pulses separated by periods of slow diversification (Stadler 2011a; Morlon et al. 2011a). For microorganisms with a good fossil record such as diatoms, diversity curves reveal apparent bursts (Spencer-Cervato 1999). If these bursts are real and not the result of a preservation artefact (Westacott et al. 2021), this indicates that diversification rates have varied through time (Knoll et al. 2007; Lazarus et al. 2014; Cermeno 2016; Wiese et al. 2016). Diversification analyses have also shown periods of fast diversification in microorganisms (Morlon et al. 2012; Gubry-Rangin et al. 2015; Perez-Lamarque et al. 2020; Pinseel et al. 2020) ([Fig. 5c,d](#)). These diversification pulses, which indicate the radiation of one or several clades within the studied

group of organisms, have often been attributed to environmental changes (Barnosky 2001) or the acquisition of key innovations (Hunter 1998).

Diversification slowdown. – Another often observed pattern is a slow-down of diversification rates (Moen and Morlon 2014a), which has also been observed for microbial species (Morlon et al. 2012; Perez-Lamarque et al. 2020). This pattern has often been attributed to the *niche-filling hypothesis* whereby as diversity accumulates, niches (see [Box 2](#)) tend to saturation leading to a diversification rate slow down, although it could also be related to other processes (Moen and Morlon 2014a).

Heterogeneous diversification. – Besides varying temporally, diversification rates vary across different clades or lineages. This heterogeneity in diversification rates can be related to different characteristics of the species that may affect different aspects of the speciation and extinction processes, including drivers of divergence among populations. In prokaryotes, some have suggested that archaea diversification may be drastically slower than in bacteria (López-García and Moreira 2008). This discrepancy has also been argued across dinoflagellates, coccolithophores and diatoms when investigating the fossil records (Falkowski et al. 2004b).

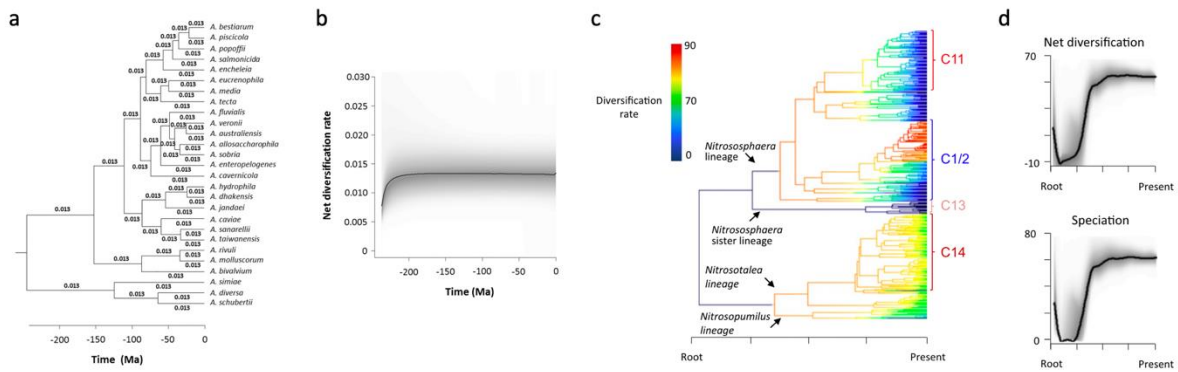


Figure 5: Bayesian Analysis of Macroevolutionary Mixtures (BAMM) analyses for *Aeromonas* (a genus of Gammaproteobacteria) from (Sanglas et al. 2017) in a) and b) and for *Thaumarchaeota* (an archaeal phylum) from (Gubry-Rangin et al. 2015) in c) and d). a) Branch-specific diversification rates estimation based on the *Aeromonas* species tree. b) Net diversification rates-through-time for *Aeromonas*. Curved black lines represent the median values obtained and the 95% confidence intervals are shadowed in gray. c) Branch-specific diversification rates in terrestrial *Thaumarchaeota* inferred from a 370-sequence *amoA* tree. The rates are not in unit of time as the phylogeny was not calibrated. The different colors represent the mean of the marginal posterior density of diversification rates. d) Plots of net diversification rate and speciation rate (y axis) through time during *thaumarchaeotal* evolution. Shaded areas denote 90% Bayesian credibility intervals.

In the following sections, we detail processes that may induce variations in diversification rates.

0.2.3. Reproduction Mode and Recombination Rates

Limitations to genetic exchange is a prerequisite to the divergence of lineages and subsequently speciation. The limitation of genetic exchange necessary for speciation to occur typically involves sexual, ecological or geographical barriers. Microorganisms encompass species with very different reproduction modes, ranging from purely clonal, purely sexual and mixtures of both, with therefore potentially very different levels of genetic exchange. Prokaryotic organisms reproduce clonally, but experience Horizontal Gene Transfer (HTG) – also called Lateral Gene Transfer (LGT) – which among closely related individuals resembles parasexual processes (i.e. homologous recombination) (Fig. 6). Micro-eukaryotes can reproduce both sexually and clonally while few of them are unable to perform sexual reproduction (Flot et al. 2013). Different modes of speciation have been suggested to operate in asexual compared to sexual species. In the next paragraphs, I will detail the main modes of speciation proposed for clonal species while relating to what has been documented for sexual ones if analogies can be made. The two main models of speciation proposed for clonal species, the *Ecotype* and *Recombination* models, differ mainly by the assumed frequency of homologous recombination among closely related lineages (Doolittle and Zhaxybayeva 2009; Straub and Zhaxybayeva 2017).

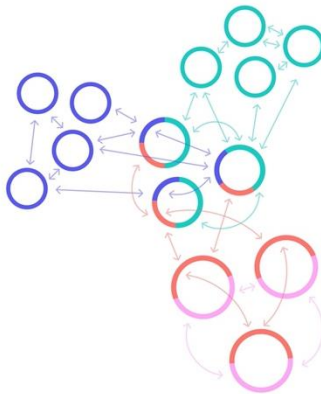


Figure 6: Lateral gene transfer and homologous recombination together can produce organisms effectively belonging to several species at once adapted from (Doolittle and Papke 2006). The all-blue, all-turquoise and red/pink circles represent genomes from three different bacterial groups that might be designated species. Each circle represents an individual genome. There is effectively no homologous recombination (arrows) between genomes or areas of different colors. LGT has, however, recently created a mosaic genome (center), with segments derived from blue, turquoise and red/pink species (itself a mosaic). Homologous recombination can occur between a segment introduced by LGT and the corresponding region of the original donor strain.

The Ecotype model (Cohan and Perry 2007). – The first model assumes that homologous recombination is rare for clonal lineages (within the same order of magnitude of mutation rate). Under this model lineages are defined as ecotypes and are reproducing purely clonally. If a

HGT or a mutation is introduced in the clonal population, the best-fit genotype will invade and purge the initial population from its diversity (i.e. periodic selection (Koch 1974)) or alternatively genetic drift will fix the newly introduced genotype (Fig. 7a). This can lead to speciation when the new genotype is able to colonize a new niche and subsequently creates a new ecotype and escapes the periodic selection from its former ecotype (i.e. ecotype formation) (Fig. 7b). In rare cases of introduction of an extremely adaptive genotype, common resources utilization among two ecotypes can lead to competitive extinction of one of the two ecotypes (i.e. speciation quashing). Under the Ecotype scenario, speciation rate is high due to low rate of homologous recombination and the extinction rate is low as extremely adaptive genotype are likely rare.

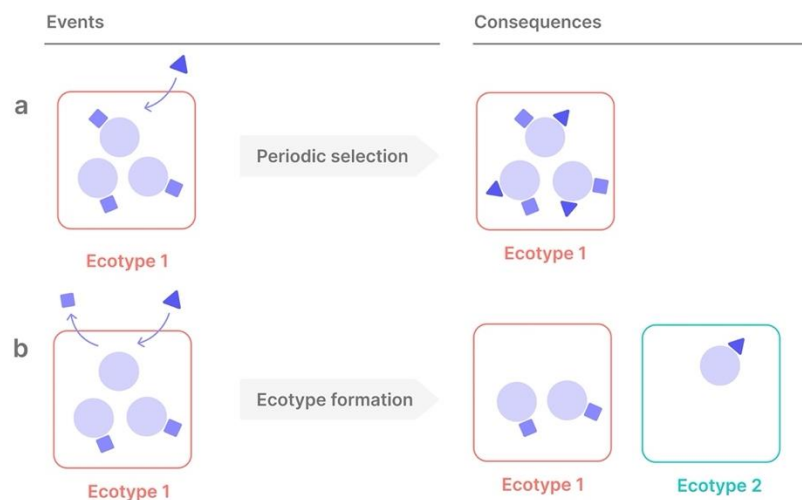


Figure 7: The consequences of a change in ecological niche following an HGT event adapted from (Wiedenbeck and Cohan 2011). Acquisition of a new ecological function by HGT (indicated by the blue triangle) can lead to either a) a periodic selection event or b) an ecotype formation event. In a) the new ecological function is added to the ecological repertoire of the recipient strain, and the resultant strain is now able to outcompete the membership of its ecotype by virtue of its greater repertoire. In b) the gain of one function is incompatible with an existing function or reduces the performance of an existing function. Thus, acquisition of the new ecological capability comes at the expense of some pre-existing function. In this case, acquisition of the new function leads to a new ecotype, which can coexist with the preexisting ecotype.

The Recombination model (Fraser et al. 2007). – The other proposed model assumes that homologous recombination is frequent among clonal lineages. Similarly to what happens in the case of sexual reproduction, the homologous recombination act as a cohesive force among closely related individuals. Speciation occurs when the mutation or the recombination with distantly related lineages is frequent enough that it can lead to lineages divergence before homologous recombination create enough cohesion among lineages. Under this scenario, the speciation rate is limited by the high homologous recombination rate and is thus lower than under the Ecotype model.

In both models HGT or mutations drive divergence among lineages but the models differ by the frequency of homologous recombination (Fig. 8). Interestingly, the recombination rate has been shown to be highly variable among clonal organisms (Doolittle and Zhaxybayeva 2009; Sakoparnig et al. 2021). Even neutral mutations can indirectly affect the recombination rate of clonal organisms (Tenaillon and Matic 2020). Hence, differences in reproduction mode and recombination rates could in part explain the heterogeneity of diversification rates among micro-organisms.

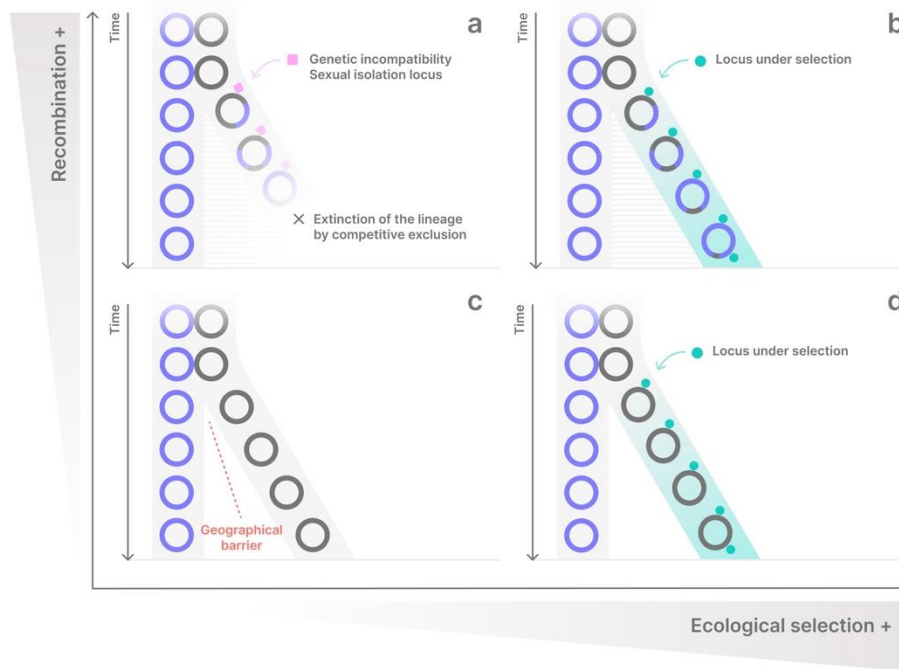


Figure 8: Models of speciation under different regimes of selection and recombination adapted from (Shapiro et al. 2016). In all models, a single population of chromosomes (circles) splits into two nascent species, distinguishable by sets of genetic differences. At each time point, the most frequent multilocus genotype is shown, but other chromosomes could be segregating in the population at lower frequencies. Different haplotypes (or clonal frames) are shown as blue or black circles. The ancestral niche is shown in grey and a new niche in turquoise. Gene flow (recombination) between species is indicated by horizontal connections between branches. In a) the simplest model of speciation with gene flow, a single mutation controlling sexual isolation (but not under selection) is the only divergent locus (pink square), with other loci experiencing gene flow between incipient species. In b) selection during speciation can produce a pattern of genetic diversity across the genome very similar to a), but species are expected to be longer-lived. Mutations under selection at early and later stages of speciation are shown as turquoise rounds. In c) allopatric speciation with a population bottleneck and neutral divergence of species. As in a), competitive exclusion should lead to the extinction of one species if they come back into contact. In d) without gene flow, the mutation under selection between species (turquoise rounds) will purge diversity genome-wide as it sweeps through one population, resulting in genome-wide divergence from the other population.

0.2.4. Ecological Divergence and Degree of Specialization

Regardless of the mode of acquisition of new genetic material, genetic novelty can confer adaptation to new environments. This has widely been observed in microbial systems (Ochman et al. 2000; Wiedenbeck and Cohan 2011; Gallone et al. 2019). Indeed, a novel acquired adaptation can allow the colonization of new environment not available to part of the population thereby limiting gene flow. A shift in tolerance range has been observed in firmicutes for example which have acquired a resilient membrane enabling them to colonize drought-prone habitats (Cohan and Koeppel 2008). When assuming a heterogeneous environment, specialization may be the result of different selective pressure inflicted by this heterogeneous environment. This local adaptation induces a higher fitness for the local population which ultimately limits gene exchange between populations. Likewise, adaptation to heat tolerance through shifts in ecological optima has been observed among strains in *Bacillus simplex* (Sikorski and Nevo 2007) and may have promoted the diversification of the group. Specialization to specific partners in mutualism, commensalism or parasitism has also been suggested to be the main driver of diversification across the tree of life (Hernández-Hernández et al. 2021) (see below for details on biotic interactions). Specialization on a resource use is also though as a major driver of speciation and common in microorganisms. A plethora of studies have shown that microbial sister clades exhibit disparate resource use such as bacterial methylotrophs which can grow on one-carbon compounds compared to their relatives growing on complex source of carbons (Chistoserdova et al. 1998). The ability of closely related marine *Prochlorococcus* to uptake nitrite versus ammonium is another example of such specialization (Rocap et al. 2003). While specialization to a completely new task (i.e. different than the parental one) may spur speciation, a finer *division* of the niche (i.e. the degree of specialization) has also been suggested to favor speciation (see [Box 2](#) for the concept of the niche). The specialization of each lineages to thrive using only part of the parental niche may foster divergence among populations and this partitioning of the niche in terms of temperature, photic level, depth and season (Johnson et al. 2006; Ward et al. 2006; Hunt et al. 2008; Weiner et al. 2012; Miller and Carvey 2019) has been observed in closely related lineages of marine micro-eukaryotes, cyanobacteria and other bacteria. The degree of specialization is also expected to influence species extinction as finer specialization alter the viability of a species under environmental fluctuations. Overall, *niche divergence* and *niche partitioning* can affect the pace of species diversification although concurrently, species ecological divergence may also be a influenced by speciation rate.

Box 2: The Concept of the Niche.

The species niche describes species ecology – also referred hereafter as ecological niche – and entails different criterion regarding which concept is adopted. The species niche is defined differently according to Grinnell (Grinnell 1917) and Elton (Elton 1927). The Grinnellian niche concept applies to variables representing environmental conditions, the so-called scenopoetic variables by Hutchinson (Hutchinson 1978). On the other end, the Eltonian niche concept refers to resource-related variables, referred to as biotonic variables by Hutchinson (1978), (see (Soberón 2007) for conceptualization). Hutchinson provided an intuitive mean to consider species niche that is a hypervolume of n dimensions (Hutchinson 1957). I refer to the realized niche definition of Hutchinson when referring to the ecological niche and consider resource related variables on which organisms can act as well as environmental variables.

0.2.5. Biotic Interactions

The Red Queen Hypothesis. – Species are constantly coexisting with one another in the same environment. The biotic interactions they experience have been postulated to be sufficient for evolution to occur despite the stability in abiotic factors as the biotic environment is constantly evolving (Van Valen 1973). In this context, species are continuously evolving as an evolutionary race but their fitness to the environment does not increase through time. This hypothesis has often been opposed to the *Court Jester Hypothesis* whereby abiotic factors rather than biotic ones are the major drivers of species diversification (Barnosky 2001). The degree of intimacy and type of biotic interactions can have a disparate effect on diversification rate.

Competition. – In this continuously evolving biotic environment, interactions can drive species to extinction in the case of competitive exclusion. In this scenario, the fitness of one species is high enough to indirectly lead to the extinction of another species competing for the same resources for example. Contrastingly, competition can also spur diversification such as in the case of character displacement (Schluter 2000). In this context, competition drives lineages to divergence for coexistence. The latter has been shown to happen experimentally on *Pseudomonas* whereby competition induced niche specialization (Rainey and Travisano 1998).

Predation and parasitism. – Predators and parasite may lead to specie coexistence and lower competitive exclusion due to control over population size. Moreover, it can spur diversification if multiple resistance strategies exist as different paths to resistance can lead to divergence among populations carrying them (Schluter 2000). It has also been suggested that predative and parasitic interactions can inhibit speciation by lowering resource competition and thus disruptive selection. Some empirical evidences seem to support this prediction in sympatry

but not in allopatry (Buckling and Rainey 2002) but no consensus has been reached (Brockhurst et al. 2004; Vamosi 2005).

Cospeciation. – As described in a previous section, the specialization in biotic interactions is thought to be a major driver of species diversification (Hembry et al. 2014). Indeed, intimate interactions among species can confer geographical or reproductive isolations as well as new capabilities such as colonizing new niches or being adapted to different resources which can substantially spurs diversification (Brucker and Bordenstein 2012). For example in the case of obligate symbiotic interactions, the divergence of the symbiont can follow the divergence of the host, a cospeciation pattern which has for example been hypothesized to shape the evolution of a small fraction of hominids gut microbiota (Moeller et al. 2016).

Facilitation. – Probably the least studied biotic interaction as a driver of species diversification is *facilitation*. Facilitation is the process by which the presence of one species enhances the fitness of another. It has been shown to spur divergence in *E. coli* as new populations arise while specializing on compounds produced by others – a process called *cross-feeding* (Turner et al. 1996; Treves et al. 1998; Doebeli 2002). Under facilitation, new species can create new niches and subsequently influence the fitness of others. This concept is termed *niche construction* and has been shown to play a critical role in the diversification of some species (Erwin 2008; Rocabert et al. 2017; Bajić et al. 2021). As an example, facilitative interactions might play an important role in the diversification of *Pseudomonas* (Day and Young 2004). In cyanobacteria, the dependency of one species towards another has been suggested to potentially arise due to a process of genomic streamlining – i.e. *the Black Queen hypothesis* (Morris et al. 2012) – which can be adaptive in a *leaky* environment (i.e. costly compounds are available from other organisms in the environment).

0.2.6. Abiotic Environment

Despite intrinsic factors influencing the propensity of a lineage to diversify, species can also diverge owing to extrinsic factors. Indeed, according to the Court Jester hypothesis, speciation and extinction can occur in response of erratic environmental changes (Barnosky 2001). In this context, extrinsic factors can englobe average temperature, solar radiation, global resource availability and may fluctuates due to movement of tectonic plates, biogeochemical feedback loops, evolutionary transitions amongst others. Changes in the abiotic environment have affected global diversity drastically such as during mass extinction events (Sepkoski 1984). Meanwhile, some abiotic changes have had disparate effect depending on clades. In diatoms for example, abiotic factors such as sea level, temperature, or silica availability has influenced contrastingly the diversification of clades (Cervato and Burckle 2003; Lewitus et al. 2018).

0.2.7. Population Size and Limits to Dispersal

Population sizes. – Drift can lead to the divergence of isolated populations. However microbial lineages are characterized by large population sizes, and thus drift is unlikely to be the main driving force of divergence among populations of the same species. Nevertheless, some organisms such as obligate parasites, commensals or mutualists acquired from one host to another may suffer from drift as the number of transmitted individuals may be scarce, an observation made in the case of the endosymbiotic bacterium *Buchnera* (Herbeck et al. 2003). Moreover, some seasonal changes in nutrients availability – in aquatic environments for example – can lead some opportunistic lineages to be relatively scarce at certain periods whilst profiting from high nutrients availability and blooming at others, creating a seasonal bottleneck. This large fluctuation in population size can allow drift to occur. Hence, different forces can act toward population divergence depending on organisms ecological strategy and the environment.

Limits to dispersal. – Limits to dispersal is assumed to be less preponderant in microbial lineages (Louca 2022) than for macro-organisms. Thus geographical barriers might not be a prevalent mechanism for microbial divergence. Still few examples of potential allopatric speciation in the microbial world (Fontaneto et al. 2008; Singer et al. 2019) exist. In the case of obligate parasites, commensals or mutualists associated to a specific host, the dispersal between individual hosts might be highly constrained and divergence can subsequently occur (as described in the previous section). In this context, the external environment of the host can be seen as a geographical barriers and divergence can happen in allopatry. This pattern has been observed in a pathogen known to have an extremely high recombination rate (Linz et al. 2007). This type of divergence is also present in the case of highly specialized organisms that exhibit a geographically restricted suitable habitat such as thermophilic archaea (Whitaker et al. 2003; Reno et al. 2009).

0.3. Approaches for Studying Diversification

0.3.1. Paleontological and Neontological Approaches

Historically, diversification has been analyzed through the analysis of the fossil record. Organisms can leave a trace of their existence in the fossil record, and these traces have extensively been used by paleontologists to shed light on past mass extinction events, the apparition of new morphologies and lineages, and more generally fluctuations in diversity and diversification rates through geological times (Benton 1997). The best-preserved organisms in the fossil record are those with a skeletal body. In microeukaryotes, numerous groups harbor

skeleton that have been fossilized in geological records. Specifically, large amount of microfossils are found in the sediments such as vegetative diatoms cells or spores, dinoflagellate cysts, ebridians, foraminifers, calcareous nannofossils and radiolarians (Prothero and Lazarus 1980; Spencer-Cervato 1999). Other well preserved microorganisms have been found in amber (Poinar et al. 1993). For soft-bodied organisms – e.g. prokaryotes – that do not leave a physical trace in the fossil record, paleontologists have used other biomarkers such as chemical ones (e.g. sterane). This has enabled them to trace the evolution main group of organisms through geological times and has also been used along with fossils records as complementary information (Schwark and Emt 2006; Kodner et al. 2008).

More recently, approaches have been developed to investigate diversification dynamics using present day data. These neontological approaches consist in reconstructing the phylogenetic trees of extant species and to use these trees to investigate diversification dynamics. While the paleontological and neontological approaches are still mostly performed separately by researchers coming from different fields, there are increasing efforts to combine the two sources of information in joint analyses (Didier et al. 2012; Silvestro et al. 2018; Gupta et al. 2020). The extraction of ancient DNA from fossils also goes in this direction, although it is limited to the recent past.

My PhD builds on the neontological-only approach. Below I introduce in more details the two key steps of this approach: reconstructing the phylogenetic trees of present-day species, and interpreting these phylogenetic trees in light of the birth-death process, supposed to represent the underlying dynamics of speciation and extinction.

0.3.2. Phylogenetic Trees

0.3.2.1. *Description*

A phylogeny is a tree (graph theory sensus) – i.e. a connected graph containing no closed loops – composed of vertices (nodes) and edges (branches) which represents the evolutionary relationship among related entities and contains a measure of their divergence. These entities can designate genes, populations or species. In this thesis, I will refer to the term *phylogeny* when dealing with *species phylogeny*. The information on the relationship among related species lies in the topology of the tree while the measure of divergence is embedded in the branch lengths. The branch lengths can be denoted in different units depending on the divergence unit of interest. In the case of diversification rate analysis, the unit of interest is time (typically millions of years (MA)). The resulting timed phylogeny is sometimes referred as a *chronogram* or a *timetree*. Conceptually, the nodes and branches can have different interpretations. I adopt the branch-based tree terminology whereby branches represent lineages and nodes represents lineages' splitting events (Martin et al. 2010). In macroevolution, it is common to use dichotomic – i.e. internal vertices are linked to three edges – and rooted trees. The complete phylogeny represents extant (i.e. present day) and extinct lineages ([Fig. 9a](#)). In a purely neontological approach, only extant lineages are present in the phylogeny and the

resulting phylogeny is called a *reconstructed phylogeny* whereby all the tips of the tree are at equidistance from the root (i.e. ultrametric tree) ([Fig. 9b](#)).

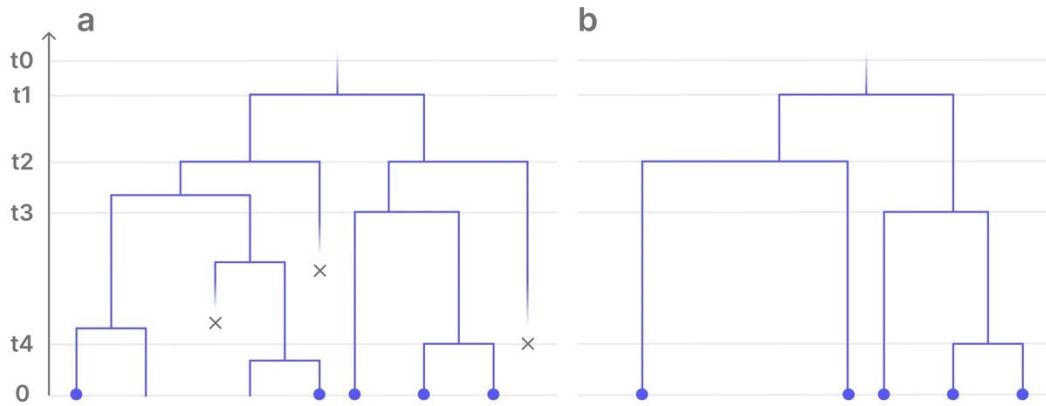


Figure 9: a) Complete phylogeny of age t_0 and b) its corresponding reconstructed phylogeny adapted from (Stadler 2013a). In the reconstructed phylogeny, all extinct and unsampled extant species are pruned, only the extant sampled species (denoted with a circle) are included. The time t_0 is called stem age, and the time t_1 is called crown age.

0.3.2.2. Representation

Full representation. – The phylogeny can be represented in different format types. The first and most commonly used format is the *Newick format* (Felsenstein 2004) – adopted in 1986 – which displays the tree information in a string. It employs commas and nested parenthesis to separate tip names recursively and can additionally incorporate distances, node and tip states information. This representation is obtained by traversing the tree in a postorder fashion. Software specific representations of phylogenies also exists such as the *phylo* object format in R language (Paradis et al. 2004). The tree information of the *phylo* object is contained as a list. The list contains a matrix depicting the tree topology and a vector displaying the associated branch lengths. In the matrix, each row represents an edge and is defined by its ascendant and descendant vertices. Another matrix representation had been proposed in the past (Ragan 1992) but is substantially less compact. In [Chapter 2](#), I present a compact matrix representation of a phylogeny that contains branch length information and tip trait states.

Summary statistics. – The phylogeny can also be summarized using metrics called summary statistics. Simple summary statistics can be derived from a phylogeny such as the number of tips or the root age. Other more complex summary statistics have been derived to represent specific characteristics of the tree and often deviations from those expected under the classical constant rate homogeneous birth-death process (see section below). A recurrent observed pattern on empirical phylogenies is their imbalance which indicates how uneven the repartition of diversity is among clades. The tree imbalance can be captured by the β statistic (Aldous 1996). Another well-known summary statistic is the γ statistic (Pybus and Harvey

2000), which characterizes the relative position of nodes in the phylogeny using the branch length information.

0.3.2.3. *Phylogenetic Reconstruction*

The shared characteristics amongst present day diversity enable us to retrace their evolutionary history. This was historically accomplished using morphological or anatomical characteristics (i.e. synapomorphies) to retrace the common ancestors of present day lineages. The current approach is based on using genetic data – which carry inheritable characters universally among living organisms – to reconstruct the evolutionary history of species. It enables to reconstruct a phylogeny of species – each species is characterized by a genetic sequence – based on the similarity of genetic sequences and will be the focus of the following sections.

Sequence alignment. – The first step of phylogenetic reconstruction consists in aligning the sequences representing each extant species. The classical Multiple Sequence Alignment (MSA) framework aim at aligning sequences in order to maximize sequence similarity. This is typically fast to perform for sequence pairs but lengthy on multiple sequences as required computation time increase exponentially with increasing number of sequences. Thus some heuristics or approximations are often applied to fasten the search (Edgar and Batzoglou 2006).

Model of DNA evolution. – Once the multiple sequences are aligned, a stochastic model of DNA evolution (Neyman 1971) is used to accurately estimate the evolutionary distance among sequences by accounting for unobserved changes along the sequences (Felsenstein 1981). This model describes the rate at which discrete changes of individual states (A, T, C, G) per sites occur, treating sites as independent and assuming that the evolutionary model does not change through time. The model is Markovian, meaning that the state of the process at any given time depends only on its current state, and the relative frequency of each state reaches a stationary frequency. The process, identical in reversible time, is referred to as the General Time Reversible (GTR) model. Several other assumptions can be made, including assuming that all state changes occur at the same rate (Jukes and Cantor 1969), specific transitions and transversions rates (Kimura 1980), and character specific rates (Kishino and Hasegawa 1989; Tamura and Nei 1993; Felsenstein 2004). These models can be selected based on their likelihoods, and the selected model is applied for subsequent phylogenetic reconstruction.

Phylogenetic inference. – For a phylogenetic tree with n tips, one can use phylogenetic algorithms to explore the $(2n - 3)!$ rooted tree space, assuming the previously selected model of evolution. This search has substantially been improved by the Felsenstein pruning algorithm (Felsenstein 1981) enabling the efficient computation of the likelihood of a tree. This development has led to a plethora of tools for performing phylogenetic inference, some based on frequentist inference (Yang 1997; Guindon et al. 2010; Price et al. 2010; Nguyen et al. 2015; Kozlov et al. 2019) and others on Bayesian inference to recover the full posterior probability distribution of phylogenetic trees (Huelsenbeck and Ronquist 2001; Lartillot et al. 2013; Höhna et al. 2016a; Bouckaert et al. 2019). In addition to the model of DNA evolution, the Bayesian

inference procedure requires assuming a tree generating process, known as the branching process prior, which is often a birth-death process (see below). All proposed tools include a metric of confidence in the branch position. The confidence is often calculated using bootstrapping (except for Bayesian inference from which a posterior probability is derived). The rationale is to resample with replacement the sequence alignment many times to get pseudo-alignments of the same size as the original one and to reconstruct a phylogeny for each pseudo-alignment. The bootstrap value is the proportion of the pseudo-alignments leading to the same branch position or clade position as in the phylogeny produced with the original alignment (Efron 1977; Felsenstein 1985). The bootstrap value cannot be considered as the probability of a position to be true, but it gives an idea of confidence. The bootstrapping procedure is computationally intensive. Faster and more reliable alternatives for confidence calculation exist (Minh et al. 2013; Hoang et al. 2018; Lemoine et al. 2018).

Phylogenetic dating. – At this step of the process, branch lengths represent the number of substitutions, and the resulting phylogeny is referred to as a non-calibrated phylogeny. To calibrate the tree – i.e. to date it – and transform the number of substitutions in time units, one needs to assume a molecular clock, i.e. a rate at which substitutions accumulate over time during the process of sequence evolution. The rate of substitution is either assumed to be constant – i.e. the strict molecular clock (Zuckerkandl and Pauling 1962) – or to vary across lineages in the tree – i.e. the relaxed molecular clock (Lepage et al. 2007). Under the latter hypothesis, substitution rates can either be sampled independently for each branch from a given distribution – i.e. the non-autocorrelated clock (Drummond et al. 2006) – or evolve as a Markov process along the tree – i.e. the autocorrelated clock (Kishino et al. 2001; Aris-Brosou 2003). The molecular clock is often informed by fossils (or secondary calibration), which can be used to constrain node ages and are typically informative on minimum age constraints (Didier and Laurin 2020).

Step by step vs full inference – The step by step procedure I have described involves some uncertainty at each step which is often neglected in downstream analyses. An elegant way to alleviate this limit and account for uncertainty is the full inference approach which considers each process – i.e. the substitution model, the molecular clock model and the tree generating process – in a joint Bayesian inference. This can be achieved in software such as BEAST (Drummond and Rambaut 2007).

Specificities of reconstructing phylogenies for microorganisms. – Genetic sequences that can be used to reconstruct a phylogeny for microorganisms range from short barcodes (e.g. coming from metabarcoding sequencing) to whole genomes (e.g. from direct sequencing of cultivable organisms, single cell sequencing, or metagenomic sequencing). In the case of short barcodes, genetic sequences need to be first clustered into operational taxonomic units (OTUs), after which a single representative sequence per lineage is used. Other methods exist which are cluster-independent (Pons et al. 2006; Puillandre et al. 2012; Zhang et al. 2013) but will not be detailed here as they are less suitable to large datasets due to their intensive computational requirement (most of the time one needs to build a phylogeny of all unique sequences). Short barcodes are useful to maximize the representability of the diversity of microbial clades, but

they typically have a weak phylogenetic signal. Short sequence alignments can easily be saturated (Xia et al. 2003) – i.e. a high proportion of the aligned sites have undergone multiple substitutions – especially when reconstructing deep divergence events. On the other hand, long sequences have more phylogenetic signal, but they are typically available for only a small fraction of the total diversity. One solution is to combine short sequences with longer ones, used to constrain the topology of the phylogeny (Talavera et al. 2022) (see [Fig. 10](#) for the full procedure). In this case short sequences need first to be aligned on the longer ones, which can be done with several tools that all rely on using the already aligned longer sequences as a reference and mainly differ in their treatment of insertions and deletions (Berger and Stamatakis 2011; Finn et al. 2011; Löytynoja et al. 2012; Katoh et al. 2019).

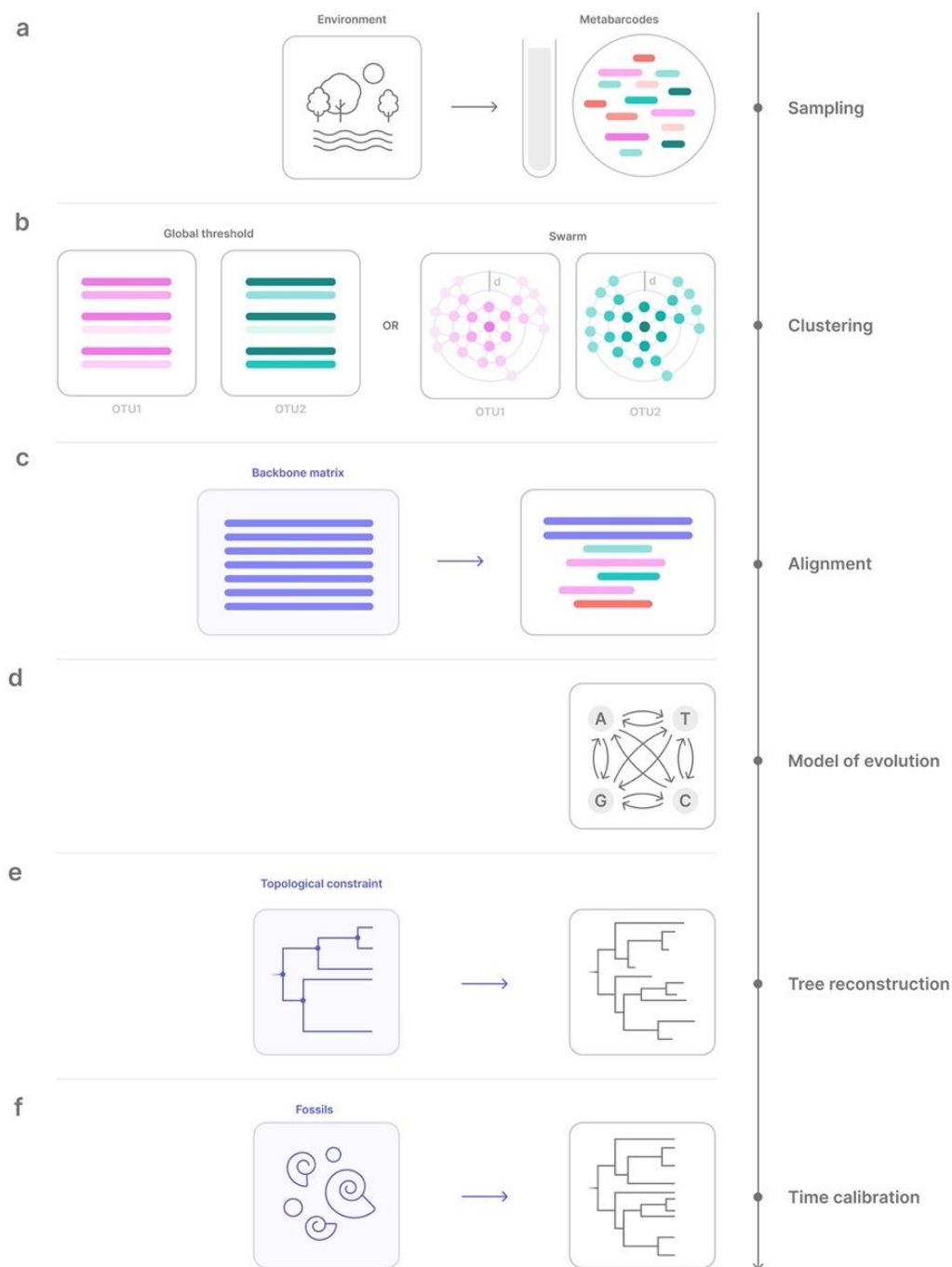


Figure 10: Conceptual framework for reconstructing a dated phylogenetic tree from microorganisms. a) describes the sampling and processing of genetic data, b) displays the clustering the barcodes in operational taxonomic units based either on global threshold clustering or local threshold clustering (swarm), c) presents the alignment step using a pre-aligned backbone matrix of longer sequences, d) illustrates the use of a model of evolution, e) describes the phylogenetic reconstruction using a backbone and f) displays the tree dating step constrained by fossils calibration.

0.3.3. The Birth-Death Process

The birth-death process (Kendall 1948) is the most commonly used formalism in macroevolution to study the species diversification process by which species speciate (birth) and go extinct (death), resulting in fluctuating diversity over time (Nee et al. 1992). The constant-rate and homogeneous birth-death process depicts a lineage-based process by which lineages evolve independently through time going through speciation events at rate λ and extinction events at rate μ . The rates are in event per million years per lineage. The process is named *Yule process* when it is purely a birth process – i.e. no extinction is occurring. The net diversification rate given by $r = \lambda - \mu$ represents the rate of species accumulation in log space, as the number of lineage is expected to accumulate exponentially through time with rate r .

Lineage through time plot (LTT) under the birth-death process – The lineage through time plot (Stadler 2008) reports the number of lineages in a reconstructed phylogeny at each time point in the past (Fig. 11), and was one of the first proposed approaches for interpreting timetrees in light of the birth-death process. Under a constant-rate homogeneous birth-death process, the slope of the LTT curve at the origin of the tree represents the net diversification rate, and the slope of the LTT curve towards the present is close to the speciation rate, as lineages which arose recently did not have time to go extinct. This apparent acceleration of the accumulation of lineages towards the present is referred as the *pull of the present*. Hence, although extinction events are not observed in a reconstructed phylogenies, extinctions leaves a signal in the repartition of branching events through time that allows estimating extinction rates (Nee et al. 1994a).

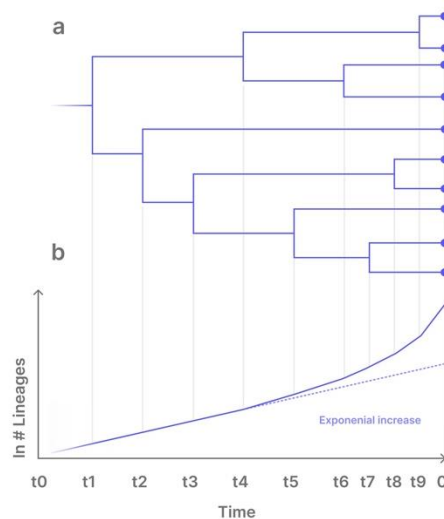


Figure 11: Lineage-through-time plot adapted from (Harmon 2019). The number of lineages in a reconstructed phylogeny increases with an increment of one at each branching event.

Likelihood of a phylogeny under the birth-death process. – By far the currently most widely used approach for interpreting phylogenetic trees in light of the birth-death process is through likelihood-based statistical inference approaches (see section 4 of the Introduction),

which require computing likelihoods, i.e. the probability of observing a given tree τ under the birth-death process. For the constant rate homogeneous birth-death process starting with two lineages at time t_1 – the *crown age* of the clade –, this probability appears in (Thompson 1975) and is given by :

$$p_{BD}(\tau|L(t_1) = 2, \lambda, \mu) = p_1(t_1, \lambda, \mu)^2 \prod_{i=2}^{n-1} \lambda p_1(t_i, \lambda, \mu) \quad [1]$$

where τ is the phylogeny with branching times $t_1, t_2, \dots, t_{n-1}$ (n is the number of lineages at present), $L(t_1) = 2$ refers to the condition to start with two lineages at time t_1 , λ and μ are respectively the speciation and extinction rates, and $p_1(t, \lambda, \mu)$ is the probability that a lineage alive at time t leaves exactly one descendant in the sample at present. The probability of the phylogeny given that the birth-death process starts with two lineages is therefore the product of the probability that the two lineages at t_1 leave one descendant each at present in the sample and the probabilities that each of the descendant lineages arising from a speciation event at time t_i leaves one descendant at present in the sample. Explicit analytical formulae for these probabilities are available.

Conditioning. – As we only observe processes that have not gone extinct before reaching the present, the birth-death process is often conditioned on survival. The first birth-death process conditioned on survival has been derived by (Nee et al. 1994c) as follows :

$$p_{BD}(\tau|t_1 = t_{crown}, \lambda, \mu) = \left(\frac{p_1(t_1, \lambda, \mu)}{1 - p_0(t_1, \lambda, \mu)} \right)^2 \prod_{i=2}^{n-1} \lambda p_1(t_i, \lambda, \mu) \quad [2]$$

where $p_0(t, \lambda, \mu)$ is the probability that a lineage alive at time t leaves no descendant at present in the sample. Equation 2 differs from equation 1 only by dividing equation 1 by the probability that the two lineages arising at time t_1 both leave at least one descendant at present. This conditioning is necessary to obtain non-biased parameter estimates (Stadler 2013a). Other conditioning exists, such as conditioning on n species being represented in the phylogeny at present (Rannala and Yang 1996), but the most often used conditioning is that on survival.

Missing species and sampling schemes. – In empirical datasets, reconstructed phylogenies often lack some present-day lineages. We refer to these phylogenies as *incomplete phylogenies*. As missing extant lineages result in missing branching events in the phylogeny, not accounting for them will bias rate estimates. In order to account for missing species, a sampling process can be added to the birth-death process, and likelihood formulae for this birth-death-sampling process have been derived. Several sampling schemes have been proposed, the first of which was the *Bernoulli sampling* scheme, under which each lineage is sampled independently and uniformly according to a sampling probability noted y (Nee et al. 1994c). In practice, y is assumed to be known and computed as the number of species in the phylogeny divided by the total number of species in the clade. Alternatives have been proposed to capture common sampling practices, in particular the *clustered* and *diversified sampling* schemes, which respectively reflect a sampling scheme where closely-related or distantly-related lineages are sampled (Hohna et al. 2011). The resulting pattern in the node depths distribution is a

prevalence of recent nodes under *clustered sampling* and deep nodes under *diversified sampling*. Another framework has been developed for *strict higher level phylogenies* – i.e. phylogeny fully resolved from the origin until a specific t_{cut} time and thenceforth fully unresolved (Stadler and Bokma 2013). As for the diversified sampling, the *strict higher level phylogenies* result in deeper nodes. Further details on this topic can be found in Chapter 1, where I present and infer diversification rates using an alternative sampling scheme, and adapt the *Bernoulli sampling* scheme to the case when the sampling probability y is not known.

Simulating the process. – Simulations are often needed to test the accuracy of a proposed model or to perform simulation-based inferences. Different approaches exist to simulate a phylogeny based on the birth-death process. The simplest is the forward in time simulation by which waiting times until the next speciation or extinction event are sampled according to given birth and death rates from the past until the present is reached. The simulation can be conditioned on having a stem age of $t_0 = T$. An elegant and fast approach to simulate a birth-death process conditioned on stem age is to use the property of the *Coalescent Point Process* (CPP). A coalescent point process is a random ultrametric tree of stem age $t_0 = T$ whose node depths are independently sampled from the same distribution noted F until a value larger than T is drawn (Lambert and Stadler 2013a). The time is here proceeding as forward (Fig. 12), (i.e. from $t = 0$ the start of the process to $T = t_0$). The constant homogeneous birth-death process is a CPP. Therefore, simulating the process can be achieved by sampling node depths from the law F and reconstructing the tree horizontally – i.e. the child lineages are grafting on the right of the tree (Fig. 12). The resulting phylogeny is pruned from its extinct species in a *post hoc* fashion to get an extant-only simulated phylogeny. Another approach is to perform simulations backward in time starting from present day species in order to condition on the number of extant species and avoid the computational expenses of simulating extinct lineages (Höhna et al. 2016b; Louca 2020).

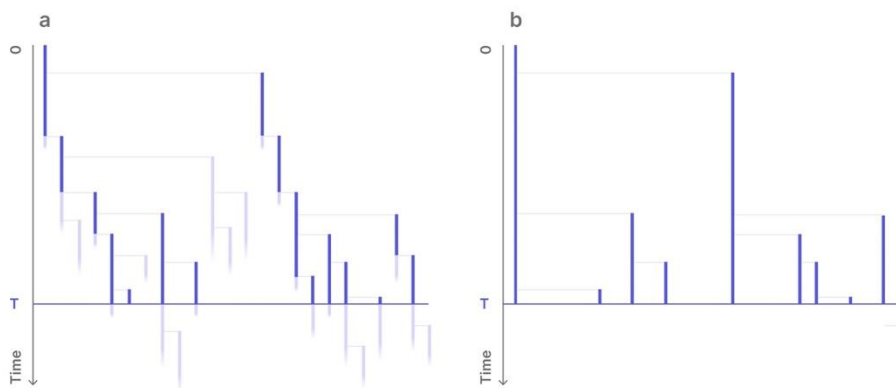


Figure 12: An oriented phylogenetic tree generated by a general birth-death model and the reconstructed tree obtained from the complete tree adapted from (Lambert and Stadler 2013a). a) represents the simulated process with extant and extinct lineages grafted on the right to the tree and b) represents the process without extinct lineages.

Model complexification. – Simple birth-death processes have been described heretofore assuming constant per lineage diversification rates as well as homogeneous rates across

lineages. A plethora of more complex models exists. In those, the diversification rates are proposed to vary through time (Stadler 2011a; Morlon et al. 2011a), along diversity accumulation (Etienne et al. 2012a), environmental gradients (Condamine et al. 2013a), species traits (Maddison et al. 2007a; Herrera-Alsina et al. 2019) or are assumed to be species specific (Fig. 13) (Höhna et al. 2019; Maliet et al. 2019a; Barido-Sottani et al. 2020; Vasconcelos et al. 2022). The current available methods allow to test different biological hypotheses such as the relative importance of the Court Jester and Red Queen Hypotheses, or the effect of specialization or species-specific traits on speciation and extinction rates. Specifically, the latter can be tested using the state-dependent birth death model (FitzJohn et al. 2009). This class of models allows the joint inference of one or several evolving traits and diversification rates through time. In its simplest form, the model is termed BiSSE for Binary State Speciation and Extinction and describes the evolution of a trait successively in one of the two possible states of the binary trait considered. Lineages associated to each state harbor specific diversification rates. Several other extensions of the state-dependent birth death model exist and allow to account for hidden states or quantitative traits for example (FitzJohn 2010; Beaulieu and O’Meara 2016). In Chapter 2, we test the performance of a likelihood-free inference approach in recovering state specific diversification rates on the simple BiSSE model.

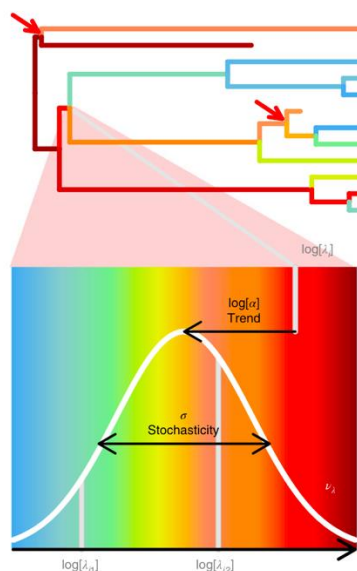


Figure 13: Illustration of the ClaDS model from (Maliet et al. 2019a). In the top, a phylogeny simulated under ClaDS, with branches colored according to their speciation rate λ_i (red, high; blue, low). Speciation rates are inherited at speciation, with a shift according to α the deterministic trend in speciation rates changes and σ the stochasticity in the rate inheritance. Red arrows indicate speciation events (and associated diversification rate shifts) that are hidden in the reconstructed phylogeny as a result of extinction.

In Chapter 3 of this PhD thesis, I use some of these birth-death models to investigate the tempo of diatoms diversification in light of their ecological strategy.

0.4. Statistical Inference

Once a probabilistic model has been established (the birth-death process in the case of phylogenetic diversification analyses), different statistical inference methods can be used to adjust the model to empirical data (e.g. a timetree of extant species), which allows to quantify the statistical support of the model and to estimate the parameters of the model (e.g. speciation and extinction rates) for this data. I will hereafter describe different inference methods; the two first are the most commonly used in macroevolution.

0.4.1. Frequentist Approach

Given a model, the frequentist approach enables the estimation of the parameter(s) θ most likely to explain the data x . The statistical rationale is that the frequency of repeated observations of empirical data can inform on the parameter value(s), without prior information on these value(s). A single observation of the data x provides a single estimation of the parameter(s) (considered to approach the true value(s)), as opposed to the Bayesian inference, which treats parameter(s) as random variable(s) and provides a distribution of parameter value(s) (see following section). The frequentist approach is commonly framed as a likelihood $L_x(\theta)$, which is a function of the parameter(s) θ defined as the probability of the data x knowing the parameter(s) θ of a given model.

$$L_x(\theta) = p(x|\theta) \quad [3]$$

In this approach, we typically want to optimize the likelihood – i.e. find the parameter value(s) that provide the highest likelihood. The maximum likelihood estimate of θ (the MLE) satisfies the following equation.

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta}(L_x(\theta)) \quad [4]$$

We obtain the one most likely parameter estimate θ using an optimization algorithm (e.g. Nelder-Mead) and a confidence interval can be drawn from the likelihood surface. This confidence interval reflects the uncertainty around the estimate.

0.4.2. Bayesian Approach

Given a model, the Bayesian approach enables the estimation of the probability that the parameter value θ is true given the empirically observed data x . The statistical rationale is to assume that some prior information on the parameter value(s) exist and need to be accounted for, along with the information given by the data x . The outcome of this inference approach is

a probability distribution of the parameter(s) θ framed as the *posterior probability* $p(\theta|x)$ (thereafter called posterior distribution) given by the Bayes theorem.

$$p(\theta|x) = \frac{p(\theta)p(x|\theta)}{p(x)} \quad [5]$$

The quantity $p(\theta)$ is the prior distribution on θ , $p(x|\theta)$ is the probability of the data knowing the parameter(s) – the so-called likelihood – and $p(x)$ is the probability of the data. The quantity $p(x)$, referred to as the normalizing constant, and which is typically obtained by integrating over the prior distribution of the parameter(s), i.e. $p(x) = \int p(x|\theta)p(\theta)d\theta$, is often impractical to compute and is typically not calculated during the *Markov Chain Monte Carlo* (MCMC) procedure (described below). The uncertainty around the estimate is computed as a credible interval – the interval in which a parameter value falls according to a given probability – from the posterior distribution. If a single estimate needs to be drawn from the posterior distribution, the common practice is to compute the *Maximum a posteriori* (MAP) – the parameter value associated to the maximum posterior probability – as an analogue of the maximum likelihood estimate (MLE) for Bayesian inference.

Markov Chain Monte Carlo. – As $p(x)$ is often intractable, one numerical approach to obtain a posterior probability $p(\theta|x)$ is to perform a MCMC. MCMC consists of a random *chain* of calculations to *sample* from the posterior distribution. At each iteration of the chain, a random draw θ' is made from the prior distribution. The ratio of posterior probabilities between the proposed and the current parameter value θ is computed, and the new parameter θ' is accepted with a probability given by the ratio (or accepted if the ratio is above 1). This procedure requires the calculation of only posterior probability ratios, which do not depend on $p(x)$ as this quantity does not depend on θ . Hence $p(x)$ is never required in the MCMC calculation (note: computing $p(x)$ may be required for model comparison). This procedure exploits the property of Markov chains, namely that the stationary distribution converges asymptotically to the Bayesian posterior distribution, to obtain the posterior distributions of the parameter(s) of interest. Several algorithms can be used to perform the MCMC, including the popular *Metropolis-Hastings* (MH) and *Gibbs sampling* algorithms. The result of a Bayesian inference is a probability distribution, hence the latter can easily be exploited for investigating the uncertainty around the estimate, especially when the likelihood contains ridges or is multimodal.

0.4.3. Simulation-based Approach

Both frequentist and Bayesian approaches are likelihood-based approaches meaning that likelihood formulae under a given model need to be derived and computed in a reasonable amount of time for parameter inference. As larger phylogenetic trees are reconstructed (in particular in the case of microbial phylogenies) and more complex models are developed to relax unrealistic assumptions, the likelihood-based methods are limiting especially with respect

to computational time. Other likelihood-free and simulation-based approaches exist but only few have been applied in the field of macroevolution.

Approximate Bayesian Computation. – *Approximate Bayesian Computation* (ABC) relies on the use of summary statistics and simulations for parameter inference and avoids exact likelihood calculations. Summary statistics can be used to compare high-dimensional objects. They aim to summarize information contained in high-dimensional data into a small amount of simple metrics – e.g. tree size, crown age, tree imbalance or relative depth node. Then a distance metric between the set of these summary statistics measures the proximity of those objects.

The procedure consists in simulating a large amount of data – phylogenies in our case – under a given model and covering a reasonable parameter space – also referred as *prior*. The simulations as well as the empirical data are then summarized into statistics.

ABC then typically uses a rejection algorithm which “accepts” or “rejects” simulations that will form the *posterior*. The simulations are accepted or rejected on the basis of a distance (such as Euclidean) between their statistics to those of the empirical tree and a chosen tolerance threshold. The parameters corresponding to the accepted summary statistics are weighted according to their calculated distance. The accepted, adjusted parameter values are assumed to be random samples of the approximated posterior distribution $p(\theta|x)$.

The chosen threshold is the result of a trade-off between accuracy and computation time. When the threshold tends to 0 the distribution of the accepted parameters tends to the exact posterior distribution, whereas when the threshold tends to infinity the accepted parameters are simply draws from the prior. To obtain a reasonable number of samples – i.e. accepted parameters – one needs to perform more simulations for a more stringent threshold as the acceptance rate will be lower. The ABC performance depends heavily on the choice of informative summary statistics. They should be designed to summarize the maximum amount of information present in the data. The choice of summary statistics is model dependent and the performance of the procedure can also be impeded by the use of too many summary statistics – i.e. limitation known as the *curse of dimensionality* (Beaumont et al. 2002) – as the probability of accepting a simulation decreases exponentially as dimensionality increases. A common practice to overcome this limitation is to perform a *regression adjustment*. It consists of performing a regression on the accepted summary statistics and their corresponding parameters in order to estimate the empirical parameters while accounting for the imperfect match between the simulations and the empirical data. This approximation method has extensively been used in population genetic (Beaumont et al. 2002; Csilléry et al. 2010a; Overcast et al. 2019) and has been applied to coalescent as well as birth-death models to estimate for example the time of divergence among lineages (Hickerson et al. 2006; Wilkinson and Tavaré 2009).

Machine learning. – Another class of simulation-based and likelihood-free inference approach – which in contrast to ABC does not rely on rejection – is machine learning. Machine learning regroups a wide variety of techniques which learn and adapt to a task at hand without being explicitly programmed for. Machine learning has been widely developed in the last two

decades thanks to the development of efficient computing systems. The main objective of machine learning is to build a model from a set of observed data (training set) to make prediction about unseen data – often referred as *inductive learning*. In the case of continuous prediction output, the task is named a *regression task* else it is a *classification task* when predicting a category. When there is prior knowledge on the observations (i.e. *labeled* observations), the task is called *supervised learning*, while when no prior knowledge exists it is termed *unsupervised learning* and the task consists in clustering observations together. I focus here on supervised regression. The procedure can be split in several steps (Zhou 2021) shortly described below.

First, one needs to obtain observations either empirically or through simulations – the latter being less common, but corresponding to the case of adjusting a probabilistic model to data. In order for the machine learning to learn from these observations, the *ground truth* needs to be known. In the case of parameter inference, the ground truth are the values of the parameters used to simulate the observations.

Next, the observations and their ground truth are given to the machine learning algorithm for *training*. The task of the machine learning algorithm is to learn the relationship between the observations and their corresponding ground truth. This task is accomplished for example by a network of connected artificial neurons organized in layers which gave rise to the term *neural network*. Each connection is characterized by a weight which is a parameter trained (i.e. updated) during the learning process. Each neuron simply multiplies each value from the previous neurons by their weight and outputs the sum. The output value is then transformed according to a chosen *activation function* – often designed to break the linearity of the data as an action potential of a biological neuron would do. The transformed output is then given to the neurons of the next layer. The network learns by minimizing a *loss function* often calculated as the error between the output of the network and the ground truth. The loss function is minimized by backpropagating the error to update weights accordingly. The resulting machine learning model is primarily characterized by its architecture and the weight values after training. Once trained, the machine learning model can be tested: it is evaluated by the error calculated on a *validation set* – a set of observations that the algorithm did not train on and for which the ground truth is also known. When training a machine learning algorithm, one main precaution to be taken is to avoid *overfitting*. Overfitting is the result of the machine learning model having learned the specificity of the training set at the cost of generalizability. This highly hampers the ability of the model to generalize to other dataset as its performance is decreased on new observations.

In the field of macroevolution, machine learning has only been used once to estimate diversification rates (Bokma 2006) without further development or application. In [Chapter 2](#), I use *deep learning*, a branch of machine learning specifically performant on complex dataset to develop a likelihood-free approach of statistical inference for phylogenetic diversification analyses.

0.5. From Model to Data: Introduction to the Empirical Systems

During my PhD thesis, I worked with two large groups of prokaryotic and eukaryotic micro-organisms, bacteria in [Chapter 1](#), and diatoms in [Chapter 3](#).

0.5.1. Bacteria and their Diversification Rates

Bacteria is surely the group of organisms for which we currently have the largest available reconstructed phylogeny. Yet, their diversification dynamic has been scarcely studied. Scientific debates have abundantly focused on the definition of species, and on the microevolutionary mechanisms that may lead to species formation. Precisely, the corpus of literature of Cohan, Lawrence and Doolittle among others have fueled the field with evolutionary concepts and hypotheses. Based on his ecotype model, Cohan predicted high speciation rates (Cohan and Koeppel 2008). According to him, no geographical or sexual isolation is required for prokaryotic speciation and the extremely high prokaryotic population size make them more prone to adaptive mutations. Cohan hypothesized that the peculiarity of prokaryotic gene exchange should foster speciation through the acquisition of ecological innovations, a widely shared idea (Lawrence 2002; Doolittle and Zhaxybayeva 2009). Others, based on estimates of recombination and substitution rates, have predicted rates relatively similar to those observed in widely studied organisms, around 0.1 splits/lineage/MA (Lawrence 2002). Finally, others hypothesized, based on the low rate of morphological change of cyanobacteria lineages (referred to as *hypobradylety*, (Schopf 1987)) and their generalism, that they should have low divergence rates (Schopf 1994; Dykhuizen 1998).

Recently, genetic resources have become available allowing a data-driven test of these hypotheses and predictions regarding diversification rates. The investigation of bacterial diversification from phylogenies under the current widely used formalism – the birth-death process – has been initiated by (Martin et al. 2004). Current studies tend to suggest a smaller net diversification rate among bacterial lineages than other commonly studied organisms such as plants, metazoan or fungi (Scholl and Wiens 2016; Louca et al. 2018), which has been attributed by some to different modes of reproduction and complexity (unicellular vs multicellular) (Scholl and Wiens 2016).

As larger phylogenies became available, more general rules on bacterial diversification have been formulated from empirical evidence. The most recent and broad studies suggest that the diversification of bacteria might be constant over time (Marin et al. 2016; Louca et al. 2018), which contrasts with diversification shifts observed for many studied clades of macroeukaryotes such as mammals (Stadler 2011a), as well as with studies on smaller bacterial clades (Morlon et al. 2012; Lorén et al. 2014).

0.5.2. Diatoms and their Diversification Rates

Diatoms are microalgae extensively studied for their high importance in the Carbon pump and are well known for their siliceous cell walls. They harbor robust frustules of silica that have ensured a remarkable preservation in the fossil record despite diagenesis-dependent alteration – i.e. alteration due to physical and chemical changes in the sediments – (Damste 2004; Katz et al. 2007). Moreover, taxonomic-specific chemical compounds left in the geological record (e.g. steroids) can be used to retrace their origin (Knoll et al. 2007).

In contrast to bacteria, diatoms diversification has been widely studied initially due to their relatively detailed fossil record. Indeed, paleontologists have extensively investigated how diatoms diversity has changed through geological times (Spencer-Cervato 1999) and the drivers of these variations (Falkowski et al. 2004a). They have shown that diatoms exhibit evident variation in diversification rates, and that diversification dynamics are clade-dependent (Spencer-Cervato 1999). Few diatom fossils have been found in the Jurassic (Rothpletz 1896) indicating their low prevalence until the Cretaceous. Observations of the fossil record establishes an early Cretaceous radiation of diatoms coinciding with increasing continental fragmentation (Fig. 14) and increasing oxidation in earth history known to have enhanced trace metals availability required for diatoms (Knoll et al. 2007).

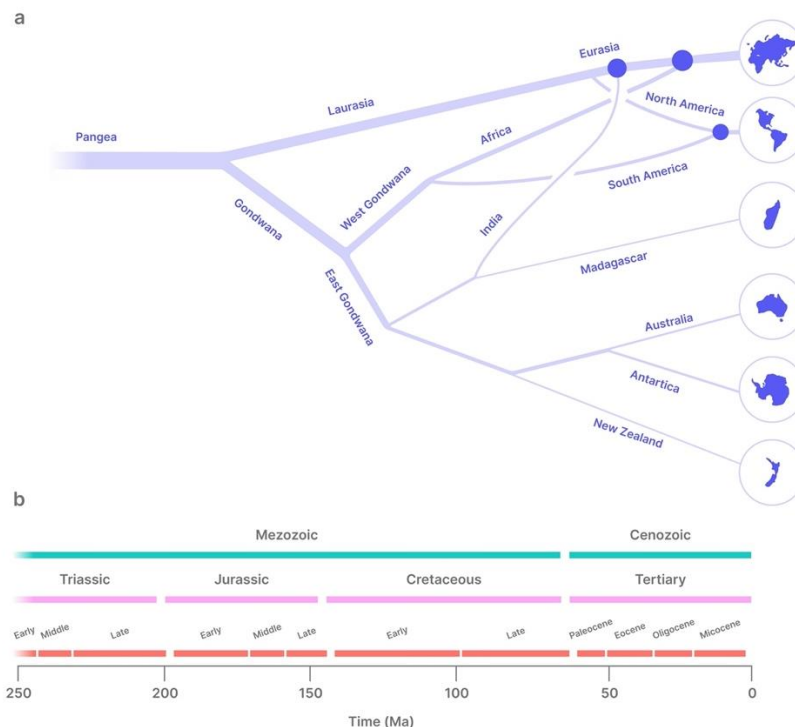


Figure 14: Simplified sketch of the break-up of the supercontinent Pangaea over the past 180 Myr adapted from (McIntyre et al. 2017). a) Continental divergences and convergences. Circles indicate continental collisions. Line thickness is a rough proxy for landmass area. b) Geological periods.

Another interesting pattern of diatoms evolutionary history is their apparent increase in diversity during the Cenozoic at a time when niche availability was decreasing due to sea level and flooded continental areas decrease (Katz et al. 2004, 2007). This period is characterized by an increase in the latitudinal thermal gradient and the vertical stratification of the water column as well as wind-mediated sporadic nutrient influx, conditions in which diatoms particularly thrive (Katz et al. 2007) (Katz et al. 2004). This period is also characterized by increased silica availability – essential for diatoms growth – from continental weathering (Falkowski et al. 2004a, 2004b) and diatoms competed against radiolarians for silica resources (zooplankton) (Kidder and Erwin 2001; Lazarus et al. 2009).

At the Eocene-Oligocene boundary, a peak of diatoms diversity was observed. This peak has been attributed to grassland expansion (Falkowski et al. 2004b) – involved in the silica loop – and competition among plankton (Kidder and Erwin 2001) yet without consensus (Rabosky and Sorhannus 2009). A more recent, middle to late Miocene pulse of diversification has been observed in the fossil record due to the expansion of bloom-forming diatoms (Katz et al. 2007).

Diatoms diversification has also been studied from extant species phylogenies under the formalism of the birth-death process, along with species trait and environmental parameters. Recent studies have shown that diatoms diversification is faster for freshwater lineages (Nakov et al. 2019), motile lineages (Nakov et al. 2018) and under specific environmental conditions (e.g. most often low sea level and high silica availability) (Lewitus et al. 2018). Their response to environmental conditions also appears to be heterogeneous across clades (Lewitus et al. 2018). Other factors have been postulated to contribute to rate heterogeneity across diatom lineages, such as their rate of division, their life style, resource acquisition strategies, cell volume, and their mode of reproduction (Kooistra and Medlin 1996).

0.6. Thesis Outline

The overall goal of my PhD was to advance our knowledge of diversification dynamics for microbial clades, using the neontological approach. There are several limitations to the application of phylogenetic diversification analyses to microorganisms. I tackled two of these limitations during my PhD. The first limitation is related to the difficulty of estimating the total diversity of microbial clade, and has given rise to the methodological development described in the first chapter of this PhD thesis. The second limitation is related to the often huge size of microbial phylogenies, which has motivated the development of a new inference framework presented in the second chapter. Finally, I explored the diversification of diatoms, a clade of micro-eukaryotes for which extensive resources are available and for which some of the limitations encountered in other microbial clades are reduced.

The phylogenies of microbial clades are typically incomplete. In the current diversification formalism, using incomplete phylogenies implies to account for the unseen present-day diversity, which can be easily achieved if robust diversity estimates are available. For microorganisms, diversity estimates are highly uncertain. In [Chapter 1](#), I develop phylogenetic diversification methods that account for under-sampling but do not require a fixed diversity estimate. I test these developments on simulated phylogenies, and apply our them to investigate the pace of bacterial diversification.

The phylogenies of microbial clades are typically large. This can induce a computational time limitation, associated to the calculation of likelihoods necessary to the statistical adjustment of birth-death models to the phylogenetic data. In [Chapter 2](#), I develop a likelihood-free deep learning framework for the inference of diversification rates, an approach that can solve such computational limitations. I develop a representation of phylogenetic data adapted to deep learning inference, as well as a deep learner trained on simulations from simple birth-death models. I assess the performance of this likelihood-free inference in comparison to maximum likelihood inference.

In [Chapter 3](#), I aim to characterize the tempo of diatoms diversification and explain their apparent rate heterogeneity. I examine the implication of disparate ecological strategies for diatoms diversification dynamics. To this end, I reconstructed a phylogeny of diatoms and characterized their ecological niche. I applied a recent diversification model to estimate lineage specific diversification rates, and tested the association between these rates and the ecological niche of the corresponding diatom species.

Finally, I summarize the main results I obtained, discuss some limitations, and provide perspectives to the field.

Chapter 1: Inferring Diversification Rates from Phylogenies when the Total Number of Species is Unknown, and Application to Bacterial Diversification

Phylogenetic diversification analyses are sensitive to under-sampling, i.e. extant species that are not represented in the phylogeny. In existing methods, under-sampling can be accounted for, however, it requires having an estimate of the total number of species in the clade. For species-rich clades such as coleopters, angiosperms or bacteria, species are still being discovered. The described species diversity often represents a minor fraction of the total diversity, and diversity estimates are highly uncertain. Studying the diversification of such largely unexplored species-rich clades is thus challenging. As a result of this limitation, among others, diversification analyses have been scarce in such groups and a comprehensive insight of their diversification dynamics is lacking.

To tackle this problem, I developed new approaches to estimate diversification rates from reconstructed phylogenies which alleviate the requirement of having a global diversity estimate of the studied clades. The proposed approaches rely on two different sampling schemes. The first is the usual Bernoulli sampling scheme, which assumes that each species in the phylogeny is sampled according to a probability, but here I integrate over this probability, which amounts to integrating over the unseen diversity. The second relies on sampling exactly the number of species observed in the phylogeny. Some models are adapted to handle single phylogenies while others handle a *forest* of phylogenies.

In this chapter, which corresponds to a project carried out with H  l  ne Morlon and Amaury Lambert and presented in the form of a manuscript is in its final stage of completion for submission, I introduced the proposed models and describe their performance on simulations. I also reevaluate diversification rates of bacterial clades using the best performing model in order to assess whether bacteria diversify at a unusual low pace.

The proposed models provides accurate net diversification and turnover rate for the best performing model. Nevertheless, it is at the expense of rather imprecise turnover rate estimates. When applied on bacterial phylogenies, low and constrained net diversification rates are retrieved suggesting that bacterial clades might diversifying at a slower pace than other organisms.

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Abstract

The birth-death process is widely used to infer the diversification dynamics of species groups through geological time from their reconstructed phylogenies. Current available methods to account for missing species in a given phylogeny require having a good estimate of the global diversity of the group, which hampers their applicability to species rich groups such as insects and microorganisms. Here, we develop and test approaches that do not require global diversity estimates. These approaches assume the constant rate homogeneous birth-death process and a uniform sampling process, using either the traditional Bernoulli sampling scheme (each species is sampled with a fixed probability) or the less-often used *k-sampling* scheme (exactly *k* species are sampled). We also develop extensions, applicable to a set of phylogenies, that assume a distribution of sampling probabilities over this set. Finally, we apply our best-performing approaches to bacterial phylogenies representing clades that have diversified from the Neoproterozoic oxygenation event 541 Million years ago to the present. We estimate an average net diversification rate of 0.01 event/lineage/Myr. Our approaches are implemented in the R package UDivEvo.

Code availability

The R package UDivEvo is available on GitHub at <https://github.com/sophia-lambert/UDivEvo/>.

1.1. Introduction

Birth-death models representing the speciation (rate λ) and extinction (rate μ) of lineages are widely used to study species diversification dynamics from species-level reconstructed phylogenies (Nee et al. 1994c). The expansion from the simple constant rate homogeneous birth-death model – where lineages diversify at the same constant rate – to more complex models, have enabled testing a variety of hypothesis related to deep time diversity dynamics (Stadler 2013b; Pennell and Harmon 2013; Morlon 2014a). These models include the possibility to account for diversification rate heterogeneities in both time (Stadler 2011a; Morlon et al. 2011a; Höhna 2013) and across lineages (Alfaro et al. 2009; Rabosky 2014; Höhna et al. 2019; Maliet et al. 2019a; Laudanno et al. 2020b; Barido-Sottani et al. 2020; Maliet and Morlon 2022), and to test the role of extrinsic (e.g. environmental, diversity) (Etienne et al. 2012; Condamine et al. 2013) and intrinsic (e.g. trait) (Maddison et al. 2007a; Herrera-Alsina et al. 2019) factors in explaining these heterogeneities.

One of the difficulties in applying birth-death diversification models to species-level phylogenies is that these phylogenies are often incomplete, in the sense that some extant species are not represented, for example because they have not been sequenced. Not accounting for missing species can severely bias diversification rate estimates, as missing species lead to missing branching events in the reconstructed phylogeny, in particular towards the present. The main approach that has been developed to overcome this difficulty consists in adding a sampling process to the speciation/extinction process during the diversification inference. The most widely used sampling process assumes a uniform Bernoulli sampling of extant species (also called *ρ -sampling*) under which each extant species is sampled with the same probability y (Nee et al. 1994c, 1994a; Yang and Rannala 1997; Stadler 2009). Importantly only two of the three model parameters (λ , μ and y) are identifiable from a single reconstructed phylogeny, such that the three parameters cannot be inferred jointly (Stadler 2009). In most empirical applications the sampling probability y is fixed – taken to be the ratio of the number of sampled lineages over the global present-day diversity N of the clade – and the inference is performed on λ and μ . An alternative sampling process (called *k -sampling* or *n -sampling*) consists in sampling exactly k species uniformly in the set of extant lineages (Stadler 2008, 2009; Höhna et al. 2011; Etienne et al. 2012a; Lambert and Stadler 2013a; Lambert 2018; Laudanno et al. 2020a). This sampling process is the one implemented in the diversity-dependent birth-death model (Etienne et al. 2012a), but its associated likelihood formula has been unpractical to compute when there are a lot of missing species (equation (18) (Lambert and Stadler 2013a)). Besides, until recently (Lambert 2018), the likelihood was conditioned on the number of extant species N , which then needs to be known (Etienne et al. 2012a; Lambert and Stadler 2013a; Laudanno et al. 2020a). Lambert (2018) derived a likelihood formula without this conditioning, but it has to our knowledge not been used. Another approach to deal with species that are not represented in the reconstructed phylogeny is to add them, for example using taxonomic information and/or polytomy resolvers (Cusimano et al. 2012; Thomas et al. 2013; Chang et al.

2020). Regardless of the approach used, all available methods require the total number of species in the clade to be known.

While the total diversity of clades is often relatively well-known for organisms such as birds, mammals, and more generally vertebrates, it is not for many plants, most invertebrates, and almost all microbial organisms (May 1988; Dykhuizen 1998; Hughes et al. 2001; Mora et al. 2011; Natarajan and Bhatt 2020). Numerous methods have been developed since the 40s (Fisher et al. 1943) to estimate species richness when a direct exhaustive inventory is impossible (Bunge and Fitzpatrick 1993; Colwell and Coddington 1994; Gotelli and Chao 2013; Bunge et al. 2014; Chao and Chiu 2016). These methods range from parametric (Curtis et al. 2002; Quince et al. 2008; Willis and Bunge 2015) to non-parametric approaches (Gotelli and Chao 2013; Chiu et al. 2014) and are based on either abundance or incidence data. Despite these efforts, global diversity estimates remain highly debated, in particular for groups such as microorganisms (Curtis et al. 2002; Quince et al. 2008; Locey and Lennon 2016; Vitorino and Bessa 2018; Louca et al. 2019). Given that applying birth-death models with incorrect sampling probability will lead to biased diversification estimates (Barraclough et al. 2009; Stadler 2009), the need to provide this probability in currently available approaches limits their applicability to well-described species groups.

In addition to the difficulty of estimating global diversity for microbial groups, other difficulties such as defining relevant taxonomic units and obtaining robust phylogenies have impeded the application of birth-death models to microorganisms. Indeed, the study of diversification dynamics is strongly biased towards macro-organisms ((Magallon and Sanderson 2001; Ricklefs 2006; McPeck and Brown 2007; Stadler 2011a; Morlon et al. 2011a; Rolland and Salamin 2016) to list a few). As a result, we possess a very limited understanding of the diversification dynamics that generated the majority of past and present diversity on Earth (Hug et al. 2016). For example, the mere question of the pace of diversification in the microbial realm is still uncertain by lack of extensive species inventory and fossil record. This represents a major knowledge gap, considering that microbial diversity has played and still plays a critical role in ecosystem functioning, such as in the regulation of major biogeochemical cycles. Despite the technical limits, there is an increasing number of attempts to study diversification dynamics in the microbial world. One of the first attempt was that of (Martin et al. 2004); since then diversification dynamics have been studied across prokaryotes (Marin et al. 2016), in bacteria (Morlon et al. 2012; Lorén et al. 2014; Sanglas et al. 2017; Louca et al. 2018), archaea (Gubry-Rangin et al. 2015), diatoms (Lewitus et al. 2018; Nakov et al. 2018; Pinseel et al. 2020) and fungi (Perez-Lamarque et al. 2020). These studies have opted for different ways of dealing with incomplete taxon sampling. Some have assumed that the number of species described in available datasets represents the global diversity (Marin et al. 2016; Scholl and Wiens 2016; Sanglas et al. 2017), others have used previously published global diversity estimates (Martin et al. 2004; Marin et al. 2016; Nakov et al. 2018), others have derived their own estimates (Lewitus et al. 2018; Louca et al. 2018; Perez-Lamarque et al. 2020; Pinseel et al. 2020), and lastly, others have checked the robustness of their results when using an arbitrarily large range of hypothetical sampling probabilities (Morlon et al. 2012; Lorén et al. 2014; Gubry-Rangin et al. 2015; Perez-Lamarque et al. 2020; Pinseel et al. 2020). While such studies rely on data that

is not as robust as for macro-organisms, they are one of the very few possibilities to study the past diversification dynamics of groups that have played a major role in Earth's history.

Patterns of diversification found in microorganisms have often been compared to those in macro-organisms. It has been hypothesized that diversification patterns may be fundamentally different in micro- versus macro-organisms, based on a variety of factors including their small size, high abundance, propensity for lateral gene transfer, and high dispersal and dormancy capabilities. Several studies have found lower net diversification rates in micro- compared to macro-organisms (Lorén et al. 2014; Marin et al. 2016; Scholl and Wiens 2016; Sanglas et al. 2017; Louca et al. 2018; Perez-Lamarque et al. 2020). An observation which also holds when comparing prokaryotes to eukaryotes (Marin et al. 2016; Scholl and Wiens 2016). Some studies have found higher relative extinction rates in micro- compared to macro-organisms (Marin et al. 2016; Louca et al. 2018), while others have found the opposite (Lorén et al. 2014; Sanglas et al. 2017). A handful of studies have investigated whether microbial diversification tends to slowdown as evolution proceeds, a pattern repeatedly found in macro-organisms (Moen and Morlon 2014a). Some studies have found evidence for such an “Early burst” pattern (Morlon et al. 2012; Gubry-Rangin et al. 2015; Perez-Lamarque et al. 2020; Pinseel et al. 2020) while others suggest that microbial diversification rates tend to be relatively constant through time (Martin et al. 2004; Lorén et al. 2014; Marin et al. 2016; Sanglas et al. 2017; Louca et al. 2018). Clearly, there is a need for many more such studies before we reach a comprehensive understanding of general microbial diversification patterns.

Here, we i) propose several approaches for estimating diversification rates from reconstructed phylogenies that do not require knowing or estimating global scale diversity, ii) test the performance of these approaches using simulations, and iii) use our best-performing approach on a set of bacterial phylogenies to reevaluate the question of whether bacteria have particularly low net diversification rate.

1.2. Methods

We consider the classical constant rate homogeneous birth-death process (Kendall 1948), which starts at time t_0 in the past with a single lineage. As time proceeds lineages experience speciation and extinction events, at constant per-lineage speciation and extinction rates, noted λ and μ . At present time ($t = 0$), lineages are sampled according to two potential sampling schemes. In the first (*Model 1*, Bernoulli sampling, also called ρ -sampling), each lineage is sampled independently with probability y (Stadler 2009). Only two of the three parameters λ , μ and y are identifiable, and it is therefore impossible to independently estimate all of them. However, we can assume that y is drawn from a prior “sampling distribution” ϕ , and integrate over this distribution to obtain estimates of λ and μ . In the second sampling scheme (*Model 2*,

k-sampling), exactly *k* lineages are uniformly sampled at present. We handle this sampling scheme by adapting recent mathematical results of (Lambert 2018) which rely on an explicit distribution of the sampling probability.

We also consider the possibility of drawing inferences from multiple phylogenies at once. The idea here is that if *y* is drawn from a sampling distribution that is common across phylogenies, the inference will be improved by this joint analysis, in which phylogenies act as replicates. We therefore consider the case of *p* independent constant rate homogeneous birth-death processes, which are sampled at present according to the Bernoulli sampling scheme, each with a specific (unknown) probability drawn from a common prior sampling distribution ϕ . We assume either that λ and μ are common to the *p* processes (*Model 3*), or that each process takes specific λ and μ values (*Model 4*).

We specify the likelihoods of these different models and use them in a Bayesian machinery to infer posterior distributions of diversification rates from reconstructed phylogenies sampled from clades of unknown total diversity. In the main text, we consider likelihoods conditioned on sampling at least one species in each sister clade of a crown phylogeny. The (least frequent) case of sampling at least one species of a stem phylogeny is considered in the [Supplementary Material](#).

1.2.1. Likelihoods for Birth Death Models when the Total Number of Species is Unknown

1.2.1.1. Likelihood under Bernoulli Sampling (*Model 1*, $BD_{\int \rho}$)

We assume here that each lineage is sampled with a probability *y* that is unknown, but sampled from a prior distribution ϕ . Integrating the likelihood formula in (Stadler 2010) over the possible values of the sampling probability *y*, we obtain the probability density of a reconstructed phylogeny (τ) with node depths noted $t_1 = t_{crown} > t_2 > \dots > t_{k-1}$, where *k* is the number of sampled tips in the phylogeny:

$$p_{BD_{\int \rho}}(\tau | t_{crown}, \lambda, \mu) = \int_0^1 \left(\frac{p_1(t_{crown}, \lambda, \mu, y)}{1 - p_0(t_{crown}, \lambda, \mu, y)} \right)^2 \prod_{i=2}^{k-1} \lambda p_1(t_i, \lambda, \mu, y) \phi(y) dy \quad [1]$$

where $p_1(t, \lambda, \mu, y)$ is the probability that a lineage alive at time *t* leaves exactly one descendant in the sample at present and $p_0(t, \lambda, \mu, y)$ is the probability that a lineage alive at time *t* leaves no descendant in the sample at present (Stadler 2010). In the [Supplementary Material](#), we detail how this expression relates to the one used in (Lambert 2018) in the case of Bernoulli sampling.

1.2.1.2. Likelihood under k -sampling (Model 2, BD_K)

Lambert (2018) derived the probability density of a reconstructed phylogeny (τ) originating at $t_0 = t_{stem}$, where k species have been sampled at present (k -sampling scheme, see Eq. 13 in that article and our [Supp Mat](#)). Here we adapt this expression to the case of a crown phylogeny $t_1 = t_{crown}$ where k_1 and k_2 species have been sampled in each sister clade:

$$p_{BD_K}(\tau|t_{crown}, \lambda, \mu, k_1, k_2) = \frac{k_1 k_2 (1-a)^2}{a^{k_1+k_2-2}} \int_0^1 \frac{1}{(1-a(1-y))^2} \prod_{i=2}^{k_1} \lambda p_1(t_i, \lambda, \mu, y) dy \int_0^1 \frac{1}{(1-a(1-y))^2} \prod_{l=k_1+1}^{k_1+k_2-1} \lambda p_1(t_l, \lambda, \mu, y) dy \quad [2]$$

where $a = 1 - \frac{1}{1 + \frac{\lambda}{\lambda - \mu}(e^{(\lambda - \mu)t_{crown}} - 1)}$ is the probability that all event times (of speciation and extinction events) are smaller than the time of the start of the process (here t_{crown}).

1.2.1.3. Likelihood under Bernoulli Sampling for Multiple Phylogenies with Common Diversification Rates (Model 3, $BD_{\int \rho}^{mult}$)

The probability density of a set of reconstructed phylogenies ($\tau_1, \dots, \tau_p$) with crown ages $t_{crown_1}, \dots, t_{crown_p}$, under the constant rate homogeneous birth-death process with rates λ and μ , Bernoulli sampling with unknown sampling probabilities independently drawn from a common prior distribution ϕ , and conditioned on survival, is given by:

$$p_{BD_{\int \rho}^{mult}}(\tau_1, \dots, \tau_p | t_{crown_1}, \dots, t_{crown_p}, \lambda, \mu) = \prod_{j=1}^p p_{BD_{\int \rho}}(\tau_j | t_{crown_j}, \lambda, \mu) \quad [3]$$

1.2.1.4. Likelihood under Bernoulli Sampling for Multiple Phylogenies with Distinct Diversification Rates (Model 4, $BD_{\int \rho}^{mult_x}$)

The probability density of a set of reconstructed phylogenies ($\tau_1, \dots, \tau_p$) with crown ages $t_{crown_1}, \dots, t_{crown_p}$, under the constant rate homogeneous birth-death process with rates $\lambda_1, \dots, \lambda_p$ and $\mu_1, \dots, \mu_p$, Bernoulli sampling with unknown sampling probabilities independently drawn from a common prior distribution ϕ , and conditioned on survival, is given by:

$$p_{BD_{\int \rho}^{mult_x}}(\tau_1, \dots, \tau_p | t_{crown_1}, \dots, t_{crown_p}, \lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_p) = \prod_{j=1}^p p_{BD_{\int \rho}}(\tau_j | t_{crown_j}, \lambda_j, \mu_j) \quad [4]$$

1.2.2. Implementation

We implemented our Bayesian inference in the R-package UDivEvo. We used the functions *fitMCMC_bdRho* to fit Models 1, 3 and 4, and *fitMCMC_bdK* to fit Model 2. These functions take a reconstructed phylogeny as input, and they output Bayesian posterior distributions of the rates of interest. We also implemented a function *sample_phylo* that samples complete phylogenies using either the Bernoulli sampling scheme or the *k-sampling* scheme conditioned on k_1 and k_2 .

1.2.2.1. Prior Sampling Distributions

For each of the models above, except *Model 2* which does not require choosing a prior distribution ϕ on y , we implemented either a uniform or a beta prior distribution.

The uniform distribution with parameters a and b ($a \geq 0, b \leq 1, a < b$) is given by:

$$\phi(y) = \begin{cases} \frac{1}{b-a} & \text{for } y \text{ in } [a, b] \\ 0 & \text{otherwise} \end{cases}$$

This distribution reflects the *a priori* hypothesis that all sampling probabilities between a and b are equally likely. a and/or b can either be fixed (e.g. $a = 0$ and $b = 1$, or $a > 0$ and $b < 1$ if prior knowledge on unrealistic sampling probabilities exists) or inferred jointly.

The beta distribution with parameters α and β ($\alpha > 0, \beta > 0$) is given by:

$$\phi(y) = \frac{(\alpha+\beta-1)!}{(\alpha-1)!(\beta-1)!} y^{\alpha-1}(1-y)^{\beta-1}$$

This flexible distribution can for example reflect the *a priori* hypothesis that sampling probabilities are concentrated around an intermediate value between 0 and 1 (when $\alpha > 1, \beta > 1$), illustrating similar sampling efforts across clades, or on the contrary that they are concentrated around 0 and 1 (when $\alpha < 1, \beta < 1$), with both well-sampled clades (e.g. cultivable organisms) and poorly sampled clades (e.g. clades made of rare species). As above, α and/or β can either be fixed or inferred jointly.

1.2.2.2. Inference

We parametrized the four models to infer the net diversification rate ($r = \lambda - \mu$) and the turnover rate ($\varepsilon = \mu/\lambda$ also known as the relative extinction rate), as these parameters are typically estimated with more precision than λ and μ (Nee et al. 1994c, 1994a) and the turnover rate is conveniently distributed between 0 and 1 (Nee et al. 1994c; Paradis 2004; Parag and Pybus 2018). We implemented the Bayesian machinery of the R package BayesianTools (Hartig et al. 2019) and DEzs sampler in our *fitMCMC_bdRho* and *fitMCMC_bdK* functions to obtain posterior distributions of the parameters r and ε . The *fitMCMC_bdRho* function provides

the additional possibility to output posterior distributions of the parameters controlling the shape of the sampling distribution ϕ (a , b for a uniform distribution and α and β for a distribution) if they are not fixed. In practice, as the diversification rates and sampling fraction are not identifiable from a single reconstructed phylogeny (Stadler 2009), we cannot hope to infer the parameters of the sampling distribution from a single phylogeny either. In this case they should thus be fixed, for example using a uniform distribution on $[0,1]$, or based on *a priori* knowledge on the shape of the sampling probability distribution. We used three MCMC chains initiated with different starting values, either given by the user or sampled in the prior distributions associated to each parameter. These priors are specified by the user. Additional options are available for parallelization of the MCMC chains and checkpointing.

1.2.3. Tests on Simulations

We assessed the ability to recover accurate r and ε values from phylogenies simulated under the different models using simulations conditioned on crown age from the R package TESS (Höhna 2013). To test *Models 1* and *2* (i.e. inferences on a single phylogeny at a time), we used 10 speciation rates values (0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1 event/lineage/million years) and 11 turnover rate values (0.01, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.99). For each simulation, we pre-assigned a sampling probability y sampled from ϕ and chose the crown age such that the expected sampled diversity at present is 500 tips. If it exceeded 4,000 Myrs, we resampled all parameters. We simulated 100 phylogenies for each parameter combination, resulting in a total of 11,000 phylogenies, and discarded simulations resulting in less than 3 tips (these represented less than 4% of the 11,000 simulated phylogenies).

For *Model 1* (Bernoulli sampling), we used for the sampling distribution ϕ a uniform distribution on $[0,1]$. We subsampled each phylogeny according to the Bernoulli sampling scheme using our *sample_phylo* function. The resulting subsampled phylogenies had between 33 and 2724 tips. We ran our *fitMCMC_bdRho* function on each subsampled phylogeny, fixing ϕ to be the uniform distribution on $[0,1]$.

For *Model 2* (*k-sampling*), in order to homogenize sample sizes for fair comparison, we set the size k of each subsampled phylogeny to that obtained for *Model 1*, that we randomized across complete phylogenies with enough tips. Next, for each phylogeny we sampled k_1 uniformly in values from 1 to $k - 1$ and set k_2 equals to $k - k_1$. We subsampled each phylogeny using our *sample_phylo* function, and ran our *fitMCMC_bdK* function on each of these subsampled phylogenies.

To test inferences on multiple phylogenies (*Models 3 and 4*), we simulated a total of 100 sets of 10 phylogenies for each model. For *Model 3* (common diversification rates over each set), for each set the common speciation rate was sampled uniformly in $[0.01, 0.1]$, and the common turnover rate was sampled uniformly in $[0, 1]$. For *Model 4* (distinct diversification rates within each set), the speciation rate was sampled uniformly in $[0.01, 0.1]$ and the turnover

rate was sampled uniformly in $[0, 1]$ independently for each phylogeny. We used for ϕ either a uniform distribution on $[a, b]$, with $b - a$ uniformly sampled in $[0, 1]$ and a uniformly sampled in $[0, 1 - (b - a)]$, or a beta distribution with β fixed to 3 and α values ranging from 1 to 7 obtained by uniformly sampling the mode of the distribution in $[0, 0.75]$. The fixed β value of 3 was initially suggested by fitting a beta distribution to estimates of the sampling probability obtained for bacteria ([Supplementary Material](#)). In both cases, we obtained sampling probabilities for each phylogeny by drawing independently from ϕ , and chose the crown age in order to have an expected sampled diversity per phylogeny at present-time of 500 tips and not exceeding 4,000 Myrs. We then subsampled all phylogenies according to the Bernoulli sampling scheme. The resulting size of the subsampled phylogenies ranged from 8 to 3097 tips for *Model 3* with the uniform sampling distribution, 6 to 4013 for *Model 4* with the uniform distribution, 7 to 3544 for *Model 3* with the beta distribution, and 6 to 2955 for *Model 4* with the beta distribution. We ran our *fitMCMC_bdRho* function on each set of 10 phylogenies, estimating both the diversification parameters and a and b in the case of the uniform distribution and α (with β fixed to 3) in the case of the beta distribution.

We ran the MCMC for 20,000 iterations for the phylogenies simulated under *Model 1*, 2 and 3 and 200,000 iterations for the phylogenies simulated under *Model 4*. We used uniform priors on $[0,1]$ for r , ε , a and $b - a$, and an exponential prior with parameter 0.01 for α . Three chains were run for each inference with the starting values of 1 out of the 3 chains calculated using the methods of moments and assuming $\varepsilon = 0$ while the others starting values were sampled from their prior. We checked that the number of iterations was sufficient for our 3 chains to reach good convergence (Gelman statistic lower than 1.05), exploration (acceptance ratio higher than 0.234), and mixing (effective sample size higher than 200). If at least one of these three criteria was not satisfied, we doubled the number of iterations. If this was still not sufficient, we discarded the corresponding phylogenies from the rest of the analysis. This corresponded to less than 2% of the single phylogenies inference and 11% of the multiple phylogenies inference under common diversification rates. We did not use the acceptance ratio as a criterion to discard multiple phylogenies inference with distinct diversification rates (i.e. *Model 4*), as this criterion is typically not used when the number of parameters to infer increases. We discarded less than 60% of these inferences. We compared the use of different metrics as our point estimates of the parameters θ (noted $\hat{\theta}$), and retained the maximum *a posteriori* probability (MAP) of the posterior distribution as the most consensual metric across parameters and models ([Supplementary Mat](#)). We also reported the posterior credible interval as the 95% highest posterior density interval (HPDI).

We measured the accuracy of all estimates (r , ε , and the parameters of $\phi(y)$) as the slope of the quantile regression and the Pearson correlation coefficient, and the precision as the absolute relative error $|\frac{\hat{\theta} - \theta}{\theta}|$ (ARE) between estimated and true (simulated) parameters. Further, we assessed the coverage probability of each parameter at a given % value X as the proportion of times in the simulations when the true parameter value falls within the $X\%$ HPDI. As the asset of Bayesian inference relies on exploiting the full posterior probability instead of a point estimate, we investigated the fidelity of the posterior distribution in representing the

distribution of the probabilities that a given parameter value is correct, by comparing the coverage probability at different X% HPDI, for X ranging from 5 to 95.

We also fitted *Model 1* on simulations under *k-sampling* and *Model 2* on simulations under Bernoulli sampling to investigate the performance of the models under model violation. Similarly, we fitted *Model 3* on simulations with specific diversification rates. Lastly, we tested the robustness of *Model 3* and *4* to model violation by assuming a prior on the sampling probability distribution different from the one used in the simulations (uniform prior on simulations obtained with a beta distribution).

For comparison with currently available inferences under the birth-death process, we compared the accuracy of estimates obtained under Models *1* and *2* (inference on a single phylogeny) to that obtained: i) when the sampling probability y is known and the Bernoulli sampling scheme is used (*Model 5*, $BD_{\rho_{known}}$), ii) when the phylogeny is assumed to be fully sampled even if it is not (*Model 6*, $BD_{\rho=1}$), iii) when using a sampling probability uniformly sampled in $[0,1]$ (*Model 7*, $BD_{\rho \in [0,1]}$), iv) when the sampling probability is unknown and the likelihood formula is reparametrized such that the inference is performed on the net diversification rate ($r = \lambda - \mu$) and the weighted speciation rate $y\lambda$ (*Model 8*, $BD_{\rho\lambda}$) rather than on r and ε . The latter option is the only alternative that does not require knowing the sampling fraction, and it allows inferring only one quantity (r) out of the two of interest (r and ε). We fitted these models using appropriate options in our *fitMCMC_bdRho* function; for these models 10,000 MCMC iterations were enough to obtain satisfactory convergence, exploration and mixing. We also compared the accuracy of *Model 3* and *4* (inference on several phylogenies) with that obtained under models where the sampling probabilities are known and diversification rates are either shared across phylogenies (*Model 5* “*mult*”, $BD_{\rho_{known}}^{mult}$) or not (*Model 5* “*mult_x*”). We applied *Model 5* “*mult*” on the set of phylogenies simulated under the uniform sampling probability distribution. Finally, we compared the performance of *Model 4* to the case of applying *Model 2* (*k-sampling*) independently to each phylogeny.

1.2.4. Bacterial Diversification

We applied our best performing models (BD_K in the case of single phylogenies and $BD_{\int \rho}^{mult_x}$ in the case of multiple phylogenies) to bacterial phylogenies obtained from (Louca et al. 2018). The authors constructed a phylogeny from partial-length 16S rRNA alignments (V4 region) from 60 distinct studies spanning various environments, clustered into Operational Taxonomic Units (OTUs) at 97% similarity. They also extracted from a 99% bacterial phylogeny the part corresponding to cyanobacteria, as this group played a major role in Earth’s evolution, in particular in its oxygenation. We applied the BD_K model to the full bacterial phylogeny as well as the full cyanobacteria phylogeny.

Next, we considered all subclades with at least 50 extant OTUs originating after the Neoproterozoic oxygenation event (NOE) 541 MA (Million years ago). This resulted in the

analysis of 42 clades from the bacterial phylogenies 97%, and 6 clades from the cyanobacteria phylogeny at 99%. We applied our $BD_{\int \rho}^{mult_x}$ model (*Model 4*) under the uniform sampling distribution to these sets of phylogenies.

1.3. Results

1.3.1. Performance of the Models

1.3.1.1. Inference on a Single Phylogeny

The net diversification rate r of the constant rate homogeneous birth-death process is well recovered with several inference procedures, even when the sampling probability is not known (*Model 1*, 2 and 8, [Fig. 1](#)). The best accuracy is reached by the reparametrized *Model 8* $BD_{\rho\lambda}$ ([Fig. 1f](#)), however this model does not allow inferring ε . *Model 1* $BD_{\int \rho}$ yields as accurate and precise r estimates as when the sampling probability is known (*Model 5* $BD_{\rho_{known}}$, [Fig. 1a,c](#) and [Supplementary Fig. 1a](#)). *Model 2* BD_K is slightly less accurate and precise in estimating r than *Model 1*, 5 and 8 with a mild underestimation ([Fig. 1b](#)) and some outliers ([Supplementary Fig. 1a](#)) while still being more accurate than when assuming complete sampling (*Model 6*, $BD_{\rho=1}$, [Fig. 1d](#)) or a randomly sampled sampling probability (*Model 7*, $BD_{\rho \in [0,1]}$, [Fig. 1e](#)). Assuming a sampling probability of 1 in the presence of missing species (*Model 6*) or assuming a wrong sampling probability (but not systematically overestimating it as in *Model 7*) leads to an underestimation of r on average ([Fig. 1d,e](#)). When the sampling probability is not known or uncertain, it is thus preferable to apply one of the proposed models (*Model 1* and 2) or *Model 8* for estimating the net diversification rate of a reconstructed phylogeny than to assume a wrong sampling probability.

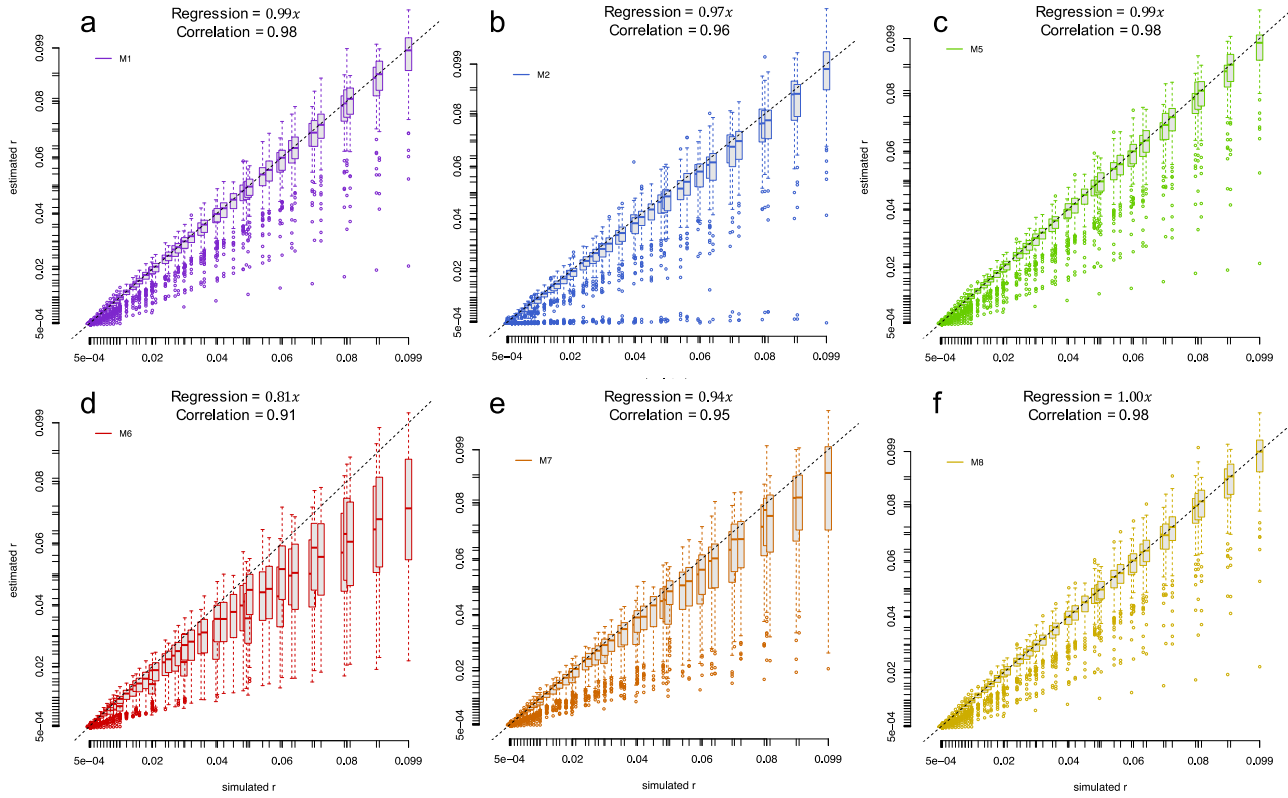


Figure 1: Accuracy of net diversification rate estimates (r) across different inference procedures.

Box plots of MAP net diversification rate estimates as a function of simulated values for 10,429 phylogenies for **a**, Model 1, BD_{ρ} , birth-death process under Bernoulli sampling scheme which assumes a uniform prior on the sampling probability **b**, Model 2, BD_K , birth-death process under k -sampling scheme **c**, Model 5, $BD_{\rho_{known}}$, birth-death process under Bernoulli sampling scheme with known sampling probability **d**, Model 6, $BD_{\rho=1}$, birth-death process under Bernoulli sampling scheme assuming complete sampling **e**, Model 7, $BD_{\rho \in [0,1]}$, birth-death process under Bernoulli sampling scheme assuming a random sampling probability and **f**, Model 8, $BD_{\rho\lambda}$, birth-death process under Bernoulli sampling scheme reparametrized. The horizontal bold line corresponds to the median of the estimates for each simulated r . The box bounds display the 75% and 25% quantiles of the estimates. Each dot is an outlier. The dashed line represents the 1:1 relationship. The value of the slope of the quantile regression and the Pearson correlation coefficient are displayed for each model in the top.

As expected, turnover rates ε are harder to estimate than net diversification rates – the median absolute relative error differs by at least two to ten folds ([Supplementary Fig. 1a,b](#)) – an observation reported as early as in (Nee et al. 1994a). Among the two models proposed when the sampling probability is unknown (Models 1 and 2), Model 2 is the most accurate, yielding an accuracy similar to the one when the sampling probability is known (Model 5) while having a lower precision ([Fig. 2b,c](#) and [Supplementary Fig. 1b](#)). Model 1 tends to underestimate ε , especially for low to moderate values ([Fig. 2a](#)). As expected, ignoring under-sampling as in Model 6 ([Fig. 2d](#)) results in large underestimates of ε . Finally, setting the sampling probability

randomly as in *Model 7* results in overestimates of ε , as well as very imprecise estimates ([Fig. 2e](#) and [Supplementary Fig. 1b](#)).

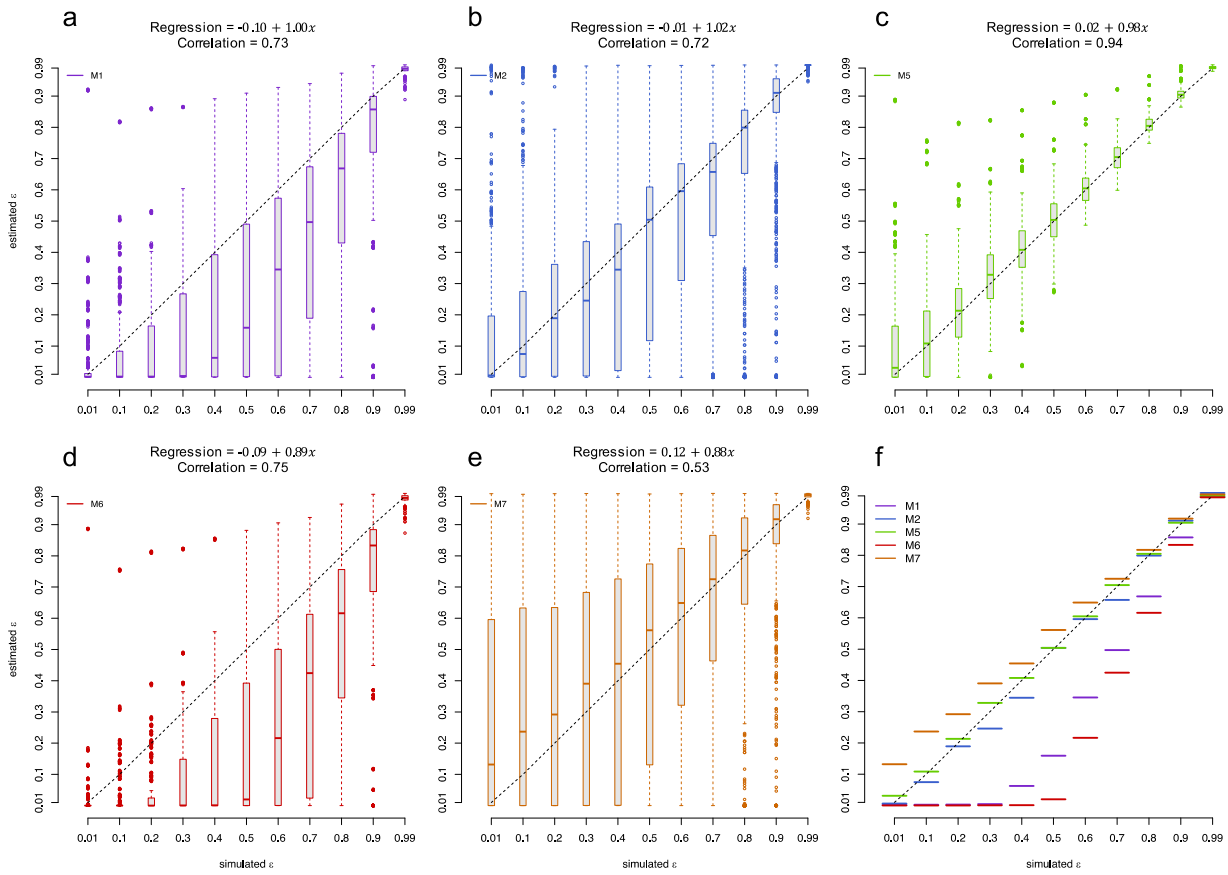


Figure 2: Accuracy of turnover rate (ε) estimates across different inference procedures.

Box plots of MAP turnover rate estimates as a function of simulated values for 10,429 phylogenies for **a**, $BD_{\int \rho}$ **b**, BD_K **c**, $BD_{\rho_{known}}$ **d**, $BD_{\rho=1}$ **e**, $BD_{\rho \in [0,1]}$ and **f**, all above mentioned models. The horizontal bold line corresponds to the median of the estimates for each simulated ε . The box bounds display the 75% and 25% quantiles of the estimates. Each dot is an outlier. For visualization purpose, the last panel **f** only displays the median of ε estimates for each simulated ε across all models. The dashed line represents the 1:1 relationship. The value of the parameters of the quantile regression and the Pearson correlation coefficient are displayed for each model in the top.

The coverage probability at different X% is close to the value X for both the net diversification rate r and the turnover rate ε (or the weighted speciation rate $\gamma\lambda$) for *Models 1, 5 and 8* ([Fig. 3](#)), meaning that in these cases, the posterior distribution provides a faithful representation of the distribution of probabilities that a given parameter value is correct. This also applies to the posterior distribution of the net diversification rate for *Model 2* and to some extent to the posterior distribution of the turnover rate, which reaches a coverage probability of 88% with the 95% HPDI ([Fig. 3a](#)). These posterior distributions (and therefore credibility intervals) are narrow for the net diversification rates, regardless of the inference procedure used ([Supplementary Fig. 1d](#)), but wide for the turnover rates, and even wider when the sampling

probability is not known (*Models 1 & 2*, [Supplementary Fig. 1e](#)). Under model violation, we expect the coverage probability at $X\%$ to be below X . For example, *Model 6* and *7* which assume an incorrect sampling probability have a coverage probability at $X\%$ substantially lower than X ([Fig. 3](#)).

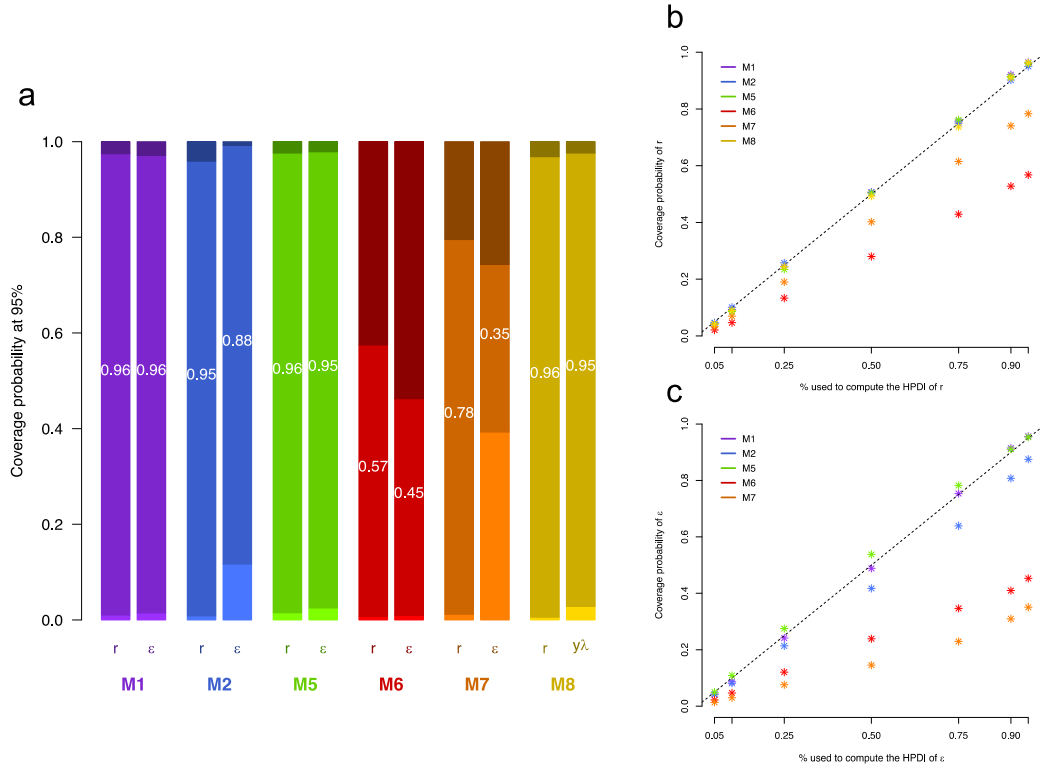


Figure 3: Coverage probabilities across different inference procedures.

Coverage probability of the net diversification rate r , the turnover rate ϵ or the weighted speciation rate $y\lambda$ on 10,429 simulated phylogenies for $BD_{\mathcal{F}\rho}$, BD_K , $BD_{\rho_{known}}$, $BD_{\rho=1}$, $BD_{\rho \in [0,1]}$ and $BD_{\rho\lambda}$. **a**, Coverage probability at 95% for r , ϵ and $y\lambda$. The value inside each barplot indicates the proportion of phylogenies for which the simulated parameter value is included inside the 95% HPDI, the lighter and darker bars respectively indicate the proportion of phylogenies for which the simulated value is below or above the 95% HPDI. **b**, Coverage probability of r at $X\%$ versus X . **c**, Coverage probability of ϵ at $X\%$ versus X . Each dot corresponds to the proportion of phylogenies for which the simulated parameter value is included inside the corresponding HPDI at (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)%. The dashed line displays the 1:1 relationship.

Model violation in terms of erroneous sampling scheme has an overall small effect on the inferences ([Supplementary Fig. 2, 3, 4 & 5](#)). Assuming a Bernoulli sampling scheme (*Model 1*) on a phylogeny sampled under a k -sampling scheme decreases the accuracy of net diversification rate estimates slightly ([Supplementary Fig. 2a](#)), while assuming a k -sampling scheme (*Model 2*) on a phylogeny sampled under a Bernoulli sampling scheme does not ([Supplementary Fig. 2b](#)). Model violation reduces error on turnover rate estimates for *Model 1*, likely because it biases ϵ in the opposite direction than the existing bias ([Supplementary Fig. 3a, 4b](#)). In contrast, applying *Model 2* to phylogenies sampled under a Bernoulli sampling

scheme reduces the accuracy and precision of turnover rate estimates ([Supplementary Fig. 3b, 4b](#)). Still, even with this increased bias, *Model 2* tends to provide more accurate turnover rate estimates than *Model 1* under all circumstances. The effect of sampling scheme violation on coverage probabilities is rather marginal ([Supplementary Fig. 5](#)).

1.3.1.2. Inference on Multiple Phylogenies

The net diversification and turnover rate estimates are well recovered by *Model 3* ($BD_{\int \rho}^{mult}$) with a similar accuracy as in *Model 5* “mult”($BD_{\rho_{known}}^{mult}$) irrespective of the prior common sampling distribution (beta or uniform) but depends on the metric used to summarize the posterior distribution ([Fig. 4a,b](#) and [Supplementary Fig. 14e & 16b](#)). The best precision is reached for *Model 3* under a beta sampling distribution (*M3Beta*), while under a uniform sampling distribution (*M3Unif*) the turnover rate estimates are less precise ([Fig. 4a,b](#) and [Supplementary Fig. 12a,b, 16b & 18b](#)). This is probably due to the fact that the version of *M3Beta* tested here took a fixed known β , thus lowering the number of parameters to infer (from four to three), which convey a slight advantage for *M3Beta* over *M3Unif*.

The parameters of the sampling distribution recovered under *Model 3* are biased when considering the MAP ([Supplementary Fig. 13a,c,d](#)). Specifically, α (*M3Beta*) tends to be overestimated, while a (*M3Unif*) tend to be underestimated, with rather imprecise estimates ([Supplementary Fig. 13a,c,d](#)). The resulting estimate of the mode of the sampling distribution for *M3Beta* is reasonably accurate ([Supplementary Fig. 13b](#)). The bias is lower when considering a statistic better fitted to summarize the peak of the posterior probabilities of the sampling probability parameters ([Supplementary Fig. 15 & 16c,d](#)). The parameter b is not accurately, nor precisely recovered independent of the metric used ([Supplementary Fig. 13d & 15c](#)).

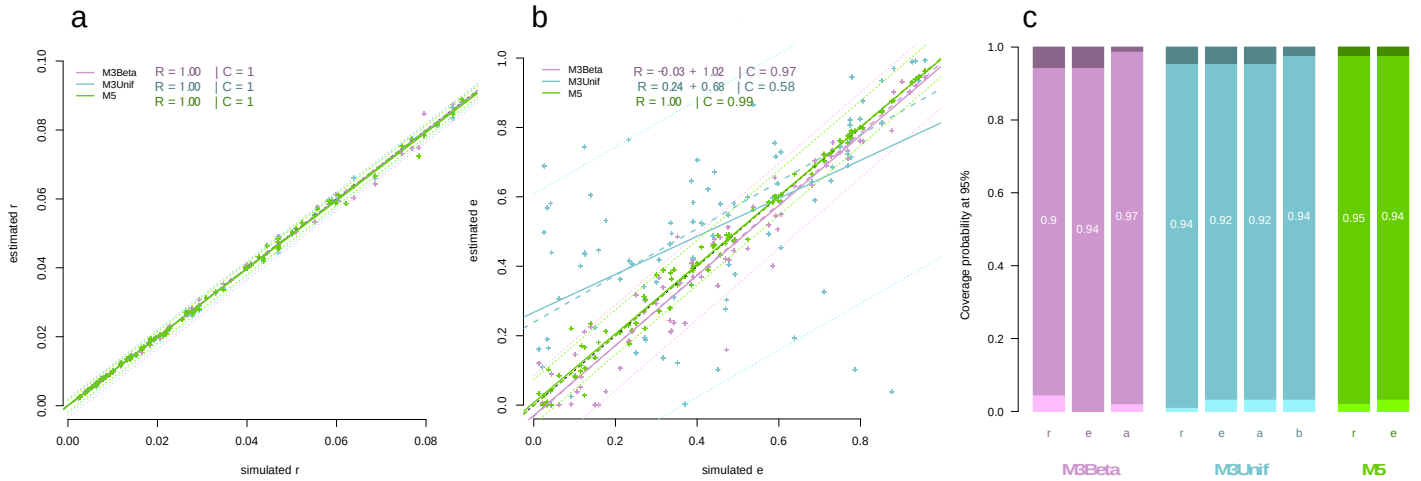


Figure 4: Accuracy of parameter estimates and coverage probabilities across inference procedures with multiple phylogenies under common diversification rates.

MAP estimates versus simulated values for 89 sets of 10 phylogenies for **a**, the net diversification rate and **b**, the turnover rate for: Model 3 (BD_{ρ}^{mult}), when assuming a common beta sampling distribution (M3Beta in light pink), Model 3 when assuming a common uniform sampling distribution (M3Unif in light blue) and Model 5 “mult”, $BD_{\rho_{known}}^{mult}$, (M5 in green). The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. The value of the parameters of the quantile regression (R) and the Pearson correlation coefficient (P) are displayed for each model in the top. The coverage probability at 95% is displayed in **c**, for the net diversification rate r , the turnover rate e and the parameters a (for a beta sampling distribution), or a and b (for a uniform sampling distribution) for: M3Beta (in light pink), M3Unif (in light blue) and M5 (in green). The value inside each barplot indicates the proportion of phylogenies for which the simulated parameter value is included inside the 95% HPDI, the lighter and darker bars respectively indicate the proportion of phylogenies for which the simulated value is below or above the 95% HPDI.

The coverage probability at different X% of Model 3 and 5 is close to X for the net diversification rate r , the turnover rate e as well as for a and a (Fig. 4c, Supplementary Fig. 17a,b,c), but not for b . The latter has a higher coverage probability than expected for intermediate X (Supplementary Fig. 17d). The higher coverage probability indicates that the confidence interval is conservative (containing the true value more than expected) and might be due to a large posterior probability. This may lead to difficulties when interpreting the results only when the confidence interval is indeed covering a large portion of the possible value. The posterior distributions (and therefore credibility intervals) are narrow for the net diversification rate, regardless of the inference procedure used (Supplementary Fig. 12c), but wide for the turnover rate, in particular when the sampling probability is not known and a uniform sampling distribution is assumed (Supplementary Fig. 12d).

The net diversification rate and turnover rate are overall correctly recovered by *Model 4* ($BD_{\int \rho}^{mult_x}$) but strictly depends on the model and the metric used to summarize the posterior distribution ([Supplementary Fig. 20a,b](#)). The best accuracy in net diversification and turnover rate estimates is recovered for *Model 4* ($BD_{\int \rho}^{mult_x}$) under uniform sampling distribution (*M4Unif*) as it yields estimates close to the accuracy of *Model 5* “*mult_x*” ($BD_{\rho_{known}}^{mult_x}$) ([Fig. 5a,b](#), [Supplementary Fig. 21a,b,c,d](#)). As observed for other proposed models, the net diversification rate is as precise and the turnover is less precise than the model under known sampling probability ([Fig. 5a,b](#), [Supplementary Fig. 21b,c](#)). The net diversification rates are slightly biased ([Fig. 5a](#), [Supplementary Fig. 21a](#)), for *Model 4* ($BD_{\int \rho}^{mult_x}$) under beta sampling distribution (*M4Beta*) and the turnover rates are particularly biased ([Fig. 6b](#), [Supplementary Fig. 21c](#)). Contrary to *Model 3*, the precision of *Model 4* under the beta sampling distribution is lower than under the uniform one for both the net diversification and the turnover rate estimates ([Fig. 5a,b](#), [Supplementary Fig. 21b,d](#)). This might be due to the fact that the simulations have resulted in smaller tree size under beta sampling in *Model 4* compared to under uniform sampling since the sampling fraction tend to be smaller in the former. When comparing *Model 4* to *Model 2* applied independently on each phylogenies of each set, the best performance is retrieved for *Model 4* under uniform sampling distribution as *M4Unif* outperforms *M2* and *M4Beta* particularly in term of precision ([Supplementary Fig. 21d](#)).

The parameters of the sampling distribution for *Model 4* ($BD_{\int \rho}^{mult_x}$) are particularly imprecise and substantially deviate from a 1 to 1 relationship between simulated and estimated values ([Supplementary Fig. 21e,f,g,h](#)). The coverage probabilities at X% are close to X for all models, unless for the *b* parameter of *M4Unif* ([Supplementary Fig. 19d](#)).

In terms of model violation, assuming a wrong shape for the sampling distribution has a negligible impact on the accuracy of parameter estimates ([Supplementary Fig. 22a,b](#)). In contrast, violating the assumption of common diversification rates (i.e. applying *Model 3* in the presence of rate heterogeneity) has a drastic impact on all parameter estimates ([Supplementary Fig. 22a,b](#)): the turnover rate is systematically overestimated and consequently the net diversification rate is widely underestimated.

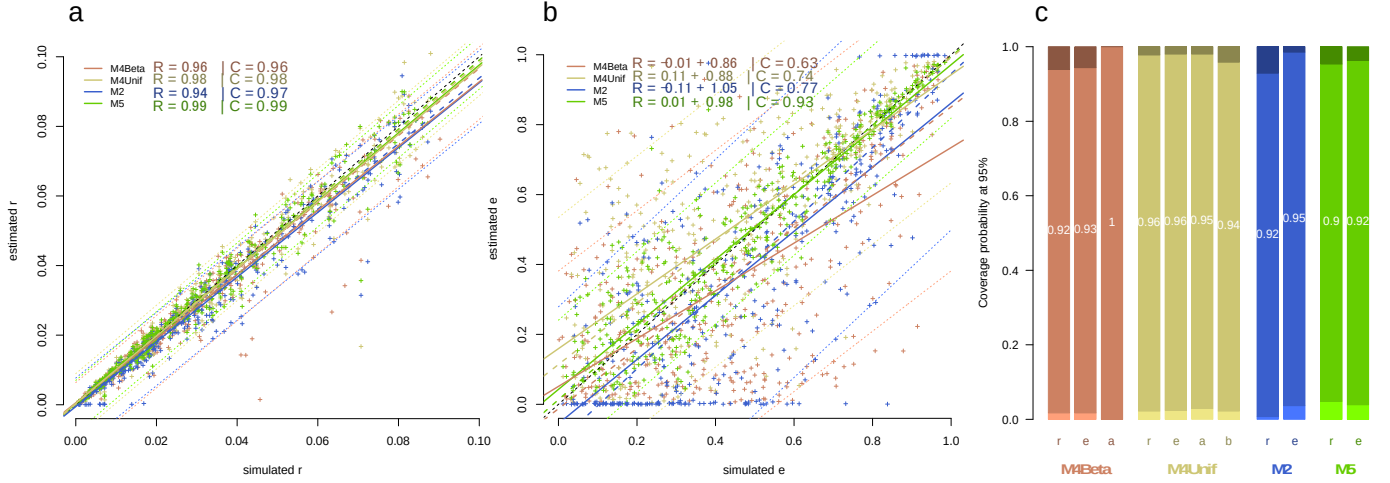


Figure 5: Accuracy of parameter estimates and coverage probabilities across inference procedures for multiple phylogenies under specific diversification rates.

MAP estimates versus simulated parameters dot plot on 41 sets of 10 simulated phylogenies for **a**, the net diversification rate **b**, the turnover rate of: Model 4 ($BD_{\int \rho}^{mult_x}$) when assuming a common beta sampling distribution (M4Beta in salmon), Model 4 when assuming a common uniform sampling distribution (M4Unif in khaki) and Model 5 “ $mult_x$ ” ($BD_{\rho_{known}}^{mult_x}$, M5 in green). Each dot is an estimate for a set of 10 simulated phylogenies. The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. The value of the parameters of the quantile regression (R) and the Pearson correlation coefficient (P) are displayed for each model in the top. The coverage probability at 95% is displayed in **c**, for the net diversification rate r , the turnover rate ϵ and the parameters α (for a beta sampling distribution), or a and b (for a uniform sampling distribution) for: M4Beta (in salmon), M4Unif (in khaki) and M5 (in green). The value inside each barplot indicates the proportion of phylogenies for which the simulated parameter value is included inside the 95% HPDI, the lighter and darker bars respectively indicate the proportion of phylogenies for which the simulated value is below or above the 95% HPDI.

1.3.2. Diversification of Bacterial Clades

In agreement with Louca et al. (2018), we found relatively low net diversification rates for bacteria in comparison to macro-organisms (Alfaro et al. 2009; Tank et al. 2015; Jetz and Pyron 2018; Fernández et al. 2018; Maliet et al. 2019a; Siqueira et al. 2020) and to a lesser extent micro-eukaryotes (Lewitus et al. 2018; Nakov et al. 2018; Pinseel et al. 2020). For the full bacteria phylogeny clustered at 97% similarity, we found a net diversification rate estimate of 0.0026 events per lineage per million year (Fig. 6a). The net diversification rate of the full cyanobacteria phylogeny clustered at 99% similarity was even lower, estimated at 0.0018 event/lineage/Myr (Fig. 6a). Particularly, low net diversification rate in cyanobacteria has previously been shown (Louca et al. 2018) and observed (Schopf 1987, 1994). This pattern

has since long been entitled *hypobradytely* “to refer to the extremely low rate of evolutionary change exhibited by cyanobacteria taxa” (Schopf and Klein 1992) – which at first referred to low morphological diversification. Net diversification rate estimates of the post Neoproterozoic oxygenation event clades are one order of magnitude higher (on average 0.01 event/lineage/Myr for bacteria and 0.007 for cyanobacteria, [Fig. 6a](#)), which remains the lower bound of what is typically found in macro-organisms. Net diversification rates appear quite constrained across these bacterial clades, with at most a two- to three-fold variation ([Fig. 6c](#)).

Turnover rate estimates should be taken with caution given our results on simulations. The posterior distribution of the bacteria turnover rate is rather flat ([Fig. 6b](#)) and therefore hard to interpret. The posterior distribution of the cyanobacteria turnover rate is much narrower, and suggests a high turnover rate (0.76 event/lineage/Myr) ([Fig. 6b](#)). The high turnover rate found in cyanobacteria corroborates a previous result (Louca et al. 2018) where high turnover rate was also retrieved globally for bacteria. Across post NOE clades, turnover rates are highly variable. Posteriors tend to peak at either low or high turnover rate values, but not intermediate values ([Fig. 6d](#)), suggesting that some bacterial clades are poorly affected by extinction while others are highly volatile. Lastly, the parameters of the sampling distribution appear particularly low (0.05 and 0.09 for respectively the lower and upper bound of the sampling distribution) for bacterial clades ([Fig. 6e](#)) which may suggest that a large portion of the global diversity is missing. Nevertheless, caution need to be taken while interpreting this result as the upper bound parameter is particularly badly estimated in the simulation analyses.

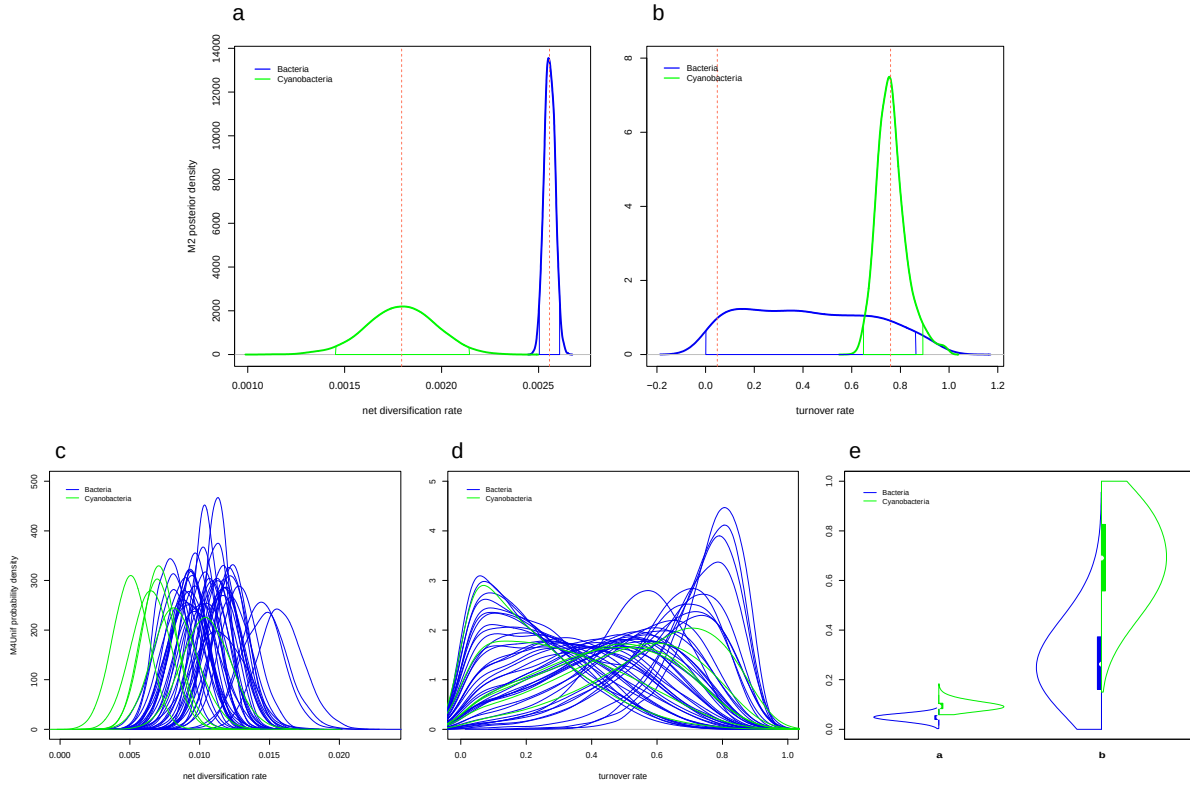


Figure 6: Bacteria and cyanobacteria diversification rates estimates.

Posterior density of **a**, the net diversification rate and **b**, the turnover rate on reconstructed phylogenies of denovo bacteria (at 97% similarity) (in blue) and cyanobacteria only (at 99% similarity) (in green) from (Louca et al. 2018) estimated using BD_K . The colored area under the density represents the 95% HPD and the horizontal dashed line represent the median for **a** and the harmonic mean for **b** of the posterior density. Posterior density of **c**, net diversification rate and **d**, turnover rate on reconstructed phylogenies of denovo bacteria (at 97% similarity) and cyanobacteria only (at 99% similarity) that diversified after the Neoproterozoic oxygenation event (NOE) as well as **e**, parameters of the sampling distribution a and b estimated using $BD_{\int \rho}^{mult_x}$, when assuming a common uniform sampling distribution (M4Unif).

1.4. Discussion

We proposed different approaches to study diversification rates from phylogenies when the total number of species is not known. We found that the net diversification rate can be estimated with confidence using several methods, and that the turnover rate can be estimated without bias but with a lot of uncertainty. Our approaches are based either on the often-used Bernoulli sampling scheme, or on the less known k -sampling scheme, and we found that the k -sampling scheme performs best. We also found that when analyzing several clades with unknown diversity, a joint analysis assuming a Bernoulli sampling scheme with a common distribution

of sampling probabilities performs better than independent analyses on each phylogeny. Applying our approaches to bacterial diversification, we found overall lower net diversification rates for bacteria than for macro-organisms, and variable turnover rates across bacterial clades.

On an empirical standpoint, the Bernoulli sampling scheme – which assumes that each species is sampled with a given probability – seems more realistic than the *k-sampling* scheme – which assumes that exactly *k* species are sampled. Undoubtedly, the number of sampled species is rarely fixed before any sampling campaign, and the representation of a given species in a clade results from a suite of events (collection, sequencing) that each occurs with a given probability. However, we found that assuming a *k-sampling* scheme when the sampling in fact followed a Bernoulli scheme does not induce a strong bias, and that *Model 2 (k-sampling)* performs better in most circumstances. We therefore recommend to use this model.

Both the Bernoulli and *k-sampling* schemes assume uniform sampling, while in practice sampling can be diversified (when trying to maximize the representation of distantly-related species), as for example in higher level sampling (Paradis 2003; Stadler and Bokma 2013), or clustered (e.g. if samples are geographically or ecologically clustered). In this regard, special care needs to be taken when applying the proposed models on empirical phylogenies that might result from a non-uniform sampling process, especially when interpreting the turnover rate as its estimate can be biased if the empirical sampling scheme is not uniform.

When working with a relatively large clade, it is often useful to consider monophyletic subsets of this clade (e.g. clades originating at a certain fixed time in the past or corresponding to lower taxonomic units). This can allow avoiding computational limits and large uncertainty in the topology and divergence times of deep branches, and renders assumptions of intra clade rate homogeneity more realistic. In this case, researchers would typically analyze each phylogeny independently, however we found that a joint analysis (*Model 4*) outperforms this approach. Our joint analysis assumes a given distribution of sampling probabilities shared across clades, and we found that assuming a uniform distribution performs best. We also found that accounting for distinct diversification rates across clades (*Model 4*) is necessary. Indeed *Model 3*, which assumes identical rates, performs particularly poorly when diversification rates vary across clades. Therefore, when estimating diversification rates across multiple clades, we recommend users to apply *Model 4* with a uniform sampling distribution.

Even under this best-performing model in the case of a joint analysis of several clades, we found that turnover rate estimates are rather uncertain. A possible option for improvement would be to assume a common turnover rate across the phylogenies, a rather common assumption (e.g. (Maliot et al. 2019a)). Further tests with simulations would be required to assess whether turnover rate estimates would then be less uncertain, and whether they are robust to model violation (i.e. output meaningful estimates when turnover rates are in fact variable across phylogenies). Another option would be to assume an *a priori* distribution of net diversification and turnover rates on the set of phylogenies, as done for the sampling probability.

We considered the case when no information on the total diversity of a clade is available. There are also cases when estimates of the total diversity can be computed from external data, but they can be highly uncertain. Large uncertainty can arise, for example, if rarefaction curves are far from reaching a plateau. In this situation, one possibility is to run analyses with a fixed sampling probability, using a range of values for this sampling probability (Morlon et al. 2012; Lorén et al. 2014; Gubry-Rangin et al. 2015; Perez-Lamarque et al. 2020; Pinseel et al. 2020). Another possibility would be to constrain the common sampling distribution used in *Model 1* or *Model 4* based on the uncertain estimates, for example by fixing (or putting informative priors on) the parameters of the sampling distribution (α and β or a and b) such that it resembles the uncertainty around the empirically estimated sampling probability.

While for simplicity and consistency we used the MAP to report all results in the main text, we recommend users to exploit the full posterior probability distribution when analyzing the results of their inference, and to adapt the choice of the point estimate metric to the shape of the posterior. Summarizing posterior distributions with a single metric can be challenging in the case of multimodal or complex shapes (Barido-Sottani et al. 2020). We compared the performance of different metrics in terms of representing the peak of the posterior, and found that the best metric depended on both the model and the parameter considered.

Our empirical finding corroborates previous ones concerning the low net diversification rate of prokaryotic organism (Marin et al. 2016; Louca et al. 2018) (here only bacteria is studied) compared to other classically studied organisms (mainly macro-organisms). Several species' intrinsic components – which may affect the rates of diversification – are drastically different between prokaryotic and eukaryotic organisms (with some exceptions in microeukaryotes). First, an apparent distinction between prokaryotes and eukaryotes is the mode of incorporation of genetic diversity into the genome. The prokaryotic lineages are reproducing clonally and horizontal gene transfer is their major source of genetic diversity while the vast majority of eukaryotes are reproducing sexually. As sexual reproduction and horizontal gene transfer act as cohesive forces among individuals through recombination, both can substantially prevent species divergence. In prokaryotes, horizontal gene transfer can additionally be a source of innovation due to the incorporation of foreign genetic content from highly distant lineages (i.e. non homologous recombination). Therefore, one might expect prokaryotes to be highly susceptible to divergence through incorporation of adaptive (non-homologous) recombination but the realization of the divergence might depend on the recombination rate among closely related individuals – which could spread the adaptive genetic content in the population through periodic selection (Koch 1974) if recombination rate is high or rather profit to the individuals harboring it, subsequently leading to divergence, if recombination rate is low. In prokaryotes, recombination rate is highly variable across lineages and can substantially be higher than substitution rate (Sakoparnig et al. 2021). Hence, according to this scenario, one might expect low diversification happening under high recombination rate among closely related individuals (Cohan and Perry 2007; Fraser et al. 2007).

Second, geographical processes seem drastically disparate between prokaryotes and eukaryotes. Indeed contrary to eukaryotes, prokaryotes are characterized by a large dispersal

rate (i.e. passive dispersal) and do not exhibit a limited geographical range (with few exceptions e.g. in extremophiles) (Louca 2022). This high dispersal rate suggests a low rate of divergence due to geographical barriers (i.e. allopatric speciation) and a high propensity to colonize new environments. On the other hand, allopatric speciation is assumed to be a major process of divergence in eukaryotes (e.g. (Kamiya et al. 1998; Pereyra et al. 2009; Imada et al. 2011; Reid et al. 2013; Amor et al. 2014; Boucher et al. 2016)).

Third, population size is commonly high in prokaryotes compared to eukaryotes. Small population size are expected to be more subjected to extinction through fixation of deleterious mutations (Ohta 1972) and more sensitive to environmental fluctuations. In contrast, large populations are expected to be less prone to extinction but rather more prone to geographical divergence as they might cover a larger geographical heterogeneity and differential selective pressures (Martiny et al. 2006). This factor may lead to enhanced diversification in prokaryotes which is contrasting with the pattern observed. However, a mixture of different mechanisms (clonal, recombination driven or macro-organisms like (allopatry, niche specialization...)) might govern diversification of microbial lineages. Incorporating microevolution processes (González-Torres et al. 2019; Sakoparnig et al. 2021) may enable us to further understand their evolutionary trajectory and disentangle the factors influencing the disparate patterns between eukaryotic and prokaryotic diversification.

In the present work, constant and homogeneous diversification rates are assumed. The assumption on relatively constant rate has been verified for some clades (Martin et al. 2004; Lorén et al. 2014; Marin et al. 2016; Sanglas et al. 2017; Louca et al. 2018) while being unrealistic for others (Morlon et al. 2012). The homogeneity in diversification rates among lineages is even more equivocal as empirical phylogenies are highly imbalanced, a signature often attributed to heterogeneity in diversification rates (Mooers and Heard 1997). Our empirical application indicates a constrained net diversification rate among bacterial clades which diversified after the NOE. Nevertheless, the turnover rate appears variable among those bacterial clades. It is thus highly possible that this variability is not restricted to clades and can extend to lineages, especially for prokaryotic organisms for which closely related lineages can have drastically disparate evolutionary properties (e.g. recombination rates). As the turnover rate estimation is often uncertain under unknown global diversity specifically for low turnover rate, we expect this observation to be exacerbated for more complex models of heterogeneous diversification rates developed in a similar framework (e.g. *k-sampling* scheme). Alternatively, model violation against heterogeneous diversification rates could be evaluated when assuming homogeneous rates to determine the magnitude of the concern.

As any diversification analysis performed on a fixed phylogeny, the confidence in the inferred diversification rates relies heavily on the assumption that the fixed phylogeny is the true phylogeny and depict accurately the evolutionary history among lineages of the studied clade. This assumption is particularly challenged in microbial diversification analysis as horizontal gene transfers can widely affect the topology of the phylogeny depending on the genetic sequence used (Sakoparnig et al. 2021). The conventional approach to deal with horizontal gene transfers is to reconstruct the phylogeny from a highly conserved marker gene,

coding for the 16S rRNA. Additionally, the simplistic dichotomic bifurcation of lineages – as represented in phylogenetic trees – is debated and a more suitable way to represent the reticulated evolutionary history of prokaryotes could be using a phylogenetic network. Extensions of the birth-death process – e.g. the birth-hybridization process which has been used as a birth-death prior in phylogenetic reconstruction (Zhang et al. 2018) – should further be examined to infer the diversification rates on phylogenetic networks.

Accounting for missing species in phylogenetic analyses of diversification is necessary to obtain meaningful inferences, and can limit such analyses in groups for which total diversity is not known. We have developed and compared several approaches that allow obtaining robust estimates of net diversification rate, and unbiased although imprecise estimates of turnover rate, even when total diversity is not known. Such approaches should extend the domain of applicability of phylogenetic diversification analyses and allow appropriate comparisons of diversification dynamic across domains of life. Bacteria apparently diversify at a slower pace than macro-organisms yet, much more applications of phylogenetic diversification analyses – on poorly known groups – are required to understand what drives such differences in diversification rates.

1.5. Appendix

1.5.1. Supplementary Methods

1.5.1.1. *Bernoulli Sampling Scheme and k-sampling scheme*

References are available at (Morlon et al. 2011a), section 2 of (Lambert 2018), *p-sampling* in (Lambert and Stadler 2013a), *sampling_p* in (Stadler 2009), *stochastic sampling* in TreeSim (Stadler 2011b), *uniform sampling strategy* in TESS (Höhna 2013) for models considering the Bernoulli sampling and at section 3 of (Lambert 2018), *n-sampling* in (Lambert and Stadler 2013a), *sampling_m* in (Stadler 2009) and sampling of the diversity-dependent birth-death model in (Etienne et al. 2012a) when considering *k-sampling* ones. To our knowledge, no models based on the *k-sampling* scheme have ever been applied with the exception of the diversity dependent birth-death model.

1.5.1.2. *Likelihoods for Birth Death Models when the Total Number of Species is Unknown*

The models described in the present article can be used on phylogenies with known stem age ($t_0 = t_{stem}$) with the following formulas:

1.5.1.2.1. Likelihood under Bernoulli Sampling (Model 1, $BD_{f\rho}$)

$$p_{BD_{f\rho}}(\tau|t_{stem}, \lambda, \mu) = \int_0^1 \frac{p_1(t_{stem}, \lambda, \mu, y)}{1-p_0(t_{stem}, \lambda, \mu, y)} \prod_{i=1}^{k-1} \lambda p_1(t_i, \lambda, \mu, y) \phi(y) dy \quad [1]$$

with

$$p_0(t, \lambda, \mu, y) = 1 - \frac{y(\lambda - \mu)}{y\lambda + (\lambda(1-y) - \mu)e^{-(\lambda - \mu)t}}$$

and

$$p_1(t, \lambda, \mu, y) = \frac{y(\lambda - \mu)^2 e^{-(\lambda - \mu)t}}{(y\lambda + (\lambda(1-y) - \mu)e^{-(\lambda - \mu)t})^2}$$

1.5.1.2.2. Likelihood under k -sampling (Model 2, BD_K)

$$p_{BD_K}(\tau|t_{stem}, \lambda, \mu, k) = \frac{k(1-a)}{a^{k-1}} \int_0^1 \frac{1}{(1-a(1-y))^2} \prod_{i=1}^{k-1} \lambda p_1(t_i, \lambda, \mu, y) dy \quad [2]$$

with

$$a = 1 - \frac{1}{1 + \frac{\lambda}{\lambda - \mu}(e^{(\lambda - \mu)t_{stem}} - 1)}$$

1.5.1.2.3. Likelihood under Bernoulli Sampling for Multiple Phylogenies with Common Diversification Rates (Model 3, $BD_{f\rho}^{mult}$)

$$p_{BD_{f\rho}^{mult}}(\tau_1, \dots, \tau_p | t_{stem_1}, \dots, t_{stem_p}, \lambda, \mu) = \prod_{j=1}^p p_{BD_{f\rho}}(\tau_j | t_{stem_j}, \lambda, \mu) \quad [3]$$

1.5.1.2.4. Likelihood under Bernoulli Sampling for Multiple Phylogenies with Distinct Diversification Rates (Model 4, $BD_{f\rho}^{multx}$)

$$p_{BD_{f\rho}^{multx}}(\tau_1, \dots, \tau_p | t_{stem_1}, \dots, t_{stem_p}, \lambda_1, \dots, \lambda_p, \mu_1, \dots, \mu_p) = \prod_{j=1}^p p_{BD_{f\rho}}(\tau_j | t_{stem_j}, \lambda_j, \mu_j) \quad [4]$$

1.5.1.3. Correspondence between Likelihood Formula in (Lambert 2018) and in the Present Article

For convenience, we will use here the likelihood formula when conditioning on survival of the tree knowing t_{stem} (see Eq. 1 of [Supplementary Material](#)) to relate to the expression Eq. 8 in (Lambert 2018), which corresponds to the Bernoulli sampling scheme.

In (Lambert 2018), the expression $\frac{1}{F_y(T)}$ corresponds to our conditioning $\frac{p_1(t_{stem}, \lambda, \mu, y)}{1-p_0(t_{stem}, \lambda, \mu, y)}$ with $t_{stem} = T$:

$$\frac{p_1(T, \lambda, \mu, y)}{1 - p_0(T, \lambda, \mu, y)} = \frac{yr^2 e^{-rT}}{(y\lambda + (r - y\lambda)e^{-rT})^2} \frac{y\lambda + (r - y\lambda)e^{-rT}}{yr} = \frac{re^{-rT}}{y\lambda + (r - y\lambda)e^{-rT}} = \frac{1}{\frac{y\lambda}{r}e^{rT} + \frac{(r - y\lambda)}{r}} = \frac{1}{1 + \frac{y\lambda}{r}(e^{rT} - 1)} = \frac{1}{F_y(T)} \quad [5]$$

Similarly, the probability of node depths $f_y(t)$ in (Lambert 2018) Eq. 7 corresponds to the expression $\lambda p_1(t, \lambda, \mu, y)$ in the present article:

$$\lambda p_1(t, \lambda, \mu, y) = \frac{\lambda yr^2 e^{-rt}}{(y\lambda + (r - y\lambda)e^{-rt})^2} = f_y(t) \quad [6]$$

1.5.1.4. Sampling Probability Distribution of Bacterial Phyla

Despite efforts in community ecology toward developing a sampling theory for microbial communities (Muscarella et al. 2019), and models to estimate the degree of sampling required to catalogue global diversity (Quince et al. 2008), we still do not know to what extent sampling fraction vary across clades. Here, we investigated the parameters of the sampling probability distribution of bacterial phyla. We used the estimated fraction of recovered bacterial OTUs from 492 studies that could be mapped to SILVA nonredundant sequences at 97% similarity analyzed in (Louca et al. 2019) as a proxy for global bacterial OTU richness covered by SILVA. We thus estimated parameters of the sampling fraction distribution of bacterial phyla in SILVA by fitting a beta distribution on the phyla's specific sampling fraction. The parameters estimated were respectively 3.2 and 1.9 for β and α . For our approach to remain flexible, we choose to fix one parameter of the distribution (β) and let the other vary (α) in our simulation framework.

1.5.1.5. Statistics of the Posterior Distribution

We investigated on simulations the fidelity of the chosen point estimate (i.e. the MAP) in describing the maximum density of the posterior distributions. We compared it to the mean, median and harmonic mean of our inferred posterior distribution per parameters and for each models. To do so, we computed the suitability of each metric to summarize the posterior probability by considering the proportion of phylogenies for which the posterior probability metric (MAP, mean, median, harmonic mean) falls within its corresponding X% HPDI at several X values.

1.5.2. Supplementary Results

1.5.2.1. Performance of the Models

1.5.2.1.1. Single Phylogenies Inference

The median of the net diversification rate (r) posterior probability summarizes best the credible interval for all the tested models across all HPDI (i.e. it has the highest coverage probability) ([Supplementary Fig. 6, 14, 20](#)). In contract, the MAP is the least suitable to represent the credible interval of r (i.e. it has the lowest coverage probability). In term of accuracy and precision, the median is also performing better than the MAP ([Supplementary Fig. 7, 8](#)) especially for *Model 2* that yields slightly biased net diversification rate estimates ([Fig. 1b](#)).

For the turnover rate ε , the metric which summarizes best the credible interval is the harmonic mean for *Model 1, 2, 6* and the MAP for the other models (*Model 6* has comparable results with the MAP and the harmonic mean) ([Supplementary Fig. 9](#)). The harmonic mean is also displaying a slightly better accuracy and performance when estimating ε for *Model 1, 2* and *6* ([Supplementary Fig. 10, 11](#)). One can expect the credible interval to diverge highly from the MAP if the posterior is fairly flat.

The choice of an appropriate metric to summarize the highest posterior density is also dependent on the parameter space. As the posterior distribution of *Model 1* and *2* is highly skewed toward $\varepsilon = 0$ for low simulated turnover rate (data not shown), the harmonic mean is well suited in this context. While this is not the case for moderate or high simulated turnover rate for *Model 1, 2* and for all *Model 5* parameter space that have posterior distribution that resembles more a normal distribution.

1.5.2.1.2. Multiple Phylogenies Inference

Similar to the single phylogenies inference, the median is the metric fitting best the credible interval of the net diversification rate r for all *multiple phylogenies models* even though the difference in accuracy and precision is, for most, negligible between the different metric ([Fig. 4a & 6a](#) and [Supplementary Fig. 14a,b,c, 16a, 21a](#)).

Concerning the turnover rate ε , the metric fitting best the credible interval is different for each model. It is the MAP, the harmonic mean and the median for respectively the *Model 3* under beta sampling distribution (*M3Beta*), the *Model 3* under uniform sampling distribution (*M3Unif*) and the *Model 5 “mult”* (*M5*) ([Fig. 4b](#) and [Supplementary Fig. 14d,e,f, 16b](#)) while being the median for all *Model 4* or the equivalent *Model 5 “mult_x”* ([Supplementary Fig. 20b, 21c](#)). The difference in accuracy and precision is negligible between the different metric for *M3Beta* and *M5* whereas it has a substantial impact on *M3Unif* and *Model 4* turnover rate estimates ([Fig. 4b, 6b](#) and [Supplementary Fig. 12b, 16b, 21c](#)). Thus as for *Model 1* and *2*, the harmonic mean should be used to summarize the posterior probability density (and the credible

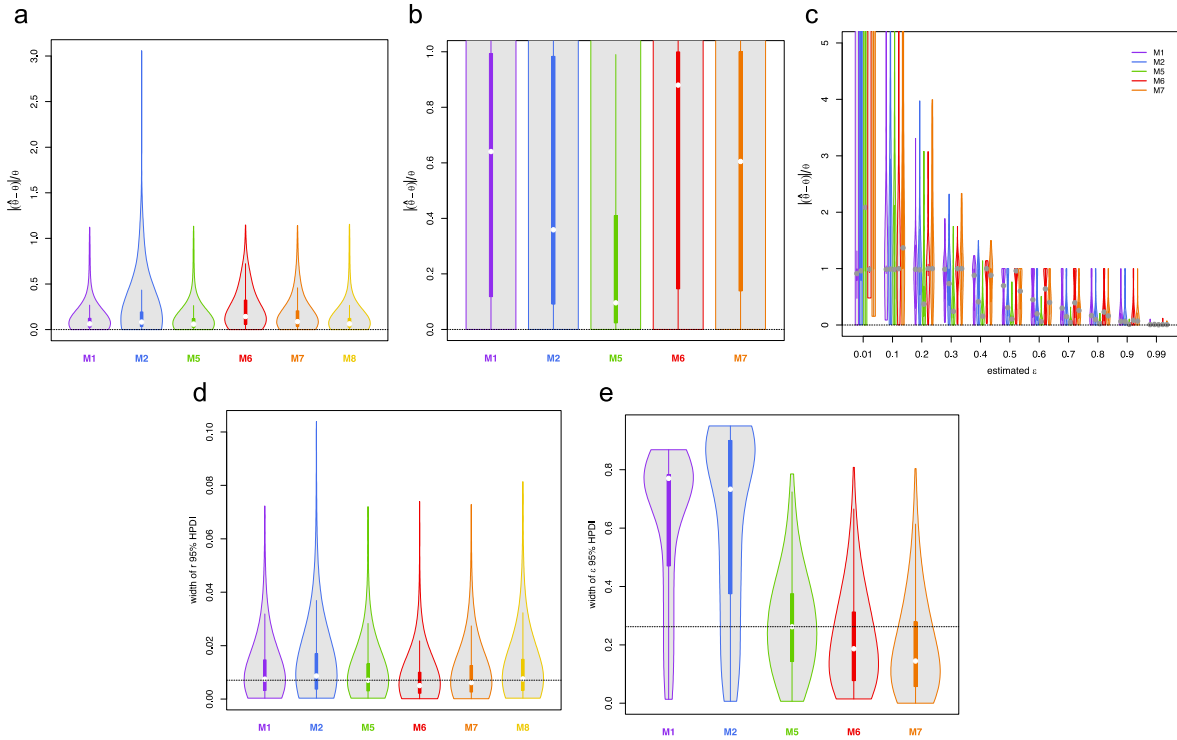
interval) of the turnover rate in the case of *Model 3* under a uniform sampling distribution and the median in the case of *Model 4*.

Concerning the sampling distribution parameters of *M3Beta* and *M3Unif*, the difference in accuracy and precision are important between the different metrics for parameter a and α which are more accurate and precise when using the best fit credible interval point estimate, the median ([Supplementary Fig. 16c,d](#)). Note that for these models, no metric is correctly summarizing the highest posterior density in the case of the b parameter ([Supplementary Fig. 15c](#)). Concerning sampling distribution parameters of *Model 4*, the use of the best fit credible interval point estimate is necessary to get accurate parameter estimates.

1.5.3. Supplementary Discussion

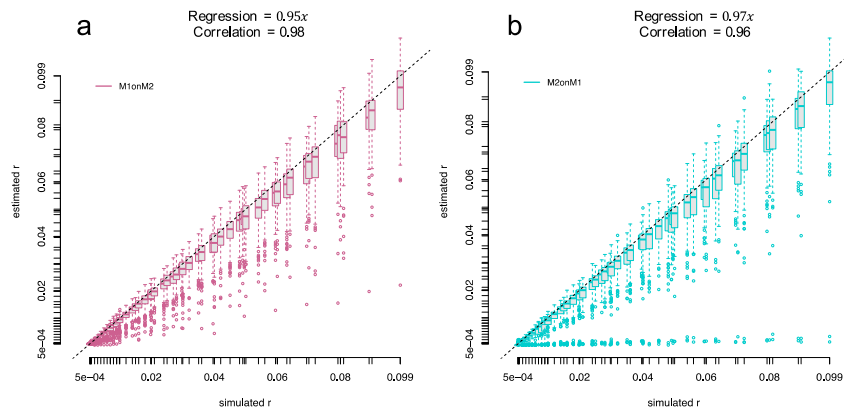
In a Bayesian framework, different statistics can be used to summarize a posterior distribution by a point estimate. One of the most commonly used metric is the maximum a posteriori probability (MAP) that resembles the maximum likelihood estimate (MLE) in a frequentist framework (if uniform priors are assumed). Other metrics such as the mean or the median of the distribution are well suited to summarize the posterior distribution depending on its form. Lastly, when dealing with rates comprised between 0 and 1 such as the turnover rate ε computing the harmonic mean as a point estimate of the posterior distribution is also suitable. Interestingly in Bayesian inference, the mode of a posterior distribution is not necessarily close to the MAP. One need to test the predictive power of these metrics as well as their adequacy to resume the posterior probability.

1.5.4. Supplementary Figures



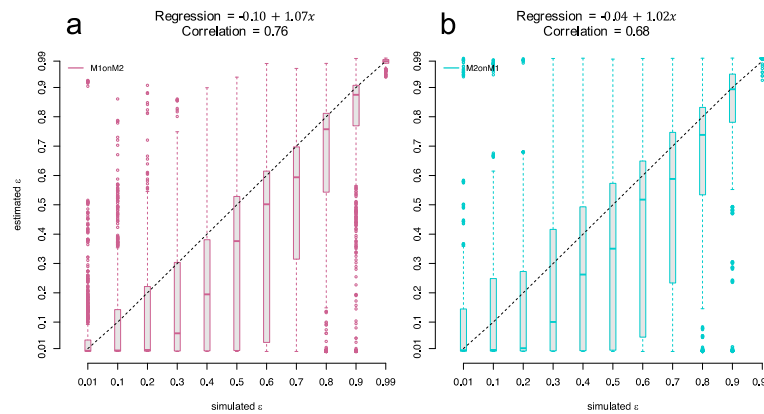
Supplementary Figure 1: Precision of parameter estimates across inference procedures on single phylogenies.

Violin plots of the error and the uncertainty distribution of MAP parameter estimates on 10,429 simulated phylogenies for $BD_{\int \rho}$, BD_K , $BD_{\rho_{known}}$, $BD_{\rho=1}$, $BD_{\rho \in [0,1]}$, $BD_{\rho \lambda}$. The distribution of the absolute relative error (ARE) is displayed for **a**, the net diversification rate **b**, the turnover rate and **c**, the turnover rate as a function of its simulated value. The horizontal dashed line displays the 0 absolute relative error for **a**, **b** and **c**. The width in parameter space of the 95% HPDI is represented for **d**, the net diversification rate and **e**, the turnover rate ϵ across different models for single phylogenies inference. The horizontal dashed line of **d** and **e** displays the median width for $BD_{\rho_{known}}$. The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The contour of the violin plot displays the kernel density plot including the outliers.



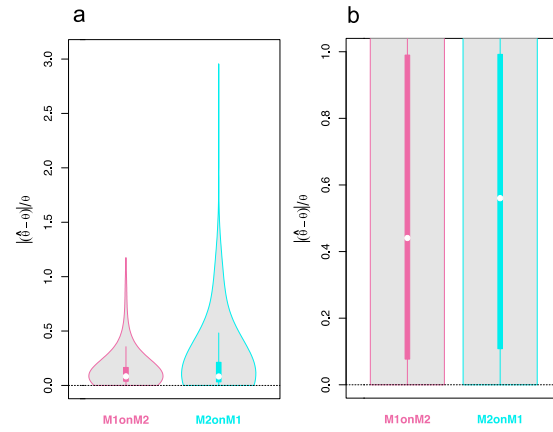
Supplementary Figure 2: Accuracy of net diversification rate estimates (r) under model violation.

Inferred versus simulated MAP net diversification rates for 10,429 simulated phylogenies for **a**, Model 1 (Bernoulli sampling) applied to simulations from Model 2 (k -sampling) and **b**, Model 2 applied to simulations from Model 1. Each grey dot is an estimate of r for a single simulated phylogeny. The horizontal bold lines correspond to the median of the estimates for each simulated ε value. The box bounds display the 75% and 25% quantiles of the estimates. Each dot is an outlier. The dashed line represents the 1:1 relationship. The value of the slope of the quantile regression and the Pearson correlation coefficient are displayed for each model in the top.



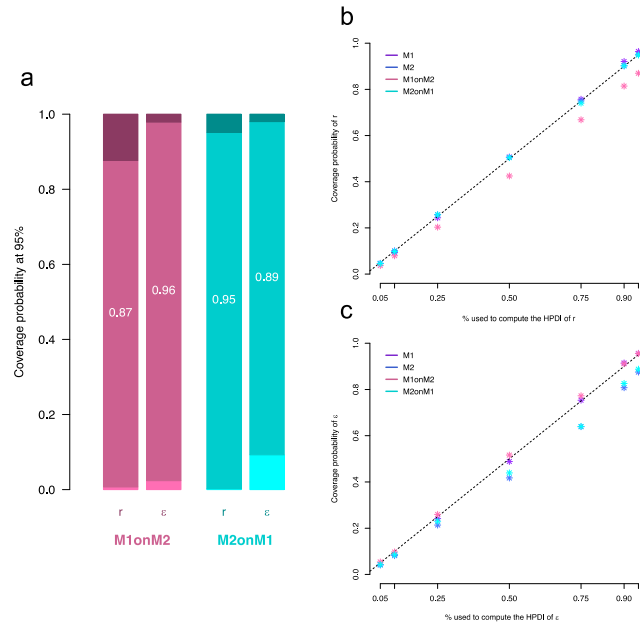
Supplementary Figure 3: Accuracy of turnover rate estimates (ε) under model violation.

Box plots of MAP turnover rate estimates as a function of simulated values for 10,429 simulated phylogenies for **a**, Model 1 (Bernoulli sampling) applied to simulations from Model 2 (k -sampling) and **b**, Model 2 on simulations applied to simulations from Model 1. The horizontal bold lines correspond to the median of the estimates for each simulated ε value. The box bounds display the 75% and 25% quantiles of the estimates. Each dot is an outlier. The dashed line represents the 1:1 relationship. The value of the parameters of the quantile regression and the Pearson correlation coefficient are displayed for each model in the top.

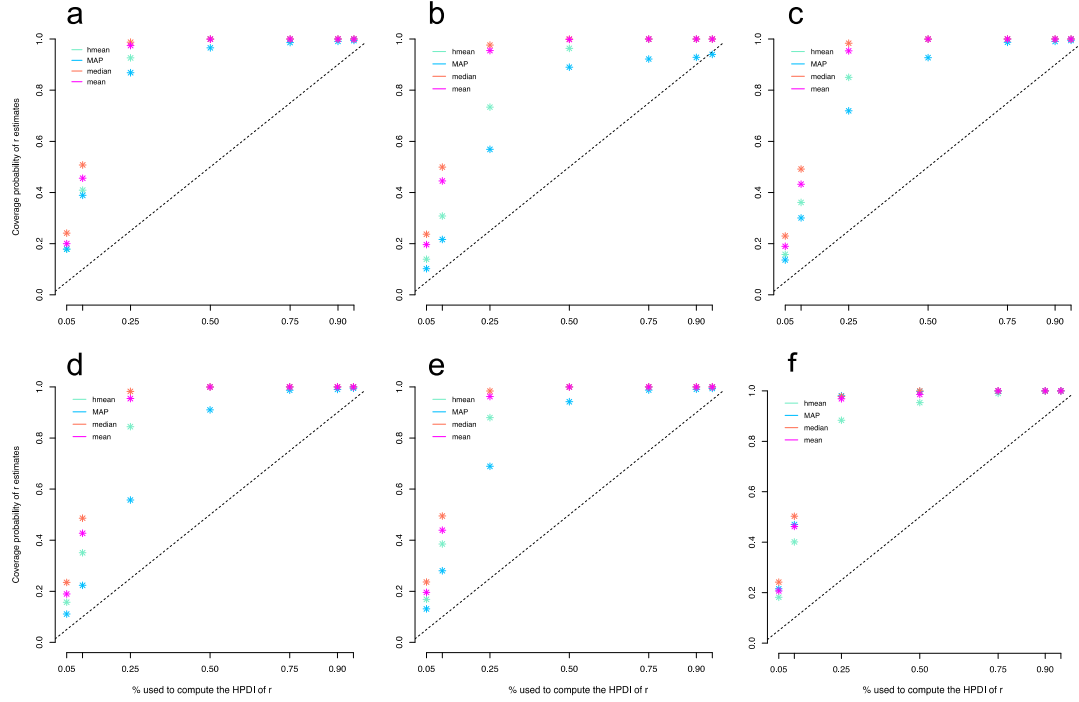


Supplementary Figure 4: Precision (absolute relative error) of turnover rate estimates (ϵ) under model violation.

Violin plots representing the distribution of absolute relative error (ARE) for **a**, the net diversification rates and **b**, the turnover rates estimated on 10,429 simulated phylogenies for Model 1 on simulations from Model 2, Model 2 on simulations from Model 1. The point estimate used here is the maximum a posteriori probability (MAP) of the posterior distribution of ϵ . The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The horizontal dashed line displays the 0 absolute relative error. The contour of the violin plot displays the kernel density plot including the outliers.

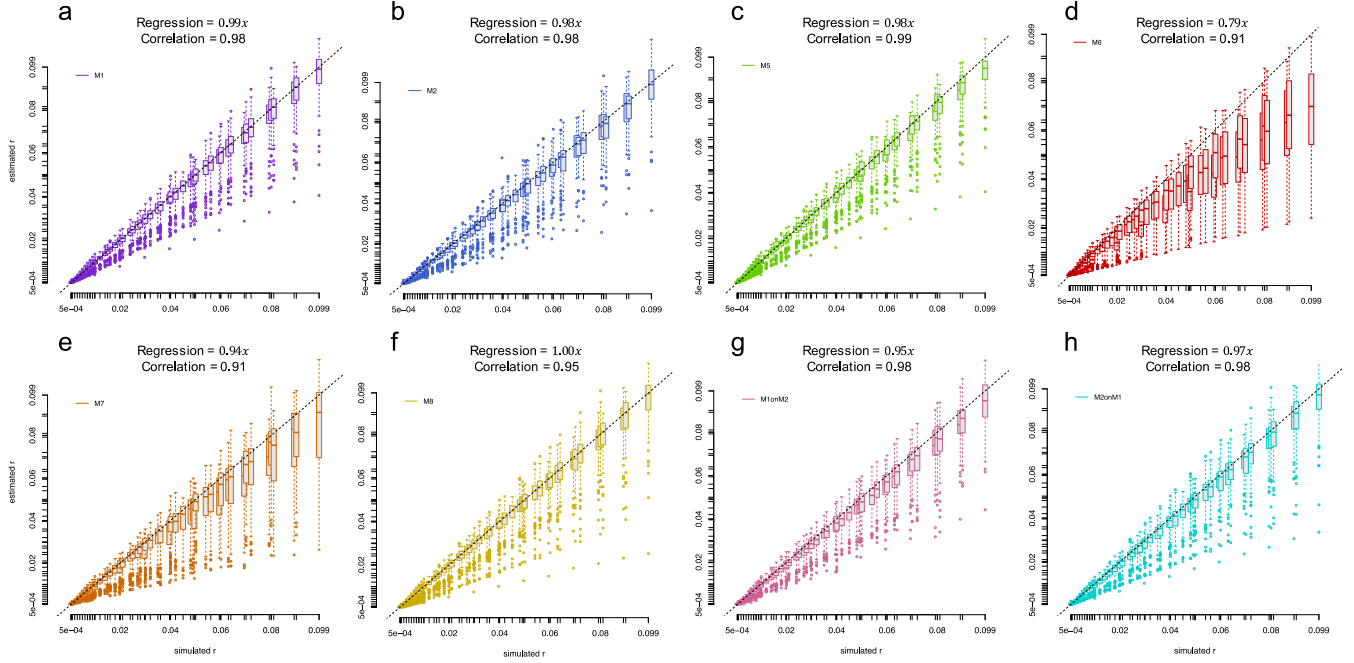


Supplementary Figure 5: Coverage probability of the parameters under model violation. Coverage probability of Model 1 on simulations from Model 2 and of Model 2 on simulations from Model 1. **a**, Coverage probability at 95% for r and ε . The value inside each barplot indicates the proportion of phylogenies for which the simulated parameter value is included inside the 95% HPDI, the lighter and darker bars respectively indicate the proportion of phylogenies for which the simulated value is below or above the 95% HPDI. **b**, Coverage probability of r at $X\%$ versus X **c**, Coverage probability of ε at $X\%$ versus X . For easier comparison the corresponding value for Model 1 and Model 2 are displayed in the panel **b** and **c**. Each dot corresponds to the proportion of phylogenies for which the simulated parameter value is included inside the corresponding HPDI at (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95)%. The dashed line displays the 1:1 relationship.



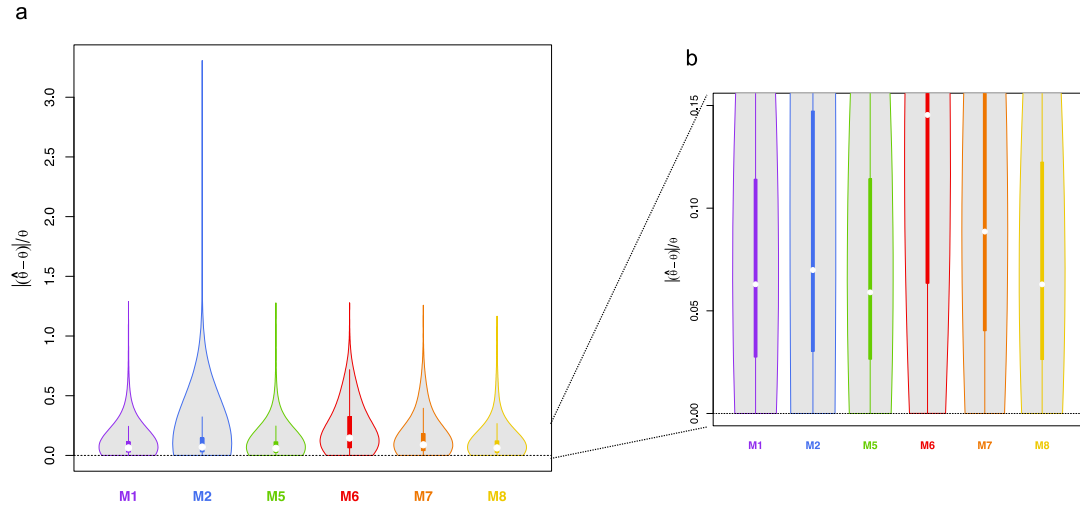
Supplementary Figure 6: Fitness of different metric to describe the highest posterior density of the net diversification rate (r) across different models for single phylogenies inference.

Coverage probability of the harmonic mean (*hmean*), the maximum a posteriori probability (*MAP*), the median and the mean versus its corresponding highest posterior density interval (HPDI) for the net diversification rate r on 10,429 simulated phylogenies for **a**, Model 1 **b**, Model 2 **c**, Model 5 **d**, Model 6 **e**, Model 7 and **f**, Model 8. Each dot correspond to the proportion of phylogenies for which the metric (*hmean*, *MAP*, median or mean) is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



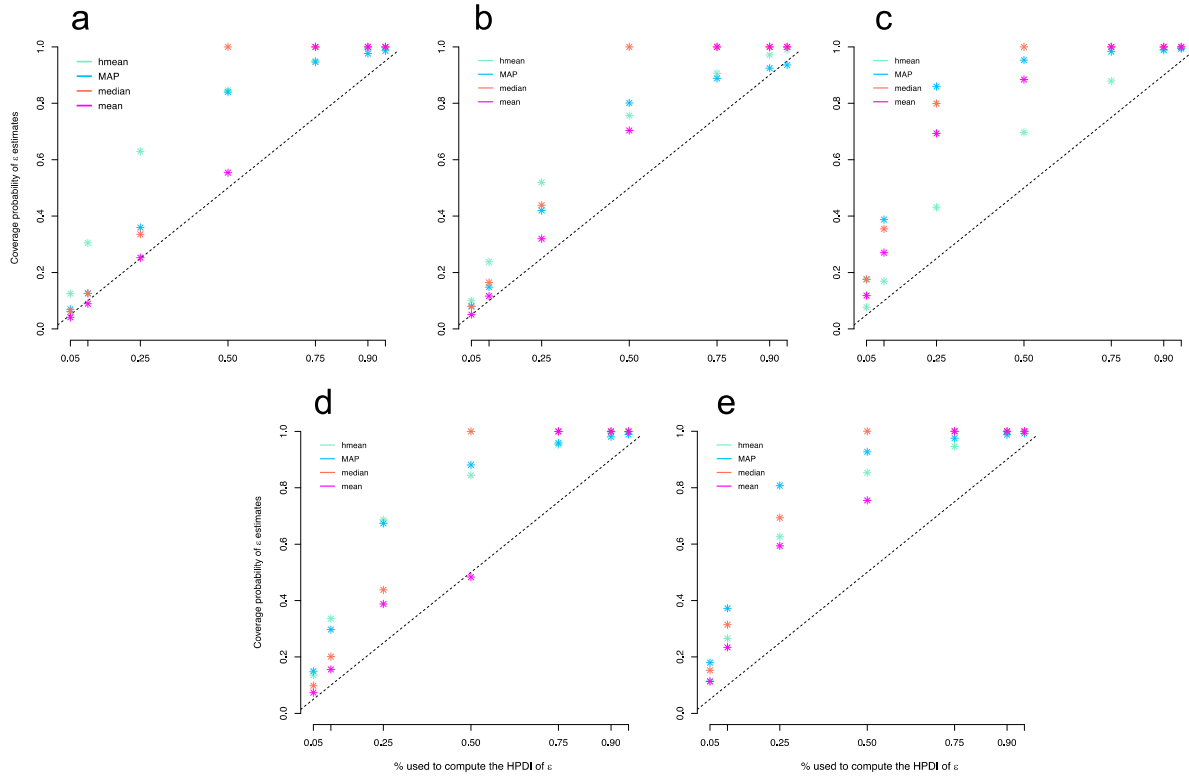
Supplementary Figure 7: Accuracy of the net diversification rate (r) inference across different models with fixed or not fixed sampling probability and the median as point estimate.

*Inferred versus simulated net diversification rate swarm plot on 10,429 simulated phylogenies for **a**, Model 1 **b**, Model 2 **c**, Model 5 **d**, Model 6 **e**, Model 7 **f**, Model 8 **g**, Model 1 on simulations of Model 2 and **h**, Model 2 on simulations of Model 1. The point estimate used here is the median of the posterior distribution of r . The horizontal bold line corresponds to the median of the estimates for each simulated r . The box bounds display the 75% and 25% quantiles of the estimates. Each dot is an outlier. The dashed line represents the 1:1 relationship. The value of the slope of the quantile regression and the Pearson correlation coefficient are displayed for each model in the top.*

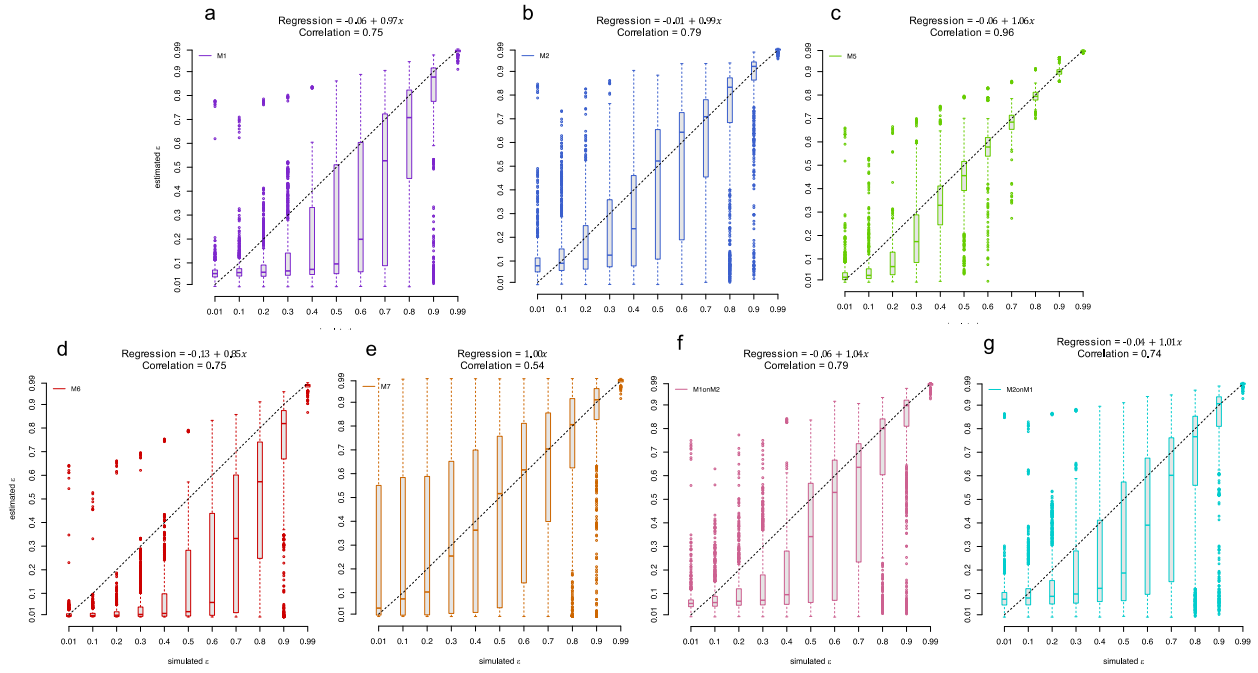


Supplementary Figure 8: Precision in term of absolute relative error on the net diversification rate (r) inference across different models using the median as a point estimate.

*Inferred net diversification rate absolute relative error (ARE) violin plot on 10,429 simulated phylogenies for Model 1, Model 2, Model 5, Model 6, Model 7 and, Model 8. **a**, Full range violin plot is shown and **b**, its zoomed version. The point estimate used here is the median of the posterior distribution of r . The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The horizontal dashed line displays the 0 absolute relative error. The contour of the violin plot displays the kernel density plot including the outliers.*

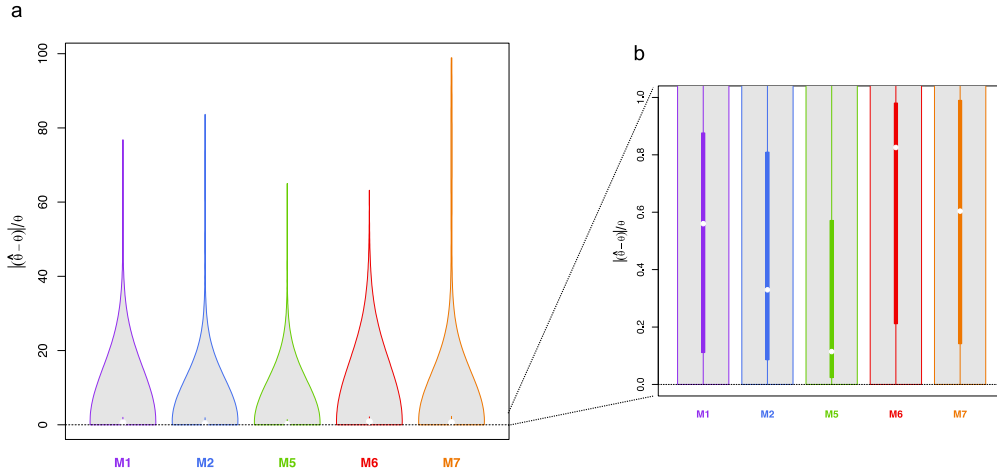


Supplementary Figure 9: Fitness of different metric to describe the highest posterior density of the turnover rate (ϵ) across different models for single phylogenies inference. Coverage probability of the harmonic mean (hmean), the maximum a posteriori probability (MAP), the median and the mean versus its corresponding highest posterior density interval (HPDI) for the turnover ϵ on 10,429 simulated phylogenies for **a**, Model 1 **b**, Model 2 **c**, Model 5 **d**, Model 6 **e**, Model 7. Each dot correspond to the proportion of phylogenies for which the metric (hmean, MAP, median or mean) is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



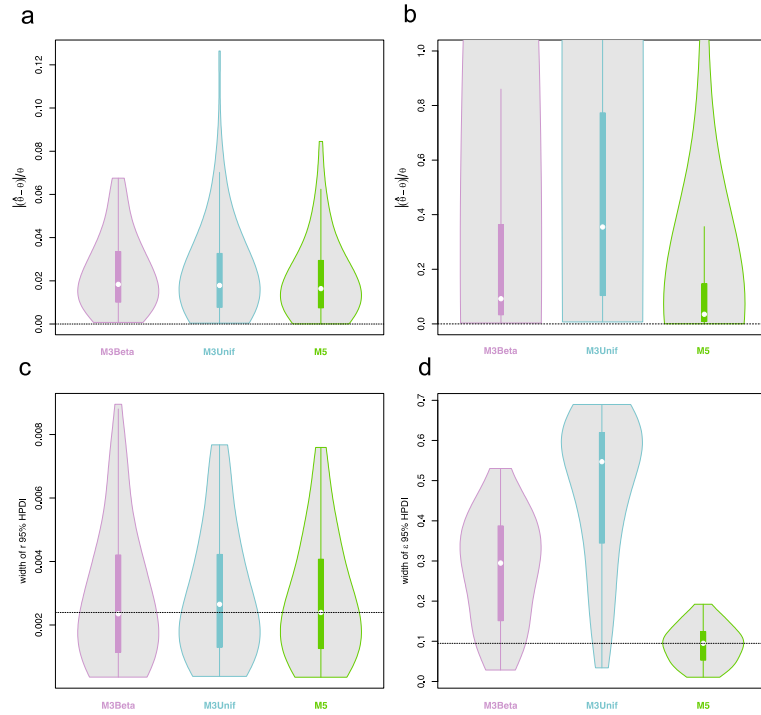
Supplementary Figure 10: Accuracy of the turnover rate (ε) inference across different models with fixed or not fixed sampling probability and the harmonic mean as point estimate.

Inferred versus simulated turnover rate box plot on 10,429 simulated phylogenies for a, Model 1 b, Model 2 c, Model 5 d, Model 6 e, Model 7 f, Model 1 on simulations of Model 2 g, Model 2 on simulations of Model 1. The point estimate used here is the harmonic mean of the posterior distribution of ε . The horizontal bold lines corresponds to the median of the estimates for each simulated ε . The box bounds display the 75% and 25% confidence interval of the estimates. Each dot is an outlier. The dashed line represents the 1:1 relationship. The value of the parameters of the quantile regression and the correlation coefficient of Pearson are displayed for each models in the top.



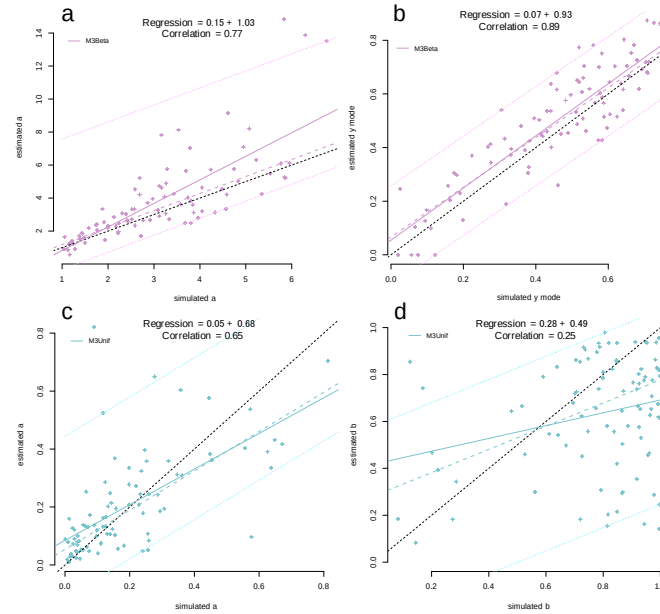
Supplementary Figure 11: Precision in term of absolute relative error on the turnover rate (ε) inference across different models using the harmonic mean as a point estimate.

*Inferred turnover rate absolute relative error (ARE) violin plot on 10,429 simulated phylogenies for Model 1, Model 2, Model 5, Model 6 and Model 7, $BD_{\rho \in [0,1]}$, birth-death process under Bernoulli sampling scheme assuming a random sampling probability. **a**, Full range violin plot is shown and **b**, its zoomed version. The point estimate used here is the harmonic mean of the posterior distribution of ε . The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The horizontal dashed line displays the 0 absolute relative error. The contour of the violin plot displays the kernel density plot including the outliers.*



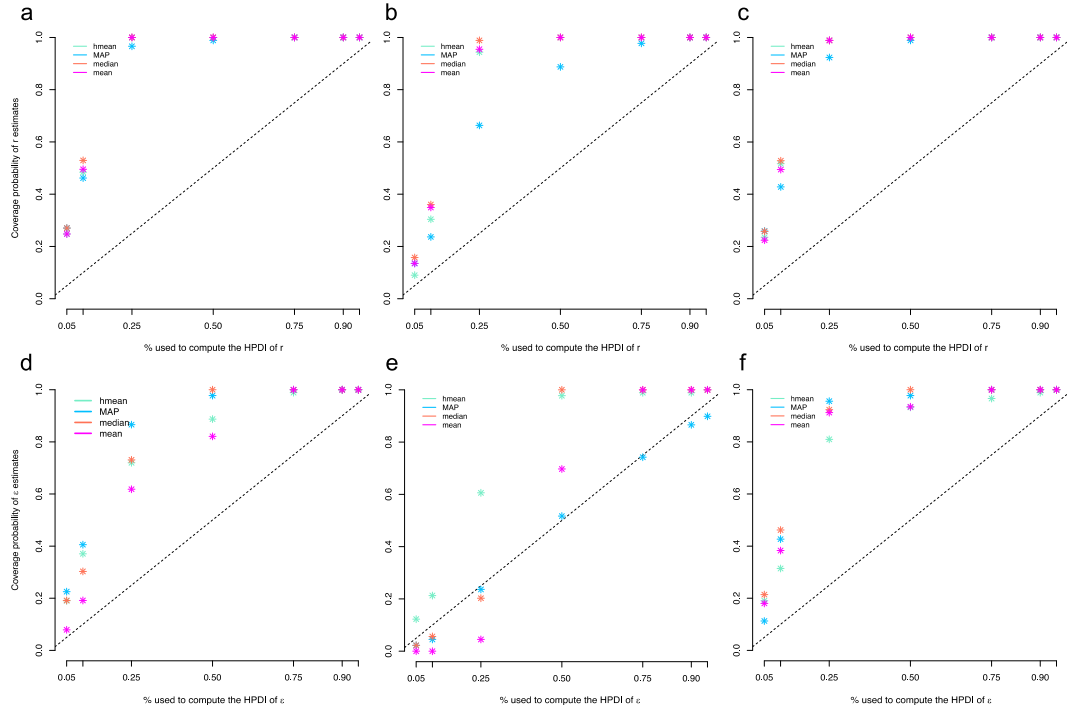
Supplementary Figure 12: Performance of parameter inference on multiple phylogenies under common diversification rates across different models.

Violin plots of the error and the uncertainty distribution of MAP parameter estimates on 89 sets of 10 phylogenies for: M3Beta (in light pink), M3Unif (in light blue) and Model 5 (in green). The distribution of the absolute relative error (ARE) is displayed for **a**, the net diversification rate and **b**, the turnover rate. The horizontal dashed line displays the 0 absolute relative error for **a** and **b**. The width in parameter space of the 95% highest a posteriori density interval (HPDI) of **c**, the turnover rate ϵ and **d**, the net diversification rate r . The horizontal dashed line **c** and **d** displays the median width for Model 5. The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The contour of the violin plot displays the kernel density plot including the outliers.



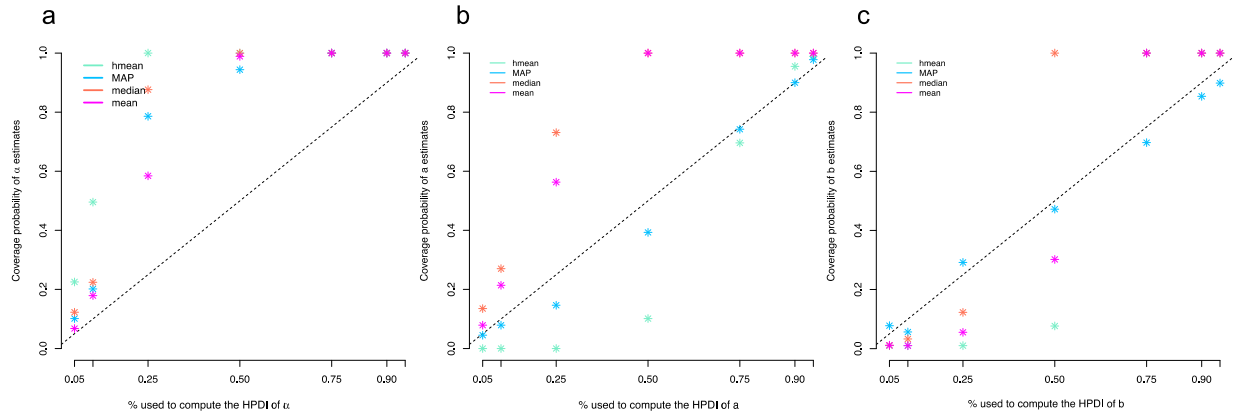
Supplementary Figure 13: Accuracy of sampling distribution parameters inference on multiple phylogenies inference under common diversification rates across different models with sampling probability fixed or not.

Inferred versus simulated parameters dot plot on 89 sets of 10 simulated phylogenies for **a**, α the parameter of the beta sampling distribution (with fixed $\beta = 3$) of Model 3, when assuming a common beta sampling distribution (M3Beta in light pink) **b**, its corresponding sampling probability mode **c**, a the parameter of the uniform sampling distribution corresponding to the lower bound of the uniform distribution of Model 3, when assuming a common uniform sampling distribution (M3Unif in light blue) and **d**, b the parameter of the uniform sampling distribution corresponding to the upper bound of the uniform distribution of Model 3, (M3Unif in light blue). The point estimate used here is the maximum a posteriori probability (MAP) of the posterior distribution for each parameter. Each dot is an estimate for a set of 10 simulated phylogenies. The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. The value of the parameters of the quantile regression and the correlation coefficient of Pearson are displayed for each models on the top.



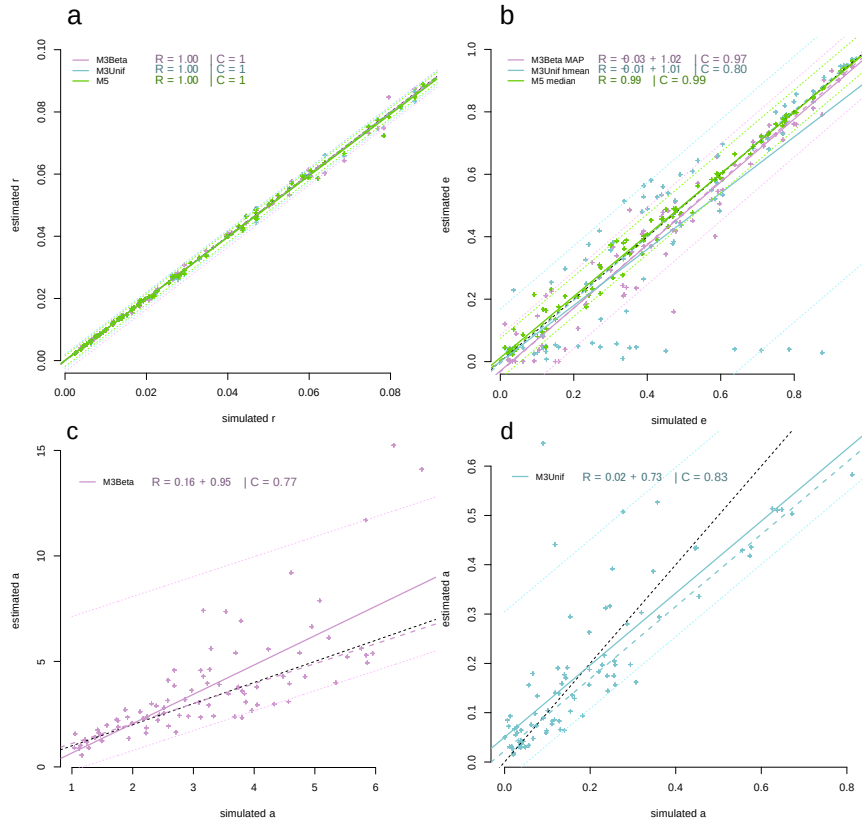
Supplementary Figure 14: Fitness of different metric to describe the highest posterior density of the net diversification rate (r) and the turnover rate (ϵ) across different models for multiple phylogenies inference under common diversification rates.

Coverage probability of the harmonic mean (hmean), the maximum a posteriori probability (MAP), the median and the mean versus its corresponding highest posterior density interval (HPDI) for the net diversification rate r (a-c) and the turnover rate ϵ (d-f) on 89 sets of 10 simulated phylogenies for **a** and **d**, Model 3, when assuming a common beta sampling distribution **b** and **e**, Model 3, when assuming a common uniform sampling distribution **c** and **f**, Model 5 “mult”, applied on phylogenies simulated under the M3Unif. Each dot correspond to the proportion of phylogenies for which the metric (hmean, MAP, median or mean) is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



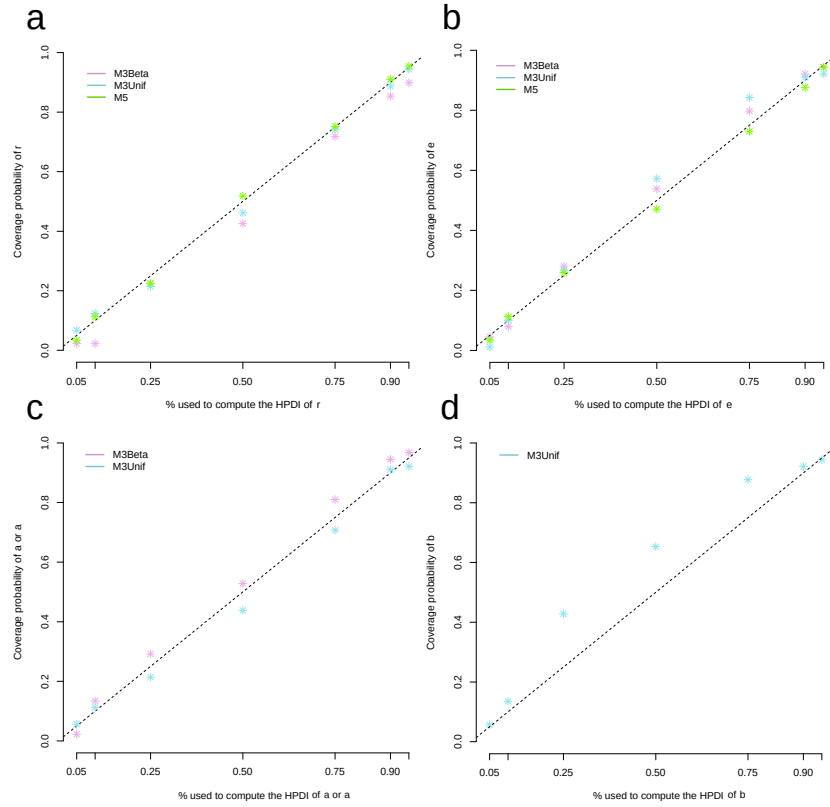
Supplementary Figure 15: Fitness of different metric to describe the highest posterior density of the sampling distribution parameters (α , a and b) across different models for multiple phylogenies inference under common diversification rates.

Coverage probability of the harmonic mean (hmean), the maximum a posteriori probability (MAP), the median and the mean versus its corresponding highest posterior density interval (HPDI) for **a**, the α parameter of the beta sampling distribution **b**, the a parameter of the uniform sampling distribution and **d**, the b parameter of the uniform sampling distribution on 89 sets of 10 simulated phylogenies for Model 3, when assuming a common beta sampling distribution (**a**) and Model 3, when assuming a common uniform sampling distribution (**b** and **c**). Each dot correspond to the proportion of phylogenies for which the metric (hmean, MAP, median or mean) is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



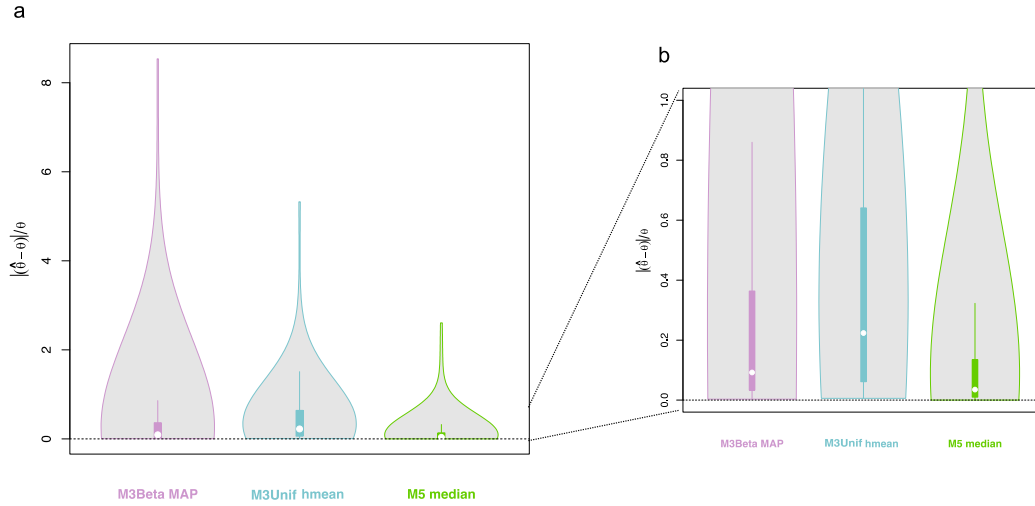
Supplementary Figure 16: Accuracy of parameters inference on multiple phylogenies inference under common diversification rates across different models with sampling probability fixed or not and best fit highest posterior density point estimate.

Inferred versus simulated parameters dot plot on 89 sets of 10 simulated phylogenies for **a**, the net diversification rate **b**, the turnover rate of Model 3, when assuming a common beta sampling distribution (M3Beta in light pink), assuming a common uniform sampling distribution (M3Unif in light blue) and Model 5 “mult” , (M5 in green) applied on phylogenies simulated under the M3Unif **c**, α the parameter of the beta sampling distribution (with fixed $\beta = 3$) of Model 3, (M3Beta in light pink) and **d**, a the parameter of the uniform sampling distribution corresponding to the lower bound of the uniform distribution of Model 3, (M3Unif in light blue). Note that **b** the parameter of the uniform sampling distribution of Model 3 is not displayed here as its best fit highest posterior density point estimate is the MAP (already displayed in the main text). The point estimates used here are the metrics corresponding to the best fit of the highest posterior density for each parameter and each model. It is the median for all net diversification estimates, the MAP, the harmonic mean (hmean) and the median for the turnover rate of respectively M3Beta, M3Unif and Model 5 “mult”, the harmonic mean for α of M3Beta and the median for a of M3Unif. Each dot is an estimate for a set of 10 simulated phylogenies. The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. The value of the parameters of the quantile regression (R) and the correlation coefficient of Pearson (C) are displayed for each models on the top.



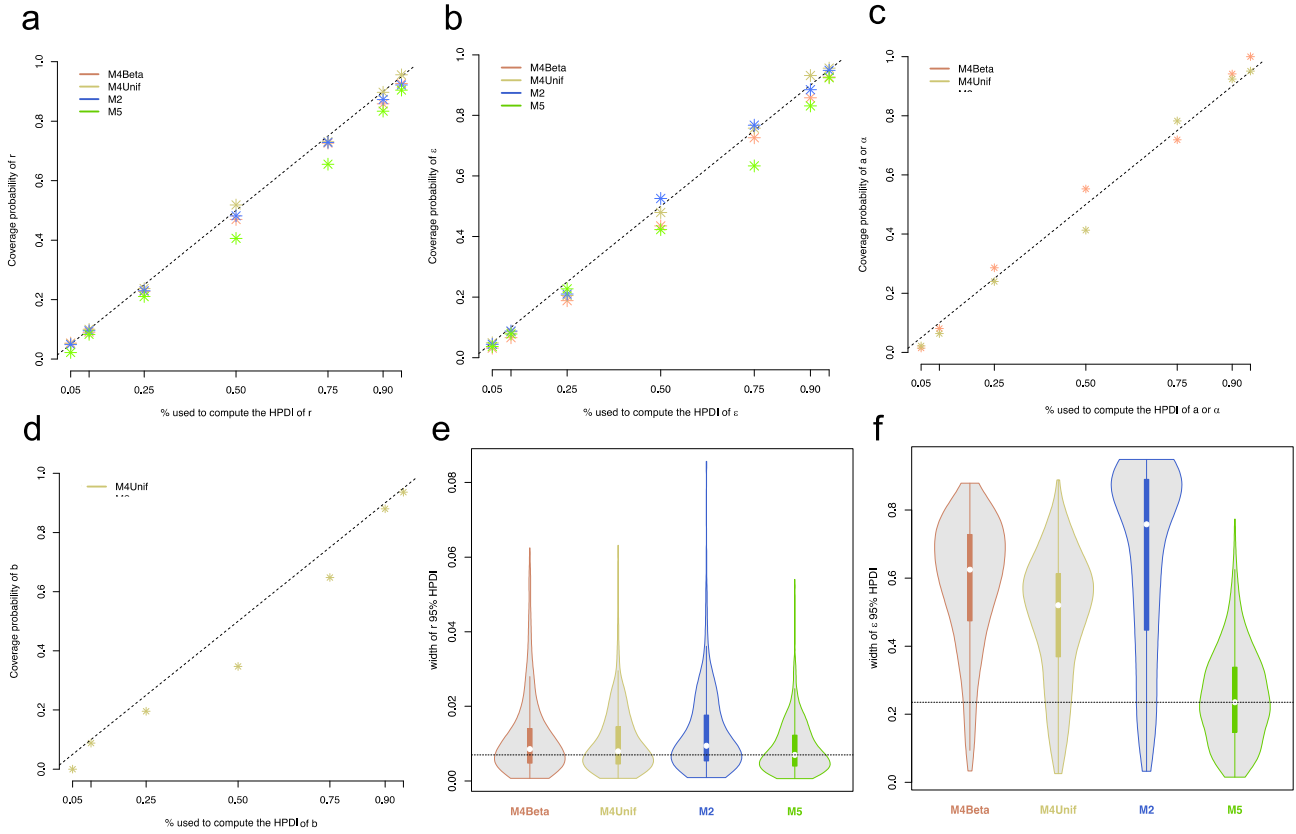
Supplementary Figure 17: Coverage probabilities across inference procedures for multiple phylogenies under common diversification rates.

Coverage probability at $X\%$ versus X of **a**, the net diversification rate r **b**, the turnover rate ε and **c**, the parameters α (for a beta sampling distribution), or a and **d**, b (for a uniform sampling distribution) on 89 sets of 10 phylogenies for: Model 3 when assuming a common beta sampling distribution (M3Beta in light pink), Model 3 when assuming a common uniform sampling distribution (M3Unif in light blue) and Model 5 “mult”, (M5 in green). Each dot corresponds to the proportion of phylogenies for which the simulated parameter value is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



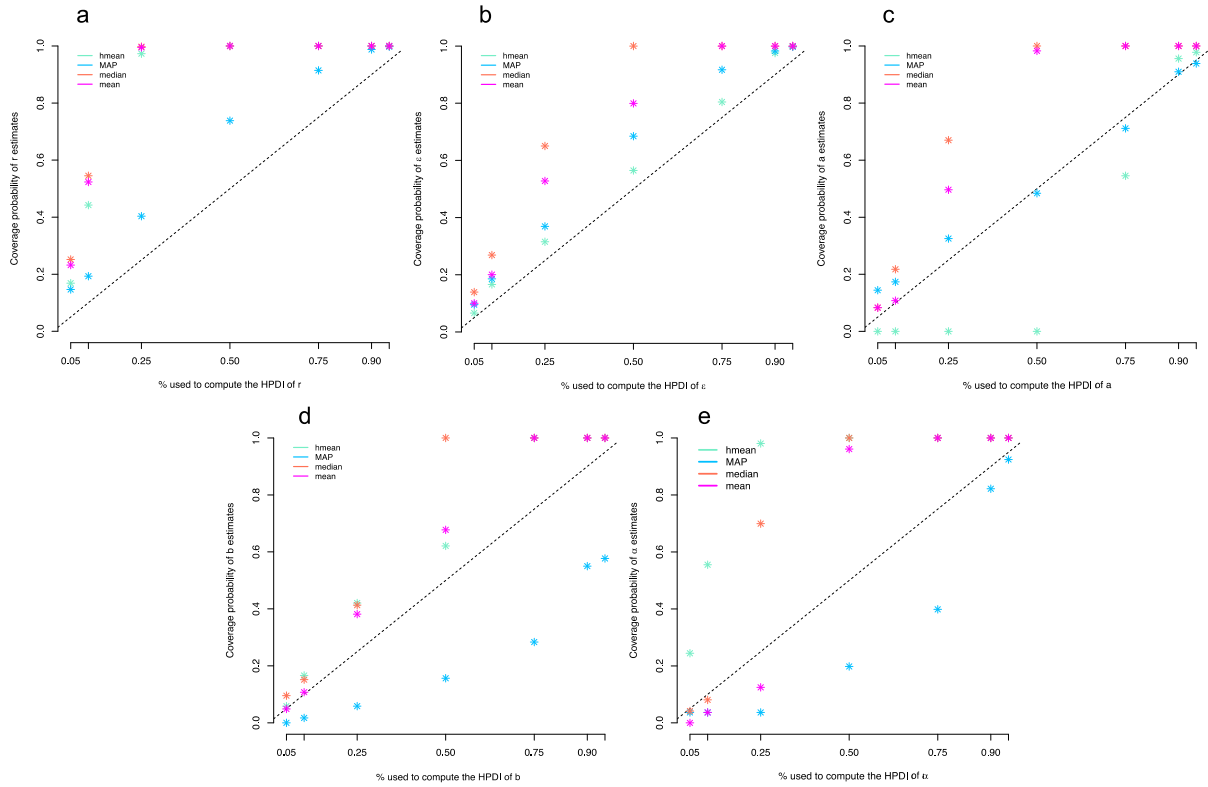
Supplementary Figure 18: Precision in term of absolute relative error on the turnover rate (ϵ) inference across different models for multiple phylogenies inference under common diversification rates using the best fit highest posterior density point estimate.

Inferred turnover rate absolute relative error (ARE) violin plot on 89 sets of 10 simulated phylogenies for Model 3, when assuming a common beta sampling distribution (M3Beta in light pink), assuming a common uniform sampling distribution (M3Unif in light blue) and Model 5 “mult”, (M5 in green) applied on phylogenies simulated under the M3Unif. **a**, Full range violin plot is shown and **b**, its zoomed version. The point estimates used here are the metrics corresponding to the best fit of the highest posterior density for the turnover rate ϵ and each model. It is the MAP, the harmonic mean (hmean) and the median for the turnover rate of respectively M3Beta, M3Unif and Model 5 “mult”. The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The horizontal dashed line displays the 0 absolute relative error. The contour of the violin plot displays the kernel density plot including the outliers.



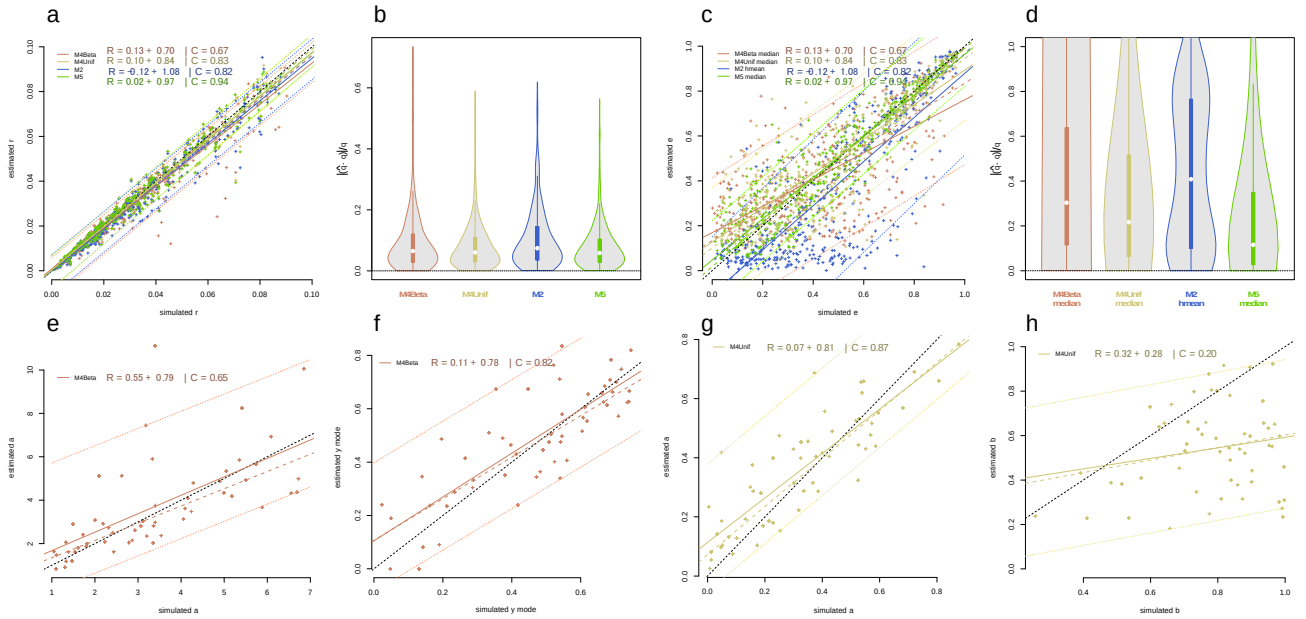
Supplementary Figure 19: Coverage probabilities and credible interval width across inference procedures for multiple phylogenies with specific diversification rates.

Coverage probability of **a**, the net diversification rate **r** **b**, the turnover rate ϵ and the parameters **c**, α (for a beta sampling distribution) or **a** and **d**, b (for a uniform sampling distribution) at $X\%$ versus X on 41 sets of 10 phylogenies for: M4Beta (in salmon), M4Unif (in khaki) and M5 (in green). Each dot corresponds to the proportion of phylogenies for which the simulated parameter value is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship. The width in parameter space of the 95% HPDI is represented for **e**, the net diversification **r** **f**, the turnover rate ϵ across different models for single phylogenies inference. The horizontal dashed line of **e** and **f** displays the median width for Model 5. The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The contour of the violin plot displays the kernel density plot including the outliers.



Supplementary Figure 20: Fitness of different metric to describe the highest posterior density of the net diversification rate (r), the turnover rate (ϵ) and the sampling distribution parameters (α , a and b) across different models for multiple phylogenies inference under specific diversification rates.

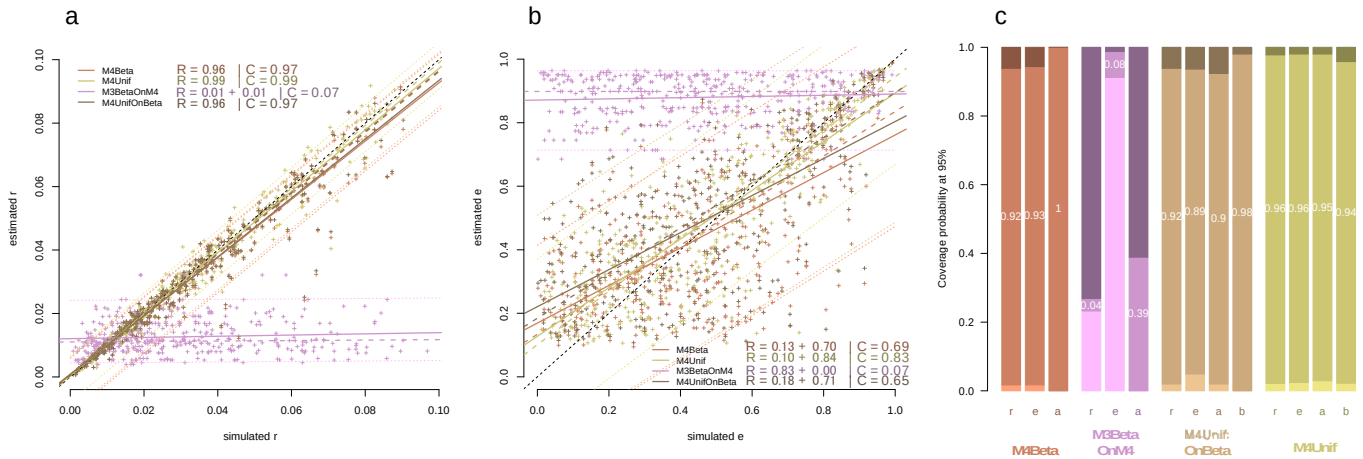
Coverage probability of the harmonic mean (hmean), the maximum a posteriori probability (MAP), the median and the mean versus its corresponding highest posterior density interval (HPDI) for **a**, the net diversification rate **b**, the turnover rate ϵ **c**, the sampling distribution parameters a and **d**, b for Model 4, when assuming a common uniform sampling distribution (M4Unif) and **e**, the sampling distribution parameters α for Model 4, when assuming a common beta sampling distribution (M4Beta) on 41 sets of 10 simulated phylogenies. Only results for all parameters of M4Unif and the α hyperparameter of M4Beta are displayed here are the results are consistent between models for a given parameter. Each dot correspond to the proportion of phylogenies for which the metric (hmean, MAP, median or mean) is included inside the corresponding HPDI (0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95). The dashed line displays the 1:1 relationship.



Supplementary Figure 21: Accuracy and precision in term of absolute relative error of parameters estimates across different models for multiple phylogenies inference under specific diversification rates using the best fit highest posterior density point estimate.

Inferred versus simulated parameters dot plot and absolute relative error (ARE) violin plot on 41 sets of 10 simulated phylogenies for Model 4, when assuming a common beta sampling distribution (M4Beta in salmon), assuming a common uniform sampling distribution (M4Unif in khaki) and Model 5 “mult_x”, (M5 in green) applied on phylogenies simulated under the M4Unif. The point estimates used here are the metrics corresponding to the best fit of the highest posterior density for each parameter and each model. It is the median for all the parameters and models except for α the parameter of the beta sampling distribution for which the harmonic mean (hmean) was the best. The inferred versus simulated violin plot are displayed for **a**, the net diversification rate **c**, the turnover rate **e**, α the parameter of the beta sampling distribution (with fixed $\beta = 3$) of Model 4 **f**, its corresponding sampling probability mode **g**, a the parameter of the uniform sampling distribution corresponding to the lower bound of the uniform distribution of Model 4 and **h**, b the parameter of the uniform sampling distribution corresponding to the upper bound of the uniform distribution of Model 4, $BD_{\int \rho}^{mult_x}$.

Each dot is an estimate for each simulated phylogenies. The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. The panel **b**, displays full range violin plot of the ARE of net diversification rate estimates and **d**, displays zoomed violin plot of the ARE of turnover rate estimates. The boxplot inside each violin plot displays the median in white, the interquartile range as the thick bar, the rest of the distribution excluding outliers as the thin line. The horizontal dashed line displays the 0 absolute relative error. The contour of the violin plot displays the kernel density plot including the outliers.



Supplementary Figure 22: Accuracy and coverage probabilities of parameters estimation across inference procedures for multiple phylogenies under specific diversification rates.

Inferred versus simulated parameters dot plot and coverage probability bar plot on 41 sets of 10 simulated phylogenies for: Model 4 when assuming a common beta sampling distribution (M4Beta in salmon), Model 4 when assuming a common uniform sampling distribution (M4Unif in khaki), Model 3, when assuming a common beta sampling distribution on simulations from M4Beta (M3BetaOnM4Beta in mauve), M4Unif on simulations from M4Beta (M4UnifOnM4Beta in light brown) and Model 5 “mult_x”, (M5 in green) applied on phylogenies simulated under M4Unif. The point estimates used here are given by the median, which are the best for these parameters models. The inferred versus simulated plots are displayed for **a**, the net diversification rate and **b**, the turnover rate. Each dot is an estimate for each simulated phylogeny. The linear regression is displayed in solid line and the quantile regression is displayed in dashed line. The area around the dashed line delimited by the dotted lines represent the 95% confidence interval around the quantile regression. The black dashed line displays the 1:1 relationship. Panel **c** displays the coverage probability at 95% for r , ϵ , α , a and b . The value inside each barplot indicates the proportion of phylogenies for which the simulated parameter value is included inside the 95% HPDI, the lighter and darker bars respectively indicate the proportion of phylogenies for which the simulated value is below or above the 95% HPDI.

Chapter 2: Deep Learning From Phylogenies For Diversification Analyses

Recently, the sequencing revolution has fed the field of macroevolution with an increasing amount of genetic data ideal to reconstruct large phylogenies and investigate the evolutionary history of species. It has provided opportunities to test hypotheses previously established by interrogating the fossil record and examining species phenotype. This has been paralleled by an increasing complexification of diversification models to better account for realistic evolutionary scenarios. The large phylogenies and the complexification of diversification models came along with a computational cost. Indeed, the widespread approach in the field consists in fitting diversification models to phylogenies using likelihood-based statistical techniques, and computing likelihoods can be time consuming, in particular for large phylogenies. Likelihoods can be fastidious to compute and may rely on numerical approximations, or can even be intractable. As a result, phylogenetic diversification models are subject to design bias, as only those with likelihoods that are tractable and efficiently computed are likely to be developed and used. There have been only few previous likelihood-free developments in the field, and those have mainly relied on approximate Bayesian computation techniques, which have several drawbacks such as relying on summary statistics.

In this chapter, I propose a likelihood-free inference machinery for the phylogenetic study of diversification based on deep learning. To that end, I first develop a compact representation of ultrametric phylogenies, which can potentially include associated tip trait data. Next, I use either this compact representation or summary statistics, along with two types of neural networks, the fully connected feed-forward neural network and the convolutional neural network. I train the neural networks to learn diversification rates from phylogenies for two simple birth-death models, the constant rate homogenous birth-death model and the binary state speciation and extinction model. I compare the accuracy of the parameter estimates obtained with deep learning to that obtained with the commonly used maximum likelihood approach. Finally, I illustrate the approach by an empirical application on the phylogeny of primates.

The deep learning inference provides as accurate and precise diversification rates as the likelihood-based approach for the simple models tested. This imply that the proposed representation and neural network architecture can reliable be used to perform diversification rates estimations.

This chapter provides a proof of concept that deep learning can provide a powerful likelihood-free inference approach for phylogenetic diversification analyses. It is presented in the form of an article written in collaboration with Jakub Voznica (co-first author) and H  l  ne Morlon. The article has been submitted to *Systematic Biology* and is available on bioRxiv

(Lambert et al. 2022). Another contribution on a related topic has been led by Ismaël Lajaaiti (M2 intern) and is available in [Appendix](#).

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Code availability

All codes underlying this article are available on GitHub, at <https://github.com/JakubVoz/deeptimelearning>.

2.1. Introduction

Phylogenetic approaches for studying species origination and extinction dynamics over deep time rely on the statistical adjustment of stochastic birth-death models (Kendall 1948) to dated phylogenetic trees representing the evolutionary relatedness of species and the dating of their divergence times (Stadler 2013b; Morlon 2014a; Harmon 2019). An increasing amount of such phylogenetic data have become available, and has been accompanied by complexification of diversification models. These models include homogeneous rate models where speciation and extinction rates are identical across lineages at any given time, and range from simple time-constant (Nee et al. 1994c) to time-dependent (Stadler 2011a; Morlon et al. 2011a; May et al. 2016), environment-dependent (Condamine et al. 2013a), and diversity-dependent (Etienne et al. 2012a) models. Diversification models also include heterogeneous rate models (Alfaro et al. 2009; Morlon et al. 2011a; Rabosky 2014; Höhna et al. 2019; Maliet et al. 2019a; Laudanno et al. 2020b; Barido-Sottani et al. 2020), with the class of State-dependent Speciation and Extinction (SSE) models that links rate heterogeneity to specific characteristics of the species (Maddison et al. 2007a; FitzJohn 2010; Goldberg et al. 2011; Fitzjohn 2012; Beaulieu and O’Meara 2016; Herrera-Alsina et al. 2019; Vasconcelos et al. 2022).

The parameters of interest of these models – mainly speciation and extinction rates – are traditionally inferred using likelihood-based techniques – Maximum likelihood or Bayesian inference. Maximum likelihood consists in finding the parameters that maximize the probability of observing the data (here the phylogeny, or the phylogeny and associated trait data in the case of SSE models); Bayesian inference uses this probability along with *a priori* information on the parameters of interest to explore the *posterior* probability distribution of the parameters knowing the data. Numerous studies have used these inference approaches to estimate speciation and extinction rates over geological times and across the Tree of Life, to investigate the processes modulating these diversification dynamics (Stadler 2011a; Etienne et al. 2012a; Pyron and Wiens 2013; Höhna 2014; Rolland et al. 2014a; Gubry-Rangin et al. 2015; Rabosky et al. 2018; Condamine et al. 2019; Stone and Wolfe 2021). While powerful, likelihood-based inference techniques are limited by potential intractability issues for complex diversification models, and by computational cost on increasingly large phylogenetic data (Hinchliff et al. 2015). Indeed, complex models do not always have a closed-form solution and/or the likelihood cannot always be evaluated in a reasonable amount of time. This renders the application of likelihood-based methods to complex models and species-rich groups (*e.g.*, insects, micro-eukaryotes and prokaryotes) difficult. As a result, there are several models of diversification in the literature which behavior has been studied with simulations but that lack a proper inference machinery (McPeck 2008; Aristide and Morlon 2019; Hagen et al. 2021). Methods based on Expectation Maximization (EM) algorithms (Dempster et al. 1977; Richter et al. 2020), data augmentation (Maliet and Morlon 2022), or composite likelihoods (Lindsay 1988; Varin et al. 2021)) can overcome some of these limitations (Raynal 2019), yet they still rely on likelihood formulae.

An alternative is the use of likelihood-free inference techniques, such as Approximate Bayesian Computation (ABC (Beaumont et al. 2002; Marin et al. 2012; Sisson et al. 2018)). In its most basic form, ABC relies on generating artificial data by simulating the process of interest along a given parameter range and compressing the data by computing summary statistics on these simulations to enable the comparison between simulated and observed summary statistics. This comparison is done by computing a distance and evaluating if this distance is sufficiently small to accept the simulated data using a chosen tolerance threshold (rejection-based approach). ABC has been useful to fit complex models in various fields, including phylogenetic diversification analyses (Bokma 2010; Janzen et al. 2015; Janzen and Etienne 2016). However, there are several limitations of ABC, including its reliance on the choice of summary statistics that should resume the information contained in data in a low number of metrics, the choice of the distance metric to compare the observed and simulated data and the choice of the tolerance threshold. While it is possible to evaluate and minimize the influence of these choices on parameter inference (Sisson et al. 2007; Beaumont et al. 2009; Blum and François 2010; Del Moral et al. 2012; Blum et al. 2013; Prangle 2017), an adjustment needs to be performed each time a new model is developed. In particular, new summary statistics must be designed to convey information relative to the problem at hand.

Deep learning offers an alternative likelihood-free inference technique. Deep learning (Goodfellow et al. 2016) is a subfield of machine learning where highly flexible statistical learning functions based on neural networks (NNs) are used to learn regression (such as parameter estimation) or classification (such as model selection) problems. The term ‘deep’ is conventionally associated with a neural network that takes raw data as input values and extracts patterns from this low level representation thus creating its own ‘summary statistics’ or high-level features, without the need of designing those. This definition is the one used in the present article. Applying a regression task with deep learning (learning model parameter values from simulations) is increasingly used in several fields including population genetics (Sheehan and Song 2016; Sanchez et al. 2020; Avecilla et al. 2022), phylogenetic reconstruction (Nesterenko et al. 2022), macroecology (Andermann et al. 2022) and physiology (Kroll et al. 2021). In macroevolution, an early progress of using machine learning (Bokma 2006) consisted in training an artificial neural network on the axes of a principal component analysis of phylogenetic branching times to infer speciation and extinction rates. A similar framework was then used for the inference of rates of phenotypic evolution (Bokma 2010). While promising, this approach was not developed further by the community.

A step forward was recently taken by Voznica et al. (2022), who developed a deep learning approach for the statistical inference of birth-death models from phylogenies in the context of pathogen phylodynamics. The authors developed a tree representation, the Complete Bijective Ladderized (CBLV) tree representation, which applies to non-ultrametric trees representing the evolutionary relationship between pathogen sequences sampled at different dates. The CBLV representation proved to be efficient for model selection and inference of transmission dynamics when combined with Convolutional Neural Networks (CNN) (LeCun et al. 1998), where it yielded accuracy at least comparable to gold-standard Bayesian

approaches. Voznica et al. (2022) also combined an extensive set of Summary Statistics with Feed-Forward Neural Networks (FFNN-SS) that yielded similar results. To our knowledge, comparable attempts to use deep learning to infer diversification dynamics from species phylogenies do not exist.

Here, we adapt the approach of Voznica et al. (2022) to birth-death diversification models used to infer diversification dynamics from phylogenies of extant species (and also potentially associated trait data). The phylogenetic data are different from the one in Voznica et al. (2022). First, the evolutionary trees are different as, if no data from fossils is included, the species are all sampled at present time and the reconstructed phylogenies are thus ultrametric, that is the tips are all at the same distance to the root. Second, we allow for possibility to include trait data associated to the tips. We begin by adapting the CBLV tree representation from Voznica et al. (2022) to ultrametric phylogenies and to ultrametric phylogenies with tip state data, in the form of ‘Complete Diversity-reordered Vector’ (CDV). We then assess the performance of deep learning inference in comparison to maximum likelihood estimation (MLE) by using the simple homogeneous time-constant birth-death (BD) model (Nee et al. 1994c), for which a closed-form expression of the likelihood exists, and the binary-state speciation and extinction (BiSSE) model, for which the likelihood is approximated by solving Ordinary Differential Equations (Maddison et al. 2007a). Finally, we illustrate the approach by applying our trained neural network for BiSSE to an empirical phylogeny of 273 primates (Fabre et al. 2009) and their associated interaction type (mutualistic or antagonistic) with plants (Gómez and Verdú 2012).

2.2. Methods

Our main goal is to develop a compact and exhaustive representation of the raw data into a matrix (the CDV), and to test the performance of a CNN combined with this representation for parameter inference (thereafter referred to as the CNN-CDV approach). We train the neural networks with data simulated under the birth-death diversification processes. For the time-homogeneous birth-death model (BD), we compare the performance of CNN-CDV to FFNN combined with a series of summary statistics (FFNN-SS), FFNN combined with the CDV representation (FFNN-CDV), and the maximum likelihood estimation (MLE) approach. For the BiSSE model, we compare the performance of CNN-CDV to FFNN-CDV and MLE. Codes used to perform the simulations, encode the phylogenetic data into the CDV representation, train the neural networks, and use the trained networks on simulated or empirical data for parameter inference, are available on GitHub (<https://github.com/JakubVoz/deeptimelearning>).

2.2.1. Tree Representations

2.2.1.1. Full Tree Representation: Compact Diversity-reordered Vector (CDV)

We adapted the Compact Bijective Ladderized Vector (CBLV) representation used in Voznica et al. (2022), originally intended for non-ultrametric trees, to ultrametric trees and to ultrametric trees with information on tip states. The encoding proceeds in the following steps ([Fig. 1](#)): 1] internal node reordering, 2] inorder tree traversal (Cormen 2009) and creation of vector representation, 3] completion to the maximum tree-size in simulations, 4] addition of tree height and sampling probability to the vector representation.

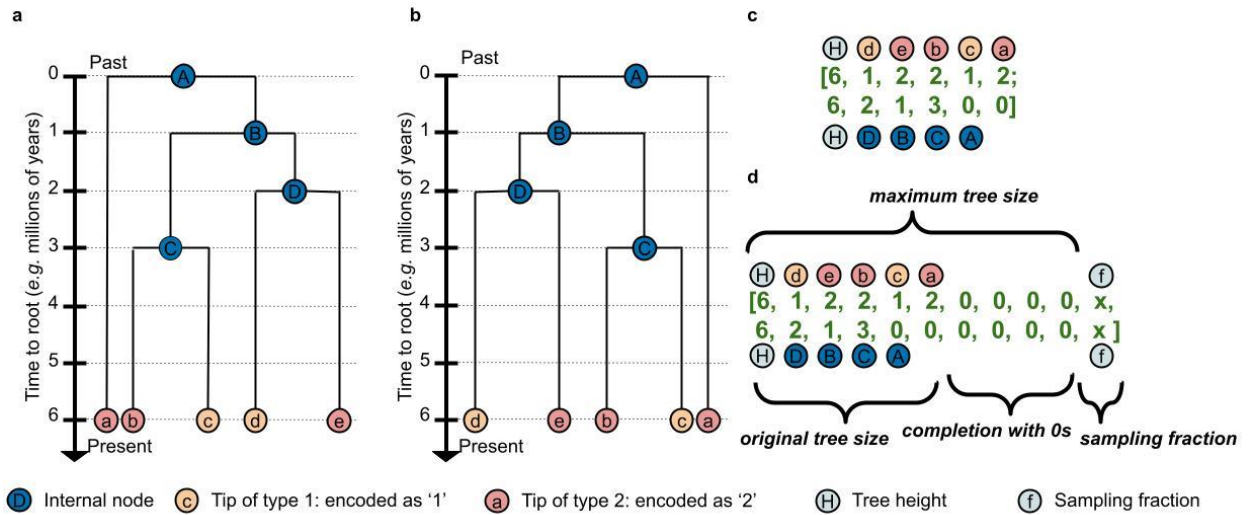


Figure 1: Representation of ultrametric trees with tip state data.

Illustration of the encoding algorithm for 'Compact Diversity-ordered Vector' (CDV) on a tree with 5 tips. (a) Ultrametric tree with tip state information (here there are two possible states for each tip, either state 1 or state 2, represented in orange and pink respectively). (b) The tree is reordered following a diversity criterion: for each internal node, the sum of the branch lengths of the descending tree is computed and the internal node with higher sum is rotated on the left. (c) We then create a 2-rows matrix filled by visiting the tree according to a tree inorder traversal algorithm, with tips represented on the top row and internal nodes on the bottom row. Tips are assigned their encoded trait state ('1' for state 1 and '2' for state 2) and internal nodes their distance to the root. We add a first column with tree height. (d) Finally, we complete the matrix with zeroes, so that its size is the one of the largest simulated tree and we add a column with the value of the sampling fraction. To obtain the representation of an ultrametric tree without tip state data, we do not include the first row on tip state information.

Before encoding, we rescaled the trees to unit average branch length and the rate parameters (e.g. diversification rate(s), turnover rate(s) etc.) accordingly. The trained neural networks should thus apply to trees with any time scale. The criterion used in Voznica et al. (2022) for tree reordering was based on the ladderization, where each internal node is rotated

so that the branch supporting the most recent tip is on the left. As this cannot apply to ultrametric trees where all tips are sampled at the same time, we used a diversity criterion: for each internal node we compute the sum of the branch lengths of the descending tree (i.e. phylogenetic diversity) and the branch with highest sum is shifted to the left. Next, we perform a tree inorder traversal: the reordered phylogeny is traversed by recursively starting with the left subtree and for each visited internal node, its distance to the root is added to a vector (Cormen 2009). For trees with tip data, we create a second vector with information on the tip data, while visiting tips during the same traversal. For binary tip data (as obtained by simulating BiSSE for example), we use the values 1 and 2 to distinguish the two states. These two vectors are then combined into a matrix. We then add a first column with the value of the tree height. We name this representation a Compact Diversity-ordered Vector (CDV). This representation could be easily extended to account for information on non-binary traits, for example using one hot encoding, which consists in encoding a qualitative variable of x states into x rows with 1 representing the presence of state and 0 its absence. Similarly, multiple traits, including quantitative ones, can be encoded by stacking additional rows to the matrix. Furthermore, we can imagine treating missing trait data by assigning a value of -1 in the CDV representation when the information is lacking.

The CDV representation is bijective (under mild assumptions, for example the absence of concomitant branching events), in the sense that we can unambiguously reconstruct any given tree from its representation, and compact: $(x+1)*n$ entries for a tree with n tips ($n-1$ values for internal nodes and 1 on tree height) and x informational values on tips ($x=0$ in the absence of tip state information and $x=1$ for information on a single trait). The created vector (or matrix when tip state information is available) is completed with zeroes to obtain a representation of the same size as the largest phylogeny in the simulations. In order to account for potential missing extant species in the phylogeny, we add a last column with the value of the sampling fraction, computed as the ratio of the number of species represented in the phylogeny divided by the total number of extant species.

In order to assess whether the CDV representation with tip data allows to properly capture tip information, we compared results obtained with this representation to those obtained with a less informative representation (coined CDV-less) where we add only two values on tip type counts (i.e. the number of tips in each state) at the end of the row summarizing internal nodes.

2.2.1.2. *Summary Statistics Representation*

We used a set of 97 summary statistics (SS) representing trees (without associated trait data as we do not implement the FFNN-SS for the BiSSE model). The summary statistics were mainly based on those published in Saulnier et al. (2017) and in Voznica et al. (2022) (see the original articles for details) and they comprise:

- 8 summary statistics on tree topology
- 25 summary statistics on branch lengths
- 49 summary statistics on the Lineage-Through-Time (LTT) plot
- 14 summary statistics on consecutive internal branches
- 1 summary statistics on number of tips

We modified several statistics so that they apply to ultrametric trees. Instead of minimal and maximal tree height (the time of first and last sampled tips in non-ultrametric trees), we use the crown age. In Saulnier et al. (2017), there are several summary statistics defined on the LTT plot which consider the maximum number of the living lineages, the time of its occurrence and the slopes of the curves before and after this time. In ultrametric trees, the maximum number of the living lineages always appears at present (the moment of sampling tips) and thus such division of LTT plot is not possible. Instead, we divide the LTT plot into three equal parts and we measure the slope for each one, together with the ratios between the first and the second slope and between the second and the third slope. The computing time of these statistics grows linearly with tree size. We added the sampling fraction to these 97 measures, thus resulting in a vector of 98 scalars.

Finally, we reduced and centered the SS by subtracting the mean and scaling to unit variance, using the standard scaler from the scikit-learn package (Pedregosa et al. 2011) fitted to the training set.

2.2.2. *Neural Networks: Architecture and Training*

A NN is organized in neural layers that in turn are organized in neurons (or ‘units’). In supervised learning, a NN can be trained to minimize the difference between an expected output (or target) and the predicted one, the measure of the difference being called a ‘loss function’. Here we used the mean absolute error as the loss function. A NN contains an input layer (by which the numerical values representing the data are passed to the following layer) and an output layer (in our case outputting parameter values), potentially separated by hidden layers. If there is at least one hidden layer, we talk about deep neural networks.

Feed-Forward Neural Networks (FFNNs) are one of the most basic forms of deep NNs. They are fully connected: for each neural layer, all neurons are connected to the neurons of the previous layer. The connections are characterized by trained bias and weights *i.e.* real values

by which individual inputs of a given neuron are multiplied, as well as an activation function by which the summed input (input values multiplied by weights to which a bias value is added) is transformed. Here, we used the exponential linear function as activation function (Clevert et al. 2015). FFNNs typically work well on structured data, such as summary statistics, where the same summary statistics are at exactly the same entry in the input vector. On unstructured data (such as images or CDV) however, the number of parameters to train increases quadratically with the size of the input, and the information is scattered along the input vector, making FFNNs potentially less efficient.

Convolutional Neural Networks (CNNs) (LeCun et al. 1998) contain a convolutional part and a fully connected one. The convolutional part consists of convolutional and pooling layers and outputs a vector used as input of the fully connected part. The convolutional part aims at learning and extracting repeated patterns in the input, that are then combined for prediction in the fully connected part. Convolutional layers transform their input with several convolutional operations, each one specified by a kernel (or ‘filter’ or ‘feature detector’) whose parameters are trained. Each kernel transforms subparts or patches of the input by applying the convolutional operation and outputs a single value for each patch. We set the stride (by how much the kernel moves on the input when traversing it) to 1. A convolutional layer is specified by the number of kernels and their size (the size of their input), and outputs a ‘feature map’ (intermediate representation of transformed input). Pooling layers transform the resulting feature maps into smaller ones by taking the maximum or average of values subsampled in the map with a given window. CNNs typically work well with raw, low-level (vector and matrix) data such as images, videos or time-series recordings, by learning and extracting repeated patterns or ‘features’ through trained convolution functions and building from them their own high-level features (such as a set of summary statistics). They do not need prespecified feature input such as summary statistics. Each kernel learns and extracts one pattern in the data that can appear anywhere in the representation. In comparison to Voznica et al. (2022), increased kernel size in the first convolutional layer performed slightly better (see below; data not shown).

The training of a given network consists in iteratively changing the parameters of the NN (*e.g.* bias, weights) that minimize the loss function. This is performed by an optimization algorithm for stochastic gradient descent. Here, we used the Adam optimizer (Kingma and Ba 2015). The networks were trained on simulated data, the targets being the parameters of the diversification models (here BD and BiSSE).

Several training ‘tricks’ were developed for efficient and robust training in practice. We used a training set of 990.000 simulations split into subsets called batches, during the training. After measuring the loss on the whole batch, the trained parameters are updated with the optimizer to minimize the loss. Splitting the training set into batches of simulations enables to update the trained values more robustly (and moving into ‘right direction’ with respect to the minimal error). We set the batch size to 8,000 simulations. When the network parameters were updated on the whole training set (we talk about an ‘epoch’), the training starts again passing through the whole training set.

To prevent overfitting, we used a dropout of 0.5, which consists in shutting down randomly half of the neurons in the network during the training phase (Srivastava et al. 2014). We also used a technic called early stopping (Bengio 2012): at the end of each epoch, the loss is computed on a validation set (here, we used a validation set of 10.000 simulations), and the training is stopped when the loss on the validation set starts to increase. Typically, during the training of our networks, there will be hundreds of epochs before the training is stopped. The test set then enables to measure the true accuracy of the network. In our case it consisted of 500 simulations for the BD model and 10,000 simulations for BiSSE.

For the BD model, we used a CNN architecture on the CDV representation (referred to as CNN-CDV), a FFNN architecture on the CDV representation (FFNN-CDV), and a FFNN architecture on the summary statistics representation (FFNN-SS). For the BiSSE model, we used only the CNN-CDV and the FFNN-CDV, as applying the FFNN-SS would require deploying a new set of summary statistics accounting for tip data. We used the same FFNN architecture as in Voznica et al. (2022) in the case of both the FFNN-CDV and the FFNN-SS (see the original article for details). The CNN architecture differed only by the size of the kernels (i.e. the size of their inputs) in the first layer; we used layers of size 5×2 and 5 for respectively the CDV and CDV-less as it performed slightly better than kernels of size 3×2 (respectively 3, data not shown). Also, unlike in Voznica et al. (2022), the function of the output layer was set to the exponential linear function (Clevert et al. 2015).

We implemented the NNs in Python 3.6 using the Tensorflow 1.5.0 (Abadi et al. 2016), Keras 2.2.4 (Chollet 2015) and scikit-learn 0.19.1 (Pedregosa et al. 2011) libraries.

2.2.3. Macroevolutionary Models and Simulations

We assessed the performance of the NNs with two widely used models, the simple homogeneous, time constant birth-death (BD) model, and the binary-state speciation and extinction (BiSSE) model.

2.2.3.1. *The Birth-Death (BD) Model*

The BD model has a closed-form expression of its likelihood, which allows us to compare the performance of the Deep Learning and MLE approach in a “best case” scenario for MLE. Here, when referring to BD, we imply the homogeneous time-constant birth-death-sampling model (Yang and Rannala 1997; Stadler 2009), as we consider the possibility that some extant species are not represented in the phylogenies. In this model, new species originate with a constant speciation rate λ and go extinct with a constant extinction rate μ , typically expressed in number of events/lineage/Myr. At present each extant species is sampled with probability f (Bernoulli sampling scheme (Stadler 2009)). We assume f to be fixed, in which case λ and μ are identifiable.

We parametrized our simulations with the turnover ($\varepsilon = \mu / \lambda$) and speciation rate (**Table 1**); the parameter values were sampled uniformly at random within parameter boundaries with standard Latin-hypercube sampling (McKay et al. 1979) using the Python PyDOE package. We performed the simulations with our own simulator, using a Gillespie algorithm (Gillespie 1977). Each simulation started with one lineage and ended when the number of living species reached $\frac{s}{f}$, where s is the number of tips in the sampled phylogeny. We then sampled s species. We thus conditioned the simulations on the number of tips. We trained the NNs to learn λ and ε .

Table 1: Parameterization of the constant-rate birth-death model with incomplete sampling.

Parameters	Symbols	Ranges
Turnover rate	ε	U(0.01,1)
Speciation rate	λ	U(0.01,0.5)
Tree size	s	U(200, 500)
Sampling fraction	f	U(0.01,1)

For each parameter, we display its symbol and the range of values used in the simulations by sampling from the uniform distribution (indicated with U) in training, validation and testing sets. Parameters indicated in bold are those that are estimated during the inference.

2.2.3.2. The Binary State Speciation and Extinction (BiSSE) Model

The BiSSE model is the simplest state-dependent birth-death model: species are characterized by a binary state (1 or 2) which can influence their constant speciation (λ_1 and λ_2) and extinction rates (μ_1 and μ_2). Species can also transition anagenetically from one state to the other (with rates q_{12} and q_{21}). As for the BD model, we can add to this diversification process a Bernoulli sampling scheme at present, which allows analyzing trees with missing extant species. We consider here a simple version of BiSSE with symmetrical transition rates $q = q_{12} = q_{21}$ as well as turnover rate and sampling probabilities at present (ε and f , respectively) shared across species irrespective of their state.

The BiSSE model allows us to illustrate the utility, and test the validity of the CDV representation with tip data. The likelihood of this model can be computed by solving Ordinary Differential Equations (ODEs), and recent efforts have been made to provide an efficient maximum-likelihood inference machinery for this model on large phylogenies (Louca and Pennell 2020a).

We parameterized our simulations with the speciation rate associated to state 1 (λ_1), the turnover rate (ε) and the ratios of λ_2 and q_{12} relative to λ_1 . We sampled these parameters within a biologically realistic parameter space, given the literature on empirically inferred parameters (e.g. (Villarreal and Renner 2013; Williams et al. 2014; Gamisch 2016)) (**Table 2**). The parameter subspace was covered with standard Latin-hypercube sampling (McKay et al. 1979)

using the Python PyDOE package. The simulations were performed using the R package castor 1.6.6 (Louca et al. 2018) and were conditioned on number of tips. We trained the NNs to learn λ_1 , λ_2 , q_{12} and ε .

Table 2: Parameterization of the Binary State Speciation and Extinction model with incomplete sampling.

Parameters	Symbols	Range
Turnover rate	ε	U(0,1)
Speciation rate 1	λ_1	U(0.01,1)
Speciation rate 2 and its ratio to λ_1	λ_2, r_{λ_2}	[10e-3,1] U(0.1,1)
Transition rate and its ratio to λ_1	$q_{12}=q_{21}, r_q$	[10e-4,0.1] U(0.01,0.1)
Tree size	s	U(200, 500)
Sampling fraction	f	U(0.01,1)

For each parameter, we display its symbol and the range of values used in the simulations by sampling from the uniform distribution (indicated with U) in training, validation and testing sets. Parameters indicated in bold are those that are estimated during inference. λ_2 is parameterized with respect to λ_1 , being at 10% to 100% of its value. The transition rates are also parameterized with respect to λ_1 , being at 1% to 10% of its value. The corresponding ranges of values are indicated in brackets.

2.2.4. Maximum Likelihood Estimation

We compared the NN approaches with MLE. For the constant-rate BD, we used a MLE based on the Nelder–Mead optimization algorithm encoded within our custom function fitMLE_bdRho available at <https://github.com/sophia-lambert/UDivEvo/tree/master/R>. The function encodes a likelihood formula conditioned on the age of the phylogeny (here $t_0 = t_{\text{crown}}$) under a Bernoulli sampling scheme, and parametrized to infer the net diversification rate ($r = \lambda - \mu$) and the turnover rate (ε) as it is easier to maximize the likelihood in this reparametrized likelihood landscape. For the BiSSE model, we used the MLE deployed under the R packages diversitree 0.9-3 (Fitzjohn 2012) and castor 1.6.6 (Louca and Doebeli 2018), both conditioned on the crown age of the phylogeny, under a Bernoulli sampling scheme and parametrized to infer λ_1 , λ_2 , μ_1 , μ_2 , and $q=q_{12}=q_{21}$. Diversitree is the traditional package used for fitting BiSSE; castor was developed more recently, and implements a faster algorithm for computing the likelihood of SSE models on large phylogenies (Louca and Doebeli 2018).

2.2.5. Performance Assessment

2.2.5.1. Accuracy of Parameter Estimation

To assess the accuracy of parameter estimation, we used 500 simulated test trees for the simple BD model and 10 000 for BiSSE. 17 BiSSE simulations for which castor and/or diversitree outputted an error message or no estimated values (15 for castor, 3 for diversitree with one simulation in common) were excluded from these analyses, resulting in 9.983 test trees.

To avoid over-penalizing the MLE approaches that, contrary to the NNs, do not have constrained parameter ranges, we added similar constraints to the MLE estimates. Indeed, to the exception of λ for constant-rate BD and λ_1 for BiSSE for which NNs can predict values outside of the parameter values initially covered by the simulations due to tree rescaling, the other parameter estimates (ε , λ_2 and $q=q_{12}=q_{21}$) are constrained by the parameter range used for ε , r_{λ_2} and r_q in our simulations. We imposed similar constraints to the MLE estimates to prevent overpenalizing MLE in accuracy comparison. For example, if the MLE for ε is above 1, we set it to 1, and if the MLE for λ_2 is lower than $0.1 \cdot \lambda_1$, we set it to $0.1 \cdot \lambda_1$ (0.1 is the minimum value for r_{λ_2} used in our simulations).

To evaluate the distribution of the bias among the test set, we calculated the bias between the true (simulated, or ‘target’) parameter values and the predicted values per predictions as follows:

$$\text{Bias } B_i = \text{predicted}_i - \text{target}_i$$

We computed three measures of error between the true parameter values and the predicted values on the test set:

- Mean absolute error $MAE = 1/n \cdot \sum_i^n \text{abs}(\text{predicted}_i - \text{target}_i)$
- Mean relative absolute error $MRE = 1/n \cdot \sum_i^n \text{abs}(\text{predicted}_i - \text{target}_i) / \text{target}_i$
- Mean bias $MB = 1/n \cdot \sum_i^n (\text{predicted}_i - \text{target}_i)$

We also report the Pearson correlation coefficient between simulated and predicted values, computed with the R package ‘stats’ (‘cor.test’ function) version ‘4.2.1’.

Finally, we assessed the influence of tree size on parameter estimation accuracy.

2.2.5.2. Time Efficiency

We compared the average time of estimation between CNN-CDV and MLE for BiSSE. For the CNN-CDV approach, we reported the average CPU time of encoding a tree (averaged over 1,000,000 trees). The estimation on itself is negligible with respect to the time of encoding. For

MLE BiSSE estimation with the *castor* and *diversitree* packages, we reported the average CPU time (average over 10.000 test trees, out of which 18 resulted in an error).

2.2.6. Empirical Illustration

Primates can have an antagonistic interaction with plants through herbivory, or a mutualistic one through frugivory and seed dispersal (Gómez and Verdú 2012). To illustrate the application of our deep learning approach, we reanalyzed the dataset of Gómez and Verdú (2012), that categorizes primate species according to the nature of their interaction with plants (state 1 for a mutualistic interaction, and 2 for an antagonistic one), using the primates phylogeny of Fabre et al. (2009). We started by pruning taxa without information on interaction type (13/273) and rescaled the phylogeny as previously described. Next, we performed some sanity checks to verify that the empirical data fell within the space covered by our BiSSE simulations; if it does not, this means that the model and/or the range of parameters used in the simulations is not well adapted to the empirical data, in which case application of the trained neural network to the data might output meaningless results. For these analyses, we used the set of phylogenies that we simulated under the BiSSE model to produce our test set. First, we checked that each of the summary statistics values calculated on the empirical data fell within the range spanned by the summary statistics values calculated on the simulated phylogenies. Then, we performed a principal component analysis (R package ‘FactoMineR’, function ‘PCA’) on the set of summary statistics described above (see Methods) with the addition of four simple summary statistics adapted to the inclusion of tips data: the number of tips in each state (1 or 2) and the phylogenetic diversity of each state (R package ‘picante’, function ‘pd’). Finally, we transformed the data into its CDV representation and fed it to the trained CNN-CDV network on the BiSSE model. We used a sampling fraction of 0.68, computed using a global diversity of 381 stable species complexes for primates, following Gómez and Verdú (2012).

2.3. Results

2.3.1. Performance Assessment

The comparison of parameter estimates obtained using deep NNs versus MLE for the BD model shows that CNN-CDV and FFNN-SS are as accurate as MLE, while FFNN-CDV has a lower accuracy, in terms of bias, absolute and relative error, for the speciation, extinction, net diversification and turnover rates ([Fig. 2](#), [Table S1](#)). The likelihood of BD model has an exact analytical solution. This implies that MLE, together with CNN-CDV and FFNN-SS, are as accurate as one can be. The good performance of FFNN-SS might be explained by the representation of the lineage-through-time plot in the summary statistics, that contains all

information available in the tree for homogeneous BD models (Nee et al. 1994c). The lower performance of FFNN-CDV was expected given that FFNN is less adapted to unstructured data. The neural networks seem to avoid cases of high negative bias on the turnover rate compared to MLE: while this bias can reach down to -0.6 with MLE, it never falls behind -0.35 with CNN-CDV and FFNN-SS (Fig. 2d). This is probably due to the fact that contrary to MLE, CNN-CDV and FFNN-SS rarely return estimates close to 0 for the turnover rate (Fig. S1d).

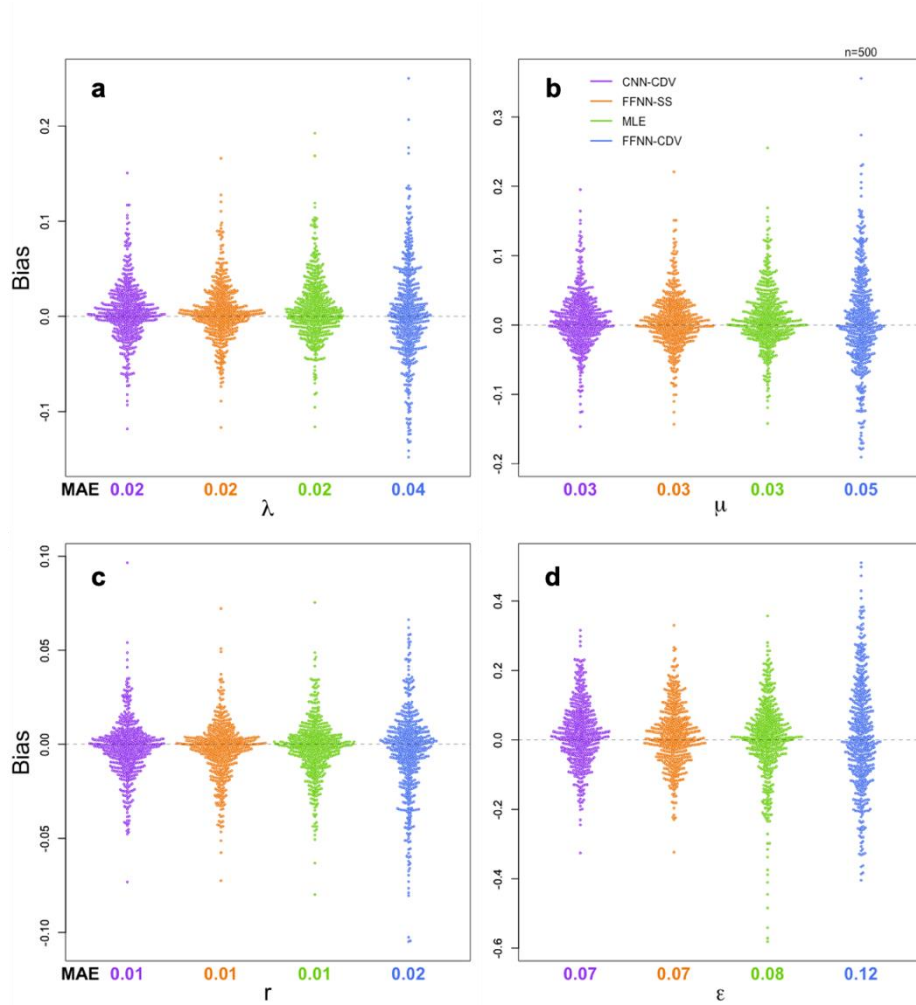


Figure 2: Comparison of estimation accuracy between pretrained CNN-CDV, FFNN-SS, FFNN-CDV and MLE for the BD model.

Swarm plots representing the distribution of estimation biases across 500 simulations. Each dot represents the bias of a single simulation. Estimated parameters were obtained with Convolutional Neural Networks- Complete Diversity-reordered Vector (CNN-CDV in purple), Feed-Forward Neural Networks- Summary Statistics (FFNN-SS in orange), Maximum Likelihood Estimation (MLE in green) and Feed-Forward Neural Networks- Complete Diversity-reordered Vector (FFNN-CDV in blue) for (a) the speciation, (b) the extinction, (c) the net diversification and (d) the turnover rate. The mean absolute error (MAE) is displayed under each swarm plot.

The comparison of parameter estimates obtained using deep learning versus MLE for the BiSSE model confirms that CNN-CDV is at least as accurate as MLE. The only exception

is for λ_2 estimates, where MLE implemented in diversitree is slightly more accurate than CNN-CDV in terms of mean absolute error. CNN-CDV is more accurate than the fast MLE algorithm implemented in castor for all parameters (Fig. 3, Table S2). The accuracy of the MLE estimation implemented in castor is similar to that of the least reliable FFNN-CDV neural network; it performs slightly better for some parameters (μ_1 and q_{12}) but slightly worse for others (μ_2). The CNN trained on a CDV without the individual tip information (CNN-CDV-less) is much less accurate, indicating that the information on individual tips states is well presented in the CDV representation, and extracted by the CNN (Table S2).

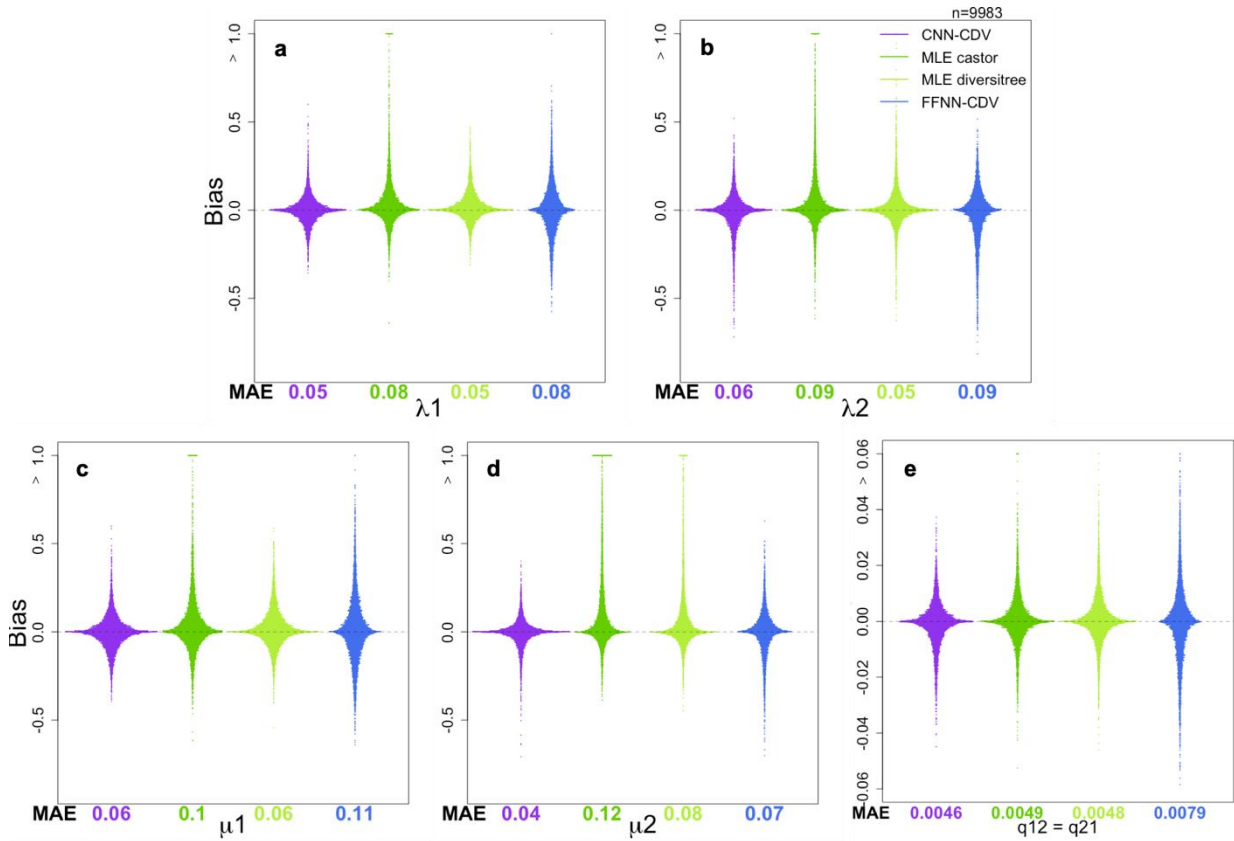


Figure 3: Comparison of estimation accuracy between pretrained CNN-CDV, FFNN-CDV, and MLE obtained with two different inference software for the BiSSE model.

Swarm plots representing the distribution of estimation biases across 9,983 simulations. Estimated parameters were obtained with CNN-CDV (in purple), castor (in dark green), diversitree (in light green) and FFNN-CDV (in blue) for a) the speciation rates 1 and b) 2, c) the extinction rates 1 and d) 2 and e) the transition rates ($q_{12}=q_{21}$, in our setting). The simulations for which castor or diversitree outputted an error message (15/10,000 for castor and 3/10,000 for diversitree) were excluded from the comparison. The MAE is displayed under each swarm plot. For visualization purposes, the upper outliers of the swarm plot are not displayed at their extreme values but instead are put at the boundaries of the chart.

For both the BD and BiSSE model, the accuracy of CNN-CDV increases as the tree size increases (Fig. 4). Similar to MLE, a tree size of at least 300 sampled extant species is required for a median relative absolute error of 6% for the speciation rate and 18% for the extinction rate

in the case of the BD model (Fig. S3). In the case of the BiSSE model, a tree size of at least 380 is required for a median relative absolute error of less than 14% on the speciation rates, 25% on the extinction rates, and 13% on the transition rate.

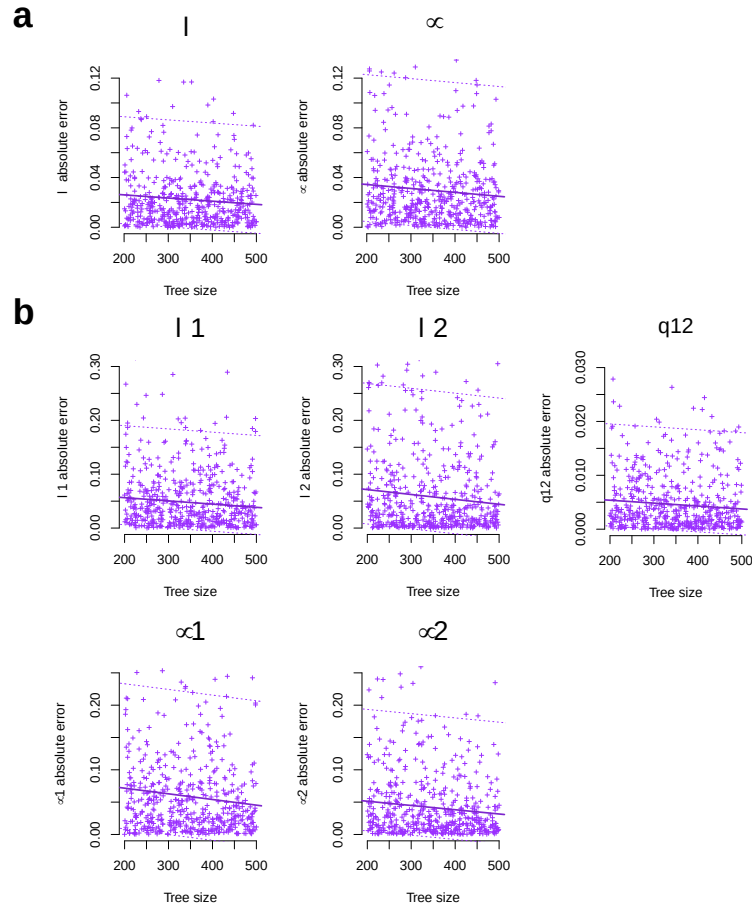


Figure 4: Effect of tree size on the absolute error of parameter estimates when using CNN-CDV under the BD and BiSSE models.

For each model a) constant-rate birth-death (BD), and b) Binary-State Speciation and Extinction (BiSSE), we display the regression on absolute error for each parameter as a function of tree size for 500 test trees (for BiSSE 500 values are shown instead of 10.000 for visualization purposes). The area around the solid line delimited by the dotted lines represent the 95% confidence interval around the regression.

The parameter estimation for the 10.000 BiSSE simulations took 629.5 CPU hours with diversitree, 55.6 CPU hours with castor and 0.4 CPU hours with CNN-CDV, which consisted in encoding the test set into CDV. Once the CNN-CDV is trained, the estimation is thus around 140 times faster than castor and over 1500 times faster than diversitree. Training the network entailed first simulating the training set (1 million trees, 40 CPU hours). The training in itself then took 8 CPU hours. Contrary to MLE for which a large number of CPU hours are required for each new empirical analysis, with deep learning the empirical analyses are very fast once the network has been trained. The same pre-trained networks should be applicable to a large

variety of empirical trees, thanks to tree rescaling which enables applications to clades of very different ages and speciation rates.

2.3.2. Empirical Illustration

The primate phylogeny with associated character state (mutualistic or antagonistic interaction with plants) passed the sanity checks on model adequacy. Indeed, all the summary statistics computed on the empirical data fell within the range spanned by the simulations. Likewise, our PCA analysis on these summary statistics showed the empirical data nested in the simulations space, when considering both PC1 and PC2 (explaining together 69% of the variance of our data, [Fig. 5a](#)) and PC3 and PC4 (that represent an additional 12 points of explained variance, [Fig. 5b](#)). The CNN-CDV analyses estimated a speciation rate of 0.295 for primate species with a mutualistic interaction with plants, and of 0.093 for those with an antagonistic interaction. The turnover rate ε was estimated at 0.234, and the transition rate at 0.0089. The resulting estimated net diversification rate is 0.225 for primates with a mutualistic interaction, and 0.071 for those with an antagonistic interaction. This simple analysis suggests that mutualistic interactions can favor diversification in primates, as found by (Gómez and Verdú 2012), although we do not interpret this result further here, as models with hidden traits should be used to reach more convincing biological conclusions (Beaulieu and O'Meara 2016). The goal of this empirical analysis is simply to illustrate how the method and trained networks can be used on empirical data.

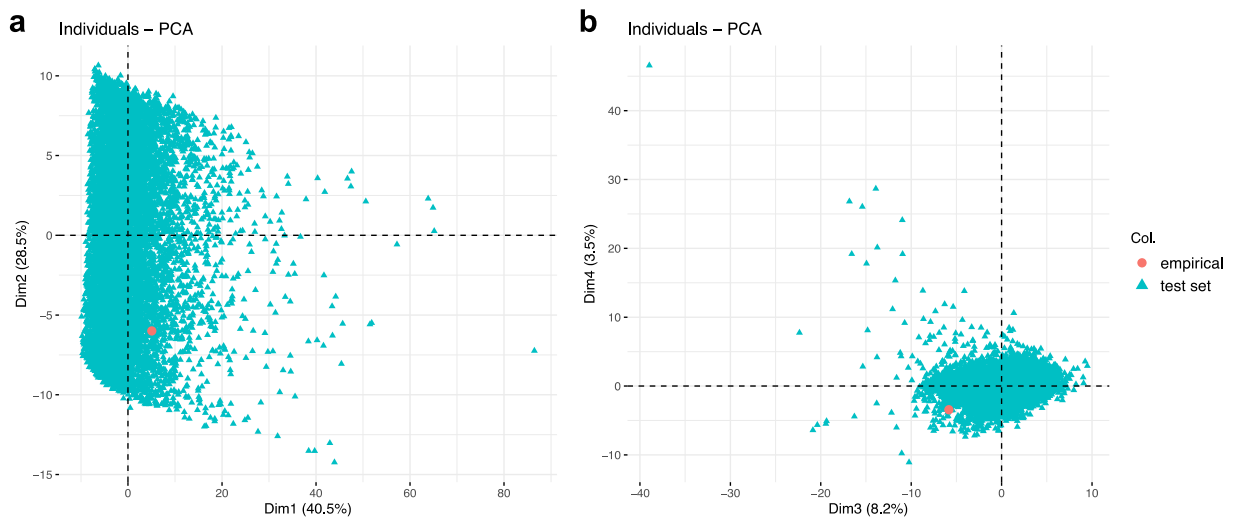


Figure 5: The primate data falls within the space of our BiSSE simulations.

Coordinates of the empirical data (in pink) and of the test set simulations (turquoise) on the a) PC1 and PC2 axes and b) PC3 and PC4 axes of a principal component analysis performed on 102 summary statistics.

2.4. Discussion

We developed, tested and illustrated the use of a deep learning based inference approach for phylogenetic diversification analyses, including the case of trait-dependent diversification. We found that both convolutional neural networks combined with a compact representation of the phylogenetic data into a matrix (the CDV) and feed-forward neural networks combined with summary statistics can reach levels of parameter estimation accuracy comparable to those obtained with the well-established maximum likelihood approach, while being faster by several orders of magnitude.

To demonstrate the potential of deep learning for phylogenetic diversification analyses, we worked with two simple diversification models on which MLE estimates can easily be obtained for comparison (the BD and BiSSE models). We also worked with phylogenies of relatively moderate size (200 to 500 extant species sampled). The real value of deep learning will be to allow rapid inferences for more complex models for which likelihoods are not tractable or long to compute and for large phylogenies of several thousands of extant species. Our analyses on simple models demonstrate that it is worth putting efforts and computation power into simulating more complex models, generating larger trees, and training neural networks on such simulations. Given our results, we can expect efficient simulators combined with the CDV representation and CNN to provide an accurate likelihood-free estimation method applicable to very large phylogenies. The CDV representation is not model-specific and can easily be enriched with information on both internal nodes and tips. It could be used for example to represent information on species multidimensional traits, geographic distributions, abundances, and genetic diversity. Combined with efficient simulation models for the evolution of biodiversity (Hagen et al. 2021), the CNN-CDV deep learning inference approach could help adjusting biologically realistic biodiversity models to multifaceted data for a better understanding of how present-day biodiversity was generated, maintained, and distributed geographically.

Parameter estimates are more meaningful when associated with a measure of confidence. With our deep learning framework, obtaining a confidence interval around the estimates can be achieved through approximated parametric bootstrapping (Voznica et al. 2022), which uses the distribution of prediction error measured on simulations. Besides parameter estimation, diversification models are widely used for model comparison, in order to test alternative hypotheses about how diversification proceeds. This problem can be treated with deep learning as a classification problem, with neural networks trained to distinguish simulations from different models. This approach has been developed in Voznica et al. (2022) and shown to perform well. In the case of the BiSSE model for example, a neural network could be trained to distinguish data simulated under a model where states influence speciation rates from data simulated under a model where they do not. Given that speciation rates can be estimated with good accuracy with deep learning, we expect that neural networks will also be

able to efficiently distinguish these models given enough differences in speciation rates between states.

We have explored the applicability of CNNs and FFNNs for phylogenetic diversification inference. Other classes of neural networks may also bring accurate solutions for parameter estimation and model selection when combined with simulated datasets. Noticeably, the Graph Neural Networks or Graph Convolutional Networks are neural networks designed for graph data that could be particularly well suited for analyzing phylogenies, encoded as directed graphs. Applications of the GNNs have been developed in the last couple of years across different fields, for instance in protein interaction prediction, drug design or social networks analyses (Zhou et al. 2020). More work is required to assess which network architecture performs best for phylogenetic diversification analyses, which has been initiated in Lajaaiti et al. (*in prep cf Annexe*).

As the ground truth is unknown, the neural networks are trained on simulations. This raises questions on how robust such approaches are to model misspecification, even though some studies suggest that machine learning approaches might be more robust to model misspecification than other inference approaches (Liang and Jordan 2008; Lee et al. 2010). We illustrated how some sanity checks can be performed to verify that the empirical data falls within the space of simulated data. If it does not, outputs of the neural networks should not be trusted. As in Voznica et al. (2022), we used summary statistics for these sanity checks. Another approach circumventing the use of summary statistics would consist in using autoencoders (Hinton and Salakhutdinov 2006), often deployed for anomaly detection (Chalapathy and Chawla 2019). The autoencoders are a family of NNs, which are trained to output their input while enforcing dimensionality reduction within their neural layers. The induced reconstruction error, *i.e.* the difference between the input and output, can then be used to check if an empirical data is well represented by simulations. For example, autoencoders could be trained on the CDV representation of phylogenetic data simulated under diversification models, and then applied to empirical phylogenetic data. If the reconstruction error is larger for the empirical data than for the simulations, this indicates departures from the simulated model.

We have considered the task of fitting birth-death diversification models to a fixed phylogeny, assumed to be known. While this is a current practice in the field, a better way to account for phylogenetic uncertainty consists in performing full phylogenetic inference, where the phylogenetic tree is inferred from molecular sequence alignments jointly with the parameters of the diversification process (Bouckaert et al. 2014, 2019). Machine learning is already used to data mine molecular sequence alignments (Yang et al. 2020), and recent progress has been made for phylogenetic reconstruction as well (Suvorov et al. 2020; Zou et al. 2020; Nesterenko et al. 2022; Solis-Lemus et al. 2022). Ultimately, these recent advances could be in the long term combined to train CNNs directly on sequences simulated from a joint process of speciation, extinction, and sequence evolution.

Deep learning is gaining popularity in biology, including in ecology and evolution (Borowiec et al. 2021). It has been used as a likelihood-free approach to fit population genetic models

(Flagel et al. 2019) to sequence data, and more recently to fit epidemiological models to pathogen phylogenies (Voznica et al. 2022). We have shown that it can also perform well as a likelihood-free approach for fitting diversification models to phylogenies of extant species. More work is needed to establish which data representation and network architecture perform best, to perform statistical inference directly on sequence alignments rather than on fixed phylogenies, and to efficiently train networks for more complex models. We hope that our article will stimulate research in this direction. Ultimately, this should foster the development of new diversification models that are not limited (or whose design is not biased) by our ability to compute likelihoods.

2.5. Appendix

Supplementary Table 1: Comparison of accuracy and precision for machine learning and maximum likelihood estimation techniques for constant-rate Birth Death model.

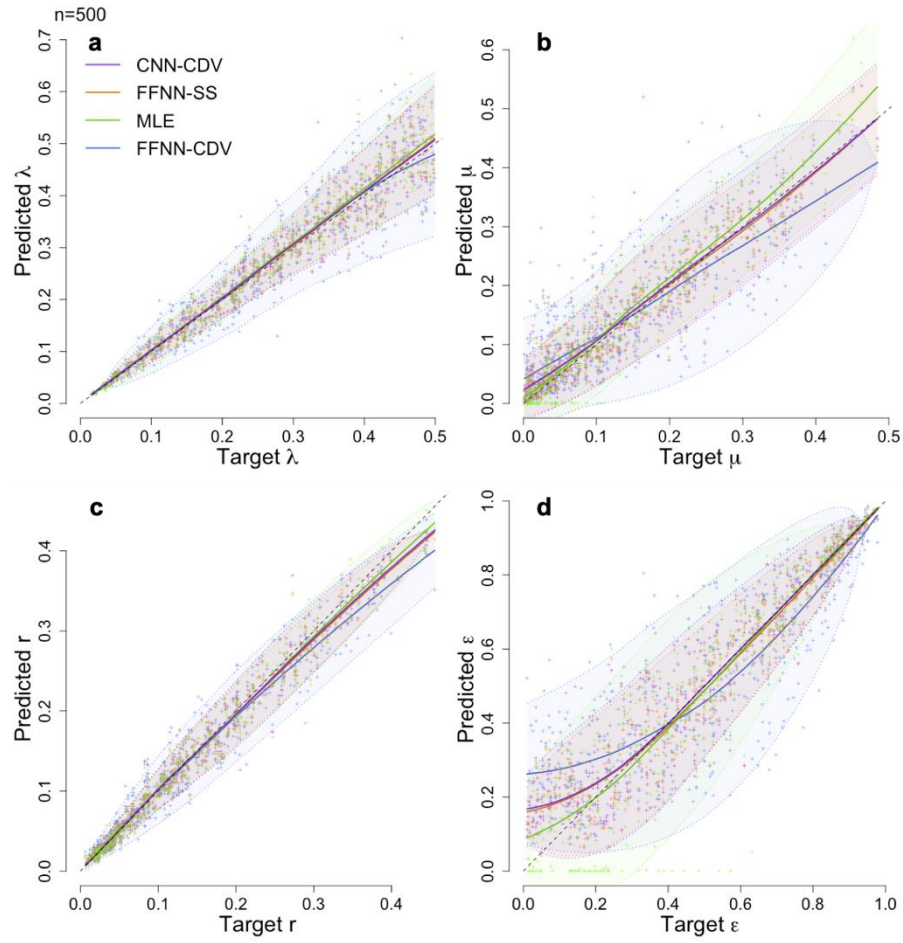
Parameters	Mean Absolute Error				Mean Relative Error			
	MLE	FFNN-Sumstats	CNN-CDV	FFNN-CDV	MLE	FFNN-Sumstats	CNN-CDV	FFNN-CDV
Speciation rate	0.02	0.02	0.02	0.04	0.09	0.08	0.08	0.14
Extinction rate	0.03	0.03	0.03	0.05	0.51	0.62	0.62	1.18
Net diversification rate	0.01	0.01	0.01	0.02	0.11	0.10	0.11	0.17
Turnover rate	0.08	0.07	0.07	0.12	0.40	0.48	0.50	0.88
Parameters	Pearson Correlation Coefficient				Mean Bias			
	MLE	FFNN-Sumstats	CNN-CDV	FFNN-CDV	MLE	FFNN-Sumstats	CNN-CDV	FFNN-CDV
Speciation rate	0.97	0.98	0.97	0.93	0.008	0.005	0.005	0.003
Extinction rate	0.93	0.93	0.93	0.78	0.010	0.007	0.008	0.006
Net diversification rate	0.99	0.99	0.99	0.98	-0.002	-0.003	-0.002	-0.005
Turnover rate	0.92	0.95	0.94	0.82	0.0008	0.013	0.019	0.015

For each estimated parameter and estimation method, we evaluated the accuracy on 500 test simulations in terms of: 1] Mean Absolute Error (MAE) in green, 2] Mean Relative Error (MRE) in blue, 3] Pearson correlation coefficient in yellow and 4] Mean Bias in grey. The compared estimation methods are a) Maximum likelihood estimation MLE-based method, b) Feed-Forward Neural Network trained on summary statistics (FFNN-SS), c) Convolutional Neural Network trained on the CDV representation (CNN-CDV), d) FFNN trained on the CDV representation (FFNN-CDV). The MRE for turnover was very high with each method, which is partially due to the covered parameter space (uniformly covered between 0.01 and 1). The CNN-CDV exhibits a higher bias for the turnover rate.

Supplementary Table 2: Comparison of accuracy and precision for machine learning and maximum likelihood estimation techniques for Binary State Speciation and Extinction model.

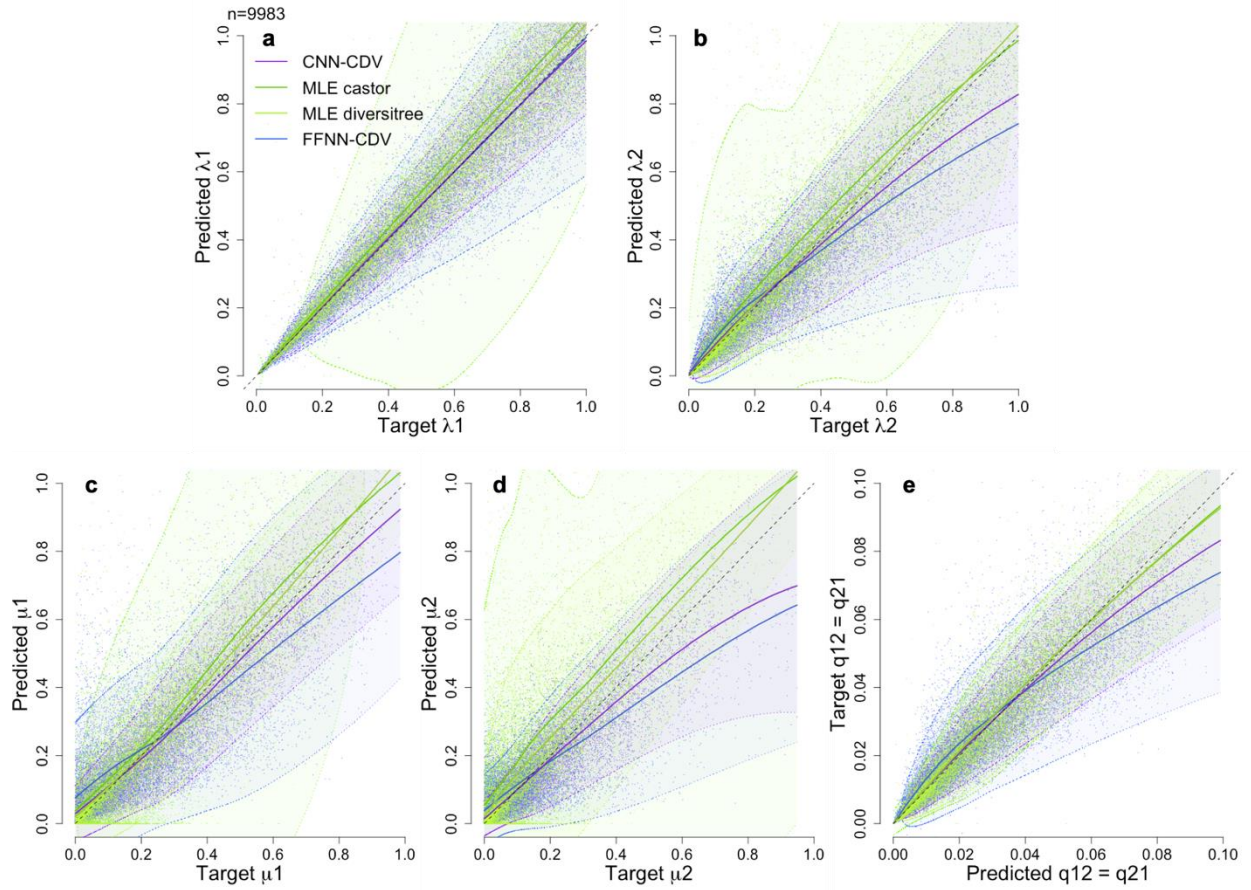
Parameters	Mean Absolute Error					Mean Relative Error				
	castor	diversitree	CNN-CDV	CNN-CDV-less	FFNN-CDV	castor	diversitree	CNN-CDV	CNN-CDV-less	FFNN-CDV
Speciation rate 1	0.076	0.048	0.047	0.150	0.084	0.16	0.10	0.10	0.31	0.17
Speciation rate 2	0.086	0.055	0.058	0.110	0.085	0.42	0.22	0.24	0.42	0.38
Extinction rate 1	0.097	0.064	0.058	0.146	0.109	3.15	1.65	1.80	4.58	4.08
Extinction rate 2	0.116	0.079	0.041	0.089	0.067	12.38	3.06	1.94	4.44	4.27
Transition rate	0.005	0.005	0.005	0.010	0.008	0.21	0.21	0.19	0.46	0.36
Parameters	Pearson Correlation Coefficient					Mean Bias				
	MLE	diversitree	CNN-CDV	CNN-CDV-less	FFNN-CDV	castor	diversitree	CNN-CDV	CNN-CDV-less	FFNN-CDV
Speciation rate 1	0.77	0.97	0.97	0.74	0.92	0.038	0.017	5e-04	0.003	6e-04
Speciation rate 2	0.67	0.93	0.92	0.75	0.84	0.045	0.012	-0.010	-0.013	-0.016
Extinction rate 1	0.65	0.92	0.92	0.48	0.73	0.040	0.016	-0.002	1e-04	0.009
Extinction rate 2	0.44	0.74	0.91	0.57	0.78	0.083	0.042	-0.006	-0.003	-0.003
Transition rate	0.94	0.94	0.95	0.71	0.85	5e-04	5e-04	-8e-04	8e-04	-4e-04

For each estimated parameter and estimation method, we evaluated the accuracy on 9.983 test simulations in terms of: 1] Mean Absolute Error (MAE) in green, 2] Mean Relative Error (MRE) in blue, 3] Pearson correlation coefficient in yellow and 4] Mean Bias in grey. The compared estimation methods are a) Maximum likelihood estimation MLE-based method using the R package castor, b) MLE-based method using the R package diversitree, c) Convolutional Neural Network trained on the CDV representation (CNN-CDV), d) CNN trained on CDV-less without individual information on tip state (CNN-CDV-less) and e) Feed-Forward Neural Network trained on the CDV representation (FFNN-CDV). The MRE for the turnover rate was very high for each method, which is partially due to the covered parameter space (uniformly covered between 0 and 1). With CNN-CDV, we obtain comparable results as with castor and diversitree.



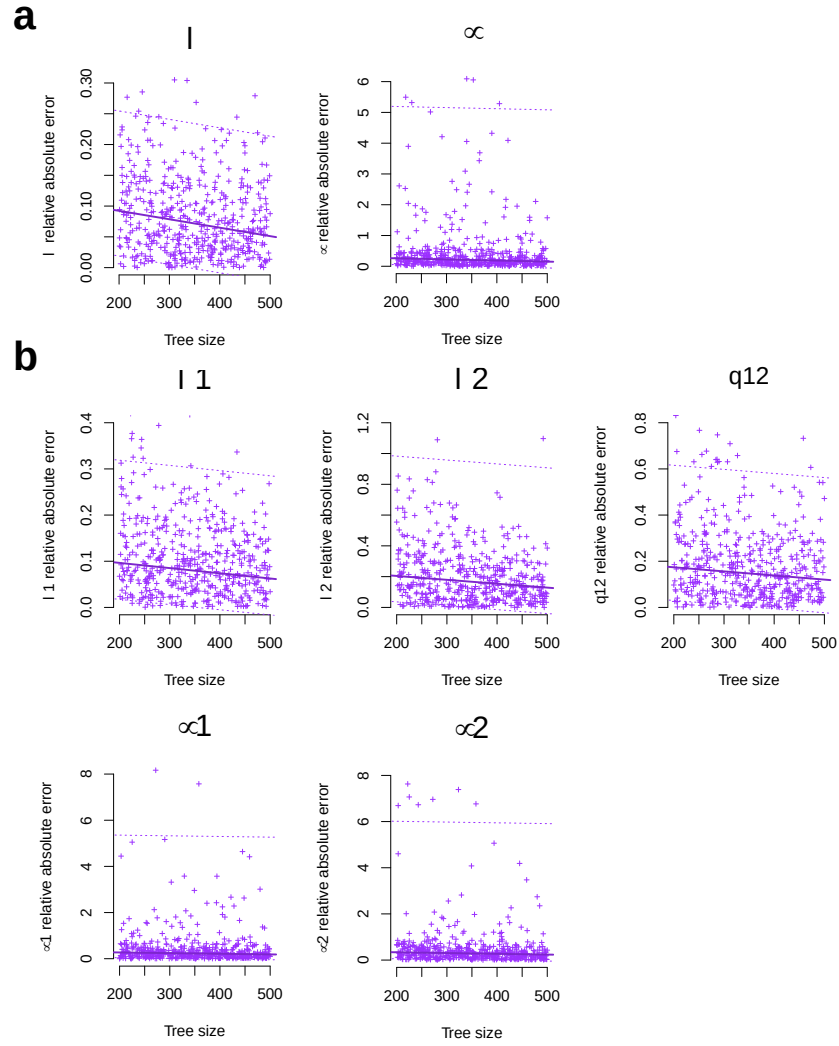
Supplementary Figure 1: Predicted versus target plots.

For each parameter predicted for BD we show a plot of predicted versus target values for 500 simulations from the test set, each dot representing one prediction: a) speciation rate λ , b) extinction rate μ , c) net diversification rate r , d) turnover rate ϵ . The dashed line represents the identity line, while the full line is the local polynomial regression with its 95% confidence interval delimited by the dotted lines.



Supplementary Figure 2 : Predicted versus target plots.

For each parameter predicted for BiSSE we show a plot of predicted versus target values for 9,983 simulations from the test set, each dot representing one prediction: a) speciation rate 1 λ_1 , b) speciation rate 2 λ_2 , c) extinction rate 1 μ_1 , d) extinction rate 2 μ_2 , e) the transition rates $q_{12}=q_{21}$. The dashed line represents the identity line, while the full line is the local polynomial regression with its 95% confidence interval delimited by the dotted lines.



Supplementary Figure 3 : Relative accuracy of CNN-CDV increases with the tree size.

For each model a) constant-rate BD, and b) BiSSE, we display the quantile regression on the relative absolute error in solid line for each parameter as a function of tree size for 500 test trees (for BiSSE, 500 values are shown instead of 10.000 for visualization purposes). The area around the solid line delimited by the dotted lines represent the 95% confidence interval around the quantile regression.

Chapter 3: Tempo of Diatoms Diversification in Relation to their Ecological Niche Strategies

A recurrent pattern when studying the tempo of speciation across clades is its heterogeneity. Some clades speciate at a fast rate such as some angiosperms, while others speciate at a slower pace – the slowest probably being in prokaryotes. A multitude of intrinsic and extrinsic factors can alone or in combination influence the tempo of speciation across lineages and clades, explaining this heterogeneity. Low dispersal rates, heterogeneous environments inducing disparate selective pressures, biotic interactions such as competition inducing disruptive selection, amongst others factors can all facilitate speciation. Species ecological niche can also influence the propensity of lineages to diversify. Indeed, a species inhabiting a wide niche – i.e. large ecological tolerance – or a niche substantially different from others – i.e. high niche marginality – or simply a niche associated to specific conditions – i.e. niche position – (all of which is referred as ecological niche strategy) may entail different tempo of diversification.

Diatoms – an evolutionary and ecologically successful planktonic group of organisms – have previously been shown to harbor heterogeneous diversification rates. Previous work has shown that factors including habitat, motility, life history traits but also sea-level and temperature fluctuations, among others, may influence diatoms diversification rates. Diatoms also appear to be extremely successful at colonizing a large spectrum of niches. They can colonize high latitude niches as well as upwelling ones. One may expect diatoms lineages that are thriving in uncrowded niche, to be more prone to speciation. Moreover, one may also expect diatoms finely specialized to a niche to have experienced higher speciation rate than others.

In this chapter, I investigate whether diatoms ecological strategies, measured in terms of characteristics of their environmental niche, affect their speciation rates, potentially explaining the disparate rates of speciation that have been found across diatoms lineages. To that end, I use the large environmental and genetic samples of *Tara Oceans* expeditions to reconstruct the present-day ecological niche of diatoms, as well as a large diatom phylogeny of more than 10,000 lineages. I estimate diatoms recent speciation rates and assess the correlation between relevant niche properties and diatoms speciation rates.

Current results indicate that diatoms occupying niches distant from the average diatoms' niche – i.e. marginal – are experiencing higher rate of speciation. On the other hand, being finely specialized does not imply higher propensity to speciate in the recent diversification dynamic of diatoms.

This chapter presents work in progress in collaboration with Richard Dorrell, Chris Bowler and H  l  ne Morlon. It is presented in the form of a preliminary article. The Bayesian inference of the appropriate diversification analyses for estimating speciation rates is still running, thus I present results using an alternative approach.

3.1. Introduction

Diatoms play an important role as the *biological pump* and are responsible for 40% of primary production in the ocean (Tréguer et al. 2018). They are globally distributed and specifically abundant and diverse in the polar region characterized by cold water, low salinity and high solar radiation. A large number of studies have focused on present-day environmental components to characterize the current diversity pattern of diatoms (Malviya et al. 2016) while few historical components have been considered in explaining the present-day diatoms diversity pattern (Lewitus et al. 2018). The Bacillariophyta (diatoms class) are thought to have originated approximately 200 million years ago (MA). Around that time the Pangea began to split, opening ocean basins. The oceanic environment shifted from an oligotrophic environment to a nutrient-rich one along with increased coastline potentially profiting to diatoms and other phytoplankton diversifying at that time (e.g. dinoflagellates and coccolithophores). As some continental areas were flooded and became shallow seas, it substantially opened new ecological niches and increased habitat heterogeneity (Katz et al. 2007).

Diatoms left a trace over geological times owing to their siliceous cell walls preserved in the fossil record. Using this record, a lot of studies have investigated their diversity dynamic over geological times. Diatoms appear to have slowly increased in diversity through the Mesozoic as well as more noticeably in the Cenozoic. The fossil-based apparent radiation of diatoms during the Cenozoic has suggested the existence of a diatoms-dominated Cenozoic ocean (Katz et al. 2004). A recent neontological study suggested that their diversification rate has remained relatively constant for a long period, with a significant upward shift at the opening of Drake Passage approximately 40 MA (Lewitus et al. 2018). It also suggested that distinct diatom clades responded differently to distinct environmental factors (e.g. CO₂, sea level, temperature (Lewitus et al. 2018)) and potentially in opposite directions (e.g. the diversification of some clades accelerated, with higher sea levels, while for others decelerated). Specific ecological characteristics of diatoms could explain this heterogeneity in diversification dynamics across diatom clades. Some authors have explored the implication of locomotion and reproductive traits on diatom diversification (Nakov et al. 2018) shedding light on the evolutionary success of raphid diatoms (Kooistra et al. 2007), which are among the most mobile single-celled eukaryotes. While characteristics of the ecological niche of diatoms also likely plays a role in their distinct diversification dynamics, this has rarely been explored.

The position of species' ecological niche, which reflects their adaptation to specific environmental conditions, can impact their diversification rates, as well as the influence of environmental conditions on these rates (Pyron and Wiens 2013; Rolland et al. 2014a; Moen and Wiens 2017). Specific niche positions which appear to be prevalent, less reachable by competitors or under stronger selective regimes will likely impact the diversification dynamic of species adapted to those. Among diatoms lineages, disparate ecological strategies exist. Some diatoms referred to as bloom-forming species are specifically adapted to nutrient pulses.

These bloom-forming species are reproducing mainly clonally, which may prevent genetic exchange and could lead to a rapid accumulation of divergence among strains. Those diatoms can be found in upwelling zones often situated on coastlines, a habitat that has increased since the break of Pangea. Diatoms also evolved a high iron-assimilation efficiency (e.g. *Pseudo-nitzschia* species (Endo et al. 2018)) which might have provided them with an excellent ecological opportunity at the Eocene/Oligocene boundary when increased oxygen concentration decreased iron availability in the open ocean (Armbrust 2009).

The breadth of species' ecological niche, which reflects their degree of specialization to specific environmental conditions, can also impact diversification rates and the influence of environmental conditions on these rates (Ackermann and Doebeli 2004; Gómez-Rodríguez et al. 2015; Rolland and Salamin 2016; Moen and Wiens 2017). Specialization can act as a driver of speciation by preventing the formation of global panmictic populations and allowing a finer partitioning of niche space. At the same time, specialization can foster extinction by reducing the ability to cope with environmental or resource-based fluctuations. Bramburger et al. (2020) showed that niche breadth is highly variable among freshwater diatoms but this has rarely been studied at a broad phylogenetic scale in marine diatoms. Ecological specialization could explain at least part of diatoms heterogeneous diversification response to changing environmental conditions.

The marginality of species' ecological niche, which reflects their ability to colonize niches far from the average niche conditions, is associated with low niche overlap and can substantially influence species diversification. Simpson's theory of ecological radiations predicts that species that are able to invade empty zones of ecological space will diversify rapidly, while those occupying already packed zones of ecological space will not. In diatoms, a clear correlation has been observed between sister-clades diversity and evolutionary novelties (Kooistra et al. 2007). The raphe for example provides motility to raphid diatoms which has been suggested to enable them to reach new niches (Kooistra et al. 2007). Other innovations linked to extreme physiological tolerance such as psychrophilic diatoms inhabiting sea ice (Mock and Junge 2007) have pushed the margins of diatoms' ecological niches in which they became dominant species.

The diversification of species can be investigated using models relying on the birth-death process. Some recently developed models allow to infer lineage specific diversifications rates (Maliot et al. 2019a; Barido-Sottani et al. 2020; Vasconcelos et al. 2022) and enable us to test for the correlation between species ecological strategies and their diversification rate. As no precise information on ancestral species ecology is available, this correlation can only be assessed for present-day species.

In this article, we use *Tara Oceans* data to characterize the ecological niche of diatoms and to evaluate if and how these characteristics have influenced diversification rates.

3.2. Methods

3.2.1. Samples

In order to investigate how diatom diversification relates to the position and breadth of their ecological niche, we used the biological samples from *Tara Oceans* expeditions 2009-2013 (englobing *Tara Oceans* 2009-2012 and *Tara Oceans* Polar Circle 2013) and their associated environmental parameters. Precisely, we used the V9 18S small sub-unit (SSU) rDNA sequences trimmed for quality, length, and fidelity of primer sequences, that were present at least 3 times in the overall expedition samples and exhibiting at least 85% of identity (PID : percentage identity) to the closest referenced V9 sequence from the Protist Ribosomal Reference (PR2 (Guillou et al. 2012)). We clustered them in Operational Taxonomic Units (OTUs) at 97% similarity by first sorting them by abundance (function *cluster_size*, *vsearch* software (Rognes et al. 2016)) and using the most abundant ribotype – defined as unique ribosomal DNA sequence – as centroid for each OTUs. We applied the same procedure on V4 18S rDNA subunit sequences to check for congruence between the two subunits. We also performed all the analysis on OTUs clustered at different similarity thresholds (99% and 95%) as well as using the swarm approach (Mahé et al. 2014) as a sensitivity analysis. Overall, the 268,295 unique V9 sequences and the 417,537 unique V4 sequences were retrieved from 1,026 biological samples in all the open oceans and the major closed seas (including the Arctic, Atlantic, Indian, East Pacific and Southern Oceans as well as the Mediterranean and Red Seas).

3.2.2. Phylogenetic Reconstruction

Backbone. – In order to reconstruct the phylogenetic relationship between diatoms OTUs and due to the restricted size of sequences available at this large scale, we used a backbone tree to help direct diatoms OTUs to their corresponding clades. The backbone was reconstructed from two independent sets of sequence alignments. The first set (set1) was retrieved from (Nakov et al. 2018) who reconstructed a phylogeny using 11 genes (including 18S rRNA present for 995 taxa, alignments available at DOI : 10.5281/zenodo.583628 under the file name 11gene.v10) and encompassing overall 1,151 diatoms taxon. The second set (set2) was retrieved from (Dorrell et al. 2021) using 18S rDNA available on PR2 and from MMETSP databases resulting in 1,036 diatoms taxon. The identical sequences were discarded which left us with overall 2,025 sequences. We conserved the alignment of set1 and aligned the sequences of set2 on the latter using the L-INS-i method available in MAFFT software (Katoh 2005) that performs sequence alignment while conserving a previous alignment (i.e. using the *-seed* option). As set2 contained introns contrary to set1, we decreased the gap extension penalty at local pairwise alignment (*-lexp -10* option) to allow large gaps in the alignment. On this alignment, we selected the appropriate substitution model using ModelFinder (Kalyaanamoorthy et al. 2017). We then used IQ-TREE 2 (Minh et al. 2020), a fast and accurate

maximum likelihood tool for the phylogenetic tree inference. We constrained the tree search with the phylogeny reconstructed from (Nakov et al. 2018). Disproportional long branches in the reconstructed backbone tree were removed as they often relate to alignment errors. Fourteen branches appeared as major outliers and were subsequently removed using TreeShrink which detects long branches based on their impact on tree diameter (Mai and Mirarab 2018). The resulting maximum likelihood backbone tree constructed using combined sequences of the two sets was then used as the backbone when reconstructing the phylogeny of V9 and V4 sequences from *Tara Oceans*.

Final tree. – The V9 and V4 sequences of each OTUs representative were aligned on the 18S set1-set2 alignment using two different methods: (i) the L-INS-i method by applying the algorithm specifically adapted to align small sequences on larger ones (*–addfragments* option) and additional high gap opening penalty and low gap extension penalty (*–op 20 –lexp -3* option) or (ii) the PaPaRa method (Berger and Stamatakis 2012) which enables the alignment of short sequence reads (i.e. the query sequences) to an existing multiple sequence alignment reference (RA) while considering the corresponding phylogenetic reference tree (RT) by applying additional high gap opening penalty and low gap extension penalty (*–p -73:-3:2:-3* option). We used the previously selected substitution model to reconstruct the final phylogenies. Similarly than for the backbone tree, we used IQ-TREE 2 for the phylogenetic tree inferences using the backbone tree as a constraint. We repeated this alignment and tree reconstruction procedure for the centroid sequences of different clustering thresholds taken as query sequences. The final phylogenetic trees were pruned to retain only tips from *Tara Oceans* samples. We dated the final reconstructed phylogeny using the TreePL program which is based on a relaxed clock model using penalized likelihood inference (Smith and O’Meara 2012) while considering several calibration points retrieved from (Lewitus et al. 2018) and (Nakov et al. 2018).

3.2.3. Environmental Conditions

For each OTUs, we used as ecological niche dimensions the following abiotic parameters measured or derived from model simulations : a set of nutrients (NO₂, NO₃, PO₄, N02/NO₃, SI, iron, ammonium) and other compounds (oxygen, chlorophyll A), the carbonate system (pH, CO₂, pCO₂, fCO₂, HCO₃, CO₃, total alkalinity, total carbon, omega aragonite, omega calcite), latitude coordinates (used without transformation here as a proxy for climatic variables such as temperature, UV radiation, nutrient concentration or climatic stability, but not as a geographical parameter), climatic and other physical properties (depth, temperature, conductivity, salinity, density, fluorescence, ice free period, sunshine duration, moon phase proportion, presence in the euphotic zone and mixed layer). Note that the iron and ammonium contents were not measured *in situ* and were thus exclusively inferred from a model simulation. After discarding the variables missing for more than 50% of the biological samples, we considered a total of 30 environmental variables.

3.2.4. Ordination Procedure to Characterize Diatoms' Ecological Niche

To characterize the main dimensions of diatoms' ecological niche, we performed an ordination procedure (Janekovi and Novak 2012). First, we imputed missing environmental variables based on a principal component analysis (PCA) (function *imputePCA*, R package *'missMDA'* (v 1.11)) which provides a value to missing measurements per sample considering the correlation among variables. We checked the robustness of the imputation procedure by repeating the analysis multiple times and discarded the variables for which the PCA coordinates were not consistent across repeats. We characterized the ecological niche characteristics of each OTU by applying the Outlier Mean Index (OMI) method to our imputed data. The OMI method is based on ordination. It consists of first performing a PCA on the samples and their associated environmental variables to get the niche hyperspace. It then allows the placement of OTUs in this niche hyperspace. Lastly, it computes metrics for OTUs niche centroid, tolerance as well as marginality, which is the distance between the OTUs centroid niche and the center of the niche hyperspace. We analyzed the distribution of the resulting niche properties (i.e. niche position, tolerance, and marginality) across the phylogeny by calculating the phylogenetic signal for each niche property using the *k statistic* (Blomberg et al. 2003) which compares the distribution of the niche properties distribution across the phylogeny to that expected under a Brownian motion.

3.2.5. Diversification Rates

We used the ClaDS model (Maliot et al. 2019) to obtain lineage-specific diversification rates, using the most recent data augmentation algorithm (Maliot and Morlon 2022). This model relies on the birth-death process and under the ClaDS2 option it assigns a specific speciation rate per lineage and a global turnover rate for the entire studied clade. At each speciation event, the lineage-specific speciation rates are sampled from a common distribution whose parameters are estimated during the inference along with the turnover rate and the speciation rate at the start of the process. It thus enables small speciation rate shifts and accounts for unseen speciation events. As with most birth-death models, ClaDS requires a global diversity estimate to account for unsampled species. We estimated the diatoms' global diversity using the frequency ratios-based method (R package *'breakaway'*) (Willis and Bunge 2015) which relies on fitting flexible non-linear models to the ratio of frequency counts. Using the ratio of frequency counts instead of frequency counts directly was suggested to be statistically more robust (Willis and Bunge 2015). We discarded the OTUs found in only one sample (hereafter named singletons) and specifically applied the frequency ratio method developed to account for this filtering (Willis 2016c) and using incidence data instead of frequency, given that frequencies (i.e. abundances) are not comparable across samples with a metabarcoding approach.

3.2.6. Correlation with the Ecological Niche

To test whether ecological niche characteristics modulate the speciation rate of present-day diatoms OTUs, we performed a Phylogenetic Generalized Least Squares (PGLS) regression to estimate the expected covariance between niche characteristics and speciation rate across diatoms OTUs while accounting for the diatoms phylogenetic structure as a source of variation.

Performing phylogenetic reconstruction from a large number of sequences is computationally intensive and only results of the phylogenetic reconstruction using V9 sequences clustered at 97% are currently available. Reconstruction using V9 sequences clustered at different thresholds, using swarm or using V4 are yet to come. The 97% V9 diatoms reconstructed phylogeny is a large phylogeny and applying ClaDS2 to it requires a long computational time. As a first and rough approximation of the diatoms speciation rate, we computed the DR statistic (Jetz et al. 2012) on the diatoms phylogeny and retrieved the tip speciation rate (Title and Rabosky 2018) hereafter called tip rate. This statistic does not account for unsampled species nor extinction events but we assume that all the lineages in the phylogeny will be affected similarly by both. Thus, we use the DR statistic assuming that it is proportional to the true speciation rates, and tested for the correlation between speciation rate and diatoms ecological strategies.

3.3. Results

3.3.1. Clustering and Diversity Estimation

The clustering of V9 18S rDNA resulted in 11,849 clusters at 97% similarity, 4,208 clusters at 95% similarity, and 99,055 clusters at 99% similarity. When performing a clustering using the swarm algorithm, the number of clusters obtained was consistent with the one obtained using the 97% clustering (13,864 clusters formed). Clustering V4 18S rDNA at respectively 97%, 95%, and 99% similarity resulted in 6,051, 3,034 and 20,013 clusters formed. Thus for an equivalent threshold, V4 resulted in fewer clusters than V9, a pattern often observed in eukaryotes (Choi and Park 2020).

The global diversity estimate recovered by breakaway indicates an overall diatom richness of 19,394 when using V9 18S rDNA and of 8,973 when using V4 18S rDNA both clustered at 97% similarity. These estimates imply that respectively 61% and 67% of the global diatoms' diversity for V9 and V4 sequences has been collected in the *Tara Oceans* expeditions. We retrieved a larger portion of the diatoms' global diversity when combining *Tara Oceans* and *Tara Oceans* Polar Circle samples compared to previous studies (i.e. estimating around 10% of the global diversity using a different diversity estimation approach (Lewitus et al. 2018; Nakov et al. 2018)).

3.3.2. Phylogenetic Reconstruction

The bootstrap values of the reconstructed backbone were above 0.7 for more than 75% of the nodes using the transfer bootstrap expectation method (Lemoine et al. 2018) which was considered sufficient for conducting the rest of the analysis. We compared the reconstructed backbone to a set2-only reconstructed phylogeny by calculating the phylogenetic diversity per genus for each tree. We found more genera with high phylogenetic diversity in the set2-only phylogeny than in the reconstructed backbone (comprising set1 and set2). Thus, as expected, constraining the backbone reconstruction with a multigene tree (set1) allows for improving the backbone topology by guiding taxa to their respective clade. When exploring the distribution of the morphotype assigned to diatoms taxa across the reconstructed backbone phylogeny ([Fig. 1](#)), we observed that the raphid pennates constitute a monophyletic clade while the polar, radial (both centric) and araphid pennates are paraphyletic as expected from the literature (Kooistra and Medlin 1996; Medlin and Desdevises 2016). Thus, we were able to confidently use the reconstructed backbone tree to constrain the final diatoms tree.

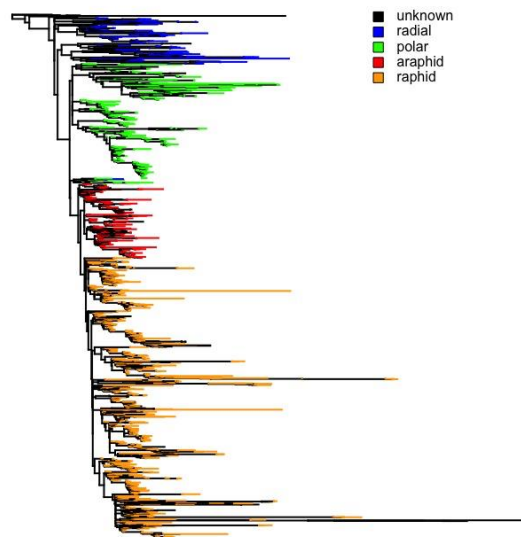


Figure 1: Diatoms morphotype distribution across the reconstructed backbone phylogeny. The morphotypes are based on the PR2 morphotype assignment. The tips of the backbone are colored according to their corresponding morphotype assignment when unknown (in black), radial (in blue), polar (in green), araphid (in red) and raphid (in orange).

Our phylogeny dating places the crown age of the diatoms at 190 MA during the Early Jurassic. This is consistent with other molecular dating estimates which put the diatoms crown age between the Early Triassic (i.e. 250 MA) and the Early Jurassic (i.e. 183 MA) (Sorhannus 2007; Lewitus et al. 2018; Nakov et al. 2018).

3.3.3. Ordination

The imputation procedure allows complementing environmental parameter values missing for some samples. The robustness of the imputation was assessed by performing multiple imputations and appears to be insufficient for four parameters related to the carbonate system (Fig. 2). Indeed, depending on the imputation the coordinates of the pH, the alkalinity, the HCO₃, and CO₃ parameters are drastically different in the PC1 and PC2 space and were subsequently removed from the rest of the analysis.

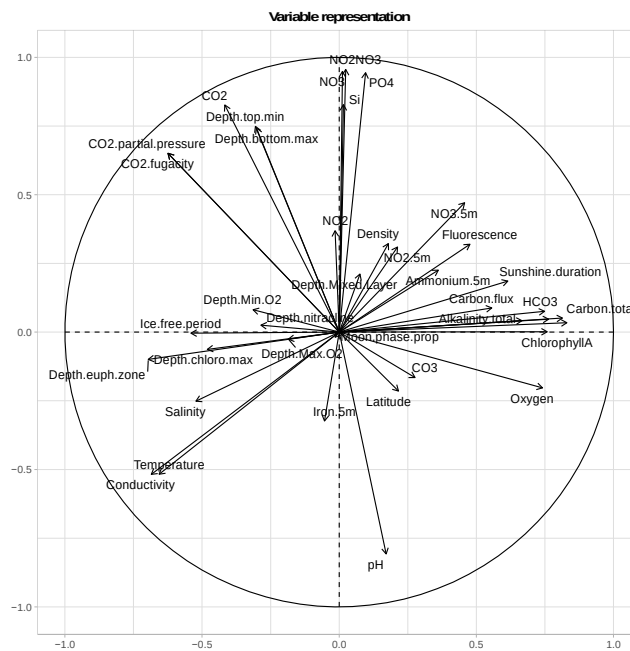


Figure 2: Principal Component Analysis (PCA) performed on the environmental variables across different Tara Oceans samples. The length of each arrow describes the relative contribution of each variable. Each variable is assigned a color and each dot represents the environmental variable coordinates in the PC1 and PC2 space for each imputation.

After removing the four parameters mentioned above, the first and second Principal Component (PC) of the OMI analysis together represent 79% of the explained variance (Fig. 3a). Accounting for the third principal component does not drastically improve the proportion of explained variance, therefore, we only focus our analysis on the first two PC axes. In our analysis, the diatoms' niche disparity is best characterized by climatic and physical properties and to a lesser extent by nutrient concentrations and latitude. Indeed, the first axis of the diatoms' niche explaining 59% of the variance is correlated to temperature, ice-free period, conductivity, and oxygen in the opposite direction (Fig. 3a,b, 4a). Meanwhile, the second axis is associated with nutrient compounds such as nitrogen, phosphorus, and silicon which is negatively correlated with latitude (Fig. 3a,b, 4b). Note that nutrients here refer exclusively to the ones cited above and do not apply to iron for example. In the niche space (i.e. PC1 and PC2 space), no diatoms' OTUs (i.e. their niche centroid) are found in warm waters as the latter are often stratified and low in oxygen. Diatoms' OTUs are distributed in the mean conditions

sampled (i.e. central coordinates), and in the extreme conditions characterized by high nutrients and oxygen (i.e. upwelling zones), as well as high latitude and oxygen (Fig. 3c).

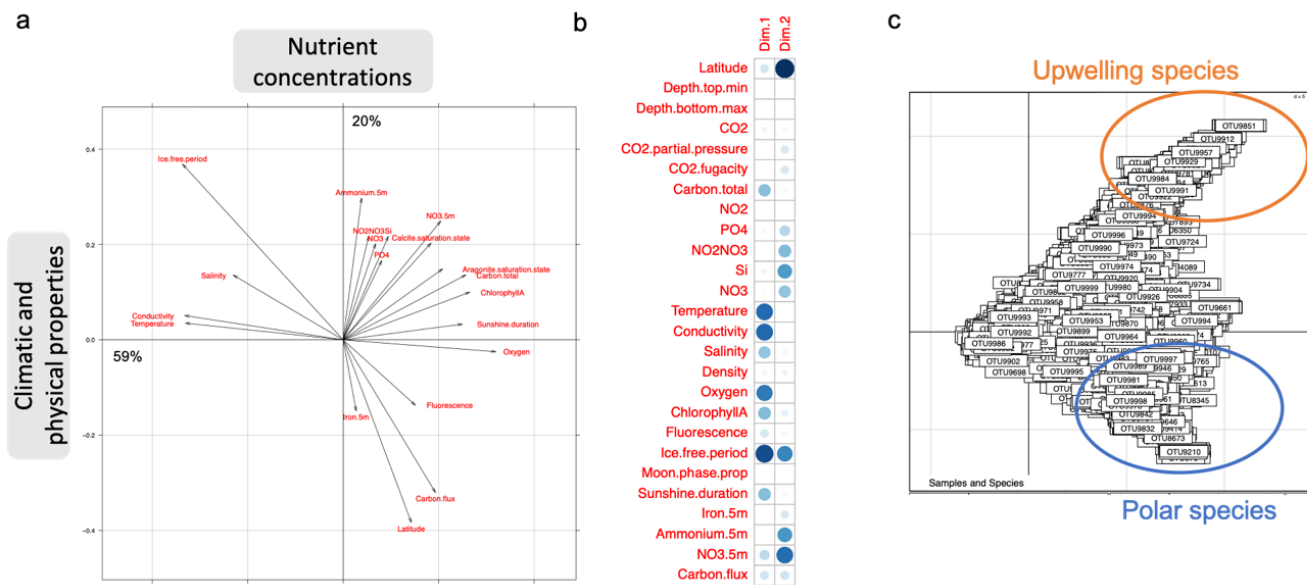


Figure 3: Outlier Mean Index (OMI) performed on the diatoms' OTUs. Environmental variables which are major contributors to the explained variance of diatoms niche separation are displayed in the PC1 and PC2 space in panel a) and b). The length of each arrow describes the relative contribution of each variable. The size and darkness of the blue points in panel b) are correlated to their relative contribution to the PC1 and PC2. The distribution of diatoms' OTUs in the PC1 and PC2 space is shown in panel c). The distinctive diatoms' species identified in the niche space are marked by a circle.

The distribution of the niche characteristics across the phylogeny shows a low phylogenetic signal ([Supplementary Table 1](#)) and no substantial niche conservatism (Losos 2008) – defined as the propensity of related species to inhabit more similar niches than expected based on their phylogenetic relatedness – with the exception of few distinct clades referred hereafter as the upwelling clade and the polar clade ([Fig. 4b](#)). Note that upwelling and polar species are not restricted to those clades and can be found throughout the phylogeny. Previous studies have shown thermal conservatism at the strain level and among closely related species in benthic marine diatoms (Stock et al. 2019) yet, consistent with our finding this appears to be clade-specific and low niche conservatism might be the rule rather than the exception (Chivers et al. 2017; Soininen et al. 2019).

Throughout our phylogeny, diatoms exhibit a low niche tolerance (i.e. breadth) and marginality (i.e. distant from the mean condition), with the exception of a few lineages particularly present in the upwelling clade that exhibits moderate tolerance and moderate to large marginality ([Fig. 4c,d](#)). Interestingly, upwelling species are dominant species which has been shown to correlate with moderate niche tolerance (Ignatiades 1994) as observed here for this particular clade. Moreover, upwelling zones are often found in coastal habitats which are characterized by high-temperature fluctuation. Therefore, species inhabiting this environment

are expected to exhibit a rather moderate to high niche tolerance. On the other hand, members of the distinct polar clade yield a substantially low niche tolerance, corroborating previous findings specifically on polar species (Stock et al. 2019).

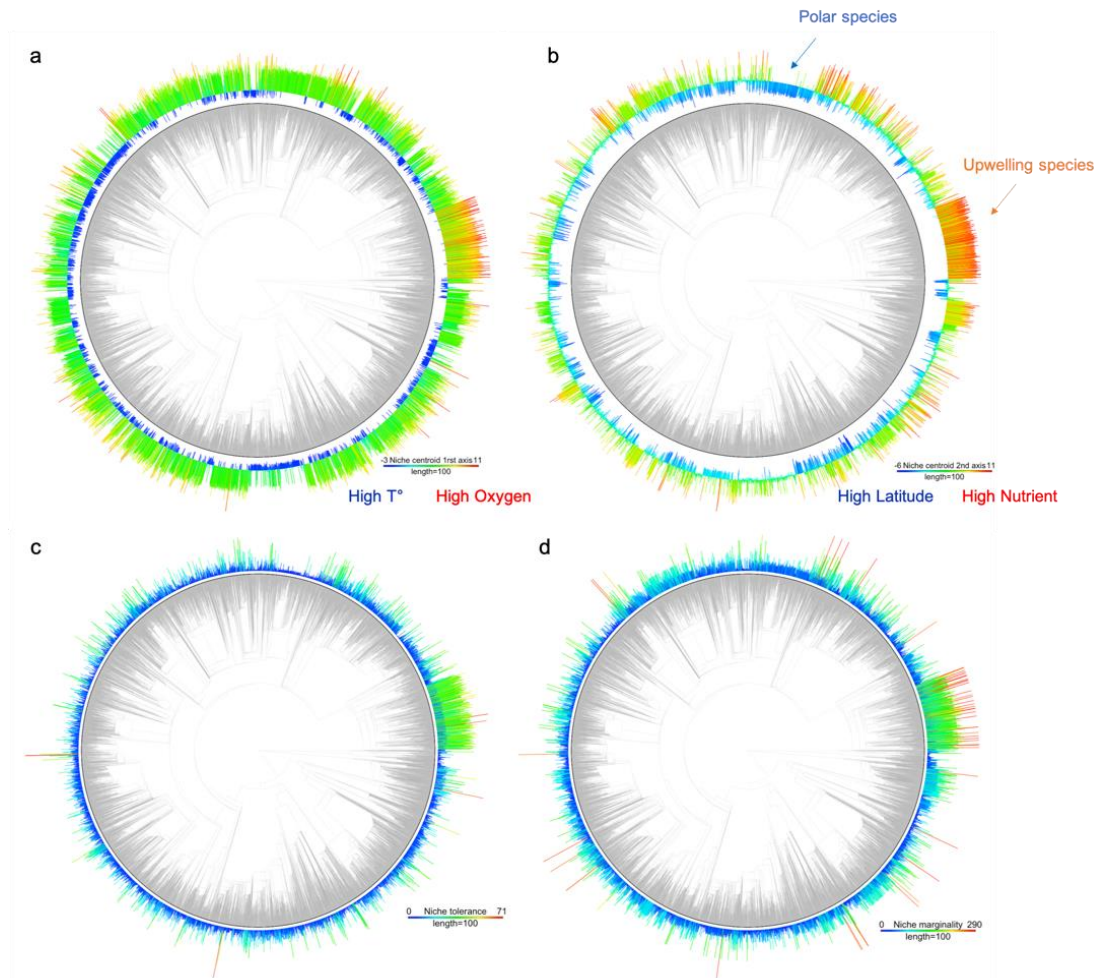


Figure 4: Ecological niche characteristics across the diatoms' phylogeny. The distribution of diatoms' niche position along the first axis of the niche space is displayed in panel a). The blue color represents warm niches and the red represents highly oxygenated waters. The diatoms' niche position along the second axis is displayed in panel b). The blue color represents polar environments and the red represents eutrophic habitats. Panels c) and d) display respectively the niche tolerance (i.e. breadth) and niche marginality (i.e. distance to the average sampled condition) across the diatom phylogeny. The blue color represents low niche tolerance and marginality while the red represents high niche tolerance and marginality. The arrows indicate clades harboring distinct ecological characteristics.

3.3.4. Tip Rates and the Implication of Ecological Niche Characteristics

The tip rates inferred using DR statistic on the diatoms phylogeny are variable and span two orders of magnitude (Fig. 5). Note that the tip rate values might be underestimating the actual

speciation rate as our diatoms phylogeny is not complete and the DR statistic does not account for under-sampling.

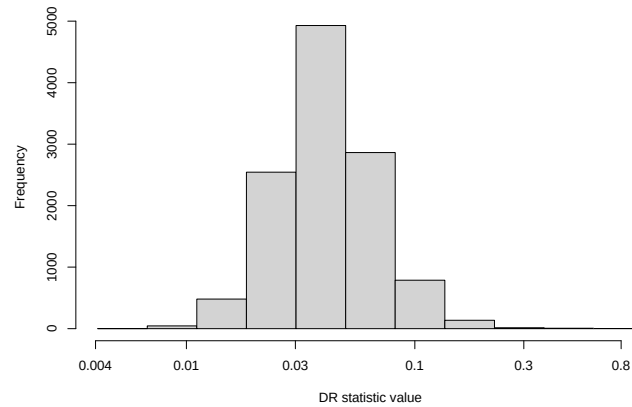


Figure 5: DR statistic histogram inferred on the reconstructed diatoms phylogeny. The x-axis displaying the DR statistic is log scaled.

When testing for the correlation between tip rates and diatoms' ecological niche characteristics – while accounting for the phylogenetic structure – we find that the niche position (in both axes) and the niche marginality correlate with tip rates ([Table 1](#)). It appears that the correlation coefficient is substantially low despite being significant, suggesting a significant correlation but with low effect. Overall, the results suggest that present-day diatoms in cold and highly oxygenated waters speciate faster than others. Moreover, the tip speciation rate is also higher for the diatoms inhabiting high latitudes regions. Finally, the diatoms able to inhabit niches that are distant from the average sampled conditions (i.e. marginal) exhibit a higher tip speciation rate. On the contrary, diatoms' niche tolerance was not correlated to their ability to speciate at a faster or slower pace.

Table 1: Results of the Phylogenetic Generalized Least Squares (PGLS) regression performed on diatoms niche characteristics to explain their tip speciation rates. The p-value below 0.05 are displayed in bold.

Niche	Centroid 1	Centroid 2	Tolerance	Marginality
p-value	0.0011	0.0002	0.1514	0.0001
Coefficient	0.0033	-0.0052	0.0002	0.0003

3.4. Discussion

The recent speciation of diatoms appears to be faster for species inhabiting high latitudes, cold waters, and ecological margins of their niche. Our study sheds light on the importance of diatoms' ecological strategies which appear to be a component of their speciation process. Hereafter we discuss potential hypothesis for the observed heterogeneous pattern of speciation.

Our analysis suggests that recent speciation of diatoms might be favored at the ecological margins of the average habitat. One might expect species adapted to extreme conditions (i.e. ecological margins) to speciate faster as the selective pressure might be higher giving rise to divergent selection. An apparent high speciation rate has also been found in diatoms colonizing a habitat considered extreme for aquatic organisms (i.e. terrestrial habitat (Pinseel et al. 2020)). Moreover, ecological margin is a proxy for low niche overlap among species and those niches might result in less densely colonized habitats. This may lower competitive exclusion and increase opportunities for speciation (Simpson 1953). Overall, this observed pattern suggests that evolution at the diatoms' niche boundary is favored.

Diatoms are particularly diverse in polar and upwelling zones (Malviya et al. 2016). Our study suggests that recent speciation rate is higher in polar regions compared to upwelling zones. In both environments, diatoms have shown diverse physiological evolution toward cold resistance or fast nutrient assimilation (Morgan-Kiss et al. 2006; Smayda and Trainer 2010). A possible explanation for the observed fast speciation in high latitudes could be due to a stronger physiological limitation in upwelling zones compared to polar regions. The polar regions might also exhibit higher selective pressure (e.g. heterogeneous solar radiation over a day) which can ultimately spur speciation. Another explanation may be related to the disparate mode of reproduction that diatoms experience in these habitats. The mode of reproduction of diatoms in upwelling zones is mainly clonal – i.e. vegetative reproduction – as they rapidly need to divide in order to profit from the sporadic pulse of nutrients (Armbrust 2009). Clonal reproduction has often been associated with a higher rate of speciation (Cohan and Perry 2007). Nevertheless, clonal reproduction in diatoms is different than in other organisms as it is limited by their siliceous cell wall and controls the clones cell size. Indeed when dividing clonally, one of the two daughter cells will be smaller than the mother along with a reduction in the thickness of the cell walls. This implies that clonal reproduction is bound by the intrinsic ability of small diatoms with thin cell walls to survive and might lower clonal divergence as sexual reproduction is necessary to occurring for cells to regain their normal size. Indeed, sexual reproduction has previously been suggested to promote speciation in the diatoms (Mann 1999; Behrenfeld et al. 2021). Interestingly, a gene responsible for sexual reproduction has been described as harboring a latitudinal pattern of divergence (Armbrust and Galindo 2001) suggesting different selective regimes on sexual reproduction along the latitude. In contrast to upwelling zones, polar regions are not particularly prone to diatoms clonal reproduction. Overall, different modes of reproduction may play a role in diatoms heterogeneous speciation. Further studies in diatoms life-history traits might eventually provide enough data on diatoms reproduction mode to test this hypothesis.

Other tips rates analyses have shown high speciation rates in high latitudes (e.g. in birds (Harvey et al. 2020), angiosperms (Igea and Tanentzap 2020)). In these studies, the main predictor of high speciation rate in high latitudes was low species diversity. This is unlikely to be the case for diatoms as polar regions do not appear to particularly harbor low diatoms species richness. Nevertheless, when considering competitors at a larger taxonomic scale, the polar regions appear to be substantially less species-rich than lower latitudes (Barton et al. 2010).

Some paleontological studies have also emphasized the potential importance of high latitudes events (e.g. Antarctica ice sheet expansion) in the recent diversification of the diatoms (Cervato and Burckle 2003). Moreover, a faster rate of speciation in cold water has been found in our study. This pattern could be explained by the fact that diatoms are specifically adapted to cold water – permanent or intermittent in polar or upwelling zones, respectively – therefore the physiological evolution necessary for them to thrive in warm water may require drastic physiological innovations limiting ecological divergence and speciation.

In contrast to expectations (Ackermann and Doebeli 2004) and what has been reported in other groups (Gómez-Rodríguez et al. 2015; Rolland and Salamin 2016), diatoms speciation does not appear to be correlated to niche tolerance (i.e. breadth). Indeed, the speciation of diatoms seems correlated to the ability to colonize new niches at the ecological margin of their niche space as opposed to a finer niche specialization spurring speciation. An explanation for this observation may be indirectly linked to the fact that diatoms' physiological strategy is mainly oriented toward the acquisition of resources and not survival – contrary to dinoflagellates for example (Smayda and Trainer 2010). Under this hypothesis, diatoms would exhibit a high niche specialization which is indeed what is shown in our analysis for the majority of species across the phylogeny. The low of heterogeneous strategy in diatoms niche tolerance results in the absence of a putative link with speciation rate. The heterogeneous strategy in niche tolerance might be present at a higher taxonomical scale and may cause a different diversification regime among diatoms, dinoflagellates and coccolithophores (Brun et al. 2015; Bramburger et al. 2020).

The patterns found in this analysis rely on the estimation of tip rates using the DR statistic. Thus, the results need to be corroborated using a finer speciation rate estimate (e.g. a parametric approach such as ClaDS) in order to confidently validate the interpretation mentioned above. Indeed, DR statistic is a non-parametric method to estimate speciation rate. It relies on the branch length information and does not account for under-sampling. It nevertheless has the advantage of correlating with actual speciation rate and being drastically faster to compute than any parametric approach.

The analysis performed concerns the speciation pattern of the present-day species and does not imply that the speciation process in the distant past relates similarly to niche characteristics. Indeed, diatoms' early speciation might have been prolific for species inhabiting eutrophic environments and warmer temperature (Kooistra and Medlin 1996). More recently, the putative increase in diversification rate around the Eocene/Oligocene boundary has been attributed to the intensification of the thermohaline circulation particularly profiting for the diversification of bloom-forming species (Behrenfeld et al. 2021). Yet, recent speciation might be attributable to different factors as discussed above.

In the present study, we only investigated the influence of the scenopoetic variables (i.e. abiotic parameters) of the ecological niche on speciation. Further investigations should consider bionomic variables (i.e. biotic parameters such as diversity and abundance of competitors like dinoflagellate, predators, evenness, functional diversity, number of assemblages) and their link

to diatoms diversification. Moreover, the investment of some abiotic variables of the niche has not been properly investigated here (e.g. iron availability) since no measurement *in situ* was available. With respect to the importance of iron in the biology of marine ecosystems, further works should investigate the iron's relevance in segregating the diatoms niche. Furthermore, iron fixation strategies have been shown to be diverse in diatoms (Marchetti et al. 2009, 2012) and crucial in the adaptation to polar regions. One strategy has potentially been inherited from cyanobacteria (Hopkinson and Morel 2009). Dating these acquisitions might be of interest when investigating the impact of innovation on diversification of diatoms.

A finer indicator of a species' ability to thrive in a particular niche is its physiology. Substantial progress has been made to characterize diatoms physiological abilities based on metagenomic or transcriptomic. Exploiting this additional relevant layer of complexity is a promising avenue to investigate the link between the diversification of diatoms and their ecology. Metagenomic along with transcriptomic data can also help us in disentangling the relative importance of realized versus fundamental niche in spurring the speciation process. Indeed, the realized niche – the niche space actually occupied by species – is often more tractable than the fundamental one and has frequently been the focus of studies linking ecological niche and diversification dynamics. Yet, other mechanisms beyond their physiological ability – can prevent species from inhabiting a niche and alter the relationship between ecological niche and diversification. Overall, the further investigation of diatoms physiological ability relies on having extensive data and on the ability to link this data to sequences used in the phylogeny reconstruction. This can be challenging as the genetic marker often used to reconstruct phylogenies is often lacking in metagenomic compilations.

The niche position, breadth and marginality of species are key components of ecological speciation. Besides, the diversification of the niche itself can impact the pace of species diversification. The ability of species to adapt rapidly to new ecological optima can lead species to diverge faster from one another. This has previously been shown to occur in some birds (Seeholzer et al. 2017), plants (Schnitzler et al. 2012) and amphibians (Moen and Wiens 2017) and could be an interesting future avenue of research. Indeed, the propensity of marginal diatoms to speciate faster as demonstrated in our study may be linked to their ability to diverge through niche space faster.

This study aims at characterizing a potential large-scale relationship. Finer analysis at narrower taxonomic resolution might shed light to different relationships between the speciation rate of species and their associated ecological strategy. For example, among closely related species phenological segregation has been suggested to correlate with divergence in *Pseudo-nitzschia* (Percopo et al. 2022). The seasonal separation of species is one mechanism of niche partitioning by which species specialize temporally rather than spatially and might be another important niche dimension segregating species at a finer taxonomic scale. We investigated the diversification of diatoms by focusing on samples retrieved from marine ecosystems. Freshwater specialists are also present in the *Tara Oceans* dataset (e.g. *Fragilaria*) but are largely underrepresented. We acknowledge the limitation of our findings to the marine diatoms' diversification. Freshwater diatoms have previously been suggested to diversify at a

faster pace than marine diatoms (Nakov et al. 2019) such that different processes may regulate diatoms diversification in the two realms.

Overall, our findings suggest that diatoms diversification may be associated with ecological divergence. It shed light on the wide diversity of ecological strategies among diatoms, emphasizing the role of adaptation to niche margins in generating diversity at a high pace. Finally, it extends the hypothesis under which high latitudes may be *cradles* of recent diversity to the marine microbial world.

3.5. Appendix

Supplementary Table 1: Results of the k statistic performed on diatoms niche properties distribution over the phylogeny. The k statistic varies between 0 and 1. The higher the k statistic is, the strongest is the phylogenetic signal.

Niche	Centroid 1	Centroid 2	Tolerance	Marginality
k statistic	0.22	0.30	0.15	0.19

Discussion

During my PhD, I tackled the question of diversification in the microbial realm. I built on preexistent models and statistical tools to explore extensions particularly adapted to the case of microbial clades. I also analyzed the diversification dynamics of two ecologically central clades (i.e. bacteria and diatoms) which reveals global distinct patterns concerning their respective tempo of diversification. In the next sections, I will summarize the methodological progress conveyed in the present work as well as the empirical findings while pointing at some promising extensions of the present work. Finally, I will discuss the remaining limitations and some potential perspectives for improvement.

4.1. Methodological Progress

As groups of microorganisms are exceptionally species-rich, describing this diversity is performed through large sampling and sequencing of cells present in the environment. This allows to reconstruct species evolutionary history and to estimate the unseen diversity. This vast but incomplete inventory poses two main issues when used for investigating diversification dynamic.

The first issue relates to the global diversity estimate as it can be highly uncertain. Yet, it is a prerequisite when investigating species diversification rates using preexistent models. In [Chapter 1](#), we have proposed to alleviate this issue by developing new models which do not require any global diversity estimate. To achieve this goal, we have extended the preexistent diversification model based on the constant and homogeneous birth-death process under two distinct sampling schemes. Under the Bernoulli sampling scheme – which assumes that each species were sampled according to a common sampling probability – we have integrated over all the possible sampling probabilities while supposing that the latter followed a specific distribution. The resulting model allows to infer diversification rates on one or a set of phylogenies assuming a common sampling distribution and common or specific diversification rates. This was motivated by the observation that current practices consist of segmenting large trees in smaller ones and applying diversification models independently on each smaller tree mainly for computational reasons. The proposed model allows to consider all the phylogenies at once which is used to infer the diversification rates and to constrain the common sampling distribution. The second model relies on the *k-sampling* scheme – which assumes that exactly

k species are sampled randomly at present. It benefits from the evidence that a tree generated under the *k-sampling* scheme is equivalent to a mixture of trees (in which node depths are independent and identically distributed) generated under a Bernoulli sampling scheme assuming an explicit distribution of the sampling probability. We tested these models and found that the Bernoulli method applied on a set of phylogenies and the *k-sampling* method result in correct estimation of diversification rates on simulated phylogenies. Nevertheless, as both methods integrates over the unknown sampling probability, it produces wide credible intervals around turnover rate estimates. Lastly, the Bernoulli method applied on a set of phylogenies outperforms the *k-sampling* method when applied independently on the phylogenies of a set. This work offers an alternative to study diversification rates on phylogenies of unknown global diversity. Moreover when applied on empirical phylogenies, we recommend to interpret the estimates while exploiting the associated credible intervals.

The present work could be extended to relax some unrealistic assumptions made. First, one could extend the proposed models to time dependent birth-death models as shown in (Lambert 2018). This would be particularly appropriate for investigating the diversification of some microbial clade which may not exhibit constant diversification rates through time (Morlon et al. 2012; Lewitus et al. 2018; Perez-Lamarque et al. 2020). One could also explore the possibility of extending the findings of (Lambert 2018) to other sampling schemes such as the diversified, clustered (Hohna et al. 2011) or higher level sampling scheme (Stadler and Bokma 2013) as they both rely on *k-sampling* referred as *n-sampling* in those cited work. Indeed, the sampling process might not be random and accounting for it – when inferring the turnover rate in particular – may lead to more accurate estimates on empirical phylogenies. Another interesting extension of the present work could be to investigate the shape of sampling probability distributions across different taxonomical scale – i.e. a phylogenetic explicit sampling theory. Some sampling theories already exist (Muscarella et al. 2019) but have never been considered in the lens of a phylogeny. Indeed, the present sampling theory is based on using the abundance or the frequency counts of species to estimate global diversity as we expect less abundant or rare species to be more numerous than others and less often sampled (Chao and Chiu 2016). These indexes might harbor a phylogenetic signal as species might be more abundant due to phylogenetically conserved life history trait or might be more frequent in samples due to their phylogenetically conserved broad geographical distribution. Subsequently, this might lead to disparate sampling probability distributions across sets of phylogenies – e.g. multimodal or unimodal sampling distributions. The potential shape could be used to further constrain the common sampling probability distribution when performing a diversification analysis – under Bernoulli sampling scheme and unknown global diversity on a set of phylogenies – as proposed in [Chapter 1](#).

The second issue relates to the intense computational requirement when performing diversification analysis on large phylogenies. This is especially limiting when assuming complex but realistic models of diversification and may hamper the application of those on microbial dataset. The critical part in applying models of diversification on large dataset is the inference part. The inference most often relies on calculating the likelihood of the model parameters and involves an optimization procedure. In [Chapter 2](#), we adopt a likelihood-free

inference approach for the inference of diversification model parameters on reconstructed phylogenies. The chosen inference approach is based on simulating phylogenies under a given model and requires training a neural network in order to *learn* the relationship between the phylogeny and its parameters. We therefore developed a compact matrix representation of a reconstructed phylogeny based on a phylogenetic diversity criterion and including node depths and potential tip trait information. We used data simulated under two widely-used birth-death models – i.e. the constant homogeneous birth-death model and the binary state speciation and extinction model both under Bernoulli sampling – and either our compact representation of the data or summary statistics to train both a fully connected feed-forward neural network and a convolutional neural network. We compared the accuracy of the parameter estimates obtained with deep learning to the gold standard maximum likelihood approach and found that the deep learning approach is as accurate as maximum likelihood while being three orders of magnitude faster. In addition of being less time consuming, this inference approach may be used for models relying on numerical approximations, or on intractable likelihoods.

Naturally, extending the inference approach to more complex and time consuming inference may be particularly appropriate. Indeed, the ClaDS model inferred using deep learning would have been particularly suited for the analysis of [Chapter 3](#) as the reconstructed diatoms phylogeny is substantially large and the assumed diversification model substantially complex. The deep learning approach can be tested on lineage specific birth-death models by training the model to estimate parameters and hyperparameters which would ultimately enable us to recover lineage specific diversification rates. Another promising extension relies on using a Graphical Neural Network (GNN) instead of the current network architecture. Indeed, it might be particularly suited as the initial representation of the data is a graph. Some preliminary work has been performed along this line by Ismaël Lajaaiti (M2 intern) but has not led to convincing results yet. Further work could also alleviate the limitation linked to the size of the phylogeny. Indeed, the current work is restricted to analyze phylogenies of 200 to 500 tips as the training set was produced in that such way since, simulating a large number of large phylogenies would have required an excessive amount of memory. Nevertheless, a previous work (Voznica et al. 2022) has shown that by extracting a nearly complete coverage of a large phylogeny by disjoint smaller phylogenies (200 to 500 number of extant species), one could retrieve an average subtree prediction for the large phylogeny estimate. Lastly, when using the proposed deep learning inference, no measure of uncertainty around the estimates is available. This measure is desirable as it represents the confidence in the estimation. One proposed way to produce this uncertainty is to implement a form of parametric bootstrapping (Voznica et al. 2022). It consists of subsampling simulations based on their distance (i.e. the closest) to the empirical phylogeny analyzed (in term of tree size, sampling probability and simulated parameter values). The error around the estimates of the selected simulations is then used to create a confidence interval around the estimate of the empirical phylogeny.

4.2. Empirical Findings

This PhD thesis also aimed at empirically characterizing the tempo of diversification of microbial clades. To that end, two major microbial clades belonging to two distinct phyla were chosen. Firstly, we have investigated the bacterial tempo of diversification on the largest phylogeny of present-day bacteria without assuming any sampling probability. Then, we have investigated the diatoms tempo of diversification and the implication of their ecological strategies in explaining the observed heterogeneous speciation rates among diatoms species.

Diversification analysis of the currently largest bacterial phylogeny were performed using the models developed in [Chapter 1](#), which assumes constant and homogeneous speciation and extinction rate while being exempted from providing a global diversity estimate. We found that overall around 3 events of diversification (speciation minus extinction) happen per lineage every billion year (Gyr) – i.e. every 1,000 million years – in the bacterial realm. When considering cyanobacteria only, this estimate was lower with overall around 2 events of diversification happening per lineage every Gyr. These estimates represents the lower bounds of the diversification rates recovered in other large clades – i.e. the often studied clades e.g. vertebrates. Moreover, our results show that cyanobacteria extinction rate is particularly high relative to their speciation rate with 3 extinction events every 4 speciation events. This suggests that extinction is a major component of cyanobacteria diversification dynamic and species turnover through time seems substantially higher for cyanobacteria compared to other large clades. When considering clades which diversified after the Neoproterozoic oxygenation event (541 MA), we found that their respective net diversification rate is highly similar and vary at most of a factor of 2. Indeed, post NOE clades exhibit from 8 to 15 events of diversification per lineage every Gyr which also represents lower bounds of observed diversification rate. The low diversification rate recovered for bacterial species has previously been suggested to be linked to some intrinsic factors such as their mode of reproduction, their dispersal abilities and their gene expression regulation among other factors.

Further work could be undertaken to investigate potential factors involved in the low and constrained net diversification patterns found among bacteria. This could include exploring the relative contribution of bacterial traits such as recombination rate, dispersal and metabolic abilities as well as gene regulation. Some of these traits are already available (e.g. dispersal rate (Louca 2022)) and others could be acquired (e.g. metabolic abilities (Machado et al. 2018)). The challenge would then be to link this data to the OTUs present in the 16S based reconstructed phylogeny and would require to establish traits from genomes containing the 16S sequence (e.g. often lacking in metagenomes). Another interesting extension of this work could be to investigate the diversification rates of archaea using our best performing model and data from online resources (e.g. Genome Taxonomy Database (GTDB) (Parks et al. 2022)). This could be as substantially compelling as previous work has suggested that archaea are diversifying at a lower rate than bacteria (López-García and Moreira 2008) and would be an opportunity to test

this assumption using appropriate diversification models. Including species traits in this analysis, such as their ecological niche, would enable to test the hypothesis that bacteria might diversify faster than archaea due to their ability to colonize a large number of different niches compared to archaea which seems more constrained to extreme niches – e.g. salt crystallizers or hydrothermal vents.

Such as bacteria among prokaryotes, diatom is an ecologically and evolutionary successful group of organisms among eukaryotic phytoplankton. In [Chapter 3](#), we investigated their evolutionary success by exploring the contribution of their ecological strategy in explaining their heterogeneous speciation rates. We characterized diatoms niche ecological properties – i.e. niche position, niche breadth and niche marginality – and found that overall diatoms niche properties are not phylogenetic conserved with few exceptions. Diatoms inhabit warm and highly oxygenated niches corroborating previous observations. Still, temperature and oxygenation appeared to be the most important characteristics to discriminate their niche. Diatoms niches also substantially differ by their biogeography (i.e. latitude) and their nutrient availability (i.e. nitrogen, phosphorous, silicon). At the species level, most diatoms exhibited a low niche breadth and marginality. When linking the characterized niche to their speciation rates, we found that species inhabiting cold niches, high latitudes or niches far from the average niche conditions (i.e. marginal) are recently speciating faster than others. The effect of latitude was the strongest to explain diatoms recent speciation rates. This pattern has been reported previously in other clades (Harvey et al. 2020; Igea and Tanentzap 2020; Morlon 2020). In our work, we proposed that this pattern might be linked to their mode of reproduction as reproduction-related traits seem to harbor a latitudinal gradient (e.g. vegetative vs sexual and sex related genes (Armbrust and Galindo 2001)) and vegetative reproduction which is more prevalent in low latitudes is not sustainable in diatoms. Diatoms inhabiting ecological niche margins were also found to speciate faster. We hypothesized that ecological margins might provide more opportunities to speciation as the selection pressure might be stronger and the niche less densely filled (Simpson 1953). Indeed, lower competition (e.g. for resources) among diatoms might be found in diatom niche ecological margins as they might represent challenging environments and require specific physiological adaptations. Species enabled to colonize these environments might have more opportunities to speciation. Overall, we showed that diatoms speciating the fastest were located in high latitudes, in extreme environment or in cold waters shading light to the potential direct or indirect role of their ecology, physiological abilities, reproduction mode and biotic interactions in spurring their divergence.

Further work could include testing explicitly the implication of antagonistic interactions (e.g. competition, parasitism or predation) in diatoms speciation process by integrating biotic variables in the characterization of their niche. This can easily be achieved by exploiting the information on the presence of other organisms (e.g. dinoflagellates or cyanobacteria) in the *Tara Oceans* samples and could be refined by the integration of interactome information (Lima-Mendez et al. 2015). Additionally, including genomic and transcriptomic data to test the implication of physiological abilities in explaining the observed patterns could substantially complement the present work. Another avenue of improvement could be to exploit the large fossil information available for diatoms in order to reconstruct a phylogeny with extant and

extinct species. Besides from potentially improving the phylogeny, it would allow to apply a fossilized birth-death model which might provide more accurate extinction rate estimates (Stadler et al. 2018) when expanding the analysis to interpret speciation as well as extinction rates.

Lastly, in [Chapter 1](#) and [Chapter 3](#), we have investigated the diversification of two major clades belonging to the same functional group – i.e. the cyanobacteria and the diatoms. When comparing our findings, besides from their different age of origination, their tempo of diversification is drastically different. Indeed, diatoms diversify 50 times faster than cyanobacteria (under a *k-sampling* model, data not shown). Further comparative analysis including the characterization of their respective ecology may be of interest to explain this pattern.

4.3. Limitations and Perspectives

4.3.1. Technical Concerns

Biased estimates. – When developing a diversification model, one needs to obtain unbiased parameter estimates. On the other hand, obtaining low variance is desirable but acknowledged rather than required. This might become problematic when applying a model – which exhibits high variance – on one single observation (e.g. one empirical phylogeny) and might provide spurious estimates. In Bayesian inference, the posterior distribution should represent the probability of the estimate to be the true value. Once this statistical property has been satisfied, one should use this posterior to inform on the uncertainty of the estimate. In [Chapter 1](#), the Bernoulli sampling model with unknown sampling probability performed on a single phylogeny (i.e. M1) provides biased turnover rate estimates and consistently underestimates the turnover rate. Yet, the posterior distribution represents correctly the probability of the estimate to be the true value (expected coverage probability). On the other hand, the *k-sampling* model (i.e. M2) provides unbiased turnover rate estimates but high variance and the posterior distribution does not precisely represent the probability of the estimate to be the true value (lower coverage probability than expected). Interestingly, the two models have complementary statistical properties. When analyzing the diversification rates of the cyanobacteria in [Chapter 1](#), a high turnover rate was retrieved using the *unbiased model* (i.e. M2). Due to the high variance and the low posterior fidelity (i.e. lower coverage probability than expected), it is substantially delicate to interpret the estimate. In this context, the *biased model* (i.e. M1) can be valuable. Indeed, as M1 is nearly systematically underestimating the turnover rate, retrieving with M1 a turnover rate close to the one estimated under M2 (i.e. high turnover rate) would provide more confidence in the latter estimate. Therefore, due to its

negative bias on the turnover rate, M1 can be used as a lower bound turnover rate estimate and can allow to confidently interpret empirical results of high turnover rates.

Phylogenetic bias and uncertainty. – Errors in the topology and the branch lengths of a phylogeny can be present due to misleading alignments or saturation of the sequences. It has been shown that failing to apply the adequate model of sequence evolution to capture unseen substitutions can lead to biases in the divergence times and consequently in the diversification rates. Precisely, this can happen when sequences are saturated (Arbogast et al. 2002) – i.e. multiple substitution at the same site that cannot be disentangled when comparing with other sequences – and may result in an underestimation of the deep branch lengths which might be interpreted as diversification slow down (Revell et al. 2005). When reconstructing the diatoms phylogeny in [Chapter 3](#), we used IQ-TREE 2 that accepts a wide range of sequence evolution models to avoid under-parametrization. Another source of error relates to appropriately considering uncertainty. Throughout this manuscript, reconstructed phylogenies were handled as *fixed* therefore assuming that they represent the true evolutionary history of species. Nevertheless, uncertainty inherent to the phylogenetic reconstruction might be present. Bayesian approaches offer the possibility to appropriately account for this uncertainty by *sampling* trees from the posterior distribution of the Bayesian phylogenetic inference and thus integrating over the uncertainty while performing diversification rates estimation. This is also handled naturally in full inference approach. In [Chapter 3](#), as the number of OTUs was substantially large, we did not consider Bayesian approaches for the phylogenetic reconstruction and instead performed a sensitivity analysis – i.e. applying the diversification model on the maximum likelihood tree, the consensus tree and 10 bootstraps trees (data not yet available).

Computational time. – The development of increasingly performant computational systems have enabled us to address increasingly complex, realistic and pertinent questions in biology and beyond. This has come with a cost that is often neglected when describing a research project. A substantial amount of computational time was needed to perform the work presented in this manuscript.

4.3.2. Biological Reasoning

Patterns and processes. – This manuscript aims at characterizing patterns of diversification in the microbial world. In [Chapter 3](#), we have either suggested or tested some potential factors *correlating* with the observed patterns of diversification. We did not test for factors *driving* the diversification process, as in the present framework, one cannot resolve the contingency between drivers and correlates. Indeed, we speculated for potential mechanisms (e.g. different selection pressure, different ecological opportunities, different mode of reproduction) explaining the observed differences in speciation rates but could not infer the processes leading to the observations. A different approach consists in explicitly considering the microevolutionary processes that may generate macroevolutionary patterns, for example using individual based models (Pontarp and Wiens 2017)), rather than testing for correlations.

While this approach is elegant, it does not necessarily elucidate the ambiguity, as different processes can lead to the same pattern. Besides, microevolutionary processes are often inferred from present day data which might show mechanisms somewhat different than the ones in the deep past (Cohan and Koeppel 2008). Therefore, when investigating common diversification patterns through time and across lineages, considerations need to be made regarding the contingency between patterns and processes. Moreover, even though global diversification patterns are described, smaller scale patterns (and processes) can substantially differ as globality does not imply universality.

The narrative. – When studying the macroevolution of microorganisms, the same framework than for the study of macro-organisms is employed. Yet, several evolutionary processes are drastically different between micro and macro-organisms in particular for prokaryotes. When developing models, one needs to recognize potential constraints in the current narrative (Otto and Rosales 2020). In this regard, some have suggested to consider phylogenetic networks when studying macroevolution. This might be particularly appropriate for prokaryotes and for some micro-eukaryotes as their evolutionary history might be substantially reticulated. It could enable the investigation of the relative contribution of *hybridization* in the generation of species.

Perspective Switch. – One cannot neglect the potential importance of some microorganisms in directly or indirectly influencing the diversification of other species (the other side of the coin). Indeed, prokaryotes have drastically shaped the chemical properties of the Earth through their involvement in biogeochemical cycles (Falkowski et al. 2008) and substantially affected the evolution of life. It has been suggested that the evolution of multicellularity might be linked to cyanobacteria diversification (Schirmer et al. 2013). Microbial diversification may influence the evolution more complex life forms, as suggested by (Knoll et al. 2007) concerning the Proterozoic and Cambrian radiation of the phytoplankton and animal evolution. More broadly, through niche construction, microbial species provide further ecological opportunities and foster their own diversification as well as that of others (Doebeli 2002; Erwin 2008; Bajić et al. 2021). Lastly, one actor of microbial diversification, particularly of prokaryotic diversification, that I have not mentioned is viruses. The place of viruses among living organisms is ambiguous as they do not harbor all the defined properties. Yet, their role as a repository of genetic information and in the exchange of this information is considerable (Goldenfeld and Woese 2007). Indeed, phages through genetic exchange and acquisition of innovations have been suggested to substantially enhance prokaryotic diversification (Weinbauer and Rassoulzadegan 2003; Williams 2013).

4.4. General Conclusion

My PhD only scratched the surface of the difficulties inherent to studying the diversification of microbial clades. At the same time, it demonstrates that the task is not impossible. Further methodological developments, both in sequencing technologies that will allow a more complete representation of microbial diversity with more phylogenetic resolution and in phylogenetic approaches for studying diversification, should render such analyses increasingly robust. We can then hope to better understand the evolutionary history of organisms that represent the first form of life and most of diversity on Earth.

Glossary

Cohesive unit: *a unit within which mechanisms limit genetic diversity.*

Diagenesis: *describes the process of physical and chemical changes in the sediments.*

Ecotype: *defines a group of organisms with ecologically relevant physiological similarities.*

Extant lineages: *lineages alive at present.*

Holobiont: *refers to the assemblage of a host and its symbiotic microbiota.*

Homologous recombination: *exchange or addition of homologous genetic material.*

Hypobradylety: *exhibiting very low rate of evolutionary change.*

Incomplete phylogenies: *phylogenies not including all the present-day descendants of the common ancestor.*

Monophyly: *property of individuals belonging to a group composed of a common ancestor and all its descendants.*

Niche conservatism: *the propensity of related species to inhabit similar niches.*

Niche construction: *process whereby organisms through their metabolism and activities modify their own and/or other's niches.*

Periodic selection: *selection of the best fit genotype in a population which leads to the fixation of that genotype and consequently purge the genetic diversity within that population.*

Reconstructed phylogenies: *phylogenies of extant only lineages.*

Sex: *genetic exchange enabled by the meiotic apparatus.*

Speciation: *generation of permantly distrinct cluster of closely related individuals.*

Synapomorphy: *existence of a derived character state shared among several taxon.*

Ultrametric phylogenies: *phylogenies in which all the tips are at equidistance from the root.*

List of Abbreviations

ABC: *Approximate Bayesian Computation*

ACE: *Abundance Coverage Estimation*

CMR: *Capture-Marking-Recapture*

CNN: *Convolutional Neural Network*

CPP: *Coalescent Point Process*

CPR: *Candidate Phyla Radiation*

DNA: *Deoxyribonucleic Acid*

DPANN: *Diapherotrites, Parvarchaeota, Aenigmarchaeota, Nanoarchaeota and Nanohaloarchaea*

FFNN: *Feed Forward Neural Network*

GNN: *Graphical Neural Network*

GTDB: *Genome Taxonomy Database*

Gyr: *Giga year*

HGT: *Horizontal Gene Transfer or Lateral Gene Transfer (LGT)*

ICE: *Incidence Coverage Estimation*

LTT: *Lineage Through Time*

MA: *Millions of years Ago*

MAP: *Maximum A Posteriori*

MCMC: *Markov Chain Monte Carlo*

MH: *Metropolis-Hastings*

MLE: *Maximum Likelihood Estimate*

MMR: *Marine Mesozoic Revolution*

MSA: *Multiple Sequence Alignment*

NOE: *Neoproterozoic Oxygenation Event*

OTU: *Operational Taxonomic Unit*

PC: *Principal Component*

PCA: *Principal Component Analysis*

rRNA: *ribosomal Ribonucleic Acid*

SAC: *Species Accumulation Curve*

SAD: *Species Abundance Distribution*

Appendix: A Comparison of Deep Learning Architectures for Inferring Speciation and Extinction Rates from Extant Phylogenies

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ABSTRACT

Estimating past speciation and extinction rates across taxa, space and time is key to understanding how ecosystems and biodiversity evolved. Currently, extant phylogenies are fitted by diversification models whose parameters are estimated by Maximum Likelihood or Bayesian inference. However, this process can become intractable or slow to calculate even for diversification models with moderate complexity. Likelihood-free inference based on deep learning algorithms may provide an alternative, but their performance relative to likelihood-based statistical methods is still poorly understood, as is the question of which deep learning architectures are preferable for this task. Here, we evaluate the performance of different neural network architectures (DNN, CNN, RNN, GNN) and phylogeny representations (summary statistics, Lineage Through Time or LTT, phylogeny encoding and phylogeny graph) for inferring rates of the Constant Rate Birth-Death (CRBD) and the Binary State Speciation and Extinction (BiSSE) models. We find that deep learning methods can reach similar or even higher accuracy than Maximum Likelihood Estimation, provided that network architectures and phylogeny representations are optimally adjusted to the respective model. For example, for the CRBD model we find that CNNs and RNNs fed with LTTs outperform other combinations of network architecture and phylogeny representation, presumably because the LTT is an optimal data representation for homogenous BD models. For the more complex BiSSE model, however, it was necessary to feed the network with both topology and tip states information to reach acceptable performance. Overall, our results suggest that deep learning provides a promising alternative for phylogenetic inference, but that data representation and architecture have strong effects on the inferential performance.

Keywords: Cladogenesis, Machine Learning, Macroevolution, Stochastic Biodiversity Model

INTRODUCTION

Species richness varies greatly across taxonomic groups (G. E. Hutchison 1959), geological times (Barnosky et al. 2011) and geographical regions (Gaston and Blackburn 2000). It is generally accepted that these patterns in species richness emerge from variation in speciation and extinction rates in addition to dispersal and migration processes. For instance, important ecological phenomena such as the ‘Latitudinal Diversity Gradient’ (Hillebrand 2004) may be partly explained by variations in species net diversification rates defined as the balance of the speciation and the extinction rate (Mittelbach et al. 2007; Rolland et al. 2014b; Pontarp et al. 2019). In general, however, our understanding of why and when these rates change is still very limited (Rabosky 2009a, 2009b; Condamine et al. 2013b; Moen and Morlon 2014b).

To better understand the drivers of variation in diversification rates and their consequences for biodiversity dynamics, it is crucial to first estimate diversification precisely (Pyron and Burbrink 2013), and then ideally separate it into speciation and extinction rates (Stadler 2013c). The latter is not trivial, as fossil data are rarely available and reconstructed phylogenies do not include information about extinct species, thus requiring statistical models to infer underlying speciation and extinction rates from a net diversification signal. Fortunately, it is possible to separate the signal of extinction and speciation in many commonly used speciation models (Nee et al., 1994; Etienne et al., 2012; Morlon, 2014; but see also Louca and Pennell, 2020; Morlon et al., 2022).

Arguably one of the simplest diversification models is the Constant Rate Birth-Death (hereafter ‘CRBD’) model, which assumes that the speciation and extinction rates are both homogeneous across lineages and constant through time. Nee et al. (1994) showed that extinction and speciation rates in this model can be estimated based on reconstructed phylogenies with Maximum Likelihood using a distinct pattern called the “pull of the present”. However, the CRBD model is obviously very simple, and the attention of the field in recent years has turned to consider more complex models to reconcile predictions made from extant phylogenies and fossil data (Morlon et al. 2011b; Etienne et al. 2012b). These models commonly allow for variations of speciation and extinction rates through time (time-dependent models), across lineages (inhomogeneous models) as well as in dependence of environmental or biotic factors (ref). For instance, the time dependent Birth-Death model (Hallinan 2012) allows to consider homogenous changes (e.g. an exponential decay) of extinction rates over time (Morlon et al. 2011b). Other diversification models allow lineage-specific shifts in diversification rates that can be expected to occur with the appearance of key innovations (ref). These shifts can be either discrete (e.g. Binary State Speciation and Extinction model, hereafter ‘BiSSE’; Maddison et al., 2007) or continuous (e.g. ‘ClaDS’ model; Maliet et al., 2019).

A constant challenge when developing new diversification models is establishing robust methods to fit them to data. For diversification models with low or moderate complexity, the likelihood function of the model, i.e. the probability to observe a reconstructed phylogeny given the model, can be computed analytically (see Nee et al., 1994 for the CRBD model and Maddison et al., 2007 for the BiSSE model). If such an analytical likelihood is available, model rates can be inferred based on that latter using either Maximum Likelihood Estimation (MLE) or Bayesian MCMC sampling (Ricklefs 2007; Silvestro et al. 2011).

For more complex diversification models, however, likelihoods often become intractable. In this case, simulation based methods such as Approximate Bayesian Computation (ABC) could be used. The ABC approach approximates the likelihood of a model by comparing model

predictions to data via summary statistics (Csilléry et al., 2010; see Saulnier et al., 2017 for an example). Although ABC can successfully infer parameters from complex diversification models, it has two major drawbacks. First, ABC typically requires a large number of simulations, which is often computationally prohibitive in practical applications. Secondly, ABC suffers from the curse of dimensionality, meaning that the prediction accuracy quickly deteriorates with the dimensions of the summary statistics, which, for sufficient statistics, must be at least as large as the number of model parameters (Beaumont 2010; Hartig et al. 2011).

Recent progress in the development of deep learning algorithms offers a promising alternative for likelihood-free inference of model parameters from reconstructed phylogenies. Contrary to ABC, deep learning can easily deal with high-dimensional data and even benefit from it (LeCun et al. 2015). Indeed, deep learning approaches based on Convolutional Neural Networks (CNNs) have outperformed ABC for inference tasks in population genomics (Chan et al. 2018; Schrider and Kern 2018). Moreover, as pointed out by (Schrider and Kern 2018) neural networks have a high flexibility for the input data structure and could even change the way we think about the data representation. Unlike for ABC (Csilléry et al. 2010b), however, there is practically no experience on using deep learning algorithms for parameter inference in diversification models, except for a recent paper (Lambert et al., 2022) that provides a proof of concept.

A first question for applying deep learning in this context is the choice of the network architecture (Pichler and Hartig 2022). The easiest option would be to use a standard fully connected Deep Neural Network (DNN). The downside of this option is that DNNs do not perform favorably with high-dimensional and unstructured data compared to other network architectures. Therefore, we can anticipate that we would have to summarize the phylogeny shape (which by default is high dimensional with neighborhood relationships being represented as a graph) using a limited number of summary statistics (Saulnier et al. 2017b) to use them as input to a DNN, rather than providing the phylogeny directly.

A second option would be the use of Convolutional Neural Networks (CNNs). CNNs apply one or several filters that slide incrementally across the input data to detect spatial patterns. CNNs have shown great success for a variety of tasks. A recent study has shown that phylogenies can be encoded in matrix representation and processed by CNNs, leading to gains in predictive performance compared to likelihood-based methods (Voznica et al. 2021). Alternatively, CNNs could also be used on the Lineage Through Time (hereafter ‘LTT’) plot, as it is known that the slope of the LTT is related to the diversification rates for homogeneous models (Nee et al. 1994b; Ricklefs 2007).

When representing the phylogeny by its LTT, another option is the use of Recurrent Neural Networks (RNNs). The neurons of RNNs include a memory and, thanks to this feature, RNNs can process sequential data (e.g., time series) more efficiently. A common and efficient RNN neuron architecture is the Long Short-Term Memory (LSTM) cell. RNNs based on LSTM cells can flexibly handle long and short term dependencies of the input data (Hochreiter and Schmidhuber 1997; Yu et al. 2019) and have been applied with success in many fields, including Natural Language Processing tasks (Huang et al. 2019), financial market forecasting (Bukhari et al. 2020) or phoneme classification (Graves et al. 2005).

Eventually, when considering the nature of a phylogeny, Graph Neural Networks (GNNs) arise as a natural choice. GNNs are a relatively new architecture that generalizes the idea of

CNNs, which require Euclidean neighborhoods, to graphs (Scarselli et al. 2009; Zhang et al. 2019; Wu et al. 2021). GNNs have outperformed alternatives for many applications with a graph structure, *e.g.* predicting molecules properties (Gilmer et al. 2017; Mansimov et al. 2019) or classifying texts (Huang et al., 2015). The potential advantage of using GNNs over CNNs is that GNNs can directly perform convolutions along the phylogeny's topology, whereas the use of a CNN requires transforming the phylogeny into an Euclidean structure, which potentially distorts neighborhood relationships.

However, these considerations are mainly theoretical. To our knowledge, no systematic comparison exists in the literature about which combination of phylogeny representation and neural network architecture works best to infer parameters from reconstructed phylogenies. CNNs feeding on encoded phylogenies were shown to infer rates with a good accuracy for different diversification models (Voznica et al. 2021); however, given that for homogeneous birth-death models, all the relevant information is in the LTT (Lambert and Stadler 2013b), one could expect that a full encoding, which provides unnecessary information to the neural network, may not always be optimal.

In this paper, we address the question of how architecture and data representation of the phylogeny affect the ability of deep neural networks to infer parameters of diversification models from extant phylogenies. To find the best combination of neural network architecture and phylogeny representation depending on the complexity of the diversification model, we tested different combinations of these for the simple homogeneous CRBD model and for the more complex inhomogeneous BiSSE model. We represented phylogenies with either: LTTs, sets of summary statistics, phylogeny encodings from (Voznica et al., 2021), or phylogeny graphs. Then, these representations are combined with the suitable neural network architecture(s): summary statistics with DNNs, phylogeny graphs with GNNs, encoded phylogenies with CNNs, and LTTs with CNNs or RNNs. To evaluate the predictive performance of these deep learning inference methods, we compare the prediction errors of each method and break them down into three terms: 1) variance, 2) uniform bias and 3) consistency bias (Smith and Rose 1995). Moreover, we evaluate the predictive performance of the MLE in the same way to compare deep learning methods with this reference.

Using this set-up, we ask three questions. The first is: can deep learning methods infer rates from diversification models, and if so, could these methods outperform MLE regarding the prediction error as suggested by (Lambert et al. 2022)? We hypothesize that deep learning methods can theoretically outperform MLE as they can trade off bias against variance to minimize the total error.

Secondly, what is the optimal representation of phylogeny data to infer rates for diversification models of different complexity? We expect, for example, that the LTT is a sufficient statistics for simple homogeneous models and is often preferable due to its simplicity. For more complex models and specifically for inhomogeneous models, on the other hand, we expect that simple representations such as the LTT will fail to provide the neural network with the necessary information. More complex representations are necessary for optimal inferential performance.

Lastly, we ask how the choice of the neural network architecture in combination with the chosen phylogeny representation affects the inference. We hypothesize that the more complex the phylogeny representation, the more important the neural network architecture becomes.

METHODS

Diversification Models

As case studies to test the performance of deep learning algorithms for phylogenetic inference, we chose two established diversification models: 1) the relatively simple and homogeneous CRBD model, and 2) the more complex inhomogeneous BiSSE model. The rationale for choosing these two models was to characterize how the complexity of the diversification model, and especially its (in)homogeneity, influences the efficiency of phylogeny representations to infer rates.

The Constant Rate Birth-Death (CRBD) model is one of the simplest conceivable diversification models. The CRBD model has two parameters, the speciation rate (λ) and the extinction rate (μ), that are homogeneously constant over time. The likelihood of this model is well-known (Nee et al., 1994) and depends only on the times of speciation events and not the phylogeny's topology.

In the Binary State Speciation and Extinction (BiSSE) model the species of the phylogeny can alternate between two states (0 and 1), where each state has its own speciation and extinction rate. The switch through time between the two states translates the apparition of key evolutionary innovations which lead to rate shifts. In total the model has 6 parameters: 2 speciation rates (λ_0, λ_1), 2 extinction rates (μ_0, μ_1) and 2 two transition rates (q_{01} for transition $0 \rightarrow 1$; q_{10} for transition $1 \rightarrow 0$). As for the CRBD model, the likelihood of the BiSSE model is known (see Maddison et al., 2007). We imposed four constraints on the model to simplify inference, thus decreasing the number of free parameters from six to two. The constraints are: 1) $\lambda_1 = 2\lambda_0$; 2) $\mu_0 = 0$; 3) $\mu_1 = 0$; 4) $q_{01} = q_{10}$. Constraint 1) ensures that states 0 and 1 are different, 2) and 3) simplify the model by making it pure birth, and 4) is an assumption of symmetry that makes the probabilities to switch from one state to another equal.

To train the neural networks, we simulated 100,000 phylogenies for each model using the Diversitree library (FitzJohn 2012) in R and we assumed a complete sampling of the phylogenies. For the CRBD model, we draw the underlying parameters as follows: 1) we draw uniformly λ in $[0.1, 1.0]$; 2) we draw uniformly the turnover rate ε in $[0, 0.9]$ from which we compute the extinction rate $\mu = \varepsilon\lambda \in [0, 0.9\lambda]$. By doing so, we avoid the critical case where $\lambda \lesssim \mu$ (i.e., speciation rate is inferior to or equivalent to the extinction rate). For the BiSSE model, we took $\lambda_0 \in [0.1, 1.0]$ to stay in the same range as for the CRBD model and $q_{01} \in [0.01, 0.1]$ to ensure that one state is not overly represented compared to the other one.

Phylogeny Representation

Most of the deep learning methods that we considered cannot directly be trained with the simulated phylogenies, but require to reformat the phylogenies into a more regular data structure such as a set of summary statistics or the LTT. Here, we explain all options that we considered (see also [Fig. 1](#)).

Summary statistics.— Arguably the most basic option is to represent the phylogeny by a number of summary statistics. We used a set of 84 summary statistics inspired from the set of (Saulnier et al. 2017b). Summary statistics can be split into three groups:

- 1) 8 statistics related to phylogeny topology (*e.g.* ratio of the width over the depth of the phylogeny);
- 2) 25 branch lengths statistics (*e.g.* median of all branch lengths);
- 3) 51 LTT statistics (binned LTT coordinates and LTT slopes).

The list of the summary statistics and the changes compared to (Saulnier et al. 2017b) are detailed in the Table S1 of the Supplementation Information.

Lineage Through Time (LTT).— LTTs illustrate the increase in lineages over time. For a phylogeny of n tips, the LTT is composed of $n-1$ points where each point is defined by two coordinates: time (t , abscissa) and the number of lineages (N , ordinate). For a binary phylogenies where each speciation even results in two outgoing lineages, the LTT can be compressed to a 1-d array without loss of information, by throwing away the number of lineages and keeping only the times of speciation events.

Phylogeny encoding (CBLV).— Yet another alternative is to encode the phylogeny in a real-values vector of length $2n-1$ named the ‘Compact Bijective Ladderized Vector’ (‘CBLV’, see Voznica et al., 2021). Each value of the vector corresponds to one node (internal node or tip), thus there are n values for the n tips and $n-1$ values for the internal nodes $n-1$, which result in a vector of $2n-1$ values. The encoding is done in two steps: 1) phylogeny ladderization, and 2) phylogeny traversal. The principle of ladderization is to order each node’s children, such that the encoding is bijective (*i.e.*, one CBLV maps exactly to one phylogeny and reversely). Here we ladderized the phylogeny such that for each node the left child is the child which is further from the root. On the ladderized phylogeny, we perform an inorder traversal using a classical recursive algorithm. If the visited node is an internal node or the first tip is visited, its distance to the root is added to the vector. Otherwise, its distance from its parent node is added to the vector. Moreover, for the BiSSE model we add to the n tip states to the vector. The tip states are ordered according to the phylogeny traversal. Note that in the original method from Voznica, the tip state information was added by adding one row to the distance vector, thus resulting in a two-row matrix (first row for distances, second row for tip states), while here we concatenate two 1-d vectors resulting in a 1-d vector.

Phylogeny graph.— The main motivation for the previous CBLV encoding is to transform the phylogeny graph into a regular Euclidean format that can be easily processed by CNNs. The disadvantage, however, is that neighborhood relationships of the graph can be distorted in this process. Recent research in the field of machine learning suggests that in such a case, it is often better to train neural networks directly with the original graphs. This is done by so-called Graph Neural Networks or GNNs, which extend the CNN idea to graphs. Here, we use the Pytorch Geometric framework (Fey and Lenssen 2019) to represent the phylogeny as graph and train GNNs. We provide the GNN with the phylogeny’s topology and 4 attributes per node: distance

to the root and the lengths of the 3 edges linked to the node (1 incoming edge: parent $\rightarrow$ node; 2 outgoing edges: node $\rightarrow$ child₁ and node $\rightarrow$ child₂). Moreover, for the BiSSE model we add as an attribute the tip states (0 or 1 for tips, -1 for nodes whose state is unknown).

Neural Network Architectures

For predicting model parameters from the formatted phylogenies, we considered four different neural network architectures: DNN, CNN, RNN and GNN (see also Fig. 1). All architectures were built within the torch framework (Falbel and Luraschi 2019) in R, except GNNs which were built with Torch in Python because the PyTorch Geometric library (Fey and Lenssen, 2019) was needed for GNNs. Hyperparameter tuning for each architecture was performed by hand (see Supplementary Information Tables S0-S5).

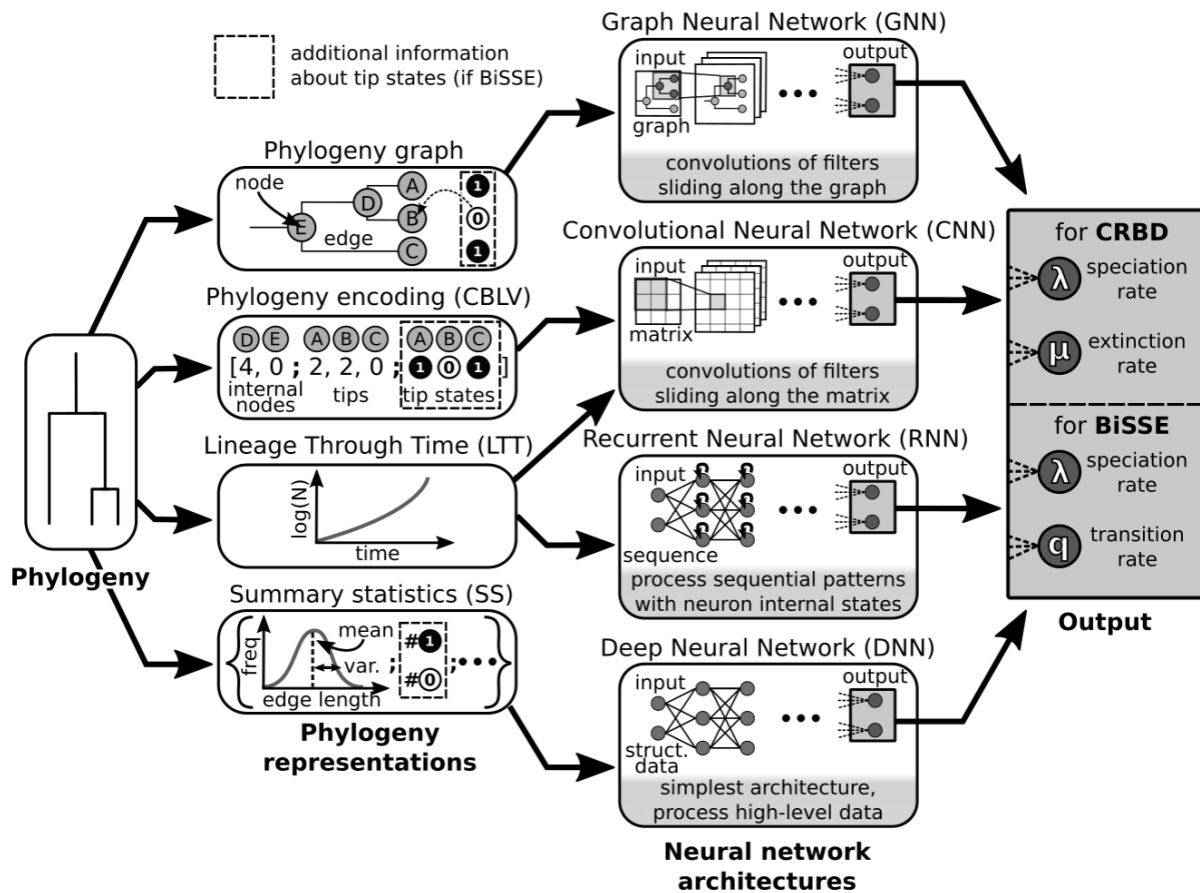


Figure 1. Combining phylogeny representations with neural network architectures. The five different combinations considered between phylogeny representations and neural network architectures are indicated by arrows from the first column to the second one. The rationales behind these five combinations are the following: 1) for phylogeny graph with GNN, GNN is the only architecture able to process graphs; 2) for CBLV with CNN, CNNs are able to detect patterns that are expected in the phylogeny encoding (Voznica et al. 2021); 3) for LTT with CNN, CNNs can detect patterns in the slope of the LTT; 4) for LTT with RNN, RNNs are designed to process time series; 5) SS with DNN, SS do not have spatial or temporal hierarchical order and therefore can be processed by a simple architecture. The third column describes the output of the inference methods for the two diversification models examined (CRBD and BiSSE).

Training Neural Networks

To train and validate the performance of the networks, the simulated phylogenies were split randomly in three groups: 1) a training set to train the neural networks (90,000 phylogenies); 2) a validation set used to quantify performance of the neural network during the training (5,000 phylogenies) and 3) a test set used to quantify the performance of the neural network after the training (5,000 phylogenies). These split sets were the same for all neural network architectures, so all of the architectures are trained and tested with the same phylogenies. Moreover, the Maximum Likelihood Estimator (hereafter ‘MLE’) was tested on the phylogenies of the test set to fairly compare neural networks and MLE.

Training of the networks followed the standard practice of dividing the training data into small batches (typically 64 phylogenies per batch, for more details see Supplementary Information). During optimization, batches are successively provided to the network and its weights are adjusted to reduce the prediction error (defined as the mean squared error). Once the network has been fed with the whole training set, the weights are frozen and the performance of the network is evaluated on the validation set. This process of training and validation is called an ‘epoch’. We repeated this process until the prediction error on the validation set stopped to improve (early stopping). This ensures that the network does not overfit the training data, which tends to improve the performance of the model on new data (Prechelt 1998). In addition to early stopping, we also set at each new epoch a small proportion of the network weights to zero (dropout, set to 1%), another standard method to avoid overfitting. After the training of the networks, we evaluated their performances on the test set.

Interpreting Predictions Error

Once the neural networks were trained, we use them to infer the parameter values of the models from the test data (the phylogenies in the hold-out). We also calculated the Maximum Likelihood Estimate for these phylogenies to establish a baseline. Thus, for each method (the neural networks or the MLE) we have the predicted model parameters (y_{pred}) vs. the true values (y_{true}).

To better understand the origin of deviations between the true and predicted parameter values, we divide the prediction error into stochastic error and bias according to Theil coefficients (Smith and Rose 1995). The idea of this approach is that the sum squared differences $U_{tot} = \sum_{i=1}^n (y_{pred,i} - y_{true,i})^2$ can be expressed as a sum $U_{tot} = U_{uniformBias} + U_{consistencyBias} + U_{variance}$ where: 1) $U_{uniformBias} = n (\overline{y_{pred}} - \overline{y_{true}})^2$ corresponds the systematic error; 2) $U_{consistencyBias} = (b - 1)^2 \sum_{i=1}^n (y_{true,i} - \overline{y_{pred}})^2$ is the slope of the linear regression of y_{pred} vs. y_{true} and $\overline{y_{pred}}$ the mean of the predicted values; 3) $U_{variance} = \sum_{i=1}^n (y_{pred,i} - y_{pred,lm,i})^2$ is the stochastic error, measured by the variance of the y_{pred} around the regression slope $y_{pred,lm,i}$. Lastly, we normalize each term by the sum squared of the true parameter values: $U_d = \sum_{i=1}^n y_{true,i}^2$. The error decomposition is illustrated in [Fig. 2](#).

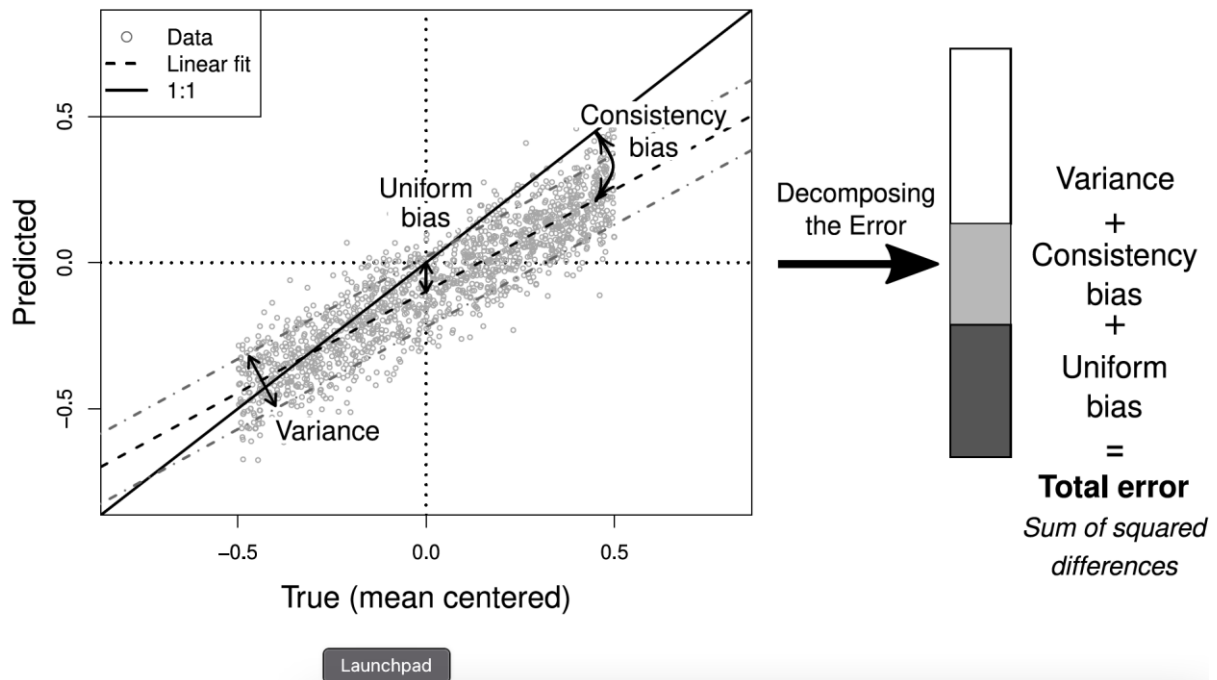


Figure 2. Decomposition of the prediction error. The total error defined as the sum of squared differences is split in three terms also known as the Theil's coefficients: 1) variance that describe the width of the points distribution across the linear fit, 2) the uniform bias which quantifies the systematic error and 3) the consistency bias which expresses how far the slope of the linear fit is from 1 (Smith and Rose 1995). For the example we generate synthetic data such that: $\text{Predicted} \sim 0.7 \cdot \text{True} + N(\mu=0.05, \sigma^2=0.1)$ with 1,500 True values uniformly distributed in $[0,1]$. This decomposition of the prediction error is summarized by a stack barplot (left, not to scale; for another example see Fig. 3). Note that the True axis has been mean centered such that the uniform bias can be measured at $x=0$.

To fairly compare neural networks and the Maximum Likelihood Estimator, we restricted the parameter space to test the inference models to an inner domain of the explored parameter space. Indeed, during the training phase the neural networks learn to not predict values outside the parameter space they have seen, whereas the Maximum Likelihood cannot do so. Thus, when estimating parameters close to the boundaries of the explored parameter space, the variance of the neural network predictions artificially decreases due to the specific set up of the phylogeny simulations.

RESULTS

Overall, we find that deep learning methods show good performances as they predict rates with an error comparable to MLE in most cases (Fig. 3). Some deep learning methods even outperformed MLE (CNN and RNN with LTTs for the CRBD model, see Fig. 3a,b). Looking at the decomposition of the error, we see that the reason does not seem to be that deep-learning methods have a greater bias than MLE, e.g. DNN with SS vs. MLE for the CRBD model (Fig. 3a,b). Thus, the bias-variance trade-off does not fully explain why deep-learning methods can outperform MLE regarding the total error. We will tackle the reasons for this behavior in the next section on phylogeny representations. The only notable exception to the good performance

of the deep learning models was the GNN, which performed rather poorly for both diversification models. For that reason, we omitted the GNN in the following analyses and tackled possible reasons for its failure in the [Discussion](#).

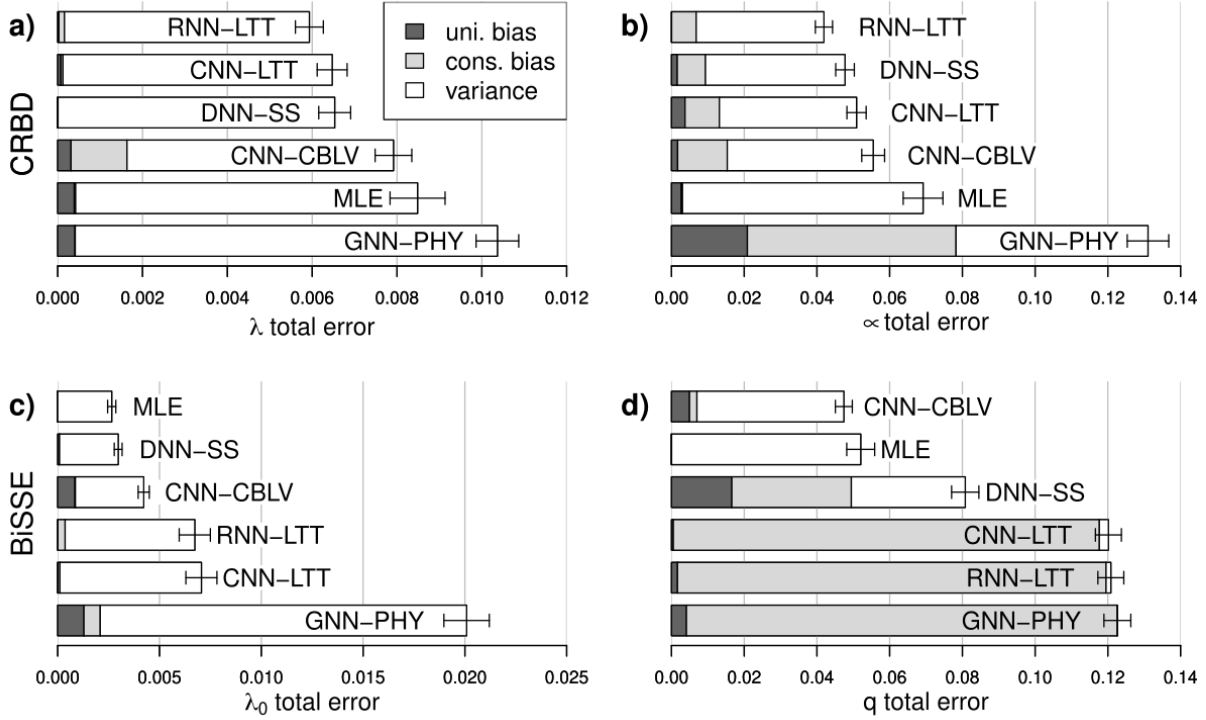


Figure 3. Prediction error of deep learning methods and MLE for the CRBD (a-b) and the BiSSE (c-d) model. From the CRBD we infer: a) the speciation rate (λ) and, b) the extinction rate (μ). From the BiSSE model we infer: c) the speciation rate of state 0 (λ_0) and, d) the transition rate (q), the other parameters are constrained (see Methods for more details). The deep learning methods evaluated are the following: DNN with summary statistics (DNN-SS), CNN with LTTs (CNN-LTT), RNN with LTTs (RNN-LTT), CNN with encoded phylogeny (CNN-CBLV) and GNN with phylogeny graphs (GNN-PHY). The total summed-squared error is split in three terms: variance, uniform bias and consistency bias (for more details see [Fig. 2](#) or Smith and Rose, 1995). The error bars correspond to the 95% confidence interval.

To understand how the optimal phylogeny representation depends on the complexity of the underlying diversification model. Our findings show that for the CRBD model, the best models are based on the LTT (CNN-LTT and RNN-LTT). Those models can even outperform the MLE ([Fig. 3a,b](#)). In contrast, inference methods relying on more complex phylogeny representations have a higher prediction error. This is especially clear for the two methods based on the most complex representations: CNN with phylogeny encodings and GNN with phylogeny graphs. In sum, the more information is added in the data, the worse the deep learning method performs, with the LTT being the optimal.

For the BiSSE model, we found a different behavior, that is, CNN with encoded phylogenies, but not GNN with phylogeny graphs, outperformed simpler LTT-based models ([Fig. 3](#)). To understand which information is missing in the LTT and thus is responsible for the increased performance of the CNN with encoded phylogenies compared to LTT-based models,

we provided neural networks with data of more fine-grained increasing complexity. Specifically, we compared a CNN fed with LTTs as a baseline (1) with a CNN and three increasingly detailed encoded phylogenies (2-4). In the latter, we varied 2 features: the number of tip states and the location of tip states in the phylogeny. Thus, the 3 options are: excluding all tip states information (2); including information about the tip states, but not their location (3); including information about the tip states and their location (4; see Fig. 4c). To sum up, we compare 4 options: (1) no information about phylogeny's topology, tip state numbers and location, (2) information about topology but not about tip states, (3) information about phylogeny topology, tip state numbers but not about their location, and (4) information about phylogeny topology, tip state numbers and their location.

This analysis confirms that the LTT alone is insufficient to infer rates accurately, as can be seen by the CNN and RNN combined with LTTs being among the deep-learning methods with the highest error (Fig. 3c,d). Moreover, the predictive performance increases with each information that was added about tip states, suggesting that the each of the intermediate steps considered loses information (Fig. 4).

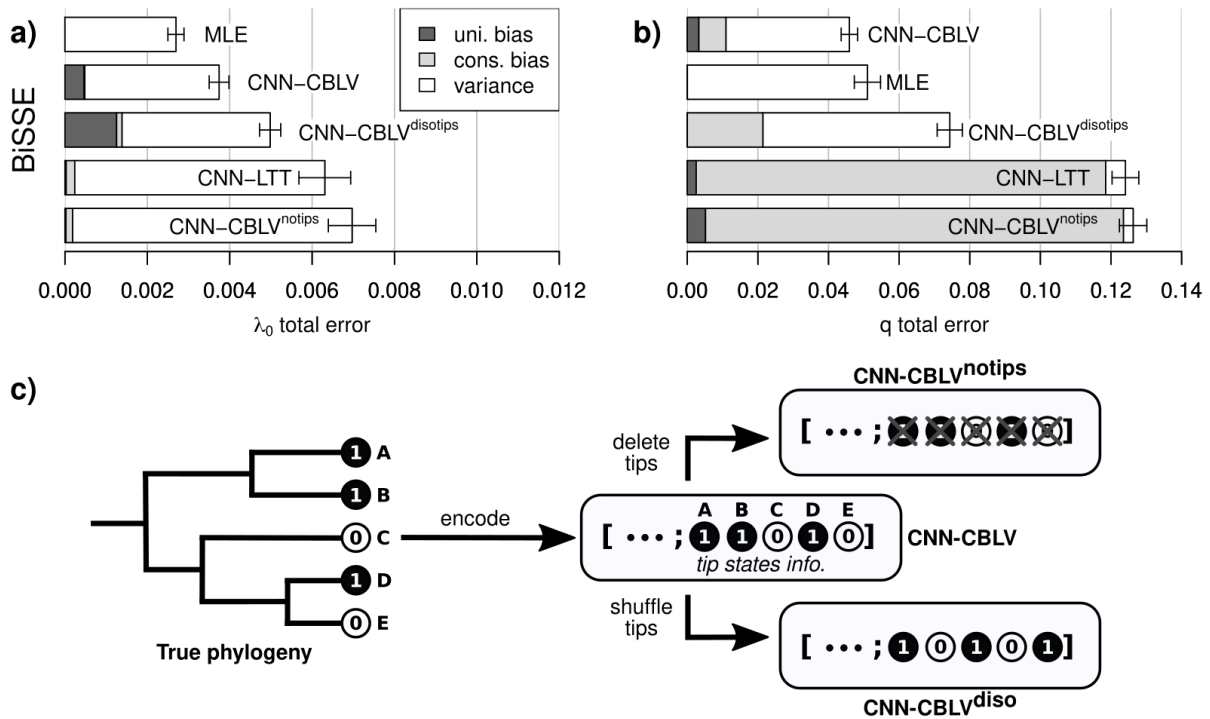


Figure 4. Prediction error of deep learning methods using different levels of tip states information for the BiSSE model. We infer: a) the speciation rate of state 0 (λ_0) and, b) the transition rate (q), the other parameters are constrained (see Methods for more details). c) The deep-learning methods evaluated are CNN with encoded phylogenies: excluding tip states information (CNN-CLBV^{notips}), containing tip states but randomly shuffled (CNN-CBLV^{disotips}), and containing tip states in order (CNN-CBLV) and are presented in panel c). Additionally, we consider CNN with LTT to have a reference of another deep learning method that does not use tip states information, and MLE which is our baseline. The total summed-squared error is split in three terms: variance, uniform bias and consistency bias (for more details see Fig. 2). The error bars correspond to the 95% confidence interval.

When looking at the influence of the neural network architecture, we see a mixed picture: for simple data representations of the phylogeny, CNN and RNN combined with LTTs achieved a similar prediction accuracy in all cases (Fig. 3), and they were overall the best option for the CRBD model, whereas for the more complicated BiSSE model, the even simpler DNN architecture with summary statistics outperformed the LTT-based architectures, but the CNN with encoded phylogenies. That latter combination outperformed all other options by a large margin for the BiSSE model, including the GNN (Figs. 3, 4). Indeed, that latter is interesting because the encoded phylogeny is a bijective representation of the phylogeny as a graph. Thus, the data provided to the CNN and GNN are equivalent. Then, the differences in performance between CNN with phylogeny encodings and GNN with phylogeny graphs can be attributed to the network architecture.

DISCUSSION

The purpose of this paper was to explore the ability of different deep learning architectures to infer parameters from diversification models. Our main findings are that deep learning methods are surprisingly accurate for this task and can even outperform the MLE in some cases. Looking in more detail at the deep learning method, we find that the optimal phylogeny representation and network architecture depend on the diversification model. For the simple homogeneous CRBD model, we found that feeding the networks with the LTT as sufficient statistics was the main factor that improved inference, presumably because this removed noise that was uninformative for the inference. For the inhomogeneous BiSSE model, it was optimal to provide the data in its most complex form, and the choice of the network architecture played a far greater role. We speculate that these results will generalize: if simple sufficient statistics exist for a model, these should be used and the choice of the network architecture is likely secondary. If no simple sufficient data representation exists for a given model, more complicated network architectures have to be used, and the choice of the network architecture becomes more critical.

Total Error and Bias

To quantify the performance of the models, we ranked the methods primarily according to the total error, consisting of bias and variance (Smith and Rose 1995). This is in line with machine learning practice, whereas in statistics, bias is often seen as more crucial than variance, because an unbiased estimator allows one to accumulate evidence (Shmueli 2010). Given the nature of this field, however, where there are usually no independent replicates of a phylogeny, we find it defensible to compare methods by their total error. We note, however, that the ranking would likely change if one would try to minimize bias only.

Our results regarding total error support earlier findings of (Voznica et al. 2021) who also reported that deep learning methods can outperform state-of-the-art methods to infer rates from phylogenies (BEAST 2 in his study, MLE in our study). Voznica et al. (2021) also noted that increasing the training data from 100,000 to 1,000,000 phylogenies still improved performance, suggesting that the performance we report in this study provides a lower bound of what could potentially be achieved by deep learning methods for this task.

Contrary to our expectations, we did not find support for the hypothesis that deep learning methods generate their performance advantage over the MLE, where it exists, by leveraging the bias-variance trade-off. Indeed, several deep learning methods had a lower total error and a lower bias than MLE (where we expected that they traded off variance against a greater bias to achieve lower error). We speculate that reason is thus that the MLE also has some small-sample bias for the CRBD model, especially for low net diversification rates (i.e. $\mu \lesssim \lambda$).

However, even if we succeeded in building deep learning methods able to outperform MLE regarding the total error (and sometimes the bias), the most promising neural network architecture for our inference task, GNNs, showed poor predictive performances. We investigate this failure in the following.

The Failure of GNNs

Given the generally positive results in the machine learning literature about GNNs (Zhang et al. 2019), the poor performance in our study was surprising to us. We speculate that it can be explained by two well-known limitations of GNNs: ‘hop neighborhood’ (Nikolentzos et al. 2020) and ‘over-smoothing’ (Oono and Suzuki 2019).

First, ‘hop neighborhood’ is a limitation resulting from how GNNs process node data. In a GNN, at each new layer, node attributes are updated by aggregating their neighbor attributes (Scarselli et al. 2009). Thus after k layers, each node has an attribute vector aggregating the attributes of its ‘ k -hop neighborhood’, i.e. the nodes that can be reached in k or less ‘hops’ (e.g., if considering a 3-nodes graph $A-B-C$, the nodes (A,B) are distant from 1 hop as well as (B,C) , while the nodes (A,C) are separated by 2 hops). Then following this aggregation process, node attributes only contain information about local graph structures but do not encode any global information (Kriege et al. 2018). Thus, in cases where global graph properties are needed for inference, the short-sightedness of GNNs might be a major limitation. The naïve way to counteract this limitation is to increase the number of GNN layers as the size of the ‘ k -hop neighborhood’ increases with k the number of layers. However, by doing so one will encounter the second limitation: ‘over-smoothing’ (Oono and Suzuki 2019).

Secondly, when stacking many layers in a GNN, node attributes become indistinguishable from one another due to the cumulative aggregations occurring at each layer, this phenomenon is referred to as ‘over-smoothing’ (Li et al. 2018). Then, in practice, adding layers in GNNs often results in a deterioration of predictive performances (Li et al., 2018). Thus the combination of the two latter limitations might explain why ‘classical’ GNNs failed to infer rates accurately from reconstructed phylogenies. Moreover, despite the failure of GNNs, several deep learning methods show good performance, allowing us to build an understanding on how to design efficient deep learning methods for rate inference.

It would be interesting for further research to test if results can be improved by considering specific GNN variants designed to overcome these limitations (for over-smoothing see Li et al., 2019; and for hop neighborhood see Nikolentzos et al., 2020).

Designing Efficient Deep-Learning Methods for Rate Inference

Our results suggest that deep learning methods are a viable alternative to infer rates from diversification models, which is particularly interesting for cases where likelihoods are not tractable or hard to compute. However, our results also suggest that the performance of the deep learning methods may strongly depend on data representation and network architecture. Regarding the complexity of the phylogeny representation, we find that it seems beneficial to

provide only the necessary data. For instance, for homogeneous models the topology of the phylogeny does not influence the likelihood (Lambert and Stadler 2013b). Thus, the LTT is an optimal representation as it only contains phylogeny information useful for the inference. For more complex diversification models, in particular for models with intractable likelihoods, it will likely be harder to generate simple sufficient statistics, and deep learning methods that use the whole phylogeny might be needed. For instance, for the BiSSE model, the encoded phylogeny (Voznica et al. 2021) containing phylogeny topology and tip states information appears to be optimal. In short, choosing the optimal representation is important: choosing an over-complicated representation for a simple diversification model costs unnecessary computational power and deteriorates the accuracy of the predictions, on the contrary choosing an over-simplified representation for a complex diversification model leads to poor predictions as the representation does not contain enough information to predict parameters properly.

Once the optimal phylogeny representation has been found, the appropriate neural network architecture has to be selected. First, the choice of the architecture is restricted by the data type e.g., a time series can go either with a CNN or an RNN but do not fit with a DNN which is not designed to detect patterns or to process temporal data. Secondly, the importance of the choice of the network architecture among the remaining possibilities is dependent on the complexity of the phylogeny representation: the more complex the representation, the more important the choice of the architecture is. In other words, when data become more difficult to process, more attention should be paid to the choice of the neural network architecture that will process the data.

Outlook and Further Research

The diversification models considered here remain relatively simple. Moreover, to keep the analysis and training times tractable, we simplified the inhomogeneous BiSSE model by assuming four constraints on the rate parameters. We hold that these choices were appropriate for the purpose of this study, which was to establish general principles of the problem, but it is clear that the practical benefit of this methods would be to perform inference on more complicated diversification models with intractable likelihoods and probably a higher number of parameters. How well the deep learning methods perform in this scenario remains to be explored by further research.

Moreover, inferring parameters with deep learning methods has a number of drawbacks compared to traditional MLE approaches. Most importantly, training networks requires generating a large set of simulated phylogenies under the considered diversification model. To avoid that each user has to undergo this process, it seems obvious that already trained neural networks must be provided, but those would be likely still much larger and possibly less portable than current statistical procedures.

CONCLUSIONS

In conclusion, our study supports the idea that deep learning methods can be used to infer diversification rates from reconstructed phylogenies. In our results, they often showed similar and sometimes even better performance than the MLE. Thus, deep learning methods offer a promising approach to likelihood-based statistical inference, in particular for models whose likelihoods are intractable or hard to compute. However, our results also provide a warning that deep learning methods are very diverse and the choice of the method (i.e. the combination of a

phylogeny representation and a neural network architecture) must be adjusted to the model considered. If that can be achieved, however, they may be instrumental in expanding our options to perform statistical inference for complicated macroevolutionary models.

SUPPLEMENTARY MATERIALS

The code is available at <https://github.com/ilajaait/phylo-inference-ml>
The phylogenies generated for this study are available here : *FigShare link here*.

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SUPPLEMENTARY INFORMATION

List of summary statistics

The changes in summary statistics compared to for (Saulnier et al., 2017) occur in the summary statistics of the LTT plot. Here, in addition to 40 statistics to bin the LTT in 20 points, we added 10 slopes of the LTT instead of 2 slopes for (Saulnier et al., 2017). To compute the 10 slopes, the LTT is first cut into 10 equal parts regarding time; then points of each part are fitted in scale independently with a linear model from which we extract the slope in semi-log scale. We added more of these slopes to the set of summary statistics because it is known that the LTT slope contains information about the speciation and extinction rates for homogeneous diversification models (Nee et al. 1994b; Ricklefs 2007). Lastly, for the BiSSE model we added the proportion of tips in state 1 (n_1 ; and the proportion of tips in state 0 being simply $n_0 = 1 - n_1$).

Name	Description	Number
Branch lengths		25
<i>height</i>	Height of the phylogeny: time from root to present	1
$\langle bl_{all} \rangle$	Mean length of all branches	1
$med(bl_{all})$	Median length of all branches	1
$var(bl_{all})$	Variance of the lengths of all branches	1
$\langle bl_{ext} \rangle, med(bl_{ext}), var(bl_{ext})$	Mean, median and variance of the lengths of external branches (i.e. linked to a tip)	3

$\langle bl_{int}^i \rangle, med(bl_{int}^i), var(bl_{int}^i)$	Mean, median and variance of the lengths of internal branches (i.e. not linked to a tip) belonging to the $i \in \{1,2,3\}$ third of the phylogeny.	9 (3x3)
$\frac{\langle bl_{int}^i \rangle}{\langle bl_{ext} \rangle}, \frac{med(bl_{int}^i)}{med(bl_{ext})}, \frac{var(bl_{int}^i)}{var(bl_{ext})}$	Ratio of the piecewise mean, median and variance of internal branches respectively over the mean, median and variance of external branches.	9 (3x3)
Phylogeny topology		8
<i>colless</i>	Sum over internal nodes of the absolute difference between number of left and right tip(s)	1
<i>sackin</i>	Sum over tips of the number of internal nodes separating the tip to the root.	1
$\frac{width}{depth}$	Ratio of the maximal width over the maximal depth. Depth is defined as the number of branches separating a node from the root, and the width of a phylogeny at a given depth ($width(depth)$) is the number of nodes having that given depth.	1
$max(\Delta width)$	Maximal width difference between two consecutive depths.	1
$max(ladder)$	Maximal ladder size, where a ladder is a chain of internal nodes each linked to a single leaf. The ladder size is the number of internal nodes in the chain.	1
$\frac{internal\ node \in\ ladder}{number\ of\ internal\ nodes}$	Proportion of internal nodes belonging to a ladder.	1
$stairecaseness_1$	Proportion of internal nodes that have a different number of left and right tips (<i>imbalanced nodes</i>)	1

$stairecaseness_2$	Mean of the ratio of the minimal number of tips on a side over the maximal number of tips on a side, for each internal node.	1
LTT		51
n_{tips}	Number of extant species	1
Binned LTT coordinates	LTT coordinates (time, number of lineages): binned the number of lineages in 20 equal parts and add the corresponding time.	40 (20x2)
LTT slopes	Split the LTT in 10 equal parts regarding time and infer the slope from the linear model of the LTT (in semilog scale) for each part	10
Additional statistic for BiSSE		1
$\frac{tips\ in\ state\ 1}{n_{tips}}$	Ratio of number of tips in state 1 over the total number of tips	1

Table S0. List of summary statistics computed to summarize phylogenies. These statistics are mainly inspired from (Saulnier et al., 2017).

Zero Padding of Phylogeny Representations

Zero padding is a method used when one has data of different sizes and wants to have data of the same size. For instance, in our case the phylogeny representation sizes often depend on the size of the phylogenies. However, most neural network architectures need to be fed with data of the same size. Thus we need to use zero padding to overcome this issue. The principle of zero padding is to: 1) set a maximum data size; 2) for all the data with smaller sizes than the maximum size, fill with zeros until the data reaches the maximum size previously set. This process is detailed more precisely for the different neural network methods, as each method has its particularity.

LTT with CNN.—The LTT is a time series of $N-1$ points where N is the size of the phylogeny (or the number of tips). Therefore, since CNNs need inputs of the same sizes, we use zero padding to bring all LTT vectors to the same size of $N_{\max}-1=999$, where $N_{\max}$ is the maximum phylogeny size and $N_{\max} = 1,000$.

Phylogeny encoding with CNN.—The phylogeny encoding has a size of $2N$ (N values for internal nodes plus phylogeny height + N values for the tips) for the CRBD model and $3N$ for the BiSSE model (+ N values for the tip states) with N the size of the phylogeny. For an arbitrary encoding vector of a phylogeny of size N , we do the following:

- 1) Extract the N first values corresponding to the N internal nodes plus height, and add zeros until having a vector of size $N_{\max}$;
- 2) Repeat for the N values corresponding to the tips;
- 3) If considering the BiSSE model, repeat for the N values corresponding to the tip states;
- 4) Concatenate the vectors resulting from the previous steps to obtain a final vector of size $2N_{\max}=2,000$ for the CRBD model and $3N_{\max}=3,000$ for the BiSSE model.

Lastly, when considering the BiSSE model the tip states are usually encoded using 0 and 1, however the use of 0 to encode for tip states can be confusing with the use of 0 for zero padding. Thus, we encode the tip states with -1 and 1 instead of 0 and 1 to avoid this ambiguity. We observed that encoding tip states as such led to a significant reduction of the total error.

LTT with RNN.—For RNN feeding on LTT, the zero padding was found to significantly deteriorate the RNN training, preventing the RNN from learning how to infer rates from LTTs. Thus we use a small trick to not use zero-padding for LTT with RNN. In the torch framework, RNNs can be fed with input data of different sizes, as long as in each batch data have the same sizes. Then for RNNs training, we created custom batches such that each batch contains LTTs of the same size. This trick might introduce a bias in the learning as the LTTs are not randomly given to the RNN, nevertheless, RNNs fed on LTT with this method have shown good inference performance and even outperform all other inference methods for the CRBD model.

Lastly, GNN with phylogeny graphs can work on input data of different sizes and summary statistics have always the same size (equal to the number of summary statistics considered), thus for DNN with summary statistics and GNN with phylogeny graphs zero padding is not needed.

Neural Network Structures and Parameters

In this section, we describe the neural network structures (number of layers, number of hidden neurons) of the architectures studied and their main parameters (early stopping patience, dropout rate).

Layer name	Layer size
Input	86 (87)
Fully Connected 1	100
Fully Connected 2	100
Fully Connected 3	100
Fully Connected 4	100
Output	2

Table S1. Structure of the DNNs combined with summary statistics used to infer rate from diversification models. In total, the network has 39,202 (39,302 if BiSSE) trainable parameters. In parentheses is the layer size if including tip state information in case of the BiSSE model.

Layer name	Layer size
Input	1,000
Convolution 1	8*996
Pooling 1 (average)	8*498
Convolution 2	16*494
Pooling 2 (average)	16*247
Convolution 3	32*243
Pooling 3 (average)	32*121
Fully Connected 1	3872
Fully Connected 2	100
Output	2

Table S2. Structure of the CNNs combined with LTTs used to infer rate from diversification models. In total the network has 390,798 trainable parameters. The kernel size is equal to 5. For convolutional and pooling layers a size of 8*996 means that the layer has 8 channels and each channel contains a 1-d vector of length 996.

Layer name	Layer size
Input	2,000 (3,000)
Convolution 1	8*1,991 (8*2,991)
Pooling 1 (average)	8*995 (8*1495)
Convolution 2	16*986 (16*1486)
Pooling 2 (average)	16*493 (16*743)
Convolution 3	32*484 (32*734)
Pooling 3 (average)	32*242 (32*367)
Convolution 4	64*233 (64*358)
Pooling 4 (average)	64*116 (64*179)
Fully Connected 1	7,424 (11,456)

Fully Connected 2	100
Output	2

Table S3. Structure of the CNNs combined with phylogeny encodings used to infer rate from diversification models. In total, the network has 769,782 (1,172,982 if BiSSE) trainable parameters. In parentheses are the layer sizes if including tip state information in case of the BiSSE model. For convolutional and pooling layers a size of 8*996 means that the layer has 8 channels and each channel contains a 1-d vector of length 996.

Layer name	Layer size
Input	1,000
LSTM 1	500
LSTM 2	500
Fully Connected	500
Output	2

Table S4. Structure of the RNNs combined with LTTs used to infer the rate from diversification models. In total the network has 3,011,002 trainable parameters. The recurrent layers (LSTM 1 & 2) are composed of LSTM cells which is the most used type of cells for RNNs (for more details see Hochreiter and Schmidhuber, 1997). The recurrent layers had to be wide (500 neurons here) otherwise the inference was failing.

Layer name	Layer size
Input	5 (6) attributes per node
GraphConv 1	50
GraphConv 2	50
GraphConv 3	50
GraphConv 4	100
Fully Connected 1	100
Fully Connected 2	50
Output	2

Table S5. Structure of the GNNs combined with phylogeny graphs used to infer the rate from diversification models. The Graph Convolutional layer used was GCNConv (Kipf and

Welling, 2017), other layer types (e.g. SAGEConv; see Hamilton et al., 2018) were tested but did not lead to any improvement.

Moreover, to avoid overfitting of the neural networks, we used early stopping and dropout. For all architectures, we used a patience of 3 epochs for the early stopping, meaning that the neural network training stops when the total error on the validation has not improved since 3 epochs. Furthermore, we introduced a dropout rate $p = 0.01$, meaning that at each epoch 1% of all network weights are reset to 0.

Evaluating Deep-Learning Methods Performance Far From Boundaries of the Parameter Space

Neural networks learn to not predict parameter values outside the parameter space it has seen during the training. Typically, if a neural network has only seen phylogenies with speciation rate in $[0.1, 1]$, the trained neural network will not predict speciation rate outside this range (see [Fig. S1](#)). This phenomenon artificially reduces the prediction error of the neural networks compared to the Maximum Likelihood Estimations that can go outside the simulated parameter space. Thus, to avoid this artifact, we restrict our analysis to an inner domain of the parameter space by excluding from the analysis the parameter values near the parameter space boundaries. The only exception to this is the lower boundary of extinction rate for the CRBD model, as in this case the Maximum likelihood is also constrained to only predict positive extinction rates. The parameter space as well as the inner domain considered to evaluate inference method performances are summarized in Table S6.

Diversification model	Rate	Parameter range	Inner range
CRBD	Speciation rate	$[0.1, 1]$	$[0.2, 0.9]$
	Extinction rate	$[0, 0.9]$	$[0, 0.8]$
BiSSE	Speciation rate	$[0.1, 1]$	$[0.2, 0.9]$
	Transition rate	$[0.01, 0.1]$	$[0.02, 0.09]$

Table S6. Parameter space and inner domain considered for the evaluation of inference method performances. The parameter range corresponds to the range in which the parameter was simulated. The inner range corresponds to the range which was selected to evaluate the method performances to avoid an artificial reduction of the prediction error for deep-learning methods.

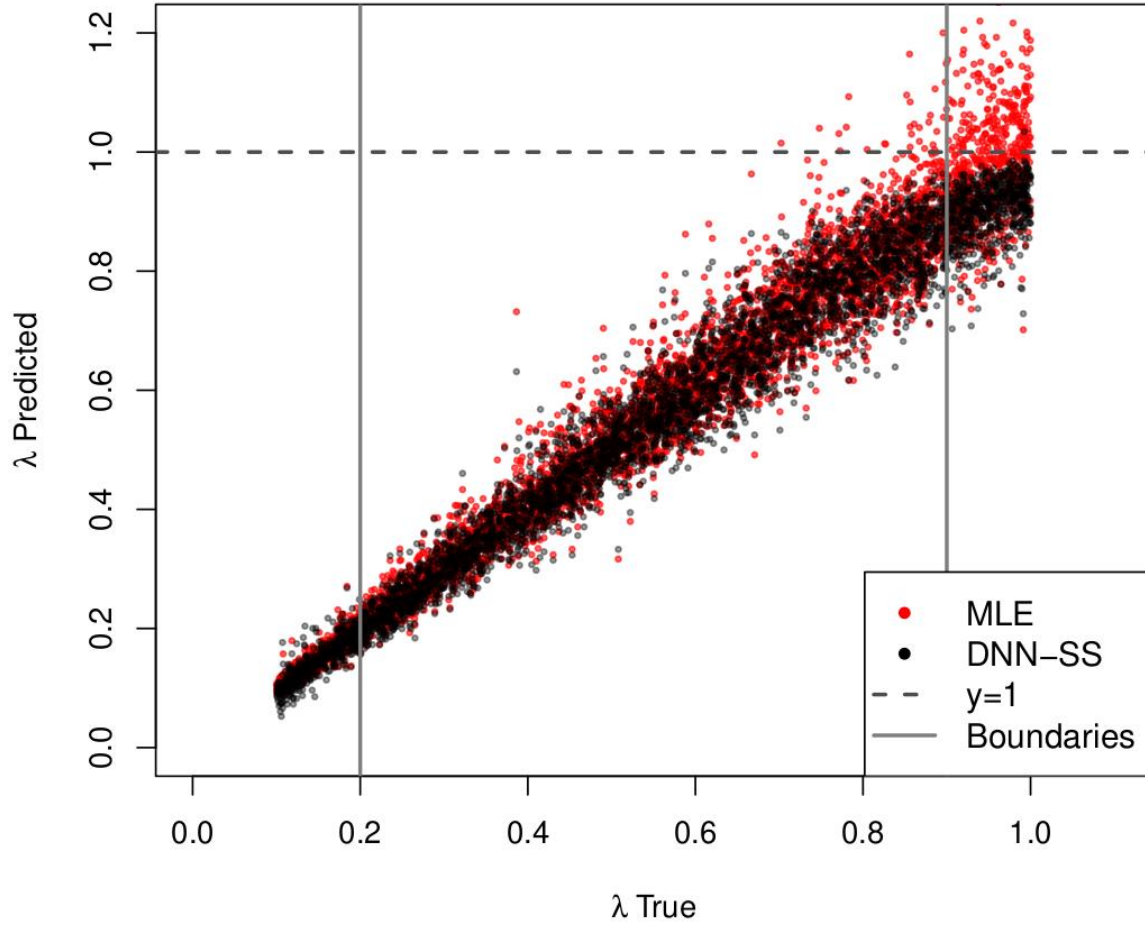


Figure S1. Artificial reduction of the prediction error of deep learning methods due to parameter space boundaries. Scatter plot of predicted vs. true values for the speciation rate (λ) of the CRBD model: predicted values by MLE are in red and predicted values by DNN fed with summary statistics (DNN-SS) in black. The true values of the speciation rate are uniformly distributed in $[0.1,1]$. We observe that the trained neural network learned to not predict speciation rate above 1 (horizontal dashed line) because it has learned the boundaries of the parameter space. This leads to an artificial reduction of the prediction error: $SSD_{DNN} = (6.2 \pm 0.3) \times 10^{-3}$ if considering the whole range ($\lambda \in [0.1,1]$) while $SSD_{DNN} = (6.5 \pm 0.3) \times 10^{-3}$ if considering an inner domain (between vertical plain lines; $\lambda \in [0.2,0.9]$).

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